MASTER OF COMMERCE

COMMERCE

(MCOM)

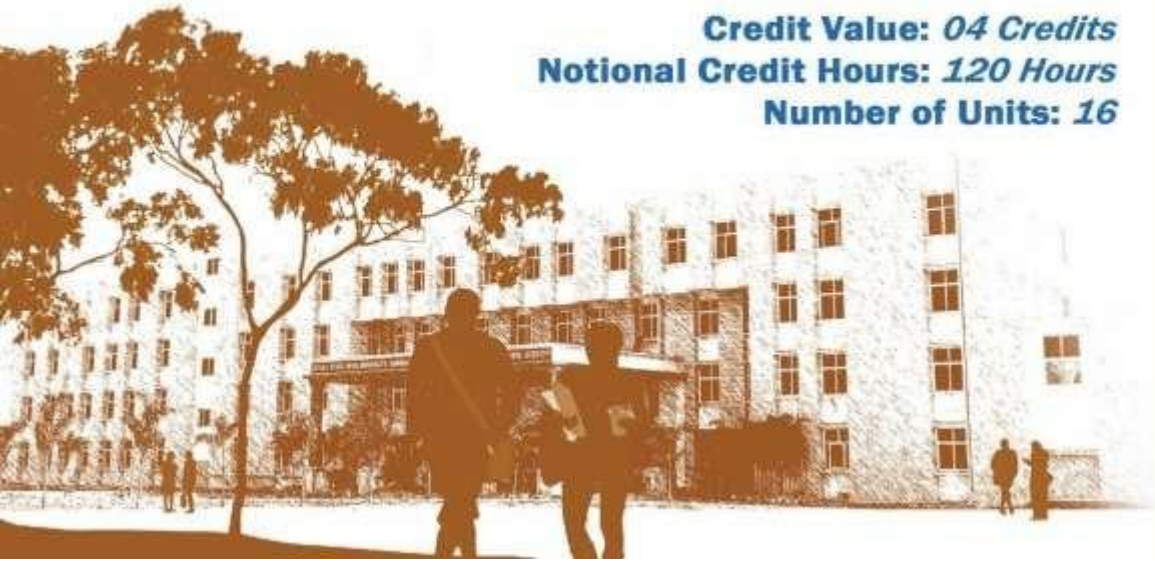
SEMESTER-I

SUBJECT SPECIFIC CORE COURSE

PCO-101:

MANAGERIAL ECONOMICS

Credit Value: *04 Credits*
Notional Credit Hours: *120 Hours*
Number of Units: *16*



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ODISHA STATE OPEN UNIVERSITY, SAMBALPUR

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ଓଡ଼ିଶା ରାଜ୍ୟ ମୁକ୍ତ ବିଶ୍ୱବିଦ୍ୟାଳୟ, ସମ୍ବଲପୁର, ଓଡ଼ିଶା
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SLM Structure

Name of the Programme	MASTER OF COMMERCE (COMMERCE)
Credit Value	80 Credits
Notional Credit Hours	2400 Hours

First Semester

Course Code	**PCO-101**
Course Title	Managerial Economics

Credit Value	04 Credits
Notional Credit Hours	120 Hours

Unit	Name of the Title	Sub Points under the Unit	Page
1	**NATURE, SCOPE AND OBJECTIVES OF BUSINESS**	1.1 Introduction 1.2 Economics and Managerial Decision Making 1.3 Economics of Business 1.4 Nature of Managerial Economics 1.5 Scope of Managerial Economics 1.6 Why Do Firms Exist? 1.7 Firms and its Objectives 1.8 Let Us Sum Up 1.9 Review Questions	12-26

2	**ECONOMIC CONCEPTS**	2.1 Introduction 2.2 Scarcity 2.3 Factors of Production 2.4 Needs and Wants 2.5 Basic Economic Questions 2.6 Opportunity Cost 2.7 Production Possibility Frontier 2.8 Let Us Sum Up 2.9 Review Questions	27-36

3	**BASIC DEMAND AND SUPPLY**	3.1 Introduction 3.2 Demand 3.3 Supply 3.4 Market Equilibrium 3.5 Let Us Sum Up 3.6 Review Questions	37-50
4	**ELASTICITY OF DEMAND**	4.1 Meaning 4.2 Types of Elasticity 4.3 Importance of Elasticity 4.4 Factors affecting Elasticity of Demand 4.5 Elasticity of Supply 4.6 Let Us Sum Up 4.7 Review Questions	51-65

5	**CONCEPT OF UTILITY AND INDIFFERENCE CURVE**	5.1 Introduction 5.2 Types of Utility 5.3 Cardinal Utility Theory 5.4 Approaches to Cardinal Utility Theory 5.5 Ordinal Utility Theory 5.6 Indifference Curves 5.7 Concept of Marginal Rate of Substitution 5.8 Let Us Sum UP 5.9 Review Questions	66-85
6	**CONSUMER BEHAVIOUR AND ITS ECCONOMIC IMPLICATIONS**	6.1 Introduction 6.2 Theories of consumer behaviour 6.3 Consumer Surplus 6.4 Economic Implications 6.5 Relations between interest rate and household saving 6.6 Let Us Sum Up 6.7 Review Questions	86-102
7	**MEANING AND CONCEPT OF CONSUMER EQUILIBRIUM**	7.1 Introduction 7.2 Budget Line 7.3 Change in Income and Price and the Budget line Consumer 7.4 Equilibrium 7.5 Corner Solutions 7.6 Let Us Sum Up 7.7 Review Questions	103-121

8	**CONSUMER EQUILIBRIUM AND DEMANDS EFFECT**	8.1 Introduction 8.2 Demand Effects 8.3 Let Us Sum Up	122-140
9	**ROLE AND SCOPE OF THEORY OF PRODUCTION**	9.1 Introduction 9.2 Some Basic Concepts of Production Theory 9.3 Production Function 9.4 Production Function with one Variable Input 9.5 Importance of Production Theory 9.6 Isoquants 9.7 Let Us Sum UP 9.8 Review Questions	141-154

10	**LAW OF VARIABLE PROPORTIONS**	10.1 Introduction 10.2 Law of Production 10.3 Law of Variable Proportions 10.4 Graphical Representation: 10.5 Applications of the Law of Variable Proportions 10.6 Let's Sum Up 10.7 Review Questions	155-167

11	**LAW OF RETURN TO SCALE**	11.1	Introduction	168-187
		11.2	Returns to scale through Isoquants	
		11.3	Economies and Diseconomies of Scale	
		11.4	Long-run cost Curves	
		11.5	Let us Sum up	
		11.6	Review Questions	
12	**MODERN THEORY OF COST ANALYSIS**	12.1	Introduction	188-215
		12.2	Types of costs	
		12.3	Concept of cost	
		12.4	Concept of long run cost	
		12.5	Modern theory of cost curves	
		12.6	Long run cost under modern theory of cost	
		12.7	Let us Sum up	
		12.8	Review Questions	
13	**MEANING AND SCOPE OF PRICING STRATEGY**	13.1	Introduction	216-230
		13.2	Types of pricing strategies	
		13.3	Let us Sum up	
		13.4	Review Questions	

14	**PRICING EQUILIBRIUM UNDER PERFECT COMPETITION**	14.1	Introduction	231-244
		14.2	Understand the significance of market forms	
		14.3	Understand price taker assumption	
		14.4	Derive short-run equilibrium	
		14.5	Derive Long-run equilibrium	
		14.6	Let us Sum up	
		14.7	Review Questions	
15	**PRICING UNDER OLIGOPOLY**	15.1	Meaning of Oligopoly	245-254
		15.2	Various Models of Oligopoly	
		15.3	Let us Sum up	
		15.4	Review Questions	

16	**PRICING UNDER MONOPOLY AND MONOPOLISTIC**	16.1 Meaning of Monopoly 16.2 Relationship between Demand and Revenue 16.3 Short-run equilibrium under Monopoly 16.4 Long-run equilibrium under Monopoly 16.5 Concept of Monopolistic 16.6 Assumption of Monopolistic Competition 16.7 Product differentiation and Demand curve 16.8 Concept of 'Industry' and 'Product group' 16.9 Derive long-run equilibrium 16.10 Let us Sum up 16.11 Review Questions	255-273

ODISHA STATE OPEN UNIVERSITY, SAMBALPUR

Programme Name: MASTER OF COMMERCE (COMMERCE)

Course Name: Managerial Economics **Course Code:** PCO-101

Semester: 1 **Credit:** 4 **Unit No.** 1 to 16 **Page: 274**

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PREFACE

This comprehensive academic resource has been meticulously developed to provide students and aspiring managers with a strong foundational understanding of Managerial Economics—a vital discipline bridging economic theory and practical business decision-making. Structured to reflect the rigors and analytical depth expected at institutions of the highest academic standards, including the Indian Institutes of Technology (IITs), this text integrates core economic principles with real-world managerial applications, enabling learners to navigate complex business environments with confidence and precision.

The material is systematically organized into fifteen chapters, each building progressively on key concepts ranging from the nature and scope of business, demand and supply dynamics, elasticity, utility theories, and consumer behaviour, to more advanced themes such as production theory, cost analysis, and pricing strategies under various market structures. Each chapter presents clear objectives, logically sequenced subtopics, graphical illustrations where appropriate, and practical examples that bring theoretical models to life. Special emphasis has been placed on concepts like opportunity cost, the production possibility frontier, and consumer equilibrium—frameworks essential for informed resource allocation and strategic planning.

Further, the curriculum highlights the interconnectedness of microeconomic and macroeconomic variables in shaping firm-level decisions. Students will gain insights into how economic tools support real-time managerial choices in production, pricing, and investment, ensuring relevance to both academic study and industry practice. The inclusion of case studies and review questions encourages critical thinking and application, aligning with Bloom's higher-order learning outcomes of analysis, evaluation, and synthesis.

Designed with both clarity and intellectual depth, this volume serves not just as a textbook but as a strategic guide for developing sound economic reasoning and evidence-based decision-making skills. It is well-suited for undergraduate and postgraduate programs in business, economics, and management, as well as for professionals seeking to enhance their economic acumen in corporate contexts.

We trust that this resource will be an invaluable companion in your academic and professional journey, fostering both theoretical knowledge and practical competence in managerial economics.

UNIT-1: NATURE, SCOPE AND OBJECTIVES OF BUSINESS

STRUCTURE

1.1 Introduction

1.2 Economics and Managerial Decision Making

1.2.1 Managerial Decision-Making Process

1.3 Economics of Business

1.4 Nature of Managerial Economics

1.5 Scope of Managerial Economics

1.6 Why Do Firms Exist?

1.7 Firms and its Objectives

1.8 Let Us Sum Up

1.9 Assignment Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the concept of managerial economics
- Learn how economics and managerial decision making is interrelated.
- Explain the Economics of a business.
- Know about the nature and scope of managerial economics.
- Discuss the objectives of the Firm.

BLOOM'S LEVEL OF LEARNING:

This unit primarily engages multiple levels of Bloom's Taxonomy of Learning. It starts at the Comprehension level, where students understand the nature and scope of managerial economics and the interrelationship between economics and managerial decision-making. As they progress, the unit involves Application, with students explaining how economic principles apply to business contexts, particularly in decision-making. In the section on the Managerial Decision-Making Process, students are expected to Analyze the steps of decision-making, evaluate alternatives, and identify constraints. This aligns with the Analysis level. Furthermore, when discussing Firms and Their Objectives, students Evaluate various firm objectives and their implications, addressing the

Evaluation level. Finally, parts of the unit challenge students to Create new models or policies based on theoretical concepts, reflecting the Synthesis level. Overall, the unit spans all six Bloom's levels, from Remembering basic concepts to Creating new frameworks for decision-making.

1.1 INTRODUCTION

This module will explain what managerial economics is and how it deals with deciding the allocation of firm's scarce resources among competing uses. The inter- linkage between economics and managerial decision making will be discussed. The major portion of the module will be elaborating the concept of a firm and the various goals of a firm.

1.2 ECONOMICS AND MANAGERIAL DECISION MAKING

Economics, as defined by Lionel Robbins (1932), is a science which studies human behavior as a relationship between ends and scarce means which have alternative uses. It is a social science that studies how individuals, governments, firms and nations make choices on allocating scarce resources to satisfy their unlimited wants. Thus, by its very nature, it requires decision making, decision concerning the optimal use of scare resources to satisfy unlimited needs and wants.

On the other hand, Management in business and organizations is defined as the function that coordinates the efforts of people to accomplish goals and objectives using available resources efficiently and effectively. The two concepts clearly highlight the significant relationship between Economics and Managerial decision making, and together they have led to the emergence of the field of Managerial Economics.

Managerial economics is the "application of the economic concepts and economic analysis to the problems of formulating rational managerial decisions". In the words of McGutgan and Moyer: "Managerial economics is the application of economic theory and methodology to decision-making problems faced by both public and private institutions".

Managerial economics attempts to optimize business decision making, given the firm's objectives and constraints imposed by scarcity of resources.

To solve the business problems and take decisions is the task of the managerial economists. Resources with the economy/ firm are scarce; therefore, the manager has to come out with optimal solutions so that the resources can be used efficiently and effectively to accomplish the objectives of the firm. Limited availability of resources acts as one of the constraints faced by the managers of the firm, the other constraints are imposed by the general economic, political and legal framework of an economy, for example, the state of economy, the laws

related to production, taxation framework, import-export policy, monetary and fiscal policy, stability of the government, competition from the rival firms, market entry and exit barriers etc. Given such complex constraints, the manager makes business decisions. Managerial decision making is a complex process, and requires the managers to make choice among the alternative courses of actions. And, every choice that a manager makes involves an opportunity cost, i.e. the cost of an alternative course of action forgone to pursue a certain action. The various decision areas covered under the gamut of managerial economics are:

- Selecting a profitable business area,
- Assessment and allocation of funds (Investment decisions),
- Choice of product, price and market,
- Choice of factors of production,
- Risk analysis,
- Demand estimation, production decision, etc.

1.2.1 MANAGERIAL DECISION-MAKING PROCESS

Decision making is crucial for successful functioning of a firm. At each stage managers have to make decisions concerning which product to produce, what price to be charged, in what quantity should the product be produced, how should the product be distributed in the market, how to grow the business, etc. Though there is no fix procedure of how managers will decide about the various issues discussed above, but still decision-making process can be generalized and consist of the following steps:

1. Establishing the objectives

The first step in the decision-making process is to establish the objective. A firm may several objectives; however, it is considered that one of the most important objectives of the firm is profit maximization. The other objectives of the firm may include wealth maximization, sales maximization etc.

2. Defining the problem

The second step requires the firm to define the problem clearly. This step requires a thorough analysis of the external and internal environment of the firm to clearly understand and define the nature of the problem. For example, a firm incurring continuous losses needs to examine various factors to understand why it is facing loss. The analysis may require the firm to look at its business strategies, the competition, customers' demands, labor- management relations etc.

3. Identifying the possible/ alternative courses of action

After having analyzed the problem, manager looks for the solution. Finding an optimal solution requires a manager to identify and consider all possible alternative solutions. Say for instance in the example mentioned above, after understanding the problem that why firm is incurring loses, may consider 2- alternative courses of actions: One, updating and installing the latest machinery to produce good quality products and second, changing its business strategies to focus more on services offered along with the product. Naturally, the choice between the alternatives will depend on which of the alternative will bring larger returns.

4. Evaluating the alternative courses of action

After narrowing down to the possible courses of actions, manager evaluates these actions in terms of cost-benefit analysis. The course of action appears to be optimum will be chosen.

5. Implementing the action/ decision

The final step is implementation of the decision. Implementation requires constant monitoring of the actions over the period of time, so that the expected results can be obtained.

1.3 ECONOMICS OF A BUSINESS

This concept discusses the economic concepts that are fundamental to understanding many of the issues faced by business firms. It relates to the key issues that affect the ability of the firm to earn a desirable rate of return on the investment made by it. These include issues like the economic perspective on the nature, scale, and organization of the firm; the role of information, competition and technology; principal-agent theory; contracting and the firm's relationships with customers and suppliers, etc. In this regard, following are some of the issues fundamentals to economics of a business:

- Business must use its resources as efficiently as possible since the resources are limited in supply and there are costs in acquiring and using them. Business must choose carefully where it wants to put its resources, as with every decision there is an opportunity cost of foregoing the alternative course of action.
- If business wants to maximize profit and grow with time, it is crucial that the firm must investing money in new capital, products and ways to produce them. Investment is crucial for business growth.
- Products must be priced competitively, be of good quality, imaginatively designed, marketed effectively and distributed as efficiently and cheaply as possible if customers are to demand them.

- A business must have long term plans to achieve its objectives. Objectives of firm keep on changing with time. It is usually said that profit maximization is one of the most important objectives of the firm, however the firm may have alternative objectives like sales maximization, market growth etc. Thus, the objectives and plans to achieve these objectives must be clearly laid down.
- Business must sense and respond to challenges coming from competitors and from the national and international business environment in which they operate.
- Businesses are becoming increasingly global in its nature and thus organizations must look across national boundaries to secure economic advantage.

1.4 NATURE OF MANAGERIAL ECONOMICS

Managerial Economics is a Science: Managerial economics is a science because it establishes relationship between cause and effect. It studies the effect of change in price of a commodity factors and forces on the demand of a particular product. It also studies the effects and implications of the plans, policies and programmes of a farm on its sales and profit.

1. **Managerial Economics is an Art:** Managerial economics may also be called an art. Because it also develops the best way of doing things. It helps management in the best and most efficient utilisation of limited economic resources of the firm.
2. **Managerial Economics is a Microeconomics:** Entire study of economics may be divided into two segment- Macroeconomics and Microeconomics.

   Managerial economics is mainly micro economics. Micro economics is the study of the behaviour and problems of the individual economic unit. In Managerial economics unit of the study is firm or business organization and an individual industry. It is the problem of business firms such as problem forcasting demand, cost of production, pricing, profit planning, capital management etc.

   Managerial economics is the economics of firms: Managerial economics largely use that body of economic concepts and principles which is known as 'Theory of the firm' or 'Economics of the firm.'

3. **Managerial Economics uses Macro economic Analysis:** Managerial economics also uses macro-economics to analysis and understand the general business environment in which the business firm must operate. Business management must have the adequate knowledge of external forces that affect the business of the firm. The important macro-factors that affect the firm are trends in the national income and expenditure, business cycle, economic policies of the government, trends in foreign trade etc.

4. **Managerial Economics is pragmatic:** It is concerned with practical problems and results. It has nothing to do with abstract economic theory which has no practical application to solve the problems faced by business firms.
5. **Managerial Economics is normative science:** There are two type of science - Normative science and Positive science. Positive science studies what is being done. Normative science studies what should be done. From this point of view, it can be concluded that managerial economics is normative science because it suggests what should be done under particular circumstances.
6. **Aims at helping the management:** Managerial economics aims at supporting the management in taking corrective decisions and charting plans and policies for future.
7. **Prescriptive rather than descriptive:** Managerial economics is a normative and applied discipline. It suggests the application of economic principles with regard to policy formulation, decision-making and future planning. It not only describes the goals of an organisation but also prescribes the mean of achieving these goals.

1.5 SCOPE OF MANAGERIAL ECONOMICS

Managerial economics is the application economics theories in the process of decision making and formulation of future plans. The management will have to analyse the business problems that are faced by the firm. Thus, the principles relating to following topics constitute the scope of subject matter of managerial economics.

I. Microeconomics applied to operational or internal Issues

1. **Demand Analysis and Forecasting**: Demand Analysis pertains to the behaviour of consumers in the market. It studies the nature of consumer preferences and the effect of changes in the determinants of demand such as, price of the commodity, consumers' income, prices of related commodities, consumer tastes and preferences etc.

   Demand Forecasting is the technique of predicting future demand for goods and services on the basis of the past behavior of factors which affect demand. Accurate forecasting is essential for a firm to enable it to produce the required quantities at the right time and to arrange, well in advance, for the various factors of production viz., raw materials, labor, machines, equipment, buildings etc. Business Economics provides the manager with the scientific tools which assist him in forecasting demand.
2. **Production and Cost Analysis**: Production theory explains the relationship between inputs and output. A business economist has to decide on the optimum size of output, given the objectives of the firm. He has also to ensure that the firm is not incurring undue costs.

Production analysis enables the firm to decide on the choice of appropriate technology and selection of least - cost input-mix to achieve technically efficient way of producing output, given the inputs. Cost analysis enables the firm to recognize the behavior of costs when variables such as output, time period and size of plant change. The firm will be able to identify ways to maximize profits by producing the desired level of output at the minimum possible cost.

3. **Inventory Management**: Inventory management theories pertain to rules that firms can use to minimize the costs associated with maintaining inventory in the form of 'work-in-process,' 'raw materials', and 'finished goods'. Inventory
policies affect the profitability of the firm. Business economists use methods such as ABC analysis, simple simulation exercises and mathematical models to help the firm maintain optimum stock of inventories.
4. **Market Structure and Pricing Policies**: Analysis of the structure of the market provides information about the nature and extent of competition which the firms have to face. This helps in determining the degree of market power (ability to determine prices) which the firm commands and the strategies to be followed in market management under the given competitive conditions such as, product design and marketing. Price theory explains how prices are determined under different kinds of market conditions and assists the firm in framing suitable price policies.
5. **Resource Allocation**: Business Economics, with the help of advanced tools such as linear programming, enables the firm to arrive at the best course of action for optimum utilization of available resources.
6. **Theory of Capital and Investment Decisions**: For maximizing its profits, the firm has to carefully evaluate its investment decisions and carry out a sensible policy of capital allocation. Theories related to capital and investment provide scientific criteria for choice of investment projects and in assessment of the efficiency of capital. Business Economics supports decision making on allocation of scarce capital among competing uses of funds.
7. **Profit Analysis**: Profits are, most often, uncertain due to changing prices and market conditions. Profit theory guides the firm in the measurement and management of profit under conditions of uncertainty. Profit analysis is also immensely useful in future profit planning.
8. **Risk and Uncertainty Analysis**: Business firms generally operate under conditions of risk and uncertainty. Analysis of risks and uncertainties helps the business firm in arriving efficient decisions and in formulating plans on the basis of past data, current information and future prediction.

II. Macroeconomics applied to environmental or external issues

Environmental factors have significant influence upon the functioning and performance of business. The major macroeconomic factors are related to:-

- the type of economic system
- stage of business cycle
- The general trends in national income, employment, prices, saving and investment.
- government's economic policies like industrial policy, competition policy, monetary and fiscal policy, price policy, foreign trade policy and globalization policies
- working of financial sector and capital market
- Socio-economic organisations like trade unions, producer and consumer unions and cooperatives.
- Social and political environment.

Business decisions cannot be taken without considering these present and future environmental factors. As the management of the firm has no control over these factors, it should fine-tune its policies to minimize their adverse effects.

1.6 WHY DO FIRMS EXIST?

It is a microeconomic concept that lays its foundation in neoclassical economics that states that firms (or corporations) exist and make decisions to maximize profits. Interaction of businesses takes place with the market and this phenomenon assists in determination of pricing and demand and then the resources are allocated according to models that look to maximize net profits.

The theory of the firm goes hand in hand with the theory of the consumer, wherein the consumers' objective is to maximize their overall utility.

Economic agents act and interact through the market. Producers employ factors of production, sell products, hire labour, raise funds, etc., all of which is done by virtue of explicit or implicit contracts. One of carrying out production is to rely on the market for each such transaction. This method is adopted because producers or businesses feel that the market is the most efficient mechanism for handling such transactions. It is capable of evaluating the cost (price paid) and the benefit (the product or return received) in the best possible manner. In theory, markets a so efficient that there is no cost assumed in the use of markets by businesses. The invisible hand of the market is supposed to be the organizer. The market is believed to be an organizer of production

'par excellence'.

If this were the case then there would be no need to employ factors of production like managers as employees. If markets are most efficient then logically speaking there is no need for a firm to exist. This is because a firm is a collection of contracts so designed to carry out economic activity within the firm.

Firms exist as an alternative system to the market-price mechanism when it is more efficient to produce in a non-market environment. For example, in a labor market, it might be very difficult or costly for firms or organizations to engage in production when they have to hire and fire their workers depending on demand/supply conditions. It might also be costly for employees to shift companies every day looking for better alternatives. Similarly, it maybe costly for companies to find new suppliers daily. Thus, firms engage in a long-term contract with their employees or a long-term contract with suppliers to minimize the cost ormaximize the value of property rights

Ronald Coase first expounded his thinking about the firm in a lecture in Dundee in 1932, when he was just 21 years old. He published "The Nature of the Firm" five years later. His central insight was that firms exist because going to the market all the time can impose heavy transaction costs. You need to hire workers, negotiate prices and enforce contracts, to name but three time-consuming activities. A firm is essentiallya device for creating long-term contracts when short-term contracts are too bothersome. But if markets are so inefficient, why don't firms go on getting bigger forever? Mr. Coase also pointed out that these little planned societies impose transaction costs of their own, which tend to rise as they grow bigger. The proper balance between hierarchies and markets is constantly recalibrated by the forces of competition: entrepreneurs may choose to lower transaction costs by forming firms but giant firms eventually become sluggish and uncompetitive.

1.7 FIRMS AND ITS OBJECTIVES

A business firm is commonly defined as a commercial organization that participates in selling goods or services to consumers in order to reap profits. It employs productive resources to obtain products and/or services which are offered in the market with the aim of making a profit. Conventional theory of firm assumes profit maximization as the sole objective of business firms; however, the firms pursue more than one objective. Some important objectives, other than profit maximization are:

(a) Maximization of the sales revenue

(b) Maximization of firm's growth rate

(c) Maximization of Managers utility function

(d) Making satisfactory rate of Profit

(e) Long run Survival of the firm

(f) Entry-prevention and risk-avoidance

Let us discuss in details the various objectives of the firm:

1. **Profit Maximisation:**

Profit is defined as the difference between the revenue received by the firm and the cost it incurs. It is the main aim of all the business firms to maximize its profit. Under this objective, firms choses to invest in projects which yield larger profits, and drop out unprofitable activities. In maximizing profits, input-output relationship is critical, and in order to maximize the profit, the firm shall either minimize the input to achieve a given amount of profit or maximize the output with a given amount of input. In this sense, profit maximization objective of the firm enhances productivity and improves the efficiency of the firm. However, in the real world, firms may pursue other objectives apart from profit maximization

2. **Long Run Profit Maximisation:**

In some cases, firms may sacrifice profits in the short term to increase profits in the long run. For example, by investing heavily in new capacity, firms may make a loss in the short run, but enable higher profits in the future. This involves sacrificing consumption in the present in Favor of long run investment. This also involves undertaking capital budgeting or long-term investment decisions by the managers. The economic calculations are different and do not depend on marginal cost and marginal revenue.

3. **Wealth Maximisation:**

The main objective of the firm should be to take decisions that maximize the value of the company for its owners. Firms usually have separation of ownership and control. People who own the company (shareholders) often are not involved in the day to day functioning of the firm, this leads to the subtle conflict between the owners and the managers, as owners want to maximize their wealth. This problem of separation between owners and managers is referred to as Principal agent problem. The main financial objective of the firm should be 'the maximization of shareholder wealth'. Shareholders receive their wealth in the form of

dividends and capital gains, and thus shareholder wealth will be maximized by maximizing the value of dividends and capital gains received over time.

4. **Sales Maximisation:**

This objective aims to maximize the volume of sales by selling as many products as possible, without incurring loss. Firm in such a case produces output so that total revenue generated from sales sufficiently covers the total costs of production. Firms often seek to increase their market share – even if it means less profit. This could occur for various reasons:

a) Increased market share increases monopoly power and may enable the firm to put up prices and make more profit in the long run.

b) Managers prefer to work for bigger companies as it leads to greater prestige and higher salaries.

c) Increasing market share may force rivals out of business. E.g. supermarkets have lead to the demise of many local shops. Some firms may actually engage in predatory pricing which involves making a loss to force a rival out of business.

5. **Market Share And Growth:**

Firms also wish to increase their share of a market and grow. Firm growth rises when more profit is retained by managers for future investment, instead of distributing it to the shareholders in the form of wealth. This objective is significant for firms operating in markets with a few large competitors. This is similar to sales maximization and may involve mergers and takeovers.

With this objective, the firm may be willing to make lower levels of profit in order to increase in size and gain more market share.

6. **Survival:**

Some firms take a short-term view and only aim to survive in the market. Survival is crucial for new firms and for firms operating in highly competitive markets. It is also become one of the common objectives when the macro-economy undergoes recession or downturn.

7. **Satisficing:**

The term was first used by Herbert Simon in 1957. Satisficing means when firms consider a number of different and competing objectives, without aiming to

'maximize' any single one. Managers wish to maintain a minimum level of profit that satisfies the owners and then try to deal with other objectives.

In many firms there is separation of ownership and control. Those who own the company (shareholders) often do not get involved in the day to day running of the company. This is a

problem because although the owners may want to maximize profits, the managers have much less incentive to maximize profits because they do not get the same rewards, (share dividends) Therefore, managers may create a minimum level of profit to keep the shareholders happy, but then maximize other objectives, such as enjoying work, getting on with other workers. (E.g. not sacking them) This is the problem of separation between owners and managers.

This 'principal agent' problem can be overcome, to some extent, by giving mangers share options and performance related pay although in some industries it is difficult to measure performance.

8. Social/Environmental Concerns:

A firm may incur extra expense to choose products which don't harm the environment or products not tested on animals. Alternatively, firms may be concerned about local community / charitable concerns.

Many companies who have adopted such strategies have been quite successful.

This has encouraged more firms to consider these over objectives, but a cynic may argue they see it as another opportunity to increase profits rather than a genuine sacrificing of profits in order to promote other objectives

9. Co-Operatives

Co-operatives may have completely different objectives to a typical PLC. The basic premise of running a co-operative to maximize the welfare of all stakeholders – especially workers. The profit made by the co-operative are shared amongst all members.

CASE STUDIES:

Case Study 1: Strategic Decision-Making in a Competitive Market

Background:

ABC Electronics Pvt. Ltd. is a mid-sized firm that manufactures home appliances. Recently, the company faced declining sales due to stiff competition and changing customer preferences. Management noticed that despite their premium quality, their products lacked modern features compared to competitors.

Problem:

Management had to decide whether to invest in R&D for new feature-rich products or reduce prices to boost demand.

Decision-Making Process:

1. Objectives Established – The main aim was to regain market share and maximize long-term profit.

2. Problem Defined – Decline in sales due to lack of innovative features.
3. Alternatives Considered – Invest in R&D or compete via pricing.
4. Evaluation – R&D investment would increase costs now but could create a sustainable product line. Pricing reduction would give short-term results but reduce margins.
5. Implementation – The firm chose to invest in R&D and introduced a new smart appliance line within a year.

Outcome:

Within 18 months, the new product line increased sales by 35%, improved brand image, and restored profitability.

Case Study 2: Resource Allocation and Investment Decisions

Background:

Green Earth Organics is a firm dealing in organic farming products. The firm had limited funds to expand and had to decide whether to invest in digital marketing or open a new store in a metro city.

Problem:

The firm needed to decide how best to allocate scarce financial resources for growth.

Decision-Making Process:

1. Objectives Established – Increase customer base and revenue in the long term.
2. Problem Defined – Where to allocate limited funds for the best return.
3. Alternatives Considered – Invest in online digital campaigns or open a new physical outlet.
4. Evaluation – Digital marketing had a lower cost and wider reach; opening a store had higher investment and operational risks.
5. Implementation – Digital marketing was chosen with performance tracking and targeted campaigns.

Outcome:

Online sales increased by 45% in six months, and customer inquiries doubled. ROI on digital marketing proved more cost-effective than expected.

1.8 LET US SUM UP

The module discussed how Managerial Economics relies on application of economic theory and concepts and helps manager make decisions that are in best interest of the firm.

- The general managerial decision-making process was discussed.

- Various economic concept and issues that are fundamental to understanding the issues faced by business firms were explained.
- The concept of a firm was described, along with the various goals and objectives a firm may pursue.

KEYWORDS:

1. **Managerial Economics** – The application of economic theory and methodologies to business management practices for effective decision-making.
2. **Opportunity Cost** – The value of the next best alternative foregone when a decision is made.
3. **Profit Maximization** – A primary objective of firms aimed at achieving the highest possible difference between revenue and cost.
4. **Transaction Costs** – The costs involved in making an economic exchange or contract, such as negotiation and enforcement.
5. **Demand Forecasting** – A technique used to predict future customer demand using historical data and market analysis.

1.9 ASSIGNMENT QUESTIONS

1. Explain the interrelationship between economics and managerial decision-making with suitable examples.
2. Discuss the objectives of firms apart from profit maximization and their relevance in today's business world.
3. Elaborate on the steps involved in the managerial decision-making process.
4. Analyze the scope of managerial economics by highlighting its micro and macroeconomic applications.
5. Describe the economic rationale behind the existence of firms as opposed to relying solely on market transactions.

OBJECTIVE TYPE QUESTIONS

1. Managerial economics is a bridge between _____ and managerial practice.
2. _____ cost refers to the value of the next best alternative foregone.
3. Profit maximization involves maximizing the difference between revenue and _____.
4. _____ theory explains the relationship between inputs and outputs in production.
5. Firms exist to reduce _____ costs associated with market transactions.

6. In the decision-making process, the first step is to establish the _____.
7. Managerial economics is largely based on _____ economics.
8. _____ analysis helps in forecasting future consumer demand.
9. The role of _____ is essential in resource allocation and investment decisions.
10. _____ is defined as a commercial entity that aims to make a profit.
11. Long-term contracts within firms help minimize market-related _____ costs.
12. Under perfect competition, a firm has no control over the _____ of its product.
13. Government policies like taxation and trade regulations are examples of _____ factors.
14. Sales maximization is considered an alternative to _____ maximization.
15. A firm's decision to invest in innovation over price reduction is influenced by _____ analysis.

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UNIT-2: ECONOMIC CONCEPTS

STRUCTURE

2.1 Introduction

2.2 Scarcity

2.3 Factors of Production

2.4 Needs and Wants

2.4.1 Satisfying needs and wants

2.5 Basic Economic Questions

2.5.1 What to produce?

2.5.2 How to produce?

2.5.3 For whom to produce?

2.6 Opportunity Cost

2.7 Production Possibility Frontier

2.8 Let Us Sum Up

2.9 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the fundamental economic concepts.
- Learn the basic questions/ problems faced by every economy.
- Explain the concept of scarcity with the help of production possibility frontier.

The content provided primarily aligns with the *Understanding* and *Applying* levels of Bloom's Taxonomy. At the Understanding level, learners are expected to grasp key economic concepts such as scarcity, opportunity cost, and the production possibility frontier (PPF), as well as distinguish between needs and wants. The unit clearly supports this level by explaining economic terminology and relationships among fundamental ideas. At the Applying level, students are encouraged to use their knowledge to interpret production schedules, evaluate trade-offs using opportunity cost, and analyze how firms make production decisions based on limited resources. The use of practical examples like

the PPF graph and real-world decisions (e.g., allocation of funds or choosing between goods) further enables learners to apply economic theories in managerial contexts. Therefore, the overall structure and learning outcomes foster both comprehension and the ability to apply foundational economic principles in decision-making scenarios.

BLOOM'S LEVEL OF LEARNING:

The content presented in this unit primarily aligns with the *Understanding* and *Applying* levels of Bloom's Taxonomy. At the *Understanding* level, learners are introduced to fundamental economic principles such as scarcity, opportunity cost, factors of production, and the production possibility frontier (PPF). This level involves explaining concepts, interpreting terminology, and recognizing relationships between basic ideas in economics. At the *Applying* level, the content encourages learners to utilize these concepts in practical scenarios, such as analyzing trade-offs using opportunity cost, evaluating resource allocation, and interpreting data from production schedules and graphs like the PPF. The inclusion of real-life case studies, such as decisions faced by a start-up and a government body, further supports the application of theoretical knowledge to realistic decision-making contexts. Overall, the unit fosters comprehension of core ideas and their practical application in managerial economic settings, preparing learners to make informed and efficient economic decisions.

2.1 INTRODUCTION

The basis of managerial economics lies in the firm and its relationship to the economy. First and foremost, we must understand that a manager needs to know about economics in two ways. First, some of the basics belong to the individual units of study in economics, like, the individual producer, individual consumer, one unit of labour, one unit of land, etc. The study at this level is one of choice and is known as micro- economics. It deals with the relative price of a commodity (Px), the wage rate of one unit of labour (w), the return on one unit of capital (interest) and the return from one unit of land (rent). At the aggregate level or the economy level economic theory deals with aggregates like National Income (Y), General Price Level (P), Aggregate Consumption (C), Aggregate Investment (I) and Aggregate Saving (S). This study is known as macro- economics. The macro economy is shaped by macro-economic policies like Fiscal and Monetary Policy. A manager, who is managing a firm, has to be aware of both micro-economics and macro- economics. Each firm is set in the context of the economy. The business conditions are defined by the macro-economic conditions and macro-economic policy. We shall now be discussing some fundamental economic concepts. These concepts are essential for further understanding of economics. The student needs to be familiar with the fundamental concepts

of Scarcity, Factors of Production, Needs and wants, basic economic problems, Opportunity cost and Production possibility analysis.

2.2 SCARCITY

In an economic context, scarcity refers to the limited resources available to meet the infinite desires and needs of individuals and society as a whole. These resources, which are essential inputs for production, include land, labour, and capital. The fundamental economic problem arises from the stark contrast between the unlimited wants of people and the finite resources that the economy possesses. This imbalance necessitates the making of various economic decisions aimed at ensuring the efficient allocation of these limited resources. Such decisions often involve trade-offs, where individuals and societies must give up or sacrifice one want in order to satisfy another. Consequently, the economy is faced with the critical task of determining the most effective ways to utilize its resources, balancing the competing demands and striving for optimal outcomes that can enhance overall welfare and satisfaction. Furthermore, the implications of scarcity extend beyond mere resource allocation; they influence policy-making, economic strategies, and the overall functioning of markets. As societies evolve, the challenge of managing scarcity becomes increasingly complex, requiring innovative solutions and adaptive strategies to navigate the intricate web of human desires and resource limitations. In this dynamic landscape, understanding the principles of scarcity is essential for fostering sustainable growth and ensuring that the needs of both current and future generations are met effectively.

2.3 FACTORS OF PRODUCTION

As discussed earlier, inputs of production or factors of production, refers to the various inputs used in the production of goods or services by the firm to earn economic profit. The factors of production include land, labour, capital and the entrepreneur.

- *Land represents* all natural resources available with an economy, such as timber, soil, crops, minerals, forests, rivers, oceans, flora and fauna etc. used in the production of a good.
- *Labor* comprises of the work done by the labourers and worker at all levels of an organization. This excludes the entrepreneur/ owner.
- *Capital* refers to all the man-made physical goods used to produce other goods and services, like building, machines, equipment, computers, tools, factories, roads etc. It also refers to the monetary resources company uses to purchase land and other capital goods. Increase in the level of capital is referred to as 'Investment'. The accumulation of investment leads to the creation of capital stock. It is the flow of capital services from this capital stock that leads to production.

- *Entrepreneur* is the individual, commonly referred to as the owner or producer who takes up idea, uses special skills to produce goods and services and attempts to earn economic profit by combining all other factors of production. All the risks and rewards of the business belong to the entrepreneur.

2.4 NEEDS AND WANTS

Our needs refer to the various goods or services that are required. This would include the needs for food, clothing, shelter and health care. Wants on the other hand refers to the goods or services that are not necessary but that we desire or wish for. For example, luxurious cars, designer clothing, furnished house. An important decision involved in economics is the allocation of resources for the production of goods so that people's needs are met, with the given scarce resources.

2.4.1 Satisfying Needs and Wants:

The process of satisfying needs and wants involves three steps, viz. production, distribution and consumption. Production refers to the process of combining all factors of production for generating goods and services. Once the goods and services are produced, firm's attempts to distribute them in the market, so that consumers can avail or consume these goods and services, in exchange for a price, which leads to a firm's revenue.

2.5 BASIC ECONOMIC QUESTIONS

As discussed earlier in this module, every economy has limited or scarce resources, but the wants and desires are unlimited. Thus, every economy has to make choice and ensure optimum allocation of resources. All economies face three basic choices/ questions that must be answered for an economy to work efficiently. This is irrespective of whether the economy is a market economy, a planned economy or a mixed economy. The three central problems are: What to produce? How to Produce?
And; for whom to produce? These remain the same for all economies.

2.5.1 What to produce?

In micro-economics we study the problems of choice. The first choice that a firm needs to answer is 'what needs and wants of the society it should satisfy given the limited resources?'. It can choose either to produce consumer goods like food, clothing etc. or produce capital goods like machinery, tools, equipment etc. This choice is made with the help of the price of each commodity (or service) with respect to the other. If the demand for commodity X is more, then the price of X (Px) would be more. The price ratio (Px/Py - knows as relative price of X) will tilt in favor of X against commodity Y. This would attract producers to produce more of commodity X. This is how the question of 'what to produce' is solved. The pricing in the product market helps solve the problem.

2.5.2 How to produce?

Goods can be produced by certain techniques- labour-intensive technique and capital-intensive technique. Firm needs to opt for the best method of production. Labor- intensive technique refers to the one in which more of labour combines with less of capital for the production of goods and services. Capital intensive technique is one in which more of machinery like forklift, assembly belts, computers, etc. are used to combine with less of labour to produce goods and services. Even services need capital to complement their productive ability. For instance, a postman provides a service but he too uses a bicycle, which is capital. You get the message even through e-mail but the capital used for such a service is much more. Depending on the cost of employing capital vs. labour the producer will decide how to produce – with the help of more capital or labour? If more of labour is available then the price of labour or the wage rate will be lower than the price of capital or the rate of interest. The factor price ratio (Pl/Pk) will tilt in favour of labour. Thus, the question of 'how to produce' is also solved by the price mechanism. Only in this case it is the factor market which solves the problem.

2.5.3 For whom to produce?

Once the price mechanism helps decide the earlier two questions then the total quantity to be produced and its price is determined. Therefore, the value of the produce is determined (Value=P*Q). The question of 'for whom to produce' needs to be understood well. In fact, the question needs to be inverted. The real question is 'who produces?' Is production of goods and services exclusively due to labour, capital or land; or is it the entrepreneur who is responsible for production. The product (its value) needs to be divided amongst all factors of production because they are jointly responsible for the product. This question, therefore, really is one of distribution of the value amongst land, labour, capital and the entrepreneur; as rent, wages, interest and profit.

2.6 OPPORTUNITY COST

Every time a manager takes a decision on the above mentioned basic economic questions, that is, what to produce, how to produce and for whom to produce, he/she makes a choice between the various available alternatives. The cost of producing one good is felt as the amount of the foregone commodity. Economists use the term Opportunity Cost to explain the same phenomenon. Opportunity cost can be defined as the value of the next best alternative forgone; whenever choice/ decision are made between several possible alternatives with the given set of resources. In other words, it is the cost of an alternative course of action forgone to pursue a certain action. For example, assume you have Rs.500/- to be spent on dinner. Opportunity cost of the money in this case is all the possible things you could have purchased or availed with the same amount of money. Say you could have bought some clothing or accessory, or

go to a saloon to avail some service or pay the registration fee at some fitness center. Therefore, in economics cost is a real resource cost. If resources are diverted for the production of a commodity (Y) in lieu of another commodity (X) the opportunity cost of producing Y is measured in terms of X.

Opportunity cost (Y) = $\Delta Y / \Delta X$

If 3 units of Y can be produced by giving up 2 units of X then:

Opportunity cost (Y) = $\Delta Y / \Delta X = 3/2 = 1.5$

2.7 PRODUCTION-POSSIBILITY FRONTIER

Production-Possibility Frontier (PPF), also referred to as Production-Possibility curve, is a graph that represents the production tradeoff that is choice of production between products, given the scarce resources. The PP-frontier graphically demonstrates the problem of scarcity, choice and opportunity cost. For the simplicity of understanding of the concept, we make the following assumptions:

- Resources available with the economy/ firm are fixed.
- With the given resources, firm only produces 2-commodities.
- The resources can be shifted/ transferred to produce more or less of 1- commodity in exchange of other.
- The economy/ firm use all the available resources.

Given the assumption, PP-frontier is a graph that depicts the various combinations of amounts of two commodities that an economy/ firm can produce, using the fixed amount of resources, i.e. factors of production. To understand the concept further, let us consider one example.

Consider the production possibility schedule given below. Production possibility schedule is a table that shows the various amounts of the two commodities that the economy/ firm can produce given the scarce resources. Let us consider that with the given amount/ quantity of factors of production, the economy/ firm choses to produce either Televisions or Laptops. Table 1 below shows the production possibilities.

Product	A	B	C	D	E
Laptops	0	200	400	600	800
Televisions	400	300	200	100	0

Table 1: Production Possibility Schedule

The table depicts that with the given factors of production, if the economy choses to produce 400- units of Televisions, it shall not be able to produce any unit of Laptop, shown as production possibility 'A'. There are no Laptops because all the resources were utilized in producing televisions. Similarly, production possibility 'E' depicts that the economy/ firm choses to exploit all resources towards production of Laptops, and thus it produces 800 units of Laptops and zero units of Televisions. Between

the two ends of the production possibility schedule, there are other alternatives available, i.e. production possibility 'B', 'C' and 'D'.

At point B, economy is producing 200-units of Laptops and 300-units of Televisions.

As it moves from 'B' to point 'C', it has to give up 100 units to televisions to produce 200-units of Laptops. However, if there is higher demand for laptops in the market, economy may choose to shift from point 'C' to 'D'.

Let us now, use the production possibility schedule to arrive at the production possibility frontier. As discussed above, the frontier or curve shows the maximum production possible, given the existing/ fixed resources. Let us depict production of Laptops on the X-axis, and production of Televisions on the Y-axis.

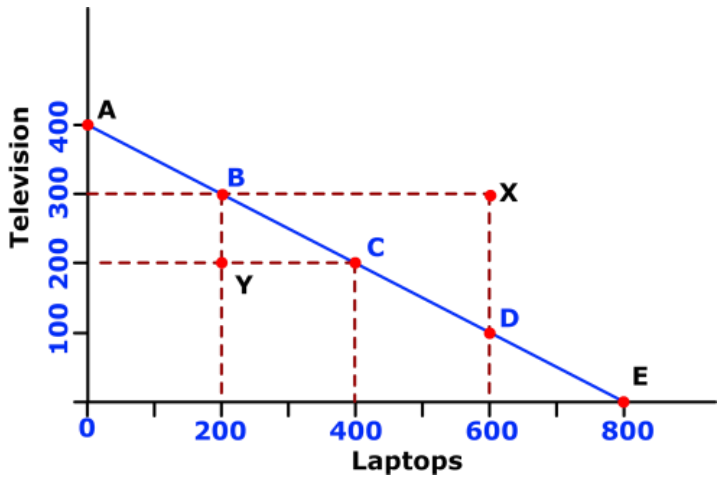

Figure 1

Given our assumptions, that the economy has limited resources, a production possibility point 'X' showing the production of 300-units of Televisions and 600-units of Laptops, as depicted in Figure 1, is impossible to achieve.

As explained above, the production possibility frontier depicts the maximum production possibility given the resources, thus any point above the curve is not viable as economy lacks resources. Also, any point under the curve, like production possibility point 'Y', shows that the economy has not fully utilized its resources, which again violates the assumption that the economy should utilize all its resources and there shall be no left over.

At point 'Y', economy is only producing 200-units each of Laptop and Television, and thus the resources are left over and not fully utilized.

As we move down from point 'A' to point 'E', we say that the resources are getting exchanged. That is at 'A' economy/ firm is producing 400-units of Televisions and Zero-units of Laptop. As it moves down the PP-frontier, and reaches point' B', the economy/ firm produces 300-units of Television and 200-

units of Laptop. Thus, we see, that in moving from point 'A' to point 'B', economy exchanges/ foregoes 100- units of television to produce 200-units of Laptops. And likewise, throughout the PP- frontier, we see that resources are exchanged or transferred to produce more or less of 1-commodity in exchange of other. In this way, production possibility frontier illustrates the range of choices and opportunity cost of each choice that an economy can make to produce 2-commodities.

Case Studies:

Case Study 1: Resource Allocation in a Start-Up

A tech start-up, "SmartEdge," is entering the wearable devices market. It has a fixed budget and limited human resources. The management must decide whether to allocate its capital towards developing a new smartwatch (consumer good) or invest in R&D for AI-based health-monitoring sensors (capital good). The founders analyze the potential demand for both products, review the production costs, and decide to focus on the smartwatch first due to immediate market demand and pricing advantages. They apply the concept of opportunity cost, knowing they forego long-term innovation for short-term gains. The decision reflects real-life application of what to produce, how to produce, and for whom to produce.

Case Study 2: Government and the Production Possibility Frontier

The local government of City X is considering how to allocate its budget between building public housing and investing in clean energy infrastructure. With a fixed amount of resources, they chart out a Production Possibility Schedule showing different combinations of houses and solar energy plants that can be built. Choosing to build more houses means fewer solar plants and vice versa. This scenario showcases scarcity, choice, and the opportunity cost of each decision. They ultimately choose a balanced mix, using a Production Possibility Frontier (PPF) to optimize public benefit.

2.8 LET US SUM UP

This module introduced you with the terminology used in the field of managerial economics and the fundamental economic concepts.

- It discussed that the key challenge is to satisfy unlimited needs and wants of the people with the limited or scarce resources.
- The problem of scarce resources and unlimited needs and wants, forces a firm to make choices from the various alternatives, and every choice made by the firm involves or imposes an Opportunity cost.
- Production possibility frontier was presented to illustrate the various production possibilities the economy/ firm faces, and how they make choices between the various production alternatives.

KEYWORDS:

- **Scarcity**: The fundamental economic problem of having limited resources to meet unlimited wants.
- **Factors of Production**: Inputs used in the production of goods and services—land, labour, capital, and entrepreneurship.
- **Opportunity Cost**: The value of the next best alternative foregone when a choice is made.
- **Production Possibility Frontier (PPF)**: A graph that illustrates the maximum feasible output combinations of two goods using available resources.
- **Needs and Wants**: Needs are essential for survival (e.g., food, shelter), while wants are non-essential desires (e.g., luxury cars).

2.9 ASSIGNMENT QUESTIONS

1. Explain the significance of scarcity in managerial decision-making with suitable examples.
2. Differentiate between needs and wants with two examples each.
3. What are the four factors of production? Describe each with one real-life example.
4. Using a production possibility schedule, explain how opportunity cost is calculated.
5. How does the price mechanism help solve the basic economic questions of what, how, and for whom to produce?

OBJECTIVE TYPE QUESTIONS:

1. Scarcity means resources are _____ while wants are unlimited.
2. The four factors of production are land, labor, capital, and _____.
3. Opportunity cost is the value of the _____ best alternative forgone.
4. Capital includes man-made goods like _____ and machinery.
5. Labor-intensive techniques use more _____ compared to capital.
6. A graph that shows the trade-off between two goods is called the _____.
7. The decision of what to produce is answered by the _____ mechanism.
8. Land includes natural resources such as soil, rivers, and _____.
9. The price paid to labor is referred to as _____.
10. The accumulation of investment results in the creation of _____ stock.

11. Consumption refers to the act of using _____ and services.

12. The economy should always aim to use all of its available _____.

13. The entrepreneur earns a _____ for taking business risks.

14. The production possibility curve shifts when there is a change in available _____.

15. Distribution is the process of delivering goods to the _____.

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UNIT-3: BASIC DEMAND AND SUPPLY

STRUCTURE

3.1 Introduction

3.2 Demand

3.2.1 Meaning

3.2.2 Law of Demand

3.2.3 Exceptions to Law of Demand

3.2.4 Ceteris Paribus Assumption

3.2.5 The Demand Curve

3.2.6 Determinants of Demand

3.2.7 Change in Demand and Shift in Demand Curve

3.2.8 Individual and Market Demand Function

3.3 Supply

3.3.1 Meaning

3.3.2 Law of Supply

3.3.3 Assumption of Law of Supply

3.3.4 Determinants of Supply

3.3.5 Change in supply and Shift in Supply Curve

3.3.6 Individual and Market Supply Function

3.4 Market Equilibrium and Producers' Price

3.4.1 Market Equilibrium

3.4.2 Producers' Price (Producers' Demand Curve)

3.5 Let Us Sum Up

3.6 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the law of demand and supply and its determinants.

- Explain the change in demand vs. shift in demand and change in supply vs. shift in supply.
- Discuss the individual and market demand and supply function,
- Know about the market equilibrium and producers' price.

BLOOM'S LEVEL OF LEARNING:

Based on the content, the level of Bloom's Taxonomy predominantly aligns with the "Understanding" and "Applying" stages. The material requires learners to comprehend fundamental economic principles such as the laws of demand and supply, the determinants that influence them, and how these factors contribute to market dynamics like equilibrium and pricing. Additionally, the learners are expected to apply this understanding to distinguish between changes and shifts in demand and supply curves, interpret graphical data, and assess real-world scenarios like brand loyalty or emergency situations that cause deviations from standard laws. Furthermore, the curriculum introduces students to analytical aspects by encouraging them to differentiate between individual and market functions, thus promoting the ability to apply concepts in varied contexts. Overall, the unit fosters cognitive skills that go beyond mere memorization, involving interpretation, comparison, and application of economic theories—hallmarks of the middle levels of Bloom's taxonomy.

3.1 INTRODUCTION

Supply and demand is perhaps one of the most fundamental concepts of economics and it is the backbone of a market economy. Demand refers to how much (quantity) ofa product or service is desired by buyers. The quantity demanded is the amount of a product people are willing to buy at a certain price; the relationship between price and quantity demanded is known as the demand relationship.

3.2 DEMAND

3.2.1 Meaning

Demand is the quantity of a good or service that consumers are willing and ableto buy at a given price in a given time period. Each of us has an individual demand for particular goods and services and our demand at each price reflects the value that we place on a product, linked usually to the enjoyment or usefulness that we expect from consuming it. In other words, demand is the relationship between the quantities of a good or service consumers will purchase and the price charged for that good.

3.2.2 The Law of Demand

The law of demand states that, all other things being equal, the quantity of a good or service is a function of price. In general, that means less is bought at higher prices, and more is purchased at lower prices.

This definition makes sense -- you only have so much money to spend, and if the price of something goes up, you can afford less of it.

There is an inverse relationship between the price of a good and demand.

1. As prices fall, we see an expansion of demand.

2. If price rises, there will be a contraction of demand.

3.2.3 Exceptions to The Law of Demand

1. ***Giffen Good:*** The good is named after Sir Robert Giffen (1873-1910). Giffen Good is necessarily an inferior good with very high negative income elasticity of demand. The good is consumed by low-paid wage earners who spend a large proportion of their income to buy it. Examples of giffen goods are bajra and jowar. In case of giffen goods, the demand curve is upward sloping.
2. ***Veblen Good:*** The good is named after Thornstein Veblen (1857-1929).
   Veblen good is a prestigious good or goods with status symbol. It promotes the social prestige of the holder. Higher the price more is the demand for them. Most suitable example of Veblen goods are diamonds.
3. ***Bandwagon Effect:***
   It refers to a kind of demonstration effect shown by a section charismatic personality. In this case the law of demand is violated because people demand more of that commodity which the upper class people are buying, even of society which tends to imitate the consumption pattern of high income groups or some popular film star or some at higher prices.
4. ***Emergency:*** In case of emergencies like war, flood, drought, the law of demand does not hold. In such case there is general insecurity and fear of shortage of necessities. Hence, consumers demand more good even at higher prices.
5. ***Snob Appeal:*** Certain items of historical, cultural or sociological importance have a Snob appeal. Examples of items of Snob appeal are classic pieces of art, original manuscript, antique sculptures etc. Whenever these itemsare auctioned in the market, their demand rises with rise in prices.
6. ***Brand Loyalty:*** This is another exception to the law of demand where consumers show some kind of attachment to the product due to brand loyalties. The consumers like the product to such an extent that their demand does not fall even if the prices rise.

3.2.4 Ceteris Paribus Assumption

Many factors affect demand. When drawing a demand curve, economists assume all factors are held constant except one – *the price of the product itself*. Ceteris paribus allows us to isolate the effect of one variable on another variable.

3.2.5 The Demand Curve (Reason for Downward Sloping Demand Curve)

A demand curve shows the relationship between the price of an item and the quantity demanded over a period of time. There are two reasons why more is demanded as price falls or in other words we can say that there are two reasons for downward sloping demand curve:

1. ***The Income Effect:*** There is an income effect when the price of a good falls because the consumer can maintain the same consumption for less expenditure. Provided that the good is normal, some of the resulting increase in real income is used to buy more of this product.

2. ***The Substitution Effect:*** There is a substitution effect when the price of a good falls because the product is now relatively cheaper than an alternative item and some consumers switch their spending from the alternative good or service.

   The relation between the consumer's optimal choice of the quantity of a good and its price is very important and this relation is called the demand function. Thus, the consumer's demand function for a good gives the amount of the good that the consumer chooses at different levels of its price when the other things remain unchanged. The consumer's demand for a good as a function of its price can be written as $q = d(p)$ it is also known as *consumer's Marshallian demand function* (named after Alfred Marshall). Where q denotes the quantity and p denotes the price of the good.

   The demand function can also be represented graphically as in the given Figure. The graphical representation of the demand function is called the **demand curve.**

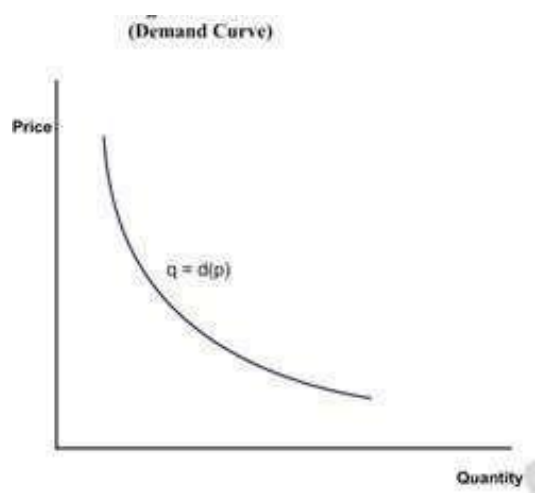


Figure 2: Demand Curve

3.2.6 Determinants of Demand

1. **Price of related goods-** These are either **complementary**, which are things that are usually bought along with the product in demand. They could also be **substitutes** for the product in demand. In case of complement as their prices increase it would also lead to decrease in the demand of the other good

of which they are complement and vice-versa. For example, if the price of petrol rises it would directly lead to the decreased demand of the petrol run automobiles such as cars & motorbikes. While in case of substitute goods if their price increases it would also lead to increase in the demand of the product of which they are substitutes and vice-versa.

2. **Income of consumers-** Income is one of the factors that bear a negative relationship with the demand of the product. Normally, we expect that as one's income rises (falls), the demand for a product will rise (fall). Because we normally expect this to be true, a good for which this statement is true is called a normal good. Occasionally, we shall encounter a good for which the statement is not true. These are called inferior goods; for these goods, as income rises (falls), the demand for the product falls (rises).
3. **Tastes or preferences of the consumers-** It involves the fact that there are certain psychological reasons for liking or disliking a particular good. Our principle is: the more (less) we like a good or service, the greater (less) is our demand for it.
4. **Expectation of future price-** The basic principle in this case is that the consumers' current demand will increase if they expect higher future prices; their demand will decrease if they expect lower future prices.
5. **Population/Number of Buyers:** The last of the factors affecting demand is the population (number of buyers). The number of buyers affects overall, or aggregate, demand. As more buyers enter the market rises, so does the quantity demanded- even if prices don't change. Therefore, if there are more buyers, there must be more market demand.

3.2.7 Change in Demand and Shift in Demand Curve

In the above sections we have discussed the law demand which establishes a negative relationship between the price and the demand of a product. If there is a change in the price of a product it will lead to change in demand i.e., movement along the demand curve. And if there is a change in any of the determinant of demand it will lead to the shift in demand (curve). This is shown in the figure given below; if the price increases from P to P1 then the equilibrium will shift from E to A and the quantity demanded will decrease from Q to Q1 and if price declines to P2 then the equilibrium will shift from E to B and it will lead to increase in quantity demanded from Q to Q2. If there is change in any of the determinant of demand it will shift the demand curve either to rightwards or leftwards. This can be referred to as increase in demand if demand curve shifts rightward which is illustrated in the given figure as demand curve shifts from D to D1 and decrease in demand if demand curve shifts to leftwards which is shown in the figure as demand curve shifts from D to D2.

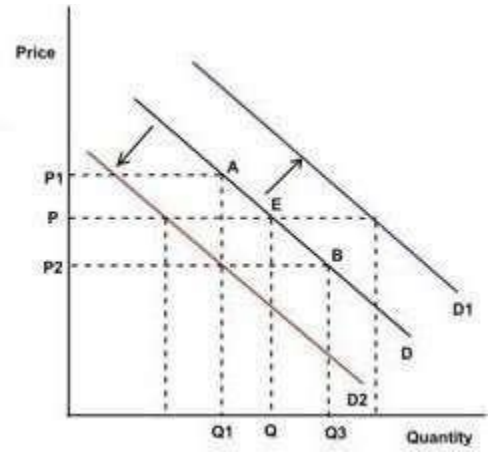


Figure 3: Shift in Demand and Movement along Demand Curve

- **Reason for Change in Demand or movement along demand curve:**

Change in the price of the product.

- **Reason for upward movement (Expansion of demand):** A fall in price of the product
- **Reason for downward movement (Contraction of Demand):** Rise in price of the product.

- **Reason for Rightward Shift in Demand in curve:**

- A rise in income of consumer.
- The price of substitute product increases.
- The price of complement product decreases.
- Expectation that the price of the product will decrease in future.
- Tastes and preferences of consumers turn in to the favor of the product.

- **Reason for Leftward Shift in Demand in curve:**

- A fall in the income of consumer.
- The price of substitute product decreases.
- The price of complement product increases.
- Expectation that the price of the product will decrease in future
- Tastes and preferences of consumers turn in to the favor of the product.

3.2.8 Individual and Market Demand

The individual demand is the demand of one individual or firm where as market demand provides the total quantity demanded by all consumers. In other it represents the aggregate of all individual demands. The individual consumer's demand for a particular good—call it good *X*—will satisfy the law of demand and can therefore be depicted by a downward-sloping individual demand curve. The individual consumer, however, is only one of many participants in the market for good *X*. The market demand curve for good *X* includes the quantities of good *X* demanded by *all* participants in the market for good *X.* The market demand curve is found by taking the horizontal summation of all individual demand curves. For example, suppose that there were just two consumers in the market for good *X*, Consumer A and Consumer B. These two consumers have different individual demand curves corresponding to their different preferences for good *X.* The two individual demand curves are depicted in Figure 4, along with the market demand curve for good *X. In this figure* D_A represents the individual demand of consumer A similarly D_B represents the individual demand of consumer B and the market demand is represented by D_M which is the obtained by the horizontal summation of the individual demand curves of both the consumers.

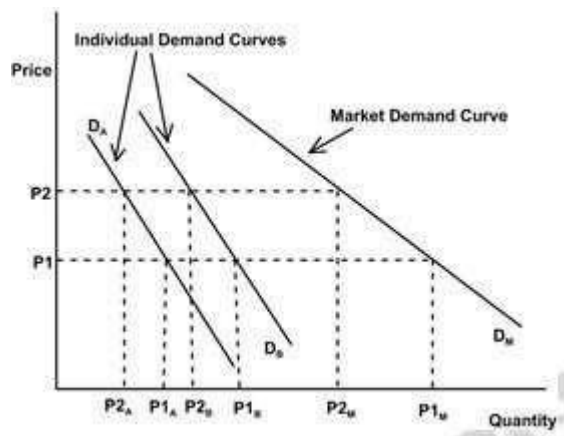


Figure 4: Individual and Market Demand

3.3 SUPPLY

3.3.1 Meaning

Supply is the quantity of a product that a producer is willing and able to supply onto the market at a given price in a given time period. The supply has certain specific features that make it unique; supply must be at a specific price, the quantity supplied has to take place at a specific time, the sellers must

have the ability to supply these quantities in the market, and the sellers must be willing to supply the commodities in the market.

3.3.2 The Law of Supply

It is observed in markets that when more price of commodities are offered to sellers. They increase the quantity supplied of these commodities and when the level of prices decreases, the sellers decrease the quantity supplied. This behavior of seller is called law of supply. Therefore we can say that there is a positive relationship between the quantity that suppliers are willing to sell and the price level.

3.3.3 Assumptions of the Law of Supply (Ceteris Paribus)

The law supply is based on the assumption that factors, other than price of the commodity, that affect the supply remain the same.

The functional relationship between the quantity supplied and the price of a commodity can be expressed as:

$Qs = f(P)$

Where Qs = quantity supplied and P = price of commodity

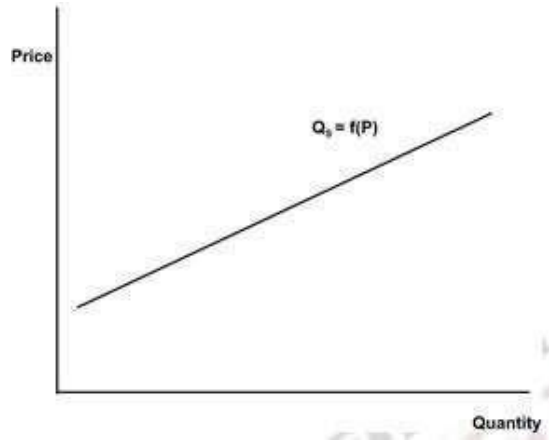


Figure 5: Supply Curve

3.3.4 Determinants of Supply

1. ***Price of the commodity:*** At a higher price, producer offers more quantity of the commodity for sale and at a lower price, less quantity of the commodity. There is a direct relationship between price and quantity supplied of the commodity as given by the Law of Supply.
2. ***Price of related good:*** Supply of a commodity depends upon the prices of its related goods, especially substitute goods. If the price of a remains constant and the price of its substitute good Z increases, the producers will find it more profitable to produce good Z.
3. ***State of Technology:*** If there is up gradation in the technique of production or new discovery, it will lead to fall in the cost of production. Thus, supply of the commodity will increase.

4. ***Cost of Production:*** a change in the cost of production i.e., prices of factors of production affects the supply of a commodity. If wages of labor or price of raw materials increase, then marginal cost (MC) of production will rise. As a result, supply of the good will fall because producers would prefer to produce some other commodities that can be produced at a lower cost.
5. ***Government Policy:*** Government levy taxes or grant subsidies to producers. If heavy excise taxes are imposed on a commodity, it will discourage producers as it will increase the cost of production and as a result, its supply will decrease and the supply curve will shift to the left. Similarly, if subsidies are granted by the government to the producers then supply will increase and the supply curve
will shift to the right.

3.3.5 Change in Supply and Shift in Supply Curve

A movement in a supply curve is a change in supply as a result of a change in price. A shift in a supply curve is a change in supply for a reason other than a change in price. This is illustrated in the figure given below. If there is a rise in price of commodity P to P1it will lead to increase in quantity supplied by the producer from Q to Q1, and if there is a fall in price of the commodity from P to P2, then it will lead to reduction in the quantity supplied by the producer from Q to Q2. On the other hand if there is any change in the determinant of supply it will lead to shift in the supply curve either left orright depending upon the nature of change in the determinant of supply.

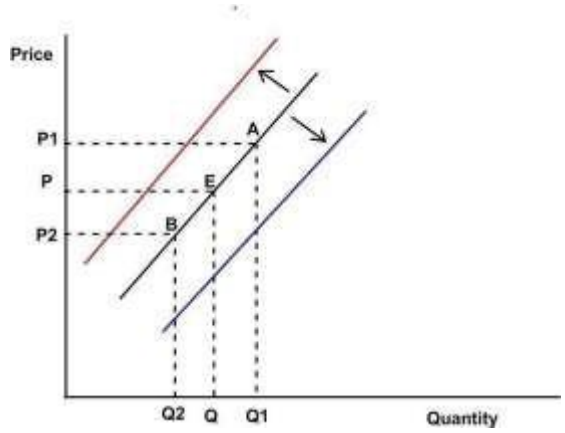


Figure 6: Shift in Supply & Movement along a Supply Curve

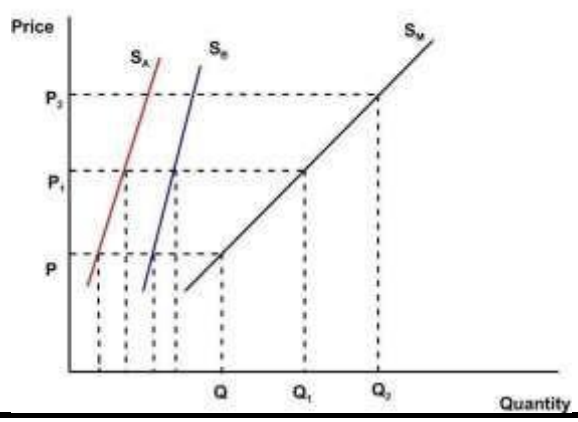
- **Reason for Change in Supply or movement along supply curve:** Change in the price of the product.
  - **Reason for upward movement (Expansion of supply):** A rise in price of the product
  - **Reason for downward movement (Contraction of supply):** A fall in price of the product.

- **Reason for Rightward Shift in Demand in curve:**
  - Reduction in average costs of production.
  - Subsidy on raw materials.
  - Subsidy on labour.
  - Improvements in technology.
- **Reason for Leftward Shift in Demand in curve:** Increase in average cost of production
  - Taxation of raw materials.
  - A switch in production to a more lucrative option.

3.3.6 Individual and Market Demand Function

An individual supply curve indicates different quantities offered for sale by an individual firm at different prices. A market supply curve reflects total quantities of the commodity offered for sale by all the individuals' firms at different prices in a particular time period. Thus, market supply is obtained by aggregating the supplies of all the firms selling that commodity at alternative prices. This can be clearly established from the given figure, where S_A is the supply curve of firm A and S_B of firm B. The market supply curve which is S_M is obtained by adding the supply curves of firm A and B.

Figure 7: Individual & Market Supply



3.4 MARKET EQUILIBRIUM AND PRODUCERS' PRICE

3.4.1 Market Equilibrium

When the supply and demand curves intersect, the market is in equilibrium. This is where the quantity demanded and quantity supplied are equal. The corresponding price is the equilibrium price or market-clearing price, the quantity is the equilibrium quantity.

From the given figure 8 this can established very clearly, here the demand curve "D" intersects the supply curve "S" at point "E" with the price level "P" where the market Equilibrium is obtained.

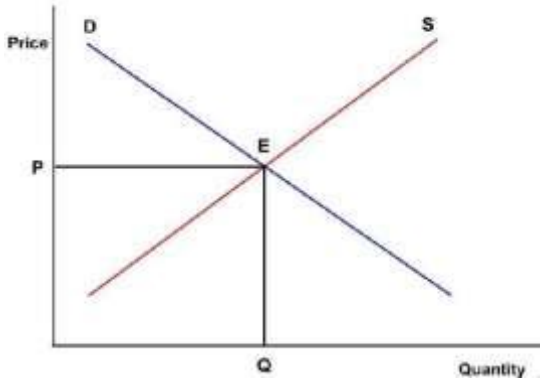


Figure 8: Producers' Price (Producers' Demand Curve)

3.4.2 Producers' Price (Producers' Demand Curve)

Producers' Price (Producers' Demand Curve) in long run will always be horizontal which **i**s nothing but the equilibrium price, where the market equilibrium is established by the intersection of demand and supply. If there are any changes in the price level due to the change in demand it will bring it to the original equilibrium price level in the long run as the supply will adjust to the changes in demand.

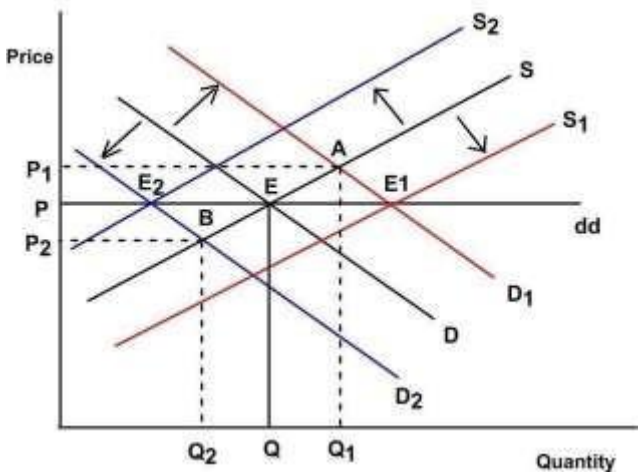
This can be explained with the help of the figure given below. Here initially the equilibrium is established at point "E" where the demand curve "D" and supply curve "S" intersect each other to give equilibrium price level "P". Now suppose there is an increase in the demand of the product which shifts the demand curve rightwards from

D to D1 giving a new equilibrium point "A" and it will lead to rise in price from P to P1. This attracts the other firms to enter in to the market and produce the same product due to which there is increase in the quantity supplied and thus the supply curve shifts rightwards from S to S1giving a new equilibrium point "E1" thereby bringing price down from P1 to original equilibrium price which is P.

Similarly if there is a decrease in the demand of the product which shifts the demand curve leftwards from D to D2 giving a new equilibrium point "B", where the price will fall from P to P2. This will result in the exit of some firms from the market, thereby the quantity supplied in the market will reduce and hence the supply curve will shift from S to S2 this will shift the equilibrium point from "B" to "E2" in order to bring down the price level from P2 to original equilibrium price level P.

If we join the three equilibrium points E2, E and E1 we will get a horizontal line which is nothing but the equilibrium price in the market. This horizontal line(dd) is the producers' demand because at this price level producer can produce and supply commodity as much he can as all the production at this price level would be consumed or purchased in the market.

Figure 9: Producers' Price (Producers' Demand Curve)



CASE STUDIES:

Case Study 1: The Effect of Price Increase on Petrol-Driven Vehicles

In 2022, the price of petrol rose sharply due to geopolitical tensions. As petrol prices increased, the demand for petrol-driven vehicles such as cars and motorbikes fell drastically. This illustrates the concept of **complementary goods**, where an increase in the price of one good (petrol) leads to a decrease in the demand for its complement (vehicles using petrol). Consumers started shifting to electric vehicles and public transportation, which represents a **rightward shift in demand** for substitute goods and a **leftward shift** for petrol-run vehicles.

Case Study 2: Brand Loyalty During a Price Hike

A leading smartphone company increased the price of its flagship model due to global chip shortages. Despite the hike, loyal consumers continued to buy the phone, highlighting an **exception to the law of demand**. This phenomenon is known as **brand loyalty**, where demand remains stable or even grows despite rising prices. This case demonstrates that psychological and emotional connections to products can override traditional economic principles.

3.5 LET US SUM UP

Law of demand clearly establishes inverse relationship between demand and price of the product due to which demand curve always slopes downwards. On the other law of supply establishes positive relationship between quantity supplied and price of the product due to which supply curve is upward sloping.

1. Shift in demand and shift in supply is caused by factors other than price. Whereas price is the only factor that leads to change in demand and change in supply which is the movement along the curve whether upwards or downwards.

2. Market demand curve and market supply curve can be obtained from the horizontal summation of the individual demand curves and individual supply curves respectively.
3. We can construct the market with the help of demand and supply curves. Putting demand and supply together, we can find equilibrium where the supply and demand curve cross that is intersect each other, which gives an equilibrium price. In the long run this equilibrium price would be the producers' price or producers' demand curve.

KEYWORDS

1. **Law of Demand**: States that, keeping all other factors constant, an increase in the price of a product leads to a decrease in its quantity demanded, and vice versa.
2. **Law of Supply**: States that, all else equal, an increase in the price of a product leads to an increase in the quantity supplied.
3. **Ceteris Paribus**: A Latin term meaning "all other things being equal," used to isolate the effect of one variable on another.
4. **Market Equilibrium**: The point where the demand and supply curves intersect, indicating the price at which the quantity demanded equals the quantity supplied.
5. **Substitution Effect**: The change in quantity demanded due to a product becoming cheaper relative to its substitutes.

3.6 ASSIGNMENT QUESTIONS

1. Explain the difference between movement along a demand curve and a shift in the demand curve with the help of examples.
2. Discuss the various exceptions to the law of demand and provide real-life illustrations for each.
3. Analyze how a change in consumer income affects the demand for normal and inferior goods.
4. How does government policy influence the supply of goods? Use diagrams to support your answer.
5. Explain market equilibrium with the help of a diagram and illustrate what happens when there is a change in demand or supply.

OBJECTIVE TYPE QUESTIONS

1. The demand curve generally slopes _____.
2. Giffen goods are a type of _____ goods with upward-sloping demand.
3. The point at which demand equals supply is known as _____.
4. The _____ effect occurs when a fall in price increases a consumer's purchasing power.
5. A rise in the cost of raw materials will lead to a _____ shift in the supply curve.

6. The _____ effect causes consumers to switch to a cheaper substitute when the price of a good falls.
7. Movement along the demand curve is caused by a change in the _____ of the good.
8. A good that consumers buy more of as their income rises is called a _____ good.
9. A _____ shift in the demand curve indicates an increase in demand.
10. The demand curve violates its usual shape in case of goods with _____ appeal.
11. A rise in the price of a substitute product leads to a(n) _____ in the demand of the original product.
12. The supply curve is generally _____ sloping.
13. When government grants a subsidy, the supply curve shifts to the _____.
14. The market demand curve is derived from the _____ summation of individual demand curves.
15. The law of supply assumes _____ conditions.

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FURTHER READINGS

- Microeconomic Theory by H.L. Ahuja
- Principles of Economics by N. Gregory Mankiw
- Modern Microeconomics by A. Koutsoyiannis
- Managerial Economics by D.N. Dwivedi
- Price Theory and Applications by Steven E. Landsburg

UNIT-4: ELASTICITY OF DEMAND

STRUCTURE

4.1 Meaning

4.2 Types of Elasticity

4.3 Importance of Elasticity

4.4 Factors affecting Elasticity of Demand

4.5 Elasticity of Supply

4.6 Let Us Sum Up

4.7 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the meaning and types of elasticity.
- Learn the importance of elasticity.
- Explain the factors affecting elasticity.
- Measure the elasticity of supply.

BLOOM'S LEVEL OF LEARNING:

Based on the content provided, the learning objectives and structure primarily correspond to the Understanding and Applying levels of Bloom's Taxonomy. Learners are expected to comprehend key economic concepts such as the meaning and types of elasticity (price, income, and cross elasticity), and how they function within demand and supply frameworks. This falls under the Understanding level, where students interpret, explain, and summarize theoretical ideas. Furthermore, the material encourages the practical application of these concepts in real-world scenarios, such as taxation policy, price discrimination, and international trade. This demonstrates the Applying level, where learners use acquired knowledge in new contexts and decision-making situations. Thus, the content not only facilitates conceptual clarity but also promotes the ability to apply theoretical principles to practical economic problems.

4.1 MEANING AND TYPES OF ELASTICITY

A firm's demand curve shows the quantity demanded of a product that consumers and firms want to buy at each possible price, ceteris paribus. A change in the price of the product leads to a movement along its demand curve. This responsiveness is known as the elasticity of demand.

4.2 TYPES OF ELASTICITY

Elasticity of demand is a measure of responsiveness of one variable, say quantity demanded of commodity (Qx), with respect to another variable, say price of commodity X (Px). A product's demand curve shows the behavior of the consumers, i.e., how much they want to buy at each possible price holding all other factors fixed that affect demand. By comparing the change in these two percentages, a consumer can decide how much to buy. The elasticity of demand can be divided into three categories, namely, price elasticity, income elasticity and cross-elasticity of demand.

We will discuss them one by one as follows.

1. PRICE ELASTICITY OF DEMAND

Given a demand curve DD (Figure10).We consider movement along DD. The elasticity of demand is a measure of the responsiveness of quantity demanded (Qx) to a change in the price of a product (Px). It is also known as own price elasticity of demand. In other words, the price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price that brought it about. This elasticity is usually symbolized by (the lowercase letter eta), η:

$$\eta = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$\frac{\Delta Q}{Q} \Big/ \frac{\Delta P}{P} = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \left(\frac{Q_2 - Q_1}{P_2 - P_1}\right) \times \frac{P_2}{Q_2}$$

There can be five possible situations while measuring price elasticity of demand. Each of which has a distinct sloped demand curve and it reflects different consumer responses to a change in price. These are discussed one by one below and table summarises the various possibilities of price elasticity of demand and the effects on revenue.

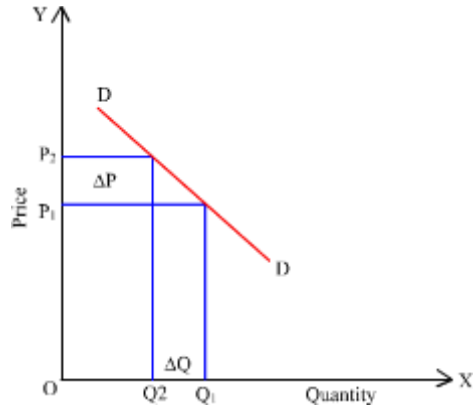


Figure 10

Perfectly inelastic demand

It is an exceptional situation where a change in the price of a product don't cause change in the quantity demanded. In this case, the elasticity of demand is equal to zero. It can be shown on graph as follows:

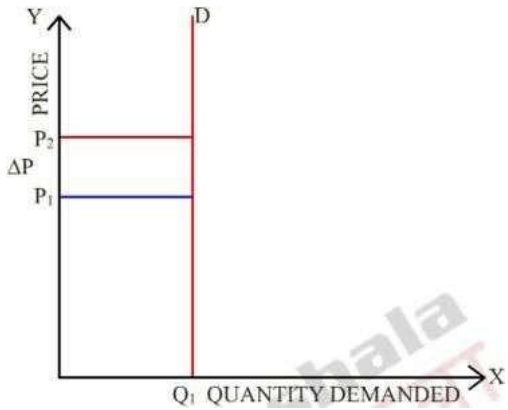


Figure 11 *Source: https://epgp.inflibnet.ac.in*

Here, the demand curve is a vertical line; showing the same quantity demanded of a commodity even when the price of the commodity increases. From P1 to P2.

Essential commodities like essential medicine are examples of this kind of elasticity.

Inelastic demand

Inelastic demand occurs when a change in the price causes a less than proportionate change in quantity demanded, i.e. change in the price is more than change in the quantity demanded. In this case, elasticity of demand would be less than 1 but greater than zero. Diagrammatically, it is shown below:

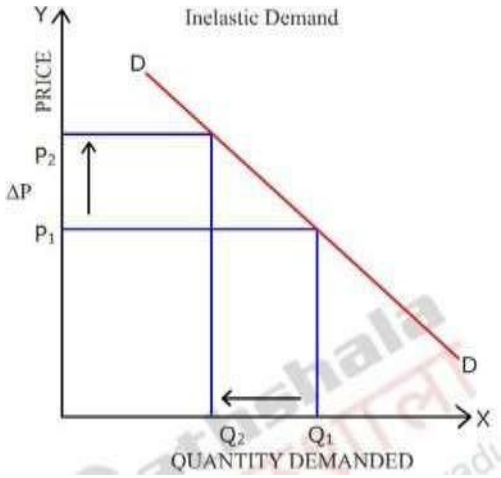


Figure 12

The demand curve is steeply sloped because the quantity demanded is not responsive.

The above diagram clearly shows that percentage change in price is much higher than percentage change in quantity demanded. In case of salt demand cannot increase even if price falls a lot. Neither can it fall if prices raise a lot.

Unit Elasticity

When change in price cause a similar change in the quantity demanded, it is known as unit elasticity of demand. Unit elasticity is always equal to 1.

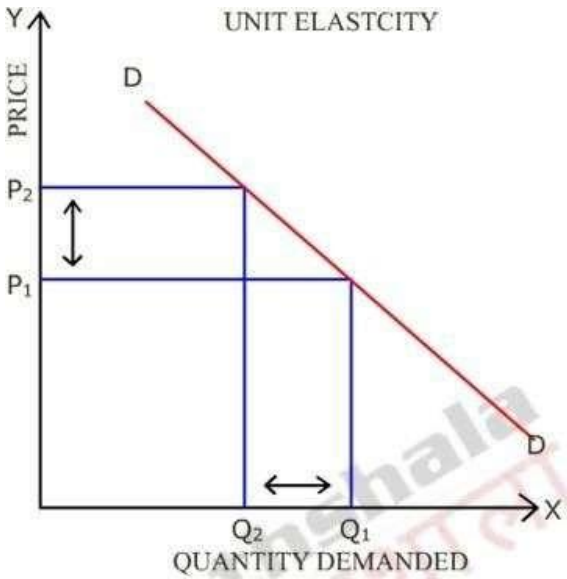


Figure 13

The demand curve for the unit elasticity is at a 45 degree angle, showing a similar percentage change in price and quantity demanded. It is quite clear from the above diagram, that change from P1 to P2 is

equal to a change that takes place from Q1 to Q2. On a straight line demand curve this is the midpoint. Ready garments are an example of unitary elasticity.

Elasticity Demand

When a small change in price leads to a more than proportionate change in quantity demanded, it is known as elastic demand. In this case, elasticity of demand will be greater than 1. Leisure goods like holiday package are an example.

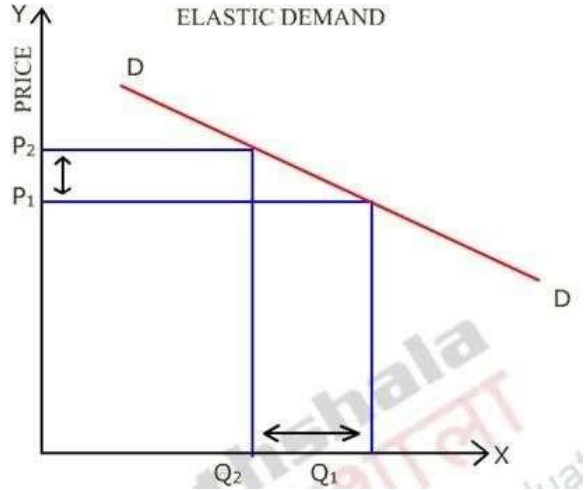
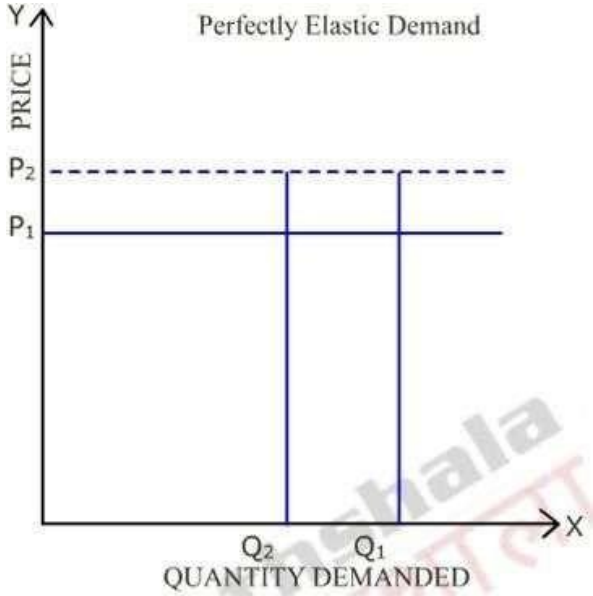


Figure 14 *Source: https://epgp.inflibnet.ac.in*

The slope of the demand curve in diagram 4 is much flatter. It clearly depicts, how a small decrease in price from P1 to P2 has led to a larger-than-proportionate change in quantity demanded from Q1 to Q2.

Perfectly Elastic Demand

When a small rise in the price leads to a zero demand situation, i.e. when demand for the product ceases, then the elasticity of demand is said to be equal to infinity. In other words, when price of the product falls by little then consumers demand an infinite quantity.



Source: https://epgp.inflibnet.ac.in

Figure 15

The above figure shows that new price line P2 does not intersect the demand curve. It indicates the ceased quantity demanded. **Summary Table 2**

Value of demand elasticity	**Description**	**Definition**	**Impact on revenue**
Greater than 1	Elastic demand	Percentage change in quantity demanded is greater than percentage change in price	Revenues increase when price decreases
Equal to 1	Unit-elastic demand	Percentage change in quantity demanded equal to percentage change in price	Revenues unchanged when price decreases
Less than 1	Inelastic demand	Percentage change in quantity demanded less than percentage change in price	Revenues decreases when price decreases

2. INCOME ELASTICITY OF DEMAND

Income elasticity of demand measures the responsiveness of demand for a product with respect to changes in income. It is defined as:

$$\eta \text{ income} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

$$= \frac{\Delta Q}{Q} \Big/ \frac{\Delta I}{I}$$

For most normal goods, income elasticity of demand remains positive. It means when there is a increase in the income level, quantity demanded also increases and vice versa. If the percentage change in quantity demanded is larger than the percentage change in income, the income elasticity of demand will be greater than unity. The relationship between quantity demanded and income can be explained with the help of following diagram:

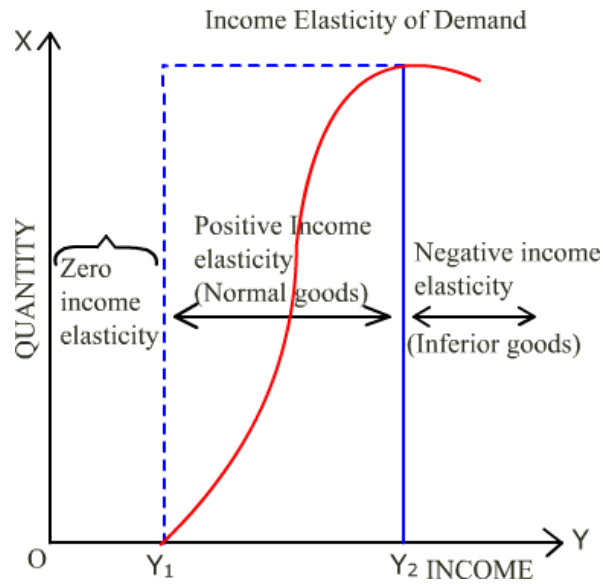


Figure 16

Normal goods have positive income elasticities; inferior goods have negative elasticities. If the income level is less than Y1 level, no quantity is demanded, so income elasticity is zero. When income rises from Y1 to Y2, quantity demanded also increases and it makes income elasticity positive. Lastly, when income rises above Y2 level, quantity demanded falls from its peak point qm, making income elasticity negative.

3. CROSS ELASTICITY

It measure the responsiveness of quantity demanded of one product to change in the prices of other products. For example, number of cars and petrol prices. Producers of large cars find their sales falling when the price of petrol rises dramatically after large oil price increases. It can be represented as follows:

$$\eta\, XY = \frac{\text{Percentage change in quantity demanded of product X}}{\text{Percentage change in price of product Y}}$$

$$= \frac{\Delta Q_X}{Q_X} \Big/ \frac{\Delta P_Y}{P_Y}$$

It can vary from minus infinity to plus infinity and substitute goods have positive cross elasticities. It means the quantity demanded of good X and price of its substitute are positively related. In case of complement goods, cross elasticity is negative. Demand for tea with respect to the price of coffee will be positive as tea and coffee are substitutes. Demand for bread with respect to price of butter will be negative as they are complements.

4.3 IMPORTANCE OF ELASTICITY

The elasticities may seem like rather boring concept to learn about and to calculate. The concept of elasticity of demand has immense importance in various sectors especially in government finance. Following points show the importance of elasticities.

1. Price discrimination by monopolist

If the monopolist finds the demand for a product, he can fix the price at a higher level in order to maximize his net profits. And in case of elastic demand, he will have to lower price. Thus, monopolist gets practical advantage form the concept of elasticity.

2. Price discrimination in cases of joint supply

The concept of elasticity becomes of practical use where the separate costs of joint products can't be measured. Here, again prices are fixed depending upon the elasticity.

3. Importance in taxation policy

It is also helpful to finance minister. He can levy high taxes on goods with inelastic demand to collect larger revenue. But if the demand is elastic, he is not in a position to increase tax rate. If he does so, the demand for the product will curtailed and the total revenue will also be reduced.

4. Importance to managers

The concept of elasticity is also of great importance to businessmen. When the demand for a product is elastic, they can increase the sales by lowering price. In case demand is inelastic, then they can charge a higher price.

5. Helpful to trade unions

The trade unions can raise the wages of the labour in an industry, where the use of the product is relatively inelastic. On the other hand, if the demand is relatively elastic, the trade union can't demand higher wages, without having to face a cut in the level of employment.

6. Useful in international trade

The term of trade between two countries are based on the elasticity of demand of the traded goods. These may be measured in value terms or barter (quantity) terms.

7. Determination of rate of foreign exchange

The rate of foreign exchange is also based on the elasticity of imports and exports of a country. If depreciation happens the domestic exports become cheaper but the value would increase only if elasticity of exports (Ex) is greater than ones i.e. the exports are elastic.

4.4 FACTORS AFFECTING ELASTICITY OF DEMAND

There are many influences that determine consumers' responses to price changes. Similarly, elasticity of demand is also affected by various factors related to the commodity, the consumers and the price itself. These are discussed below:

1. Availability of substitute goods

Substitute goods are the goods that can be used in place of other goods for the same purpose. For example, porridge and corn flakes are considered substitute as breakfast cereal. These products cause a change in the buying habits of consumers. So when price of one product increases, consumers tend to switch to another product. The greater the number of substitute products are available, the greater the elasticity of demand.

The consumer response may be different for price change in the market as a whole and to the individual producers in the market. If the retail price increases, then the consumer does not have much choice but to pay the higher price. But if the individual producer increases the price, the consumer will shift to other cheaper brands.

2. Essential commodities

The extent to which the commodity is a necessary is another factor that determines its elasticity of demand. For example, bread, milk and essential medicines are all necessary goods. These are needed regardless of their prices.

So, demand is said to be inelastic for these products.

3. Habit or desirability

This is also one of the factors that determine the elasticity of demand. When the consumer is habitual of some commodities like tobacco products, alcohol, and drugs, then demand for such commodities tend to be inelastic. Consumers tend to buy such products through force of habit and pay quite high price to get the desired satisfaction. These products even tend to attract a significant amount of tax as a way of collecting revenue for government.

4. Total outlay and income

When the outlay on a good is only a small proportion of income, its demand is expected to be inelastic. In such a case, a price increase will not affects its quantity demanded. It is due to inelastic demand.

When the outlay is large, a small price increase could affect the demand. For example 10% increase on an Rs.10, 00,000 Car is an extra Rs. 1, 00,000. In this instance, demand would be elastic.

5. Complementary products

While substitute goods can be used interchangeably, complementary goods are used together. To consume one product, you must consume both. It means separately each product is virtually useless.

For example demand for a CD is only a small fraction of the cost of buying a CD player. Generally, one can afford to buy many CDs who can afford to buy a CD player.

6. Time

The time can also affect the purchasing patterns of the buyers. For example, we tend to buy more woolen cloths and heaters because it is essential to have these products. So the demand for these products in winter will be inelastic. During summer season, demand for these products will be elastic because the consumers will defer the purchase of these products.

4.5 ELASTICITY OF SUPPLY

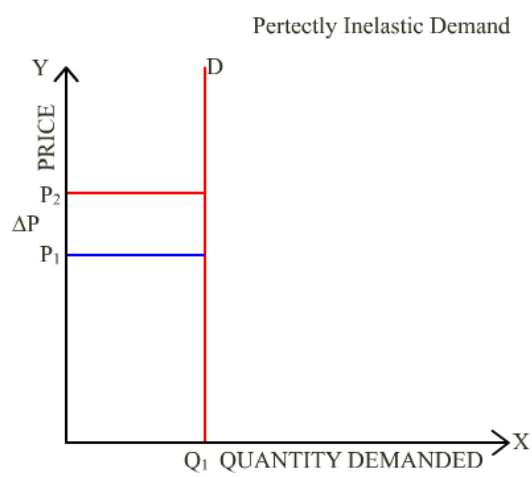
It is also termed as price elasticity of supply. It measures the responsiveness of producers to any price changes in terms of quantity supplied. It shows whether the producers will supply more or less to the market when there is a price change. It also measures how much more or less will be supplied. If the producers are very responsive to the price changes, the supply is elastic. Alternatively, if producers are not very responsive to price changes, the supply is inelastic.

As for elasticity of demand, there are five distinct elasticities of supply, each case has its unique supply curve that reflects the producers' response to price changes.

1. Perfectly Inelastic Supply

Where a change in the price does not lead producers to supply more goods, the elasticity of supply is said to be zero and the supply curve is vertical as shown below:

Figure 17



This holds good for natural resources that are non-renewable.

2. Inelastic supply

When the quantity supplied is relatively unresponsive to price changes, it is a situation of inelastic supply. The supply curve is inelastic (0<Es<1). in this case, the supply curve is steeply sloped.

Figure shows that a considerable increase in price will not lead producers to supply a large increase in quantity.

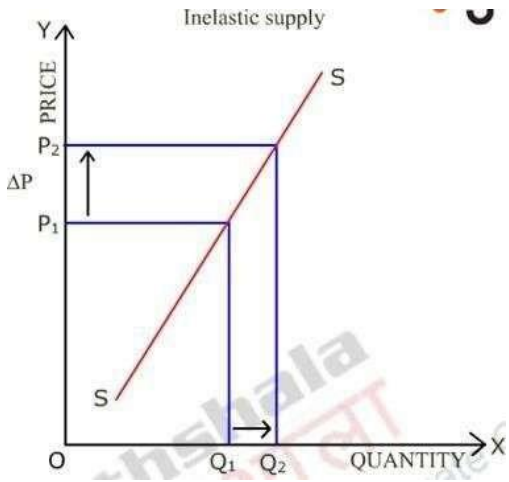
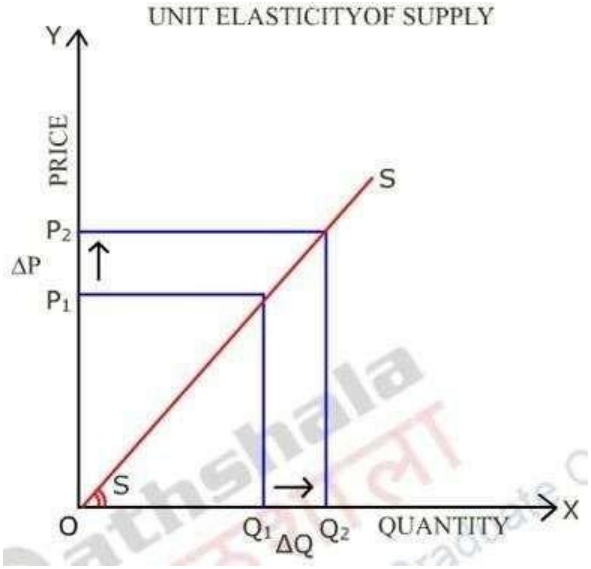


Figure 18 *Source: https://epgp.inflibnet.ac.in*

This generally the case with extractive industry likes mines and minerals. Their supply cannot increase much

3. Unit elasticity

When a price change leads to a proportionate change in quantity supplied, it isa case of unit elasticity of supply. Unit elasticity is equal to one (Es=1). The supply curve is on a 45-degree angle (figure). Any straight line supply curve pass if through the origin happens Es equal to one.



Source: https://epgp.inflibnet.ac.in

Figure 19

4. Elastic supply

When the quantity supplied is more than proportionate to the change in price,it is a situation of elastic supply. The elasticity of supply is greater than one and supply curve has a flatter slope. (figure)

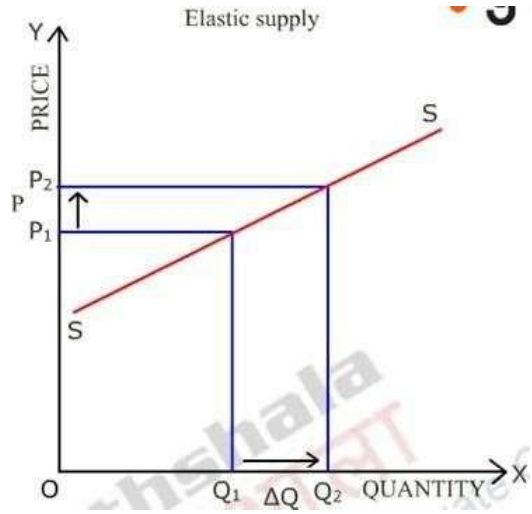
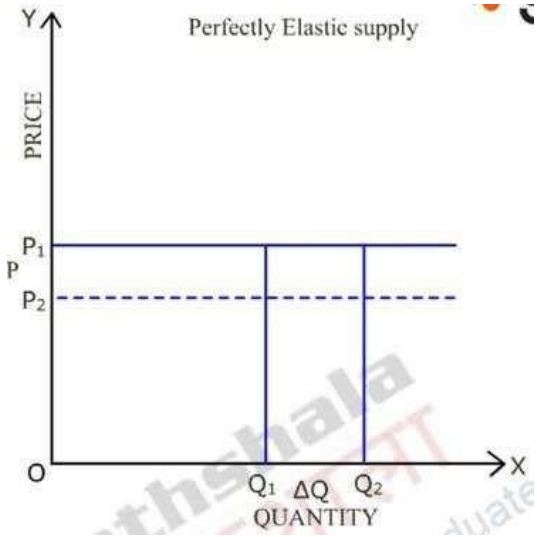


Figure 20 *Source: https://epgp.inflibnet.ac.in*

In the case of low-cost goods like plastics a small increase in the price will prompt products to increase the supply.

5. Completely elastic supply

It is a case where a change in price causes a dramatic change in supply. In other words, a change in price causes supply to cease completely. Elasticity of supply is said to be infinite and is shown as horizontal line (figure). If the price decreases from P1 to P2, supply of the goods ceases. This price is considered to be too low by producers to sustain in the market.



Source: https://epgp.inflibnet.ac.in

Figure:21

CASE STUDIES

Case Study 1: Price Elasticity of Demand and Taxation Policy

The government of a country introduces a heavy tax on cigarettes with the aim of discouraging consumption and increasing public revenue. However, after the tax implementation, it is observed that the demand for cigarettes drops only slightly. This is due to the inelastic nature of demand for cigarettes, as many consumers are habitual users and continue purchasing despite price hikes. The policy ends up being successful in generating revenue, though less effective in reducing smoking habits.

Learning Insight: This case illustrates how understanding price elasticity helps governments in shaping tax policies on inelastic goods for effective revenue collection.

Case Study 2: Cross Elasticity and Business Strategy

A company producing butter observes a sharp fall in sales after the price of bread increases significantly. Since bread and butter are complementary goods, the decrease in bread demand directly affects the demand for butter. The company responds by launching a marketing campaign and discounts on combo packs to boost sales.

Learning Insight: This example demonstrates the role of cross elasticity in influencing business decisions, especially for complementary goods.

4.6 LET US SUM UP

The elasticity of demand measures the responsiveness of change in price of one variable with change in another variable, keeping other factors constant. Elasticity of demand can of three types namely, price elasticity of demand, income elasticity of demand and cross-elasticity of demand. Price

elasticity of demand measures the responsiveness of change in price of a product with the change in quantity demanded of that product. Income elasticity measures responsiveness of quantity demanded with respect to income. Cross elasticity measures the responsiveness of price of one product with the quantity demanded of another product. The concept of elasticity is an important tool for decision making. It is used by monopolist, producers, international trade, foreign exchange rate determination, useful to finance minister and many more.

- There are many factors that affect elasticity of demand like availability of substitute goods, necessity of goods, habits of the consumers, total outlay and revenue, complementary goods, and time.
- The elasticity of supply measures the responsiveness of change in price to the quantity supplied. There could be various conditions that affect the nature of supply curve and its elasticity. We have discussed in any cases or examples of elasticity of demand and supply.

KEYWORDS

1. **Elasticity of Demand**: A measure showing how the quantity demanded of a good responds to changes in price, income, or related goods' prices.
2. **Price Elasticity of Demand (PED)**: The percentage change in quantity demanded resulting from a one percent change in price.
3. **Income Elasticity of Demand**: The responsiveness of quantity demanded to a change in consumer income.
4. **Cross Elasticity of Demand**: Measures the responsiveness of demand for one good when the price of another good changes.
5. **Elasticity of Supply**: Indicates how much the quantity supplied of a good changes in response to a change in its price

4.7 ASSIGNMENT QUESTIONS

1. Explain the different types of elasticity of demand with suitable examples.
2. Discuss the importance of elasticity of demand in business and government decision-making.
3. Describe the factors affecting price elasticity of demand.
4. Compare and contrast perfectly elastic and perfectly inelastic supply with diagrams.
5. How does the concept of income elasticity help in classifying goods into normal and inferior?

OBJECTIVE TYPE QUESTIONS:

1. The elasticity of demand is the _____ of quantity demanded to a change in price.
2. If demand does not change with price, it is called _____ elastic demand.

3. Income elasticity of demand for normal goods is usually _____.
4. Cross elasticity for substitute goods is _____.
5. If a small change in price causes a large change in demand, the demand is said to be _____.
6. Salt is an example of a good with _____ elasticity.
7. The demand for luxury items is generally _____ elastic.
8. The price elasticity of demand formula involves percentage change in quantity demanded divided by percentage change in _____.
9. When elasticity of demand is equal to 1, it is called _____ elasticity.
10. A vertical demand curve shows _____ elasticity.
11. The availability of _____ increases the elasticity of a product.
12. The elasticity of demand helps monopolists in deciding the _____ of a product.
13. In case of complementary goods, cross elasticity is _____.
14. The elasticity of supply is zero in case of _____ supply.
15. A flatter supply curve indicates _____ supply.

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- Koutsoyiannis, A. (2003). *Modern Microeconomics*. Palgrave Macmillan.

FURTHER READINGS

- Microeconomics by Robert S. Pindyck & Daniel L. Rubinfeld
- Price Theory and Applications by Steven Landsburg
- Intermediate Microeconomics by Hal R. Varian
- Managerial Economics by H.L. Ahuja
- Economics for Business by Begg and Ward

UNIT-5: CONCEPT OF UTILITY AND INDIFFERENCE CURVE

STRUCTURE

5.1 Introduction

5.2 Types of Utility

5.3 Cardinal Utility Theory

5.4 Approaches to Cardinal Utility Theory

5.5 Ordinal Utility Theory

5.6 Indifference Curves

5.7 Concept of Marginal Rate of Substitution

5.8 Let Us Sum UP

5.9 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Know about the concepts of Utility theory.
- Discuss the Law of Equi-Marginal utility and Law of Diminishing Marginal Utility
- Understand the concept of ordinal utility and indifference curve.
- Learn how to measure the utility by giving preference.

BLOOM'S LEVEL OF LEARNING:

The content provided in the utility theory unit addresses multiple levels of Bloom's Taxonomy, promoting a comprehensive understanding of consumer behavior. At the Remembering level, students recall key concepts like total utility, marginal utility, and cardinal versus ordinal utility. The Understanding level is evident as learners grasp the principles behind these theories and their assumptions, such as the Law of Diminishing Marginal Utility and the Law of Equi-Marginal Utility. The Applying level is targeted through real-world scenarios, where students apply utility concepts to make decisions about consumer choices. At the Analyzing level, learners evaluate the relationship between total utility and marginal utility, comparing and contrasting different utility approaches. The

Evaluating level involves critically assessing the assumptions and practical relevance of utility theories in explaining consumer behavior. Finally, at the Creating level, students are encouraged to synthesize their knowledge by developing new insights or applying theories to hypothetical situations. Thus, the module effectively engages students at various Bloom's cognitive levels, fostering both foundational and advanced economic reasoning.

5.1 INTRODUCTION

From time to time, various theorists have attempted to explain consumer's demand for a product and develop a valid demand theory. Cardinal utility theory explained in this module is the oldest demand theory, which explains the consumer's demand for a product and derives the law of demand, which states that there exists an inverse relationship between the price of the product and quantity demanded. Ordinal approach could be measured by giving order, ranks or preferences. Hence the ordinal utility means consumers preferences or choice for one commodity or for a basket of goods over the other. This would be explained with the help of budget line and indifference curve later in this chapter.

The Utility Theory

The utility theory aims to explain the situation of consumer behavior in regard to the satisfaction that a consumer gets from the consumption of a commodity. Utility theory was developed and introduced in 1870 by a British Economist, William Stanley Jevons. The term utility refer to the satisfaction or benefit that a consumer derives consumption of the commodity. The utility can be measured in **utils.**

Utility Analysis

Utility analysis, attempts to explain consumer behavior, on the basis of satisfaction derived from the consumption of commodities. Of course, the utility derrived from the consumption affects the consumer's purchase and consumption decision. The concept of utility can be explained with the help of various examples:

1. A person who is on fasting for two days when offered food will get utility (satisfaction)
2. A kid when crying is offered toys to play also gets satisfaction.

Thus, both examples referred above offers satisfaction owing to the satisfaction of needs and wants.

5.2 TYPES OF UTILITY

There are two types of utility:

1. Total Utility
2. Marginal Utility

1. Total Utility

The term total utility means the total satisfaction derived from the consumption of commodities.

2. Marginal Utility

Marginal utility be defined as the additional utility derived from the consumption of an additional unit of a commodity. Therefore , we can say that Marginal utility is the extra satisfaction gained from an one more additional unit of that particular commodity.

Marginal utility may be calculated as follows:

$$\text{Marginal Utility} = \frac{\text{Change in Total Utility}}{\text{Change in quantity demanded}}$$

Relation Between Marginal Utility and Total Utility

The relation between marginal utility and total utility as indicated by the above chart and graph can be stated as follows:

Marginal Utility	*Total Utility*
1. Declines	1. Increases but at a diminishing rate
2. Reaches zero	2. Reaches maximum
3. Becomes negative	3. Declines from the maximum

5.3 CARDINAL UTILITY ANALYSIS

Cardinal utility analysis is based on the cardinal measurement of utility which assumes that utility is measurable and additive. This theory was developed by neo-classical economists like Marshall, Pigou, Robertson etc. It is expressed as a quantity measured in hypothetical units which called utils. If a consumer imagines that one mango has 8 utils and an apple 4 utils, it implies that the utility of mango is twice than of an apple.

Assumptions of Cardinal Utility Analysis:

1. **Rational consumer**

   Cardinal utility analysis assumes that consumer is rational. He makes every effort to maximize his total utility under the income and price constraint. while going for the purchase or consumption of good, a consumer will act rationally to maximize his level satisfaction.

2. **Cardinal measurement of Utility**

   The utility of each commodity is measurable and quantifiable i.e. Utility is cardinal concept. The most convenient measure is money. Thus, utility can be measured quantitatively in monetary units or cardinal units. Therefore, whenever a person consumes any commodity, he can express or measure his satisfaction in cardinal term. For example, a person having a glass of milk can say that he got 10 utils from it.

   Moreover, cardinal measurement of utility enables the comparison of utilities derived from two different goods. For example, a person can say that the utility he gets from the consumption of milk is twice the utility he gets from the consumption of Juice.

3. **Constant Marginal Utility of Money**

   The utility derived from commodities are measured in terms of money. According to Marshall, money is the measuring rod of money. So, money is a unit of measurement in cardinal approach. Marshall argues that that the amount of money which a person is prepared to pay for a unit of a good rather than go without it is a measure of utility he derives from that good. According to him measurement of marginal utility of good in terms of money is only possible if the marginal utility of money itself remains constant. Hence, marginal utility of money should be constant.

4. **Diminishing Marginal Utility**

   If the stock of commodities increases with the consumer, each additional stock or unit of the commodity give him lesser and lesser satisfaction. Every additional intake of good will yield less utility as compared to the utility obtained from the previous unit of the good. It means utility increases at a decreasing rate.

5. **Independent Utilities**

   It means utility obtained from one commodity is not dependent on utility obtained from another commodity. In other words, it means that the utility which a consumer derives from

the consumption of that commodity is the function of the quantity of only that good. It is not affected by the consumption of other commodities.

5.4 APPROACHES TO CARDINAL UTILITY

The two approaches of Cardinal utility analysis are as follows:

1. Law of Diminishing Marginal Utility 2. Law of Equi-Marginal Utility 1. Law of Diminishing Marginal Utility

The fundamental assumption of utility theory of demand is that the satisfaction that a person derives in consuming a particular product diminishes or declines as more and more of a good is consumed. In other words, as successive quantity of goods is consumed, the utility we derive diminishes. This is called the law of diminishing marginal utility.

Law of DMU has universal applicability and applies to all goods and services. This law was first given by a German economist H.H. Gossen. That is why, it is also known as

'Gossen's first law of consumption'.

Utility refers to the amount of satisfaction a person gets from consumption of a certain item and marginal utility refers to the addition made to total utility, we get after consuming one more unit. An individual's wants are unlimited in number yet each individual's want is satiable. Because of this, the more we have a commodity, the less we want to have more of it. This law state that as the amount consumed of a commodity increases, the utility derived by the consumer from the additional units, i.e marginal utility goes on decreasing. The marginal utility of a commodity diminishes at the consumer gets larger quantities of it. Marginal utility is the change in the total utility resulting from one unit change in the consumption of a commodity per unit of time.

Assumptions of Law of Diminishing Marginal Utility:

The law of DMU operates under certain specific conditions. Economists call them the

'assumptions' of this law. Following are the assumptions of the law of diminishing marginal utility:

1. **Cardinal measurement of utility:**

   It is assumed that utility can be measured and a consumer can express his satisfaction in quantitative terms such as 1, 2, 3, etc.

2. **Monetary measurement of utility:**

   It is assumed that utility is measurable in monetary terms.

3. **Consumption of reasonable quantity:**

It is assumed that a reasonable quantity of the commodity is consumed. For example, we should compare MU of glassfuls of water and not of spoonful's. If a thirsty person is given water in a spoon, then every additional spoon will yield him more utility. So, to hold the law true, suitable and proper quantity of the commodity should be consumed.

4. **Continuous consumption:**

It is assumed that consumption is a continuous process. For example, if one glass of juice is is consumed in the morning and another in the afternoon, then the second glass may provide equal or higher satisfaction as compared to the first one.

No change in Quality:

Quality of the commodity consumed is assumed to be uniform. A second cup of ice-cream with nuts and toppings may give more satisfaction than the first one, if the first ice-cream was without nuts or toppings.

5. **Rational consumer:**

The consumer is assumed to be rational who measures, calculates and compares the utilities of different commodities and aims at maximising total satisfaction.

6. **Independent utilities:**

It is assumed that all the commodities consumed by a consumer are independent. It means, MU of one commodity has no relation with MU of another commodity.

Further, it is also assumed that one person's utility is not affected by the utility of any other person.

7. **MU of money remains constant:**

As a consumer spends money on the commodity, he is left with lesser money to spend on other commodities. In this process, the remaining money becomes dearer to the consumer and it increases MU of money for the consumer. But, such an increase in MU of money is ignored. As MU of a commodity has to be measured in monetary terms, it is assumed that MU of money remains constant.

8. **Fixed Income and prices:**

It is assumed that income of the consumer and prices of the goods which the consumer wishes to purchase remain constant.

9. **Goods are divisible:**

It is assumed that whatever the goods are consumed they must be divisible. If the goods are not divisible the comparison of utility is not possible.

10. Homogeneity:

Another important assumption is that goods consumed should be homogeneous in regard to quality size, taste, flavors, colour etc.

Table 1: Law of Diminishing Marginal Utility

Units of Ice Cream	Total Utility (in utils)	Marginal Utility (in utils)
1	20	20
2	36	16
3	46	10
4	50	4
5	50	0 (Point of Satiety)
6	44	-6

In the diagram, units of ice-cream are taken on the X-axis and MU on the Y-axis. MU from each successive ice-cream is being represented by the points A, B, C, D and E. From the diagram it is clear that, level of satisfaction as represented by the rectangles become smaller and smaller with every increase in consumption of ice-creams.

MU falls from 20 to 16 and then to 10 utils, when consumption is increased from 1st to 2nd and then to 3rd ice-cream. 5th ice-cream has MU= 0 i.e. no utility at all and this point is known as the 'Point of satiety'. When 6th ice-cream is consumed, MU becomes negative. MU curve slopes downwards showing that MU of successive units is falling.

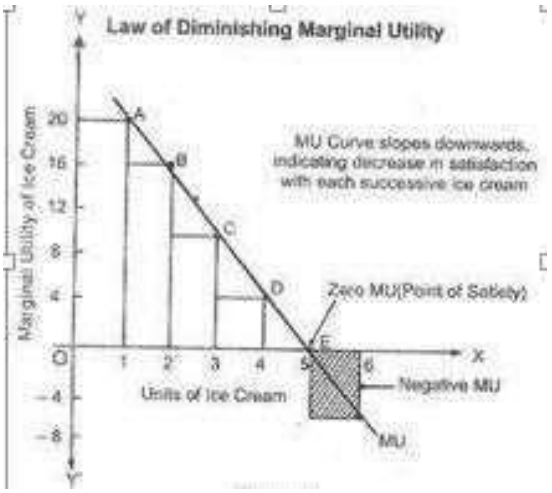


Figure 1 *Source: https://epgp.inflibnet.ac.in*

2. Law of Equi Marginal Utility

The **law of equi marginal utility** was presented in 19th century by an Australian economists H. Gossen. The Law of equi marginal Utility is another fundamental principle of Economics. This law is also known as the **Law of substitution or the Law of Maximum Satisfaction.**

A consumer has number of wants. He tries to spend limited income on different things in such a way that marginal utility of all things is equal. When he buys several things with given money income he equalizes marginal utilities of all such things. The consumer can get maximum utility by allocating income among commodities in such a way that last rupee spent on each item provides the same marginal utility.

Assumptions of the law:

1. Consumer has limited income or limited stock of a given commodity.
2. The marginal utility of money is constant.
3. Consumer has perfect knowledge of utility obtained from goods.
4. Consumer is normal person and rational, so he tries to seek maximum satisfaction.
5. The utility is measurable in cardinal terms.
6. The consumer has more than one want to satisfy. This he can do either by purchasing the required number of commodities out of a given income or putting a given commodity to various uses to satisfy his different wants.
7. The goods have substitutes.

Explaination:

The law of equi-marginal utility states that, other things being equal, a consumer gets maximum total utility from spending his given income, when he allocates his expenditure to the purchase of different goods in such a way that the marginal utilities derived from the last unit of money spent on each item of expenditure tends to be equal, that is, to say that the consumer maximizes his satisfaction, which he obtains equi- marginal utility from all the goods purchased at a given time. The Law of Equi-Marginal utility can be explained with the help of table and diagram: The law of substitution can be explained with the help of an example. Suppose consumer has six rupees that he wants to spend on apples and bananas in order to obtain maximum total utility. The following table shows marginal utility (MU) of spending additional rupee of income on apples and bananas:

Table 2

Money (Units)	**MU of apples**	**Mu of bananas**
1	**10**	**8**
2	**9**	**7**
3	**8**	**6**
4	**7**	**5**
5	**6**	**4**
6	**5**	**3**

The above schedule shows that consumer can spend six rupee in different ways:

1. Re.1 on apples and Rs.5 on bananas. The total utility he can get is:

   [(10) + (8+7+6+5+4)] = 40.

2. Re.2 on apples and Re.4 on bananas. The total utility he can get is:

   [(10+9) + (8+7+6+5)] = 45.

3. Re.3 on apples and Re.3 on bananas. The total utility he can get is:

   [(10+9+8) + (8+7+6)] = 48.

4. Re.4 on apples and Re.2 on bananas. This way the total utility is:

[(10+9+8+7) + (8+7)] = 49.

5. Re.5 on apples and Re.1 on bananas. The total utility he can get is:

   [(10+9+8+7+6) + (8)] = 48.

Total total utility for consumer is 49 utils that is the highest obtainable with expenditure of Re.4 on apples and Re.2 on bananas. Here the condition MU of apple = MU of banana i.e 7 = 7 is also satisfied. Any other allocation of the last rupee shall give less total utility to the consumer.
The same information can be used for graphical presentation of this law:

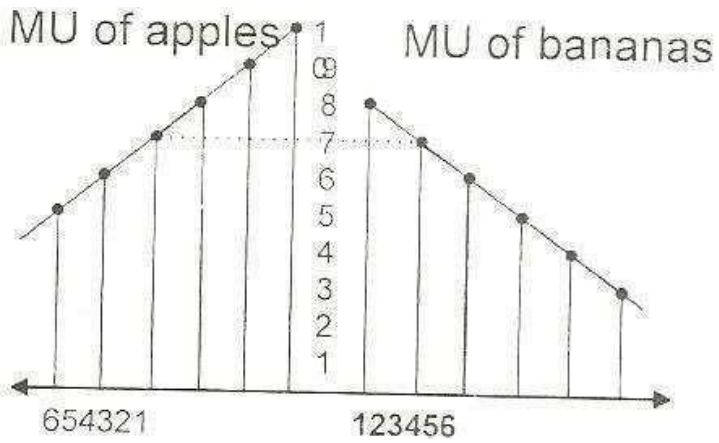


Figure 2

The diagram shows that consumer has income of six dollars. He wants to spend this money on apples and bananas in such a way that there is maximum satisfaction to the consumer.

5.5 ORDINAL UTILITY THEORY

The theory of consumer behaviour is also based on the ordinal approach was propounded by J.R. Hicks and R.G.D. Allen in 1934. According to this approach, utility cannot be measured in any quantifiable number. It could only be measured by giving order, ranks or preferences. Hence the ordinal utility means consumers preferences or choice for one commodity or for a basket of goods over the other. Here, the preferences could be expressed in terms of 'more' or 'less' preferable. Moreover, since the consumers have a limited income which they can spend on their consumption, therefore, in this approach, a consumer will prefer a basket of goods over the other given the prices of the goods and the income of the consumer.

Assumptions of Ordinal Utility Theory:

1. Rationality

A consumer is always rational i.e. he always prefers more of goods and services to derive maximum utility. Thus he always buys the commodity which gives him maximum utility first and then he buys the least utility giving commodity at the end.

2. Finite Money Income

The consumers have limited money income which they spend on the purchase of all the goods and services for their living. Thus they allocate this income as their consumption expenditure on all goods and services.

3. Ordinal Utility

The utility derived from the consumption of each good or a basket of goods could be measured ordinally by giving preferences for each good over the other

4. Transitivity and Consistency of Choice

If the stock of commodities increases with the consumer, each additional stock or unit of the commodity give him lesser and lesser satisfaction. Every additional intake of good will yield less utility as compared to the utility obtained from the previous unit of the good. It means utility increases at a decreasing rate.

Consumer's preferences are always transitive i.e., if a consumer prefers good X over good Y and the same consumer prefers good Y over good Z then according to this assumption of transitivity, he must prefer good X over good Z also,

If, $X>Y$

If, $Y>X$

Therefore, $X>Z$

Whereas as per consistency of choice, if a consumer prefers good X to good Y in one period, then he must not prefer good Y to good X in another period or must not treat both the goods as equal. Symbolically,

If, $X>Y$ in one period

Then, $Y>X$ or $Y\neq X$ in other period.

5. Non-Satiety

According to this, a consumer always prefers more of a good or larger quantity of all the goods because he has not reached the saturation level nor he is oversupplied with all the goods.

6. Diminishing Marginal Rate of Substitution

Marginal rate of substitution refers to the rate at which a consumer substitute one good X for the other good Y so that the level of utility/ satisfaction obtained from it remains the same to him. Symbolically, MRS = $\Delta X/\Delta Y$ or $\Delta Y/\Delta X$. According to this assumption, as a consumer continues to substitute X for Y or Y for X, his MRS diminishes/decreases. This would be explained in the later sections in more detail.

5.6 INDIFFERENCE CURVE

Indifference curve (IC) is defined as the locus of the points of the different combinations of different goods which gives an equal level of satisfaction to the consumer. Therefore a consumer is indifferent between any of the combination which lies on the same indifference curve. This happens because the consumer has a power to substitute among various goods, given their prices and his income level.

For simplicity let's take the example of two goods x and y. Now given the income of the consumer and the prices of x and y, the consumer can choose different combinations of x and y to be consumed in order to maximize his satisfaction. All those combinations of x and y which gives an equal utility to the consumer forms an indifference curve (as in the following fig). Moreover, the consumer can substitute between x and y and thus can either consume more of x or more of y but the main point is that the level of satisfaction arising from the different combination of x and y remains the same to him.

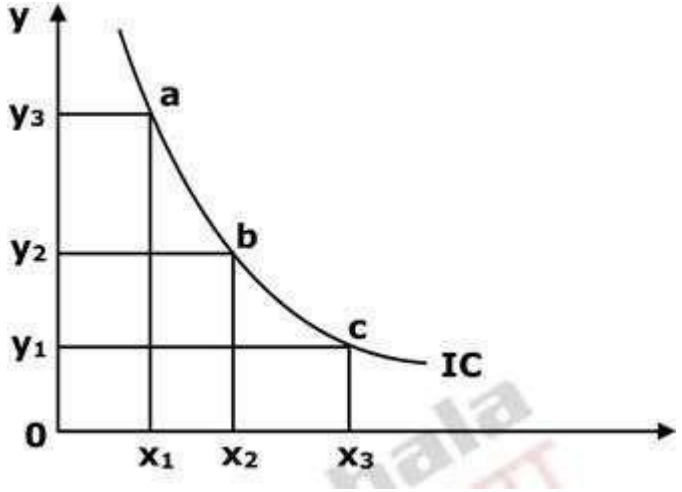


Figure 3: Indifference curve

As has been depicted in the above figure, that at point 'a' the consumer consumes y3 and x1 combination of goods, whereas at point 'b' he substitutes y with x and the combination at point 'b' becomes x2 and y2, similarly at point 'c' the consumer consumes x3 and y1 combination of both the goods, given his income and Px and Py.

However, all the points, a,b and c, gives the consumer equal level of satisfaction. Hence, he is indifferent among a,b and c. Therefore, the curve joining a, b and c is known as indifference curve, where the consumer is indifferent of the different choice which he has as they all gives him equal level of satisfaction. Here only the combinations of x and y are different at each point but the level of satisfaction which the consumer obtained from them is the same.

Now if the consumer either consumes more of a good or of both the goods then he would be able to increase his level of satisfaction and thus a new IC would be drawn above this IC which represents the different combinations of x and y (higher than the previous combinations) which gives him the same level of satisfaction (higher than the previous level of satisfaction).
Similarly, a lower IC would represent the combinations of x and y (lesser than the previous combinations) which gives him equal satisfaction (lesser than the previous level of satisfaction).
Hence the higher an IC, the higher is the level of satisfaction and the lower the IC, the lesser is the level of satisfaction obtained by the consumer.
Representation of the different combinations of goods and different levels of satisfaction on a graph is known as an Indifference map.

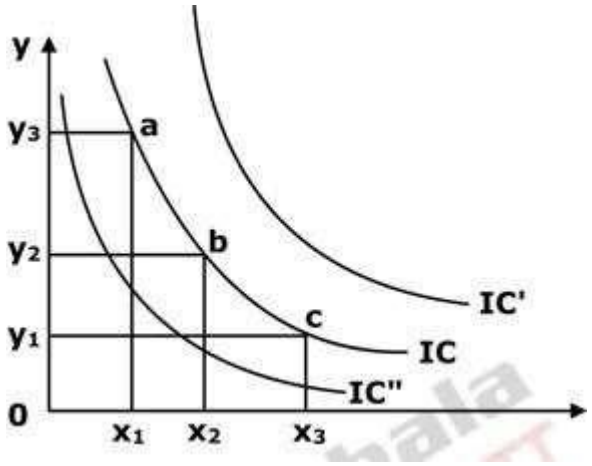


Figure 4: Indifference Map *Source: https://epgp.inflibnet.ac.in*

5.7 CONCEPT OF MARGINAL RATE OF SUBSTITUTION

Marginal rate of substitution (MRS) is defined as the rate at which a consumer substitutes one good for the other obtaining the same level of satisfaction. In this case, it is the rate at which a consumer substitute good y with good x in order to get the same level of satisfaction.

Symbolically, MRSyx = Δy/Δx or MRSxy = Δx/ΔyMRS is also known as the slope of the IC.

As we can see from the diagram below, that as the consumer consumes more ofx, he reduces the quantity of y from y3 to y2 in order to get the same level of satisfaction. Similarly, he further substitute y from y2 to y1 with good x in order to get more of x and the same level of satisfaction. This rate of change in y due to the increase in the quantity of x by one unit is known as Marginal rate of substitution (MRS).

As we can see, an IC has a diminishing MRS, means that the consumer substitute x for yat a diminishing rate. In other words, as the consumer increases the consumption of x byone unit, their consumption of y reduces at a decreasing rate. Hence we can say that anIC has a diminishing slope.

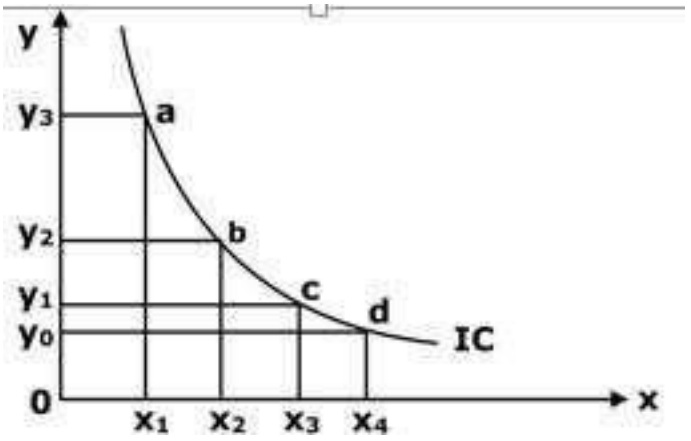


Figure 5: Diminishing marginal rate of substitution

As we can see that $\Delta x1 = \Delta x2 = \Delta x3 = \Delta x4$ but $\Delta y0 < \Delta y1 < \Delta y2 < \Delta y3$.This MRS falls because of the following reasons:

- Diminishing MU
- The decline in the ability of the consumer to sacrifice a commodity whose quantity goes on decling.

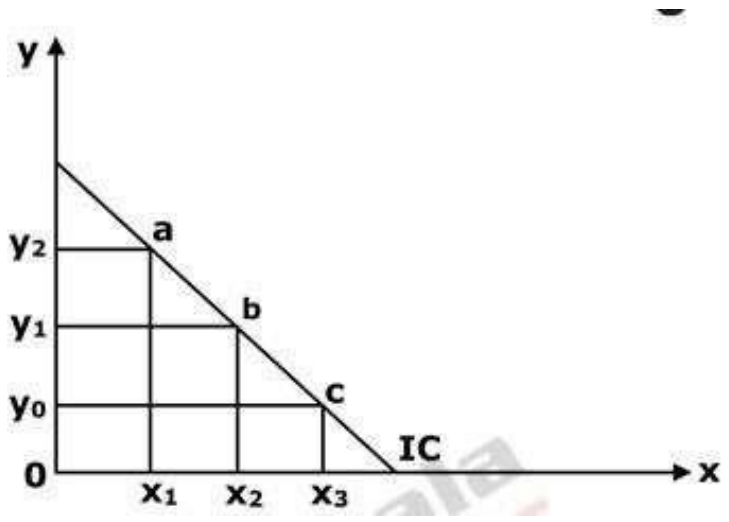
PROPERTIES OF INDIFFERENCE CURVE

- IC's have a negative slope because of the diminishing MRS.
- IC's never intersect with each other else they will break the assumption of transitivity.
- IC is convex to origin for the imperfect substitute goods.
- A higher IC represents higher level of satisfaction.

DIFFERENT SHAPES OF INDIFFERENCE CURVE

- Substitute goods: Substitute goods are those goods which give the same utility to the consumers i.e. they can be substituted in place of their related good as they posses almost same characteristics like their related good. In other words, two goods are considered as substitute goods for one another when the utility derived from both of them is same. For instance wheat and rice, coke and pepsi etc. Therefore the MRS between substitute goods is constant as $\Delta y=\Delta x$, so MRS =

  $\Delta y/\Delta x$. in this case we can write MRS = $\Delta x/\Delta x$ (because $\Delta y=\Delta x$), therefore, MRS = 1. Graphically



Source: https://epgp.inflibnet.ac.in

Figure 6: IC of substitute goods

- Complimentary goods: Complimentary goods are those goods which are consumed in fixed proportions like pen and refill, left socks and right socks, car and petrol etc. In such a case if we even increase the quantity of one good the quantity of other good does not increase, hence, MRS = 0.

  Symbolically, MRS = $\Delta y = \Delta x$

  But in such a case $\Delta y = 0$, $\Delta x = 1$

  Thus, MRS = 0/1 = 0

  Graphically,

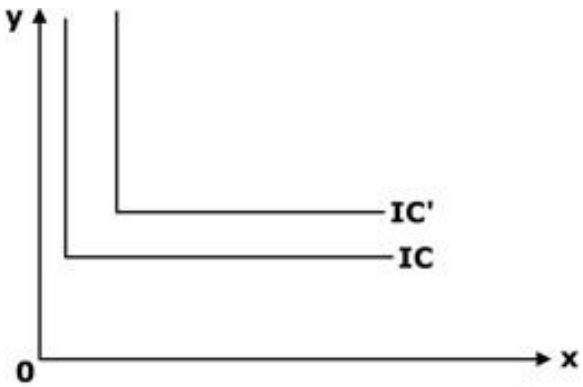


Figure 7: IC of Complimentary goods

- One good is necessary: In case of one good is a necessary good without which we cannot survive like water, then in such a case the shape of the IC would be like follows

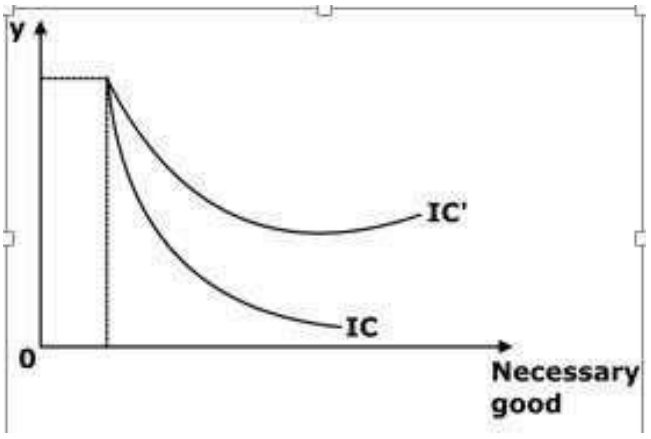


Figure 8: IC of necessary goods

- In case of a BAD good: A bad good is defined as a good whose excess is bad for the consumption i.e. if the consumer will consume more of it then his utility would decrease, for instance cigarette (for a mediocre or non smoker).

  Insuch a case the shape of the IC is as follows:

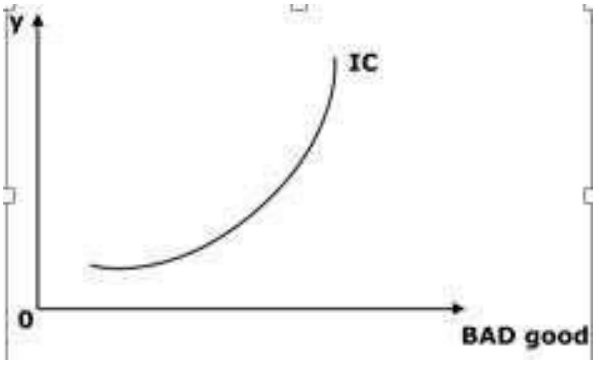


Figure 9

Hint: in order to draw an IC always find out the worst point of both the goods and then bend the IC towards that common worst point. As in, in this case, the worst of y is 0 however the worst of BAD is the excess, therefore the IC bend away from 0 towards x axis.

- In case of a neutral good: Neutral goods are those goods which neither yield utility nor disutility to its consumers. Thus a consumer is indifferent to its consumption. In such a case the IC will be a straight line as shown below.

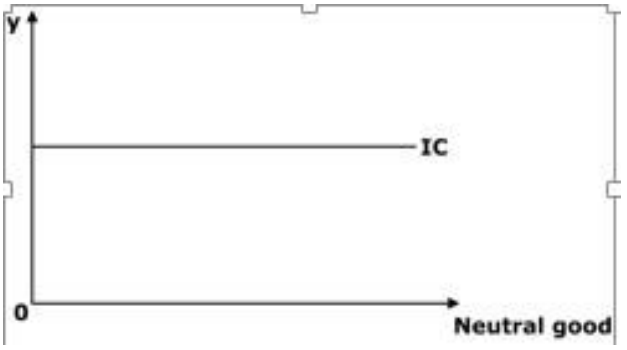


Figure 10

- A good turning bad after some point: If a good is a normal good till some point and then after that point it becomes bad for its consumer i.e. it provides utility to

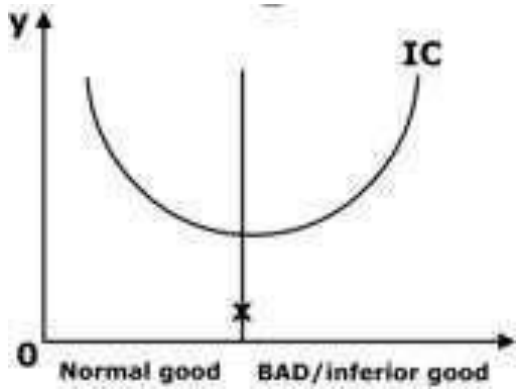


Figure 11

its consumer till some point but after that point if the consumer still consumes it then it starts providing disutility to him then in such a case the IC will be like:

CASE STUDIES:

Case Study 1: Ravi's Ice-Cream Experience and the Law of Diminishing Marginal Utility

Ravi, a college student, visits an ice-cream parlor on a hot summer day. Extremely thirsty and craving sugar, he consumes one ice-cream and experiences immense satisfaction. Encouraged by this, he eats a second one, which also satisfies him but not as much as the first. He continues to eat more ice-cream, and by the fifth serving, he no longer enjoys it. On attempting a sixth, he feels uneasy and slightly ill. Ravi's experience is a real-world illustration of the Law of Diminishing Marginal Utility, where satisfaction decreases with each additional unit consumed, eventually reaching a point of zero and even becoming negative.

Case Study 2: Sneha's Budget Allocation and the Law of Equi-Marginal Utility

Sneha has ₹100 to spend on two goods: chocolates and juice. She gains different levels of satisfaction (measured in utils) from each rupee spent on these items. Initially, the utility per rupee for chocolates is higher, so she spends more on them. As she keeps buying chocolates, their marginal utility decreases. At one point, the utility per rupee from chocolates equals that from juice. To maximize her total satisfaction, Sneha starts spending alternately on juice and chocolates,

maintaining equal marginal utility per rupee across both goods. This scenario is a practical application of the Law of Equi-Marginal Utility, highlighting optimal resource allocation.

5.8 LET US SUM UP

The concept of utility and understood that utility analysis is the core concept of the theory of consumer's behaviour.

- Cardinal utility theory which explain the consumer's demand for a product and derives the law of demand, and states the price of the product and quantity demanded are inversely related.
- Marginal rate of substitution is defined as the rate at which a consumer substitutes one good for the other obtaining the same level of satisfaction.
- Various assumptions of the cardinal utility analysis were explained.
- The entire chapter has explained how the utility can be measured in terms of indifference curve.

KEYWORDS:

1. **Utility**: The satisfaction or pleasure derived by a consumer from consuming a good or service.
2. **Indifference Curve**: A graph showing different combinations of two goods that provide equal satisfaction to the consumer.
3. **Marginal Utility**: The additional utility gained from consuming one more unit of a good.
4. **Cardinal Utility**: A measurable form of utility expressed in numerical terms (utils).
5. **Marginal Rate of Substitution (MRS)**: The rate at which a consumer is willing to substitute one good for another while maintaining the same level of satisfaction.

5.9 ASSIGNMENT QUESTIONS

1. Explain the assumptions and implications of the Law of Diminishing Marginal Utility.
2. Discuss the Law of Equi-Marginal Utility using a real-life example.
3. Compare and contrast Cardinal Utility and Ordinal Utility approaches to consumer behaviour.
4. What is an Indifference Curve? Explain its properties and different possible shapes.
5. Define Marginal Rate of Substitution. Why does it diminish along the indifference curve?

OBJECTIVE TYPE QUESTIONS

1. Utility refers to the _____ derived from consuming a good.

2. Marginal utility is the _____ in total utility from consuming an extra unit.
3. According to the Law of Diminishing Marginal Utility, the MU _____ with each additional unit consumed.
4. Total utility is the _____ of satisfaction obtained from all units consumed.
5. Cardinal utility is measured in imaginary units called _____.
6. Indifference curves are always _____ sloping due to diminishing MRS.
7. At the point of satiety, marginal utility becomes _____.
8. Law of Equi-Marginal Utility is also known as the Law of _____.
9. An indifference map shows different levels of _____.
10. Substitute goods have a _____ MRS.
11. Complementary goods are consumed in _____ proportions.
12. In the case of a bad good, the indifference curve bends towards the _____ axis.
13. The slope of an indifference curve is referred to as _____.
14. Indifference curves never _____ each other.
15. A higher indifference curve represents a _____ level of satisfaction.

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FURTHER READINGS

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- Principles of Microeconomics – N. Gregory Mankiw
- Managerial Economics – D. Salvatore
- Economics – Paul A. Samuelson and William D. Nordhaus
- Consumer Behaviour and Managerial Decision Making – Frank R. Kardes

UNIT-6: CONSUMER BEHAVIOUR AND ITS ECONOMIC IMPLICATION

STRUCTURE

6.1 Introduction

6.2 Theories of consumer Behaviour

6.3 Consumer Surplus

6.4 Economic Implications

6.5 Relation Between interest rate and household saving

6.6 Let Us Sum Up

LEARNING OUTCOMES

After studying this unit, you will be able to:

Learn about the consumer behaviour.

Learn about implication of consumer behaviour theory

BLOOM'S LEVEL OF LEARNING:

The level of Bloom's Taxonomy of learning that seems most relevant is the "Understanding" and "Application" levels. The content provided explains various consumer behavior theories, such as Cardinal and Ordinal Utility Approaches, as well as the concept of Consumer Surplus, demonstrating a foundational understanding of economic principles. The reader is expected to comprehend and explain these theories, showing the ability to interpret economic models like the indifference curve and supply-demand shifts. Additionally, the application of these theories to real-life situations such as the behaviour of Giffen goods and labour supply, including the decision-making process around savings and interest rates, implies that the learner is expected to apply these concepts to practical economic scenarios. The content focuses not only on knowledge recall but also on understanding how these economic concepts function in the real world, demonstrating a higher level of cognitive skill development as described in Bloom's Taxonomy.

6.1 INTRODUCTION

The behaviour of individual demand depends on consumer behaviour. Consumer behaviour is the study of consumer while engaged in the process of consumption. It tells us how a consumer with his limited resources purchases different varieties of goods and services in the market.

6.2 THEORIES OF CONSUMER BEHAVIOUR

The basic objective of a consumer is to get maximum satisfaction from consuming goods and services. This is possible when he reaches the position of equilibrium. It is necessary to find out how a consumer allocates his income or various goods so as to get maximum satisfaction or to reach equilibrium.

Various theories (or approaches) have been developed to explain the behaviour of consumers. There are three approaches to study the consumer behaviour. They are:

1. Cardinal utility approach
2. Ordinal utility approach
3. Revealed preference theory of Paul Samuelson.

The above both theories (Cardinal utility and Ordinal utility) has been explained in detail in Unit 5.

Cardinal Utility Approach (Cardinal Analysis)

The cardinal utility theory was developed by classical economists (Gossen of Germany, William Stanley Jevons of England, Leon Walras of France and Karl Menger of Austria). Neo-classical economists, particularly Alfred Marshal made significant refinements in the cardinal utility theory. Hence cardinal utility theory is also known as neo-classical utility theory or Marshallian utility theory.

The cardinal utility theory states that utility is measurable just as height, weight, length, temperature etc. According to cardinal utility theory utility is measurable cardinally or quantitatively. I means utility can be measured in cardinal numbers like 1, 2, 3 and so on. Boulding suggested another unit of measurement of utility. This unit of measurement is called 'utils'. Thus utility can be measured in utils. For example, an apple possesses 10 utils and an orange possesses 5 utils. Here the utility of an apple is twice that of an orange. In short, cardinal utility approach states that utility is measurable in cardinal numbers.

Defects or Limitations of Marshallian Utility Analysis (Cardinal Utility Approach) The cardinal utility approach has the following limitations or weaknesses:

1. Utility is a subjective and psychological concept. Hence, it cannot be measured cardinally.
2. The cardinal approach assumes that the utility depends upon that commodity alone (ie., utilities are independent). This assumption is not correct. In actual life, utility depends upon the availability of substitutes and complements.
3. The assumption that the marginal utility of money remains constant is wrong. As the consumer's money stock increases, the marginal utility of money decreases. So, money cannot be used as a measuring rod.
4. Money is not a correct and perfect measure of utility. This is because the value of money often changes.
5. The assumption that man is rational is not correct. No consumers compare the utility and disutility from each unit of a commodity while buying it. Further, due to ignorance or advertisement, the consumer may be forced to choose a wrong them.
6. The cardinal approach does not study income effect, substitution effect and price effect.
7. The approach fails to clarify the study of inferior and Giffen goods.
8. The cardinal theory is based on a number of unrealistic assumptions. In other words, the theory assumes too much but explains too little.

6.3 CONSUMER SURPLUS

The concept of Consumer Surplus stems from the demand curve. It was originally developed by the French engineer-economist A.J.Dupuit in 1844. He tried to measure Consumer surplus that would accrue to people as a result of the construction of a bridge across the river. But Marshall refined the concept in his book "Principles of Economics". Marshall developed the concept and gave definite shape to it. Recently, Prof. Boulding named it as "Buyer's Surplus". Today most economists believe that the concept of Consumer's Surplus is a product from the fertile brain of Marshall.

There are certain commodities which are highly useful but they are relatively cheap. Newspaper, salt, match box, post card etc. are some classic examples. For these commodities we are often prepared to pay a higher price than what we actually pay. Thus we get extra or surplus satisfaction over and above the price we pay. This is called Consumer's Surplus. For instance, a post card is priced 50 paise, but we are prepared to pay R. 1.50 for it rather than go without it.

Then if we actually pay 50 paise and purchase it we may say that we get a surplus satisfaction to the extent of Rs.1 (1.50-0.50). So Rs. 1 is termed as consumer's surplus by Marshall.

According to Marshall, "Consumer's Surplus is the excess of the price which we would be willing to pay rather than go without the thing over what we actually pay is the economic measure of this surplus satisfaction".

Prof. Samuelson defines it thus: "The gap between the total utility of a good and its total market value is called consumer's surplus".

In short,

Consumer's surplus = What we are prepared to pay - What we actually pay

Or

Total utility - Total amount spentor

Total utility - (Market price x Quantity purchased)

The concept of consumer's surplus can be explained with an illustration and a diagram. The following table shows how the consumer gets surplus satisfaction. The market price is Rs.2 per orange.

Table: 3

Orange (Number)	Marginal Utility (Price willing to pay)	Market Price(Amt. spent	Consumer's Surplus
1	6	2	4
2	5	2	3
3	4	2	2
4	3	2	1
5	2	2	0
	20	10	10

Consumer's Surplus = Total utility - Total amount spent

= 20 -10 = 10

It is clear that the consumer is willing to pay Rs. 6 for the first orange since he gets utility equal to Rs.6. But he actually pays Rs.2 for it because the price prevailing in the market is Rs.2. Therefore, he gets extra satisfaction of (6-2=4) Rs. 4 from the first orange. Similarly, we can calculated the consumer's surplus from the second, third, fourth and fifth oranges. The second orange brings a consumer surplus of Rs. 3 but it is less than the first one does. This is so because the marginal utility diminishes. As he buys more and more the marginal utility goes on falling and hence the consumer's surplus also diminishes.

The consumer is willing to pay in all Rs. 20 for all 5 oranges because he gets total utility of 20 from the five oranges. But since the market price is Rs. 2 per orange, the consumer pays the same price for all the five oranges. Thus, he pays (5x2) Rs.10. Therefore, the surplus satisfaction is equal to Total utility - (Market Price x Qty)=20 - (2x5) = 10.

The above illustration is shown in the following diagram.

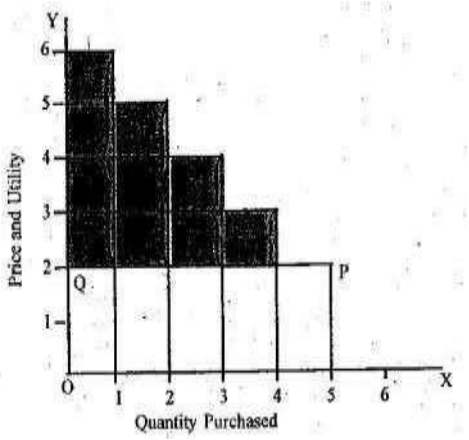


Figure 12

MP or OQ shows the market price per orange. It is same for all units (i.e Rs.2). OM is the quantity purchased when price is equal to marginal utility. The total amount paid is therefore OM x MP, i.e, OMPQ (total price paid for all five oranges). But total satisfaction derived will be equal to the total area OMPQ + the shaded area. Therefore, the shaded area represents consumer's surplus (4+3+2+1). For the fifth orange, there is no consumer surplus. This is because the marginal utility (price willing to pay) is equal to the actual price.

The concept of consumer's surplus is based on the law of DMU. As we go on purchasing more and more oranges, the marginal utility goes on diminishing. We stop purchasing when the

price we have to pay is equal to the marginal utility. The marginal utility of the earlier oranges will be more than the price. Therefore, the consumer gets more satisfaction than the price that he pays for them. This is how the consumer's surplus arises.

Assumptions of the Theory of Consumer Surplus

Marshall's concept of consumer surplus is based on the following assumptions:

1. Utility can be measured.
2. The marginal utility of money to consumer remains constant.
3. The commodity has no substitute.
4. The income, taste and fashion of consumer remains constant.
5. The utility of a commodity depends upon the quantity of that commodity alone. For example, the utility derived from Coca Cola depends only on its quantity. The quantity of Pepsi does not affect it.
6. This theory is applicable only if the law of DMU is valid.

Ordinal Utility Approach (Ordinal Analysis)

The modern economists have discarded the concept of cardinal utility. Instead, they have used the concept of ordinal utility for analyzing consumer behaviour. The ordinal utility approach to the theory of consumer behaviour is based on the idea that utility is not measurable in monetary terms. It is only comparable by ranking the level of satisfaction. This means it is always possible for a consumer to say whether a commodityis more or less or equally useful as compared to another. For example, a consumer may not be able to say that an ice cream gives5 utils and chocolate gives 2 utils. But he or she can always say whether chocolate gives more or less utility that ice cream. This assumption forms the basis of the ordinal theory of consumer behaviour. In short, the ordinal approach involves the use of ordinal numbers to measure the utility.

Concept of Ordinality: The numbers 1, 2, 3 etc. are cardinal numbers. The number 2 for example is twice the number 1, the number 9 is thrice the number 3 and so on. In contrast, the 1st, 2nd, 3rd etc. are ordinal numbers. Such numbers are ordered, or ranked. There is no way of knowing, just from the ranking, what the size relation of numbers is.

The 2nd may or may not be twice as big as the 1st.

Ordinal Utility Approach (Indifference Curve Analysis or Technique)

The modern economists believe that utility is subjective. It is a psychological concept. Hence it is not possible to measure utility quantitatively. The modern economists developed an alternative approach to analyse consumer behaviour. This alternative approach is Indifference curve Theory.

The 'Indifference Curve Technique' or the 'Ordinal Utility Analysis' was first introduced by English economist Edge worth in 1881. An Italian economist Pareto refined Edge worth's concept in 1906. It was further refined by Russian economist, Slustky in 1915. It was given perfection by Prof.J.R. Hicks and R.G.D. Allen of the Cambridge University in 1934. The most detailed presentation of the technique was made by Hicks in his book 'Value and Capital' published in 1939.

According to Indifference Curve Analysis, utility is not measurable in cardinal numbers. It is measurable ordinally. This means utility is measured as the level of satisfaction and not the amount of satisfaction. Consumers can only say whether a good or a combination of goods give him or her greater, less, or equal satisfaction (cannot say 'how much greater or less). One can simply rank the goods or combinations of goods in the order of preferences.

For example, there are two goods, say, apple and orange. Once can only say that apple is preferred to orange but, he or she cannot say by 'how much'. It may also happen that there can be some combinations of two or more goods that give the same level of satisfaction. This means the consumer is indifferent (no combination gives greater or less satisfaction i.e., all combinations give equal satisfaction).

6.4 ECONOMIC IMPLICATIONS

In this section we will discuss the basic implications of consumer Behaviour Theory. The major areas of application focused here are:

i) Identifying Giffen Goods and understanding how they violate the law of demand?

ii) How a labourer decides the number of hours he should work for, given the utility he derives from leisure and wage income,

iii) How to derive the supply curve of labour?

iv) How household savings responds to the interest rate?

6.4.1 DEMAND FOR GIFFEN GOODS

If an increase in the price of a good results in an increase in the quantity demanded, than the demand curve for such goods are upward sloping. This is due to the dominance of income effect over substitution effect. Thus, Giffen goods violate the basic law of demand. Following Figure shows the derivation of demand curve in the case of a Giffen goods

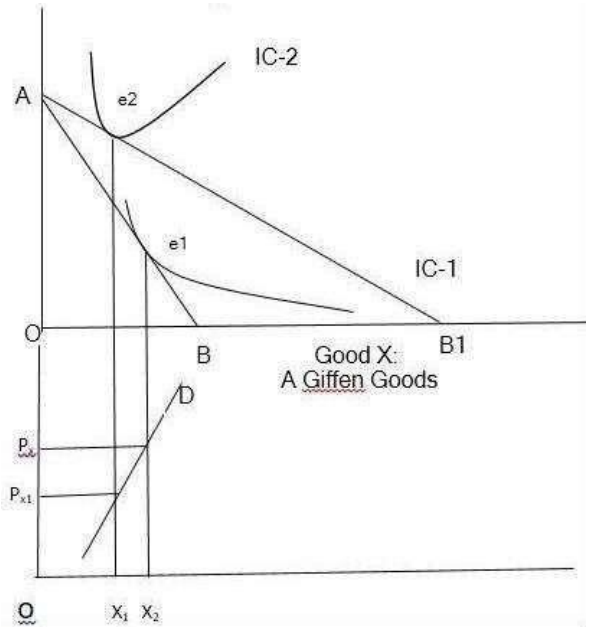


Figure 13

In above figure, the consumer is in equilibrium at e1 where IC-1 is tangential to the budgetline AB. You may notice what happens when price fall from Px to Px1.

As a result of a fall in the price of good X (a Geffen goods) from Px to P**x1** the budget line shift from AB to AB1. This is because given the income and with a fall in the price of X, it would now be possible to buy more of X, by spending entire money income on the purchase of X. Thus, the extra quantity of X can be purchased is BB1, using the entire income. You may note that the new equilibrium for the consumer is at the point where the budget line AB1 is tangential to a higher indifference curve IC-2. At equilibrium e1, when price of X was Px, the corresponding quantity demanded was OX2. However, with reduction in price from Px to Px1, the resultant new equilibrium shifts to the point e2. Corresponding to the new equilibrium e2 the quantity demand for X is

OX1. Cases like this are known as ***"Giffen's Paradox***" where quantity demanded of X falls as a result of the fall in the price of X.

Giffen Goods & Empirical Evidence

Evidence of Giffen goods are rare, though not non-existent. A study by Jensen and Miller (published in American Economic Review in 2008), provides evidence for the existence of Giffen goods. During their five months field experiment in the Chinese province of Hunan, food vouchers to subsidize the purchase of rice- the main diets in the locality were provided to some randomly selected households. Further, they conducted surveys to measure how consumption of rice responded to changes in the price. Their study finds evidence of poor households exhibiting Giffen behaviour. Lowering the price of rice with the subsidy voucher caused households to reduce their consumption of rice, and removing the subsidy had the opposite effect.

6.4.2 WAGES AND LABOUR SUPPLY

Workers get utility from leisure as well as from their work by earning income. Accordingly, they allocate 24 hours available to them between work and leisure to maximise utility.

Let's take an imaginary worker who decides to work for "h" hours a day. This implies he has taken a simultaneous decision to enjoy the balance h* hours as leisure. Since h+h*=24, the number of hours (H) to be allocated for work would be h which is equal to (24-h*). Given the wage rate per hour w, he will earn a wage income W which equals (24-h*)w = 24w – wh*. Mathematically, the maximum possible wage that he can earn is 24w, if he allocates no time for leisure (h*=0), given the wage rate w.

Similarly, if his decision is not to work at all, he will earn nothing (i.e. his wage income will be zero) and he will have the entire 24 hours available as leisure. At h* =24, the wage earned would be zero. Labour faces a trade-off between work and leisure to arrive at an optimal combination i.e. how much hours to work and how much time to be allocated for leisure.

In diagram below the indifference curve represents various combinations of wage and hours of leisure giving same level of satisfaction.

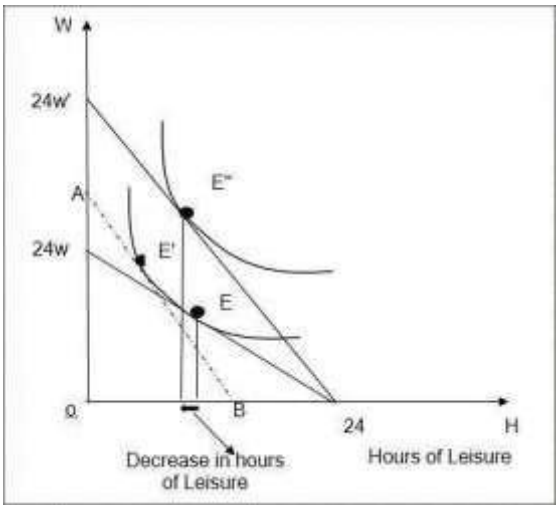


Figure 14

Suppose the wage rate per hour increases from w to w'. This will have no impact on H intercept but will shift the W intercept upward from 24w to 24w'. This would unable the labourer to move to a higher indifference curve i.e. from income-leasure combination E to E". AB line is parallel to W= (24-h*) w' and tangent to the initial Indifference curve at point E'. AB line is drawn after hypothetically withdrawing the wage income to the extent that the labourer is back to his initial indifference curve, enjoying the same level of utility as he was before the change in wage. In figure 14, it can be seen that, an increase in wage rate induces the labourer to work more number of hours as leasure becomes more expensive in comparison to work, thus resulting in an increase in the supply of labour hours. Here the movement from E to E' reflects substitution effect and the movement from E' to E" represents income effect.

According to Hal R. Varian, “For most people, the supply of labour would drop when their money income increased. In other words, leisure is probably a normal good for most people: when their money income rises. People choose to consume more leisure”. By considering different wage rates and labour supply corresponding to these wage rates, it is possible to generate the labour supply curve for our representative labourer. A case where a higher wage induces a labourer to enjoy more leisure and work less, thereby exhibiting a labour-supply curve sloping “backward.” is presented in figure

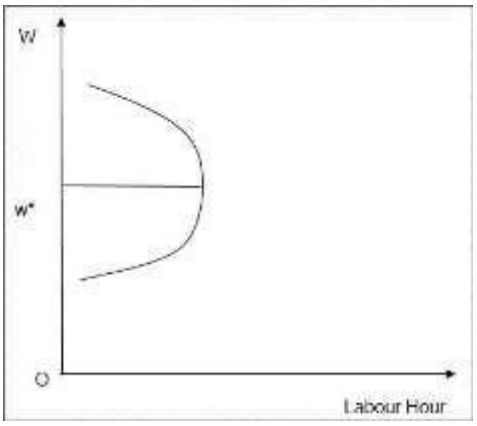


Figure 15

It can be noted from the figure that the supply of labour hour is increasing along with the wage rate, till the wage rate reaches w*. Any increase in wage rate beyond w*is lowering supply of labour hour. This results in a backward bending labour supply curve. Thus, the labour supply curve becomes downward slopping for wage rate above* indicating that lesser and lesser labour hour is supplied as wage income increases beyond w*.

In fact the choice of more leisure for wage rate beyond w* is an outcome of people earning more per hour and therefore preferring to enjoy more leisure in the form of going for outings on vacations, spending more time with family or in clubs, watching moves, paying visits more frequently to parks, taking dinner at restaurant during weekends, perusing their hobby and the like.

Minimum Wage Legislation: Rational and Implications

The rationale behind minimum wage legislation undertaken by various government especially in underdeveloped countries, is its inherent potential to pull up wages in the low-paid and informal sectors by enhancing the bargaining power of the workers. Minimum wage legislation results in generally prescribes a wage rate which is higher than the market determined wage rate. Under such a situation the sellers (in this case the labourer) cannot sell what they want to sell (i.e. number of hours of labour), but the buyer can buy what they want to. This in turn would result in a situation of excess supply labour over its demand.

6.5 RELATION BETWEEN INTEREST RATE AND HOUSEHOLD SAVINGS

An important decision that every person need to take is how much income to consume today and how much to save for future consumption. The decision is in no way straight forward and is sensitive to several factors including the following.

i) The interest rate savings will earn ii) The age group the consumer belongs to.

For simplicity we have classified the age group into two categories- young and old. In period 1, our representative consumer is assumed to be young, working and earning an income say R1. Period 2 is his post retirement period when he is old. His consumption in period 2 is financed by his saving in period 1 plus the interest that he would earn from his savings. Savings allows the consumer to postpone his consumption from Period-1 to Period 2. Accordingly, the two extremes of his budget line would be:

a) Zero savings in period-1 implying that he consumes the entire R1 when he is young and leaves nothing to consume when he is old.
b) Saving the entire amount earned for the consumption during his old age (when retired). In this case, he would have at his disposal an amount of R1 plus the interest earned on R1. Thus the amount that he would be having for consumption in period-2 is (1+i)R1.

The budget line that joins R1 and (1+i) R1 in figure, provides various combinations of income allocated between the two time periods. Income allocated for Period 1 is to be consumed in period 1. Income allocated for Period-2 is saved during period 1 and therefore includes the interest component.

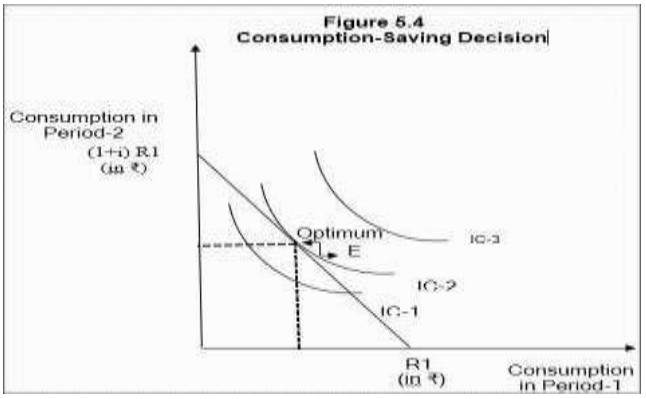


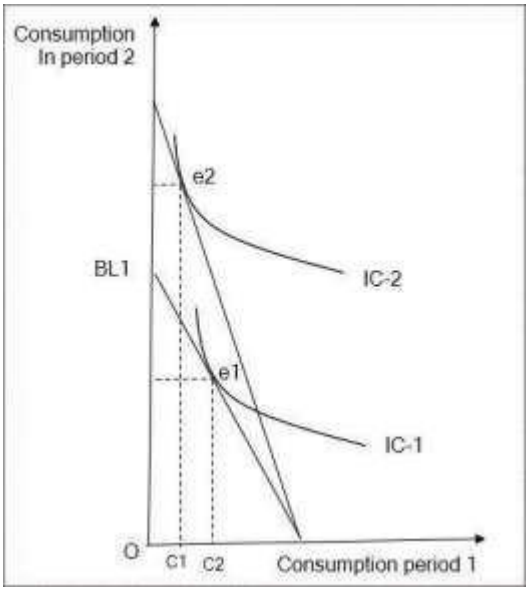
Figure 16

It can be seen from figure 16 that the optimum allocation of amount to be spend for consumption in the two periods is the point E where the indifference curve IC-2 intersect the budget line.

Suppose the interest rate increases from i to i'. With the higher interest rate i', the consumer can now spend more amount in period-2 (when he is old) for every rupee of consumption that he gives up in period-1 (i.e when he is young).Thus, a higher interest rate is likely to result in a higher savings as it induces people to trade-off their current consumption in favour of future

consumption. However, consumption when young due to the increase in the interest rate may either increase or decrease depending on the relative strength of income and substitution effect. Figure 17 and 18 show two different ways the consumer might respond to an increase inthe interest rate. It may be noted that, in both cases, consumption in period 2 when the consumer is old & retire rises. However, in figure 17 it could be seen that an increase in the interest rate from i to i' reduces consumption when young, thereby resulting in an increase in savings. In figure 18 increase in the interest rate from i to i' induces him to consume more when young,

Figure 17: Higher Interest Rate: Increased Savings & Reduced Consumption in



Period-1

From figure 17, it can be observed that, an Increase in the interest rate from i to i' rotates the budget constraint line outward enhancing the feasible region. This allows our representative consumer to move to point e2 – the tangential point of the highest possible Indifference curve IC-2 attainable given the new budget line BL2. It may be noted that, in the new equilibrium e2 is an optimal solution where he consumes more during his old age but reduces his consumption during young age. The reduction in consumption in period 1 is from OC2 to OC1. As the substitution effect of a higher interest rate is greater than the income effect, he saves more in period-1. With the rise in interest rate consumption in period 2 becomes less costly relative to consumption in period-1. Therefore, the substitution effect induces him to consume more when old and less whenyoung leading to increase in the savings in period-1.

Similarly with the rise in the interest rate he would be better off and therefore, moves to a higher indifference curve. Assuming, the consumption in both periods consists of normal goods, he tends to allocate this increase in well-being to enjoy higher consumption in both periods as shown in figure 18. Thus, in line with the consumer choice theory "an increase in the interest rate could either encourage or discourage savings" depending on the relative strength of the income effect and substitution effect.

Figure 18:

Higher Interest Rate: Reduced Savings and Increased Consumption in Period-1

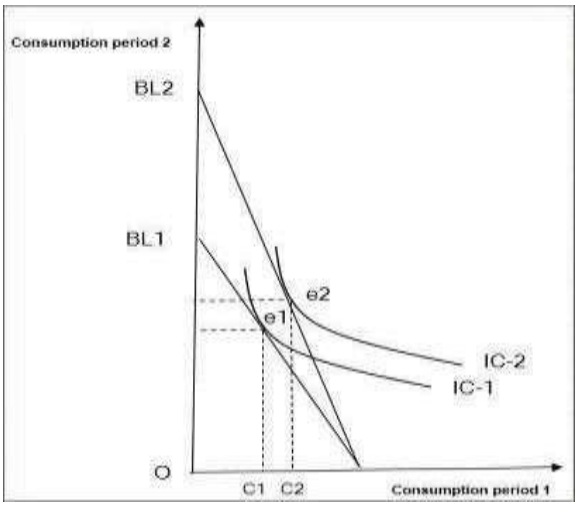


Figure 18 shows that the consumer attends a new equilibrium e2 as a result of an increase in interest rate. Thus, the income effect of the increase in interest rate induces him to save less and allocate his increased income between the two periods to increase consumption.

CASE STUDIES

Case Study 1: Giffen Goods in Rural China

Context:

In the Chinese province of Hunan, rice forms the staple diet of low-income households. In a controlled field experiment by economists Jensen and Miller (2008), food subsidies were provided to selected households to analyze the consumer response.

Observation:

Instead of increasing their rice consumption with reduced prices, some households actually decreased their rice consumption. The subsidized price allowed them to afford other desirable goods, leading to a decline in demand for rice despite the price drop.

Economic Insight:

This anomaly is consistent with the Giffen Goods concept, where the income effect outweighs the substitution effect. The study proves that in rare but specific cases, consumer behaviour can contradict the fundamental Law of Demand.

Case Study 2: Labour Supply Response to Wage Change

Context:

A mid-level software professional in an urban area earns hourly wages and chooses between leisure and work hours. When his wage rate was ₹300/hour, he worked 10 hours a day. Upon receiving a raise to ₹500/hour, he began working only 8 hours a day.

Observation:

Despite higher income, the person preferred to enjoy more leisure (such as travel or hobbies) rather than working additional hours.

Economic Insight:

This represents the backward-bending labour supply curve. The income effect dominated the substitution effect, leading to fewer working hours as the individual's preference shifted towards more leisure over additional income.

6.6 LET US SUM UP

The basic objective of a consumer is to get maximum satisfaction from consuming goods and services. This is possible when he reaches the position of equilibrium. It is necessary to find out how a consumer allocates his income or various goods so as to get maximum satisfaction or to reach equilibrium.

Various theories (or approaches) have been developed to explain the behaviour of consumers. There are three approaches to study the consumer behaviour. They are:

1. Cardinal utility approach
2. Ordinal utility approach

Revealed preference theory of Paul Samuelson

The major areas of application focused here are:

i) Identifying Giffen Goods and understanding how they violates the law of demand? ii) How a labourer decides the number of hours he should work for, given the utility he derives from leisure and wage income,

iii) How to derive the supply curve of labour?

iv) How household savings responds to the interest rate?

KEYWORDS

1. **Consumer Surplus:**
   The difference between the amount a consumer is willing to pay and what they actually pay for a good.
2. **Giffen Goods:**
   Inferior goods for which demand increases as the price increases, due to a strong negative income effect.
3. **Indifference Curve:**
   A graph showing different combinations of two goods that provide equal satisfaction to the consumer.
4. **Labour Supply Curve:**
   A curve that represents the relationship between wage rates and the quantity of labour a worker is willing to offer.
5. **Interest Rate Effect on Savings:**
   A concept explaining how variations in interest rates influence consumer decisions to save or spend current income.

6.7 ASSIGNMENT QUESTIONS

1. Explain the difference between Cardinal and Ordinal utility approaches with examples.
2. Describe the concept of consumer surplus and explain its economic significance using a diagram.
3. Discuss how the labour supply curve can bend backward and under what conditions.
4. Analyze the implications of an increase in interest rates on household savings with the help of diagrams.
5. What are the major limitations of the Cardinal Utility Theory? How does the Ordinal Utility Theory address these?

OBJECTIVE TYPE QUESTIONS

1. Consumer behaviour studies how individuals allocate their _____ among various goods and services.
2. The theory of consumer surplus was refined by _____.

3. In Cardinal Utility theory, utility is measured in hypothetical units called _____.
4. Ordinal utility suggests utility can be ranked but not _____.
5. The Law of Diminishing Marginal Utility is the basis for _____ surplus.
6. Giffen goods violate the basic _____ of demand.
7. A backward-bending labour supply curve is explained by the dominance of the _____ effect.
8. The point of consumer equilibrium occurs when the budget line is _____ to the indifference curve.
9. The income that is postponed from present to future is known as _____.
10. Indifference curves are always _____ to the origin.
11. Revealed Preference Theory was proposed by _____.
12. An increase in wage may result in more leisure if the _____ effect dominates.
13. Minimum wage laws are enacted to enhance the _____ power of workers.
14. Utility derived from leisure and wage income determines a labourer's _____.
15. Higher interest rates can lead to either more or less saving depending on the _____ of income and substitution effects.

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FURTHER READINGS

- Microeconomic Theory by Andreu Mas-Colell
- Economics by Paul A. Samuelson and William D. Nordhaus
- Consumer Behavior by Leon G. Schiffman and Leslie Lazar Kanuk
- The Economics of Welfare by A.C. Pigou
- Behavioral Economics by Edward Cartwright

UNIT-7: MEANING AND CONCEPT OF CONSUMER EQUILIBRIUM

STRUCTURE

7.1 Introduction

7.2 Budget Line

7.3 Change in Income and Price and the Budget line

7.4 Consumer Equilibrium

7.5 Corner Solutions

7.5.1 Convex Indifference Curves

7.5.2 Concave Indifference Curves

7.5.3 Case of Perfect Subsitutes

7.5.4 Case of Perfect Complements

7.6 Let Us Sum Up

7.7 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- The concept of Budget Line
- The effects of a change in income and prices on the budget line
- The concept of consumer equilibrium
- The meaning of Corner solutions
- Consumer equilibrium in case of concave indifference curves, perfect complements and perfect substitutes

BLOOM'S LEVEL OF LEARNING:

Based on the content outlined, the level of Bloom's Taxonomy being addressed can be categorized as Application, Analysis, and Synthesis. The topics engage students in applying theoretical concepts such as the budget line, consumer equilibrium, and changes in income and prices to real-world scenarios. Additionally, the analysis of consumer behavior through marginal utility and indifference curves demonstrates a deeper understanding, where students are required to analyze the effects of different market conditions on consumer choices. The concepts of corner solutions and different types of indifference curves (e.g., convex, concave, perfect substitutes, and perfect complements) call for a synthesis of knowledge to evaluate and predict outcomes in various situations. Thus, learners are expected to not only recall and understand economic theories but also apply them in problem-solving contexts, analyse different market conditions, and synthesize this information to derive conclusions about consumer equilibrium. This level involves higher-order thinking, as it necessitates both practical and theoretical comprehension.

7.1 INTRODUCTION

It is very necessary to understand the concept of budget constraint for understanding the theory of consumer equilibrium. The concept of Indifference curves discussed in earlier modules provide a hypothetical ranking of various commodity combinations. However, a consumer cannot determine the optimal combinations of goods on the basis of utility or indifference curves alone. You may recall from our discussion in the previous unit that a higher indifference curve indicates a higher satisfaction level compared to a lower indifference curve. Therefore, to maximize satisfaction the consumer will always try to reach the highest possible indifference curve; but a higher indifference curve can be achieved only by consuming a *greater* quantity of goods.

You may recall from our discussion in the previous unit that a higher indifference curve indicates a higher satisfaction level compared to a lower indifference curve. Therefore, to maximize satisfaction the consumer will always try to reach the highest possible indifference curve; but a higher indifference curve can be achieved only by consuming a *greater* quantity of goods.

In order to buy more of a commodity the consumer will face two constraints: first, his money income and second, the prices of the commodities. Thus the extent to which the consumer can maximize his satisfaction crucially depends on these two factors. These budget constraints are generally represented by budget lines.

The tastes and preferences (represented by indifference map) and the budget constraint (represented by the budget line) of the consumer are brought together to completely characterize the consumer's

choices i.e. to determine what quantities to buy of each good to maximize satisfaction. The consumer will be in equilibrium when his satisfaction is maximized subject to his budget constraint.

7.2 BUDGET LINE

In economic theory, it is generally assumed that a consumer chooses the best bundle of commodities he can afford. For this the consumer needs to factor in the budget constraints that he faces due to his limited income. In a two-commodity world, the budget line shows all the combinations of the two commodities which the consumer can buy, given his money income and the prices of the two commodities. The budget line also represents a trade-off between the two goods; i.e., the rate at which one good can be exchanged for the other in the market.

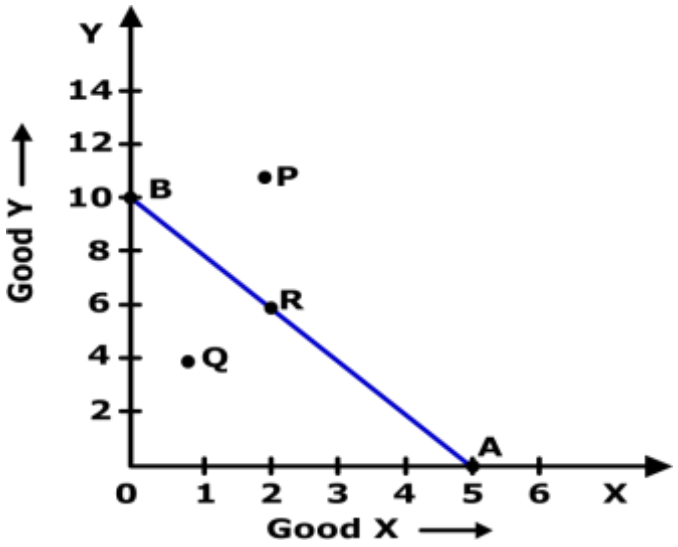


Figure 1a: A Budget line

Figure 1a shos a graphical representation of a budget line for two commodities X and Y. Let us assume a consumer has an income of Rs. 100 to spend on the two commodities. Let the market price of good X be Rs. 20 and market price of good Y be Rs. 10. In this situation the consumer can buy 5 units (OA) of commodity X if he spends his entire income on commodity X. On the other hand, if the consumer spends all his income on good Y he can purchase 10 units (OB) of Y.

The consumer can also choose any combination on the line joining A and B (e.g., point R in Figure 1a), where he spends his income partly on commodity X and partly on commodity Y. The straight line AB is called the budget line or price line. *The budget line shows all combinations of goods X and Y that the consumer can buy if he spends all his given money income on the two goods, at their given prices.*

For example, the consumer can buy a combination of 8Y and 1X, or 6Y and 2X, or 4Y and 3X, or 2Y and 4X; given his income of Rs 100 and prices of X and Y being Rs. 20 and Rs. 10 respectively. You should check that to acquire each of these commodity combinations, the entire money income of Rs.100 has to be spent by the consumer.

The Budget line also indicates that any combination that lies outside the budget line (e.g., point P in Figure 1a) will be beyond the reach of the consumer. While, to acquire any combination lying below this line (e.g., point Q in Figure 1a), his income will not be fully spent. Thus, under the assumption that all of the consumer's income is spent on the two goods at their given prices, the consumer *must* choose from all those combinations that lie on the budget line.

To understand the properties of the budget line, we must consider the budget equation of the consumer; algebraically, a budget line can be expressed as follows:

$Pxx + Pyy = M$ ------- I

Where Px and Py denote the prices; and 'x' and 'y' represent the quantities of commodities X and Y respectively and M stands for money income. In equation I, the term on the left represents the expenditure (given by price times quantity) on the two goods while the right hand side denotes the income. Thus, this equation of the budget line gives us a set of combinations of the two goods that can be purchased by spending the entire money income of the consumer. In case of ncommodities the budget equation is be given by:

$P1x1 + P2x2 + P3x3 + \ldots\ldots\ldots + Pnxn = M$ ----------II

where P1, P2, P3,…..,PN represent the prices; x1, x2, x3,....., xn are the quantities of the goods x1,
X2, X3,… ..... , Xn respectively.

Solving (I) for y we get the following alternative form of the budget equation, expressed as the equation of a straight line:

$$Y = \frac{M}{Py} - \frac{Px}{Py} X \quad \text{--------} \quad \text{III}$$

The first term of equation III i.e. M/Py shows the amount of commodity Y that can be purchased when x = 0 (quantity of good X purchased is zero). In other words, M/Py is the vertical intercept

of the budget line (figure 1b). The second term of the equation has a coefficient equal to –Px/Py which represents the slope of the budget line. Further, the negative sign of the term indicates that the budget line is downward sloping.

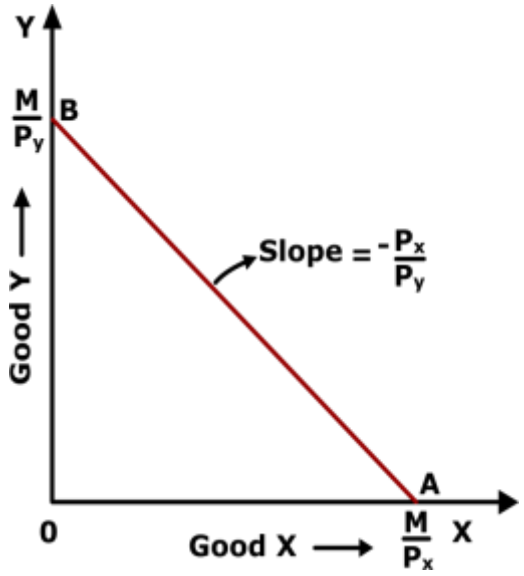


Figure 1b: A Budget line

As shown in the Figure 1b, the slope of the budget line is equal to the negative of the ratio of the prices of the two commodities. It measures the rate of substitution of commodity X for commodity Y i.e. the rate at which the two commodities can be substituted for each other, keeping total expenditure on the two goods constant (and equal to M).

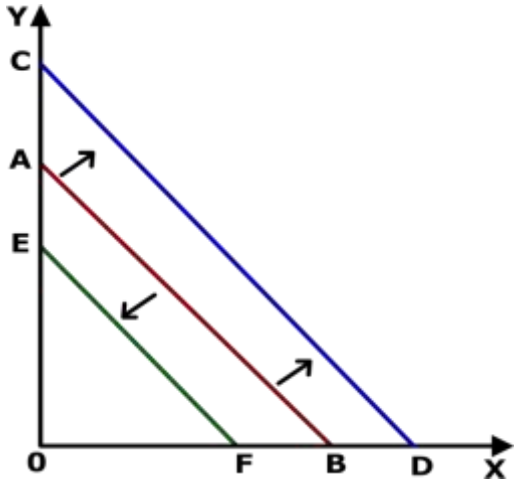
Using equation I, it is now possible to define the budget space. The budget space refers to the set of all combinations of the two goods for which the total expenditure on good X and good Y do not exceed the given money income. Algebraically, this can be expressed as the following inequality:

$$PxX + PyY \leq M \qquad \text{----------} \qquad IV$$

7.3 CHANGE IN INCOME AND PRICES AND THE BUDGET LINE

Changes in Income

Let us first consider a change in the total income of the consumer. With changes in



income the set of affordable bundles also changes. If the consumer's income increases while the prices of both goods remain unchanged, the budget line shifts parallelly upward to CD (Figure 2). This is because with increased income the consumer is now able to purchase proportionally larger quantities of both commodities than before, by spending all the income. On the other hand, if income of the consumer decreases, with price of both goods remaining unchanged, the budget line shifts downward to EF **Figure 2: Effect of a change in Income**

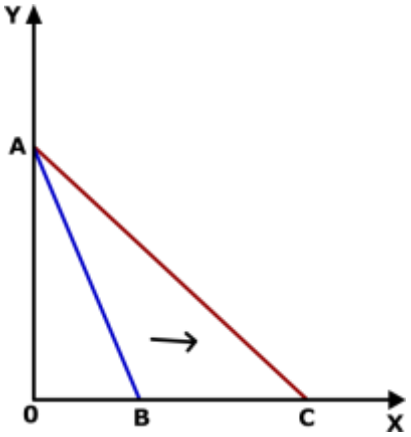
This can be easily understood from equation III. An increase in income will increase the vertical intercept (M/Py), leaving the slope of the line AB (-Px/Py) unchanged. Thus, this increase in income will result in a parallel outward shift of the budget line to CD.

Similarly, a decrease in income will shift the budget line parallelly inward to EF.

Change in Prices

Let us now consider the case of changes in prices of the goods. Suppose price of commodity X falls, income of the consumer and price of good Y remaining unaltered. With a lower price of X, the consumer is now able to purchase more of X and the same quantity of Y with his given income. As a result, the budget line will rotate from AB to

AC leaving the vertical intercept unchanged at OA (Figure 3a). The vertical intercept



will not change because a consumer who consumes only Y will not be affected by any change in the price of X. According to equation III,a fall in Px will decrease the value of Px/Py (slope of the budget line) making the budget line flatter, without changing the vertical intercept. Similarly, a decrease in Py will make the budget line steeper and increase the value of vertical intercept M/Py, with the horizontal interceptremaining unchanged (Figure 3b).

Figure 3a: Effect of a fall in Price of X

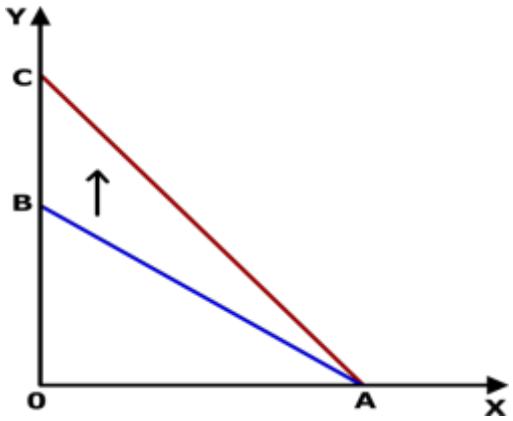


Figure 3b: Effect of a fall in Price of Y

Now, consider a case in which the prices of X and Y and income, all double. In this case the priceratio and vertical intercept both will remain unchanged (as -2Px/2Py = Px/Py and 2M/2Py = M/Py) leaving the budget line unchanged. So budget line will be unaffected in the situations in which the all prices and income change proportionally.

7.4 CONSUMER EQUILIBRIUM

Let us assume that a consumer is consuming only two goods X1 and X2. The utility which she receives from consuming X1 and X2 is given by the utility function U=f (X1, X2) and satisfies the property of eventual diminishing marginal utility. The consumer has a given money income to be spent on these two goods during the period we are analysing her behaviour. She cannot influence the prices, P1 and P2,, of the goods, through her own action. Prices are given as parameters in decision-making (consumption) as this consumer is one of the numerous consumers demanding X1 and X2. Thus, she has no market power. Since she is required to spend her entire income on X1 and X2, the budget equation is given as,

$$M = P1X1 + P2X2$$

Where M is her nominal income. Since the consumer is a utility maximiser, her consumption problem can be formulated as follows:

Maximise

$$U = f(X1, X2)$$

subject to the budget constraint

$$M = P1X1 + P2X2$$

By using the ***Marshallian Equi-Marginal Principle*** (which is based on the Lagrange

Multiplier Technique), we get the equilibrium condition,

$$MU1/P1 = MU2/P2 = \Gamma$$

This is the first order condition (necessary) for achieving equilibrium. The second order condition (sufficiency) of equilibrium is given by the law of eventual diminishing utility. The second order condition will be automatically fulfilled so long as the marginal utility schedules for each good MU_1 and MU_2 are both downward sloping. It must be noted that whenever a consumer maximises utility, equilibrium is said to be attained. What the equilibrium condition says is that to maximise total utility [$U = f(X_1,$

$X_2)$] an individual consumer must equalise the ratio of marginal utility to price for each and every good andwhich in turn must be equal to constant marginal utility of money. In other words, to obtain maximum total utility a rational consumer must equalise the marginal utility per rupee of expenditure on each and every line of expenditure (that is, on each and every good).

This relationship represents the consumer's equilibrium condition. A consumer attains equilibrium when she maximises total utility from consuming X_1 and X_2. The equilibrium condition can also be stated in an alternative form:

$$MUI / MU2 = P1 / P2 \quad ...(1)$$

The ratio of marginal utilities of goods X_1 and X_2 must equal the price ratio of the same two goods. This in turn, must equal marginal utility of money, which is constant by assumption.

Condition (1) above is the famous ***Marshallian law of the equi-marginal utility***. It can be shown that if the ratio of marginal utility of the two goods is not equal to the price ratio, then without spending more in the aggregate, just by re-allocating the given amount of money income as between the two goods X_1 and X2, the consumer can increase her total utility from consumption.

The combination of goods which provide maximum satisfaction must lie on the budget line. This is because at any point (like P) below the budget line AB the consumer will not spend all his income (Figure 4). This unspent income can then be used to increase his satisfaction which would also bring the consumer on the budget line AB. Also the consumption bundles above the budget line AB are not affordable to the consumer. Thus we are left with the combinations of goods on the budget line to determine the optimal choice.

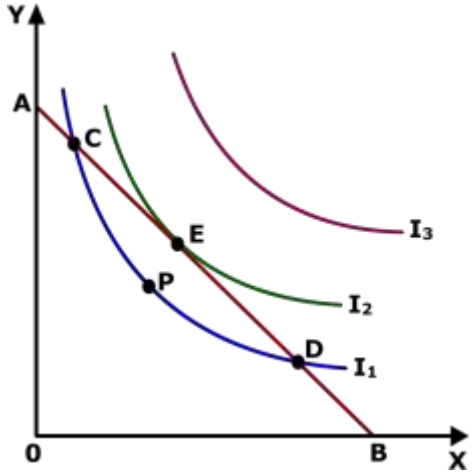


Figure 4: Consumer's Equilibrium

Let us consider the following diagram to determine the consumer's most preferred combination of commodities. The budget line AB and the indifference map (shown by

I1, I2 and I3) of a consumer are shown in Figure 4. In order to reach the equilibrium, the consumer will try to maximize his satisfaction subject to the given budget constraint. Out of the three indifference curves, I3 provides highest level of satisfaction to the consumer since it is the outermost indifference curve and I1 provides the least amount of satisfaction. However, combinations on the indifference curve I3 are not affordable for the consumer as I3 lies beyond the budget line. As we have already discussed, the consumer equilibrium must lie on the budget line. The consumer will choose that combination which lies on the highest possible indifference curve.

Now consider the points C, D and E on the budget line AB of the consumer. Point C and

D will not provide the highest possible satisfaction since these points lie on the lowest indifference curveI1 and the consumer can move to the higher indifference curve I2 by choosing point E on the budget line AB. Thus, given the money income the combination E yields greatest amount of satisfaction to the consumer. All other combinations on the budget line AB yield lower satisfaction levels as they lie on a lower indifference curve.

You may note that the combination which maximizes the consumer's satisfaction lies on the highest indifference curve that touches the budget line.

We conclude therefore that with given money income and prices (shown by the budget line AB) the consumer will maximize his satisfaction at a point (equilibrium point E) at which the budget line is

tangent to the indifference curve (I2). Further, the budget line can be tangent to only one indifference curve. At this tangency point E, the slope of the budget line AB is equal to the slope of the indifference curve I2. We also know that the slope of the indifference curve indicates the marginal rate of substitution of X for Y (MRSX,Y) while the slope of budget line shows the ratioof the prices of the two commodities. Therefore, at equilibrium point E, we have

MRSX,Y = PX/PY ---------- IV

When utility is maximized, the ratio of prices between the two commodities is exactly equal to the marginal rate of substitution. When MRSX,Y is greater than (or less than) the price ratio between the two commodities, the consumer can increase utility by substituting one of the goods for the other. For example at point C, we have MRSX,Y > PX/PY and the consumer will substitute the commodity X for commodity Y to increase his satisfaction. He will continue to consume more of X at the expense of Y until the MRS becomes equal to the price ratio and move from point C to E.

7.5 CORNER SOLUTIONS

In this Section we will discuss a few cases involving 'corner solutions' where consumer equilibrium is characterized by a slightly different set of conditions from the one discussed above. You will see that in case of corner solutions only one commodity is purchased at consumer equilibrium.

7.5.1 Convex Indifference Curves

In case indifference curves are convex and either steeper or flatter than the budget line everywhere, then the consumer's equilibrium is represented by a corner solution. The situation is shown below in the Figures 5a and 5b. In this case the consumer will spend all his income either only on commodity X or only on commodity Y.

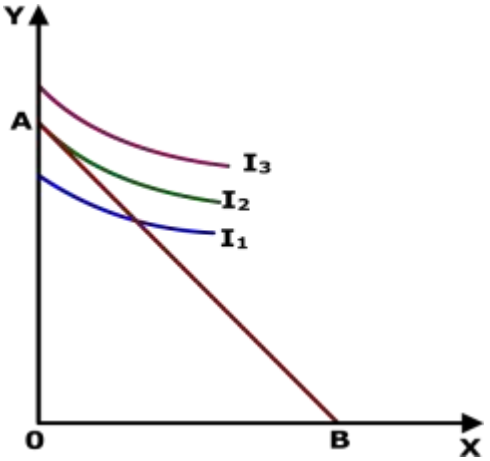


Figure 5a: Corner equilibrium (only Y is bought)

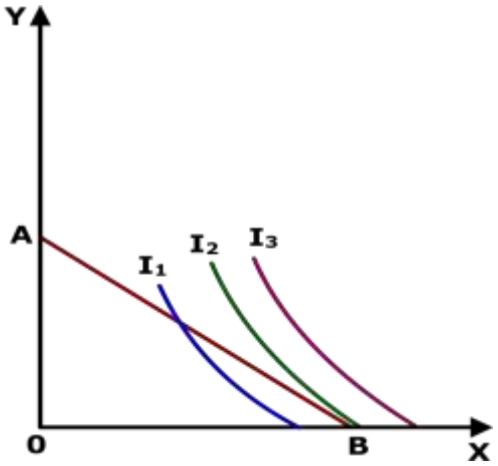


Figure 5b: Corner equilibrium (only X is bought)

In Figure 5a, the budget line AB is steeper than the indifference curve between the two goods so that the MRSX,Y is less than PX/PY for all combinations on the budget line AB. This means that the relative price of commodity X (PX/PY) is always more than his willingness to pay for X (MRSX,Y). In this case the consumer will not purchase commodity X and maximizes his satisfaction by purchasing only commodity Y. Thus a corner solution is achieved at point A. On the other hand if the budget line is always flatter than the indifference map then MRSX,Y will be greater than PX/PY at all possible combinations on the budget line AB (Figure 5b). In this case the consumer maximizes

satisfaction by purchasing only commodity X and none of Y. In this case theconsumer equilibrium will be attained at a corner solution at point B.

7.5.2 Concave Indifference Curves

Concavity of indifference curve implies that the marginal rate of substitution of X for Y increases as more and more X is substituted for Y. In case of concave indifference curves, highest satisfaction will not be achieved at the point of tangency between budget line and indifference curve. In this case an interior solution will not exist and satisfaction will be maximized at a cornersolution. In Figure 6 we have concave indifference curves.The indifference curve IC1 is tangent to the budget line AB at point P. However, a consumer with this kind of preferences can increase his satisfaction further by selecting other combinations (like points Q and B) on the budget line AB which lie on a higher indifference curve. Therefore P cannot be the consumer equilibrium in this case. The consumer's utility will go on increasing until he reaches the corner point B on the budgetline AB.

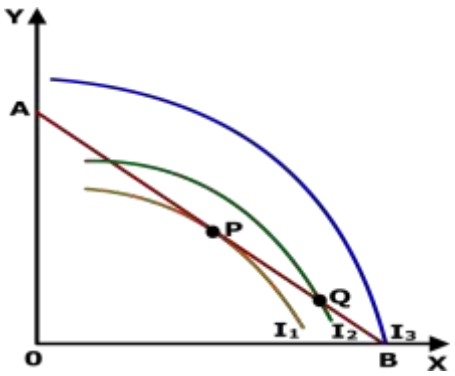


Figure 6: Corner equilibrium in case of concave indifference curves

7.5.3 Case of Perfect Substitutes

Indifference curves for perfect substitutes are straight lines with a constant marginal rate of substitution. In this case the tangency condition for consumer's equilibrium cannot be fulfilled since the budget line cannot be tangent to a straight line indifference curve. As shown in Figure 7, in this situation the budget line will cut the straight line indifference curves. If the indifference curves are steeper than budget line (MRS > PX/PY) the consumer will maximize his satisfaction atpoint B and will buy only X. On the other hand if the slope of the budget line is greater than the slope of in indifference curves (MRSX,Y < PX/PY) the consumer will buy only Y. Thus we will get corner

solution for consumer equilibrium in case of straight line indifference curves. **Figure 7: Corner equilibrium in case of perfect substitutes**

7.5.4 Case of Perfect Complements

The indifference curves of perfect complements are illustrated in Figure 8. In this case the indifference curves are L-shaped (right angled) because these commodities are consumed only in a fixed proportion. On the horizontal segment of the indifference curves, the marginal rate of substitution is zero, while on the vertical segment the marginal rate of substitution is infinity.

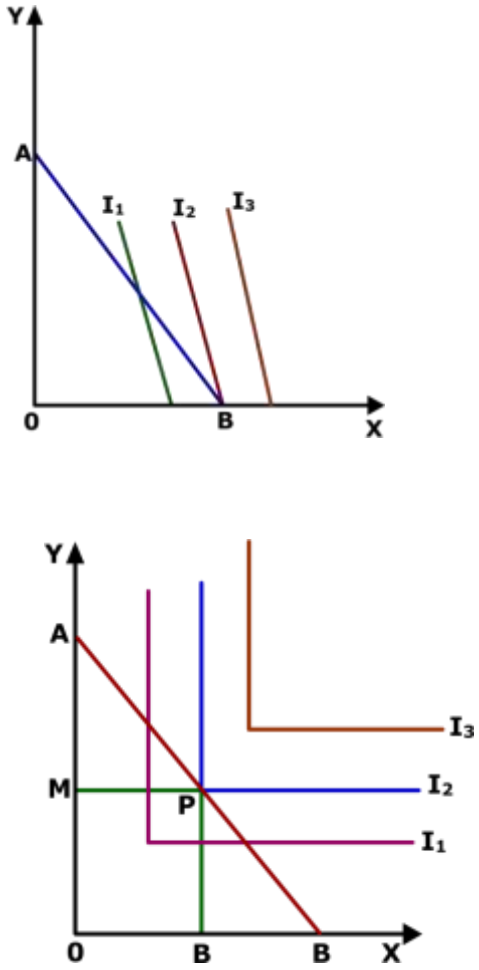


Figure 8: Equilibrium in case of perfect complements

In such a case the consumer's equilibrium will be determined at the vertex of the indifference curve IC2 which is just touching the budget line AB at point P. At this equilibrium point P, the equilibrium condition (MRSX, Y = PX/PY) is not satisfied since the slope is not defined at the corner point of an

L-shaped indifference curve. IC2 is the highest possible indifference curve which the consumer can achieve given his budget constraint. In Figure 8, the consumer will be in equilibrium at point P on IC2 and will be consuming OL amount of good X and OM amount of Y.

Consumer's Equilibrium in terms of the Utility Analysis

The objective of every consumer is to get maximum utility or satisfaction from spending his or her limited income. He would like to spend his or her limited income on a commodity or several commodities in such a way as to get maximum satisfaction (total utility).When he gets maximum satisfaction, he is said to be in equilibrium. In short, the position of maximum satisfaction implies consumer's equilibrium. In short, the position of maximum satisfaction implies consumer's equilibrium position. This can be determined with the help of three fundamental laws-(1) Law of diminishing marginal utility, (single commodity), (2) Law of equi-marginal utility (two or more commodities), and (3) Concept of consumer surplus.

In fact, the above three laws fall under cardinal utility approach to the theory of consumer behaviour. Here we discuss only law of diminishing marginal utility and concept of consumer surplus.

The above three laws has been explained in the above unit (Unit 5 &6).

CASE STUDIES

Case Study 1: Consumer Equilibrium and Changing Income

Ravi, a college student, receives a monthly allowance of ₹3000 to spend on two goods: books (₹300 each) and snacks (₹150 each). Initially, he spends half on books and the other half on snacks. After a festival bonus, his allowance increases to ₹4500, but the prices of books and snacks remain the same. As per the concept of the **budget line**, Ravi's budget constraint shifts outward, allowing him to consume more of both goods. After plotting his indifference curves, Ravi finds his equilibrium where the budget line is tangent to a higher indifference curve, indicating a higher level of satisfaction.

Case Study 2: Corner Solution with Perfect Substitutes

Anita prefers either tea or coffee and considers both as perfect substitutes with a 1:1 satisfaction ratio. If tea costs ₹10 per cup and coffee costs ₹15 per cup, she will spend all her money on tea to maximize utility. The budget line is flatter than the indifference curves, leading to a **corner solution** where Anita's equilibrium occurs at the endpoint of the budget line where only tea is consumed.

7.6 LET US SUM UP

- The extent to which the consumer can maximize his satisfaction crucially depends on his money income and the prices of the commodities.
- In a two-commodity world, a budget line shows all the combinations of the two commodities which an individual can purchase spending his entire income on the two goods, at their given prices.
- A budget constraint can also be expressed in the form of a budget equation which usually takes the form: Pxx + Pyy = M.
- An increase in income results in a parallel outward shift of the budget line.

  Similarly, a decrease in income shifts the budget line parallelly inward.
- A fall in the price of commodity X makes the budget line flatter without changing the vertical intercept. Similarly, a decrease in the price of Y makes the budget line steeper and also increases the value of the vertical intercept.
- The budget line remains unaffected in situations in which all income and prices changes in same proportion.
- The combination of goods which provide maximum satisfaction must lie on the budget line. The consumer is in equilibrium where his satisfaction is maximized subject to his budget constraints.
- At given money income and prices consumer satisfaction is maximized at a point at which the budget line is tangent to the highest indifference curve. The budget line can be tangent to only one indifference curve. Therefore, at equilibrium the marginal rate of substitution is equal to the price ratio (MRSX, Y = PX/PY).
- If the convex indifference curves are either steeper or flatter than the budget line everywhere, then the consumer's equilibrium is represented by a corner solution.
- In case of concave indifference curves and perfect substitutes the consumer satisfaction is maximized at a corner point.

KEYWORDS

1. **Budget Line** – A graphical representation showing all combinations of two goods a consumer can buy given their income and the prices of those goods.
2. **Indifference Curve** – A curve that shows different combinations of two goods that give the consumer the same level of satisfaction.

3. **Marginal Rate of Substitution (MRS)** – The rate at which a consumer is willing to substitute one good for another while maintaining the same utility level.
4. **Consumer Equilibrium** – The point at which a consumer achieves maximum satisfaction from goods consumed, given their budget constraint.
5. **Corner Solution** – A situation in consumer choice where the consumer chooses to consume only one of the goods, rather than a mix of both.

7.7 ASSIGNMENT QUESTIONS

1. Explain the impact of a rise and fall in consumer income on the budget line using diagrams.
2. Describe consumer equilibrium with the help of the indifference curve and budget line approach.
3. Differentiate between convex and concave indifference curves and discuss their impact on consumer equilibrium.
4. What is a corner solution? Explain it with suitable diagrams under different market conditions.
5. Discuss consumer equilibrium in the case of perfect complements and perfect substitutes.

OBJECTIVE TYPE QUESTIONS

1. A _____ shows all combinations of two goods a consumer can buy with a given income.
2. The slope of the budget line is given by _____.
3. A higher indifference curve represents _____ satisfaction.
4. The equilibrium occurs when the budget line is _____ to an indifference curve.
5. In case of perfect substitutes, the indifference curves are _____.
6. If the price of one good increases, the budget line rotates _____.

7. The condition MU1/P1 = MU2/P2 represents _____.
8. Indifference curves that are _____ result in interior solutions.
9. The _____ of substitution measures a consumer's willingness to exchange one good for another.
10. A decrease in consumer income will cause the budget line to shift _____.
11. At equilibrium, MRSX,Y equals the _____ of two goods.
12. The concept of utility maximization is associated with the _____ approach.
13. In case of perfect complements, the indifference curve is shaped like the letter _____.
14. Corner solutions occur when the _____ of the budget line differs greatly from the MRS.
15. The combination on the budget line that touches the highest indifference curve is known as the _____ point.

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FURTHER READINGS

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- Consumer Behaviour – Leon G. Schiffman
- Principles of Economics – Gregory Mankiw
- Managerial Economics – D.N. Dwivedi
- Introduction to Positive Economics – Richard G. Lipsey

UNIT-8: CONSUMER EQUILIBRIUM AND DEMANDS EFFECT

STRUCTURE

8.1 Introduction

8.2 Demand Effects

8.3 Let Us Sum Up

LEARNING OBJECTIVES

After studying this unit, you will be able to:

- Understand the price consumption curve
- Discuss about backward sloping PCC

BLOOM'S LEVEL OF LEARNING:

The content provided in the unit about "Consumer Equilibrium and Demand Effects" offers a comprehensive understanding of the concept of consumer equilibrium, the effects of changes in income and price on consumer behaviour, and how various goods respond to these changes. The main learning objectives highlight the importance of understanding the price consumption curve (PCC) and backward sloping PCC. The introduction covers the basic assumption that a rational consumer aims to maximize satisfaction within the constraints of limited income and fixed prices. It introduces consumer equilibrium, where the consumer chooses the highest possible indifference curve tangent to their budget line. The concept of the Income Consumption Curve (ICC) is explored, showing how changes in income influence the consumption of normal and inferior goods, with the income effect either positive, negative, or neutral depending on the nature of the goods. The price effect, along with the substitution and income effects, explains how price changes influence demand, and how these effects are captured by the Price Consumption Curve (PCC). The unit also discusses different types of PCCs, such as upward sloping, downward sloping, and backward sloping curves, and examines the role of income elasticity. Furthermore, it covers how substitution and income effects are separated using Hicksian and Slutsky methods, and addresses the specific case of Giffen goods, where the price and demand move in the same direction. Overall, this unit provides a thorough understanding of

consumer behavior in response to income and price changes, making it an essential part of microeconomic theory.

8.1 INTRODUCTION

Consumer's Equilibrium

As discussed earlier in this previous unit that the aim of every consumer is to get maximum satisfaction from the goods to be purchased. Therefore, he wants to reach the highest possible indifference curve. But there are certain limits. These limits are relating to his money income and the prices of the goods. The budget line describes these limits.

Before explaining the consumer's equilibrium, let us make the following assumptions: Every consumer has scale of preferences

He has a given amount of income to spend Prices of the goods are given and are constant

Goods are homogenous and divisible

The consumer is rational and tries to maximize his satisfaction

Where consumer get maximum satisfaction, he is said to be in equilibrium. When the indifference map and budget line are combined together, we can find out the consumer's equilibrium. Let us explain the consumer's equilibrium with the help of the graph (Fig.15) given on next page.

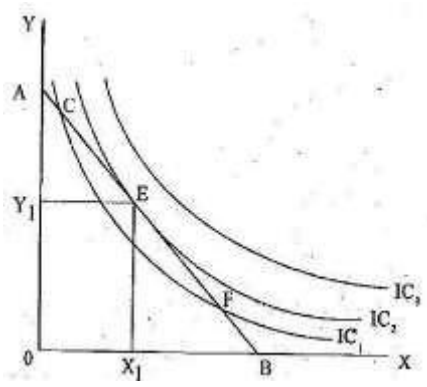


Figure 1

In the graph three indifference curves are given. They are IC_1, IC_2, and IC_3. These show combinations of two goods X and Y which give different levels of satisfaction to a consumer. AB is the budget line.

It shows various combinations of X and Y which can be bought with his fixed income (given the prices of X and Y). Given the money income to spend and the prices of the two goods, the consumer can buy and combination of X and Y which lies on the budget line AB. All the combinations on AB cost the consumer the same amount of money. In order to maximize his satisfaction, the consumer will try to reach the highest possible indifference curve. But he cannot reach IC_3. This is because he has only limited income. His income is represented by the budget line AB. So he has to choose that combination which lies on his budget line. In the graph we can see that combinations C, E and F lie on the budget line AB. These are affordable. He has to choose from these combinations only. He will test each point. Take C first. We find that C lies on a lower indifference curve (IC). So it is not the point of equilibrium. This is because he can get more satisfaction with the same amount of money. Similarly, F also lie on lower indifference curve (IC_1). It also does not give maximum possible satisfaction. Now look at point E. This lies on the budget line and also on the highest possible indifference curve (IC_2). So, E is the point of equilibrium. A point E the budget line AB is tangent to indifference curve IC_2. Thus, we can say that with given income and price of goods, the consumer will get maximum satisfaction at point E where the budget line AB is tangent to indifference curve IC_2. At the equilibrium point E, the slopes of the budget line AB and the indifference curve IC_2 are equal. That is why budget line AB is tangent to IC_2. Thus, when the consumer buys OX_1 quantity of commodity X and OY_1 quantity of commodity Y, he will be in equilibrium.

8.2 DEMAND EFFECTS

Income Consumption Curve and the Nature of the Commodity

The effect of changes in income on the consumption (income effect) of different kinds of commodities is different. It may be positive or negative or even neutral. In fact, the nature of a commodity determines whether the effect is positive or negative. When the ICC slopes upward, the income effect is positive (the ICC has positive slope) to both commodities. This means that the consumer increases the consumption of both commodities with the increase in his income. The income effect is positive (ICC is upward sloping) in case both goods are normal goods are normal goods as shown in Fig.

16. But sometimes, the income effect may be positive for one commodity and negative for another. The income effect of a commodity is positive when the consumer purchases more of that commodity with the increase in income. This is a normal good. If income effect of a commodity is negative, it

will mean that consumer purchases less of that good as the income increases. Such goods are inferior goods. Examples are maize, jowar, bajra, low quality clothes etc. When the income of the consumer increases he likes to spend more on normal goods and less on inferior goods. That is, he substitutes superior(normal) goods in place of inferior goods. This may be discussed below. **Negative Income Effect: The Case of Inferior Goods**

(a) When Good X is Inferior: This may be explained with the help of a graph (Fig.17). In the graph, the ICC begins to move towards Y axis on which we measure good Y. This shows that after a certain point, the consumer purchases less of good X and purchases more and more of good Y as the income increases. Here good X is inferior good and good Y is normal or superior good. In this case income effect is positive for good Y (superior or normal) and income effect is negative for good X (inferior). Thus, in case of inferior good, the ICC will at first be upward rising and then backward bending (slopes backward, i.e., slopes upward to the left or bends back towards Y axis). This is shown in figure.

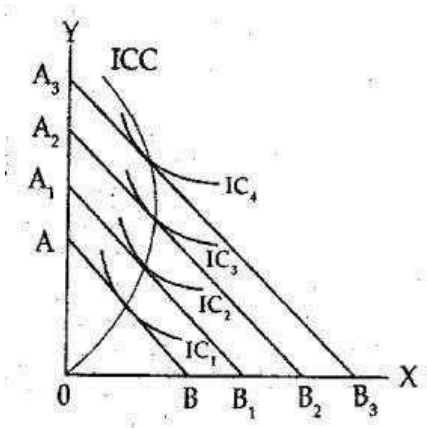


Figure 2

(b) When Good Y is inferior: When good X is normal and good Y is inferior, the ICC will slope towards good X. This is shown in Fig. 18.

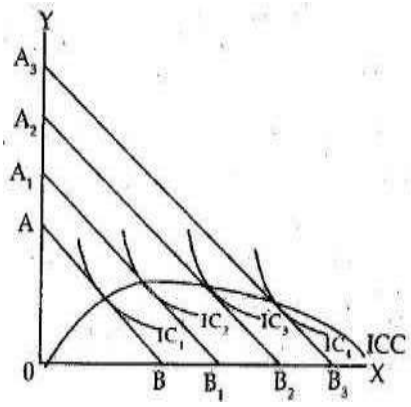


Figure 3

In the given graph, ICC slopes downwards to the right and bends towards good X. It shows that good X is purchased more and good Y is purchased less with the increased income. Here, income effect is positive (because both income and demand are increased) in the case of good X and negative in case of good Y.

The slope of the ICC will, however, depend upon the income elasticity for these two goods. If income elasticity for Y is high and that of X is low, the ICC will be sloping upwards to the left more nearer the Y axis. If income elasticity for X is high and that of Y is low, the ICC will be sloping upwards to the right more nearer the X axis.

The slope of the income consumption curve gives an indication of income elasticity. If the curve is relatively flat, income elasticity is high. If it has an angle of 45°, income elasticity is unity. If the curve is steep, income elasticity is low. If the curve is vertical line, income elasticity is zero. If the curve has negative slope, income elasticity is negative.

It may be noted here that both the commodities cannot be negative at the same time. The reason for this is that, when one commodity is inferior, the consumer spends the excess of his income on the consumption of the other commodity. So the consumer will not be able to spend his total income when both commodities are inferior. In other words, if both the goods are inferior, there will be decrease in expenditure on the two goods resulting in disequilibrium. Here the consumer will be below the budget line. He will not be reaching in optimum position of equilibrium. This is based on the assumption that the consumer spends his entire income on the purchase of two commodities. It may be noted that the consumer may consume two normal or luxury goods.

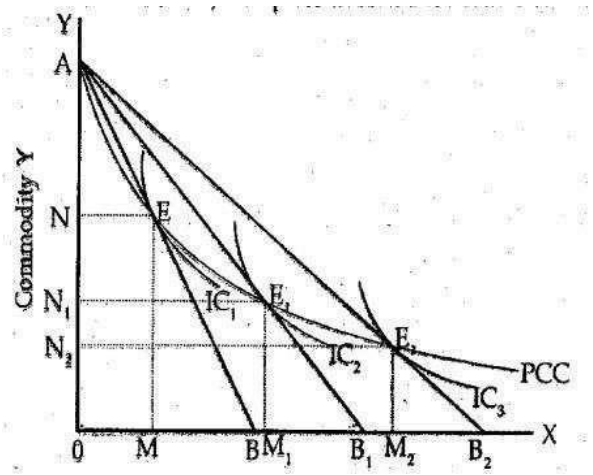
Zero Income Effect (Neutral Income Effect)

The income effect is zero for those commodities which the consumer purchases in fixed quantities. Examples are drugs, salt etc. He wouldn't buy more when income increases. Similarly,

he wouldn't buy less when the income decreases. This means that he would buy such commodities in fixed quantities whether the income increases or falls. In the case of such commodities there is no income effect. **The Price Effect (or Change in Price) and Price Consumption Curve**

It is possible that the income of the consumer remains the same, but the prices of commodities may change. When the price of one commodity changes (say X), the equilibrium of the consumer also changes. The slope of the budget line of the consumer also shifts. When the price of a good falls the real income of the consumer increases. Then the consumer would be better off (he becomes 'rich'). Accordingly, he would shift to a higher indifference curve. On the other hand, if the price of a good rises, the real income of the consumer falls. Then he would be worse off. Accordingly, he would shift to a lower indifference curve. The effect of a change in price of a good on the quantity of its purchase is known as price effect (the price of other good and the income of the consumer remain the same). The price effect is shown in the following diagram:

Figure 4



In the diagram, commodity X is taken on X axis and commodity Y is taken on Y axis. Money income of the consumer remains constant. The price line is AB. The consumer reaches at the equilibrium point E. At that point the price line AB is tangent to indifference curve IC_1. In other words, the equilibrium occurs at a point where the price or budget line touches the highest possible indifference curve. At this point, he buys OM quantity of X and ON quantity of Y. Suppose the price of X falls. Now the real income of the consumer increases (the price of commodity Y and the money income of the consumer remain the same). The consumer shall be able to buy more quantity of X. Whenever there is change in the price of commodity X, the budget or price line changes its slope. That is, with every change in the price of commodity X, the price line shifts. As already stated, the

money income of the consumer remains unchanged and the price of Y also remains unchanged. Hence, the price line shall not be parallel to the old price line (only a shift in price line but not parallel). It starts from point A. This indicates that with the money income of the consumer and the price of Y remaining constant, the consumer shall not be able to buy more quantity of Y. He shall buy more of X only because the price of X has fallen. Here, when the price of X falls, his real income increases. hence, his price line shift to AB_1. Now the consumer reaches at a new equilibrium point E_1. At this point, he buys OM_1 quantity of X and ON_1 quantity of Y. Suppose the price of X falls further. Again, the real income of the consumer increases. The price line again shifts to AB_2. The new point of equilibrium is E_2 where the budget line AB_2 is tangent to the higher indifference curve IC_3. At this point, the quantity of purchase of X is OM_2 and that of Y is ON_2. When we join all successive points of equilibrium, we get a curve. This curve is called price consumption curve (PCC). It is also called *price offer curve*. Thus, the curve connecting different equilibrium points when the price of only one commodity changes and the price of other commodity and the money income remain the same is known as

PCC. It shows how the change in price of good affects the consumer's purchases or equilibrium. From the graph it is clear that the consumer will get higher level of satisfaction from higher indifference curve and feel better off.

PCC may have different shapes and slopes. If price of X falls, the demand for Y may increase, may decrease or may remain constant. If the demand for Y increases the PCC will be upward sloping. If the demand for Y decreases, the PCC will be downward sloping. If the demand for Y remains constant, the PCC is horizontal. These different shapes of PCC may be explained below.

Downward Sloping PCC

In the given graph (Fg.23) the PCC is a downward sloping curve. In this case, the consumer purchases larger quantity of X and smaller quantity of Y when the price of X falls. Thus, the demand for X is elastic. Elasticity is greater than one.

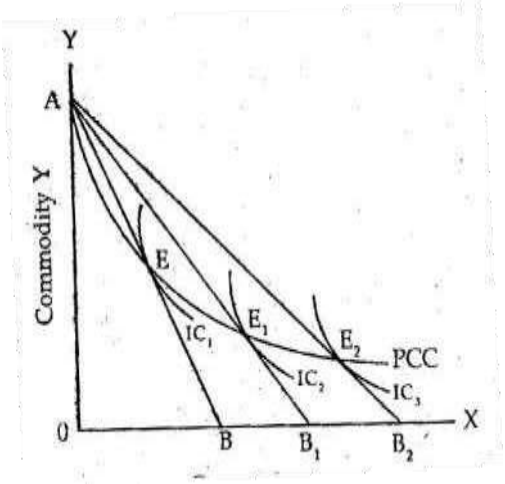


Figure 5

Upward Sloping PCC

In the case of upward sloping PCC, we can see that the quantity purchased of both X and Y rises due to fall in the price of X. We get upward sloping PCC for commodity X when the demand for X is less elasticity (less than one) as shown in figure.

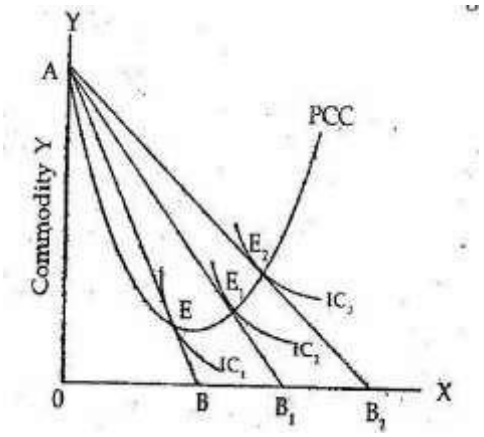


Figure 6

Backward Sloping PCC

When the PCC for commodity X is backward sloping (as shown in figure), we can find that the consumer purchases less and less of X and more and more of Ydue to fall in the price of X. This happens in the case of Giffen goods.

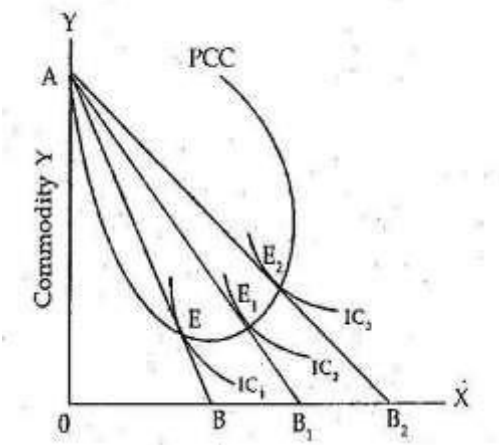


Figure 7

Horizontal PCC

Horizontal PCC for X indicates that when the price of X falls, its quantity purchased increases proportionately, while quantity purchased of Y remains at thesame level. In this case, price elasticity of demand for X = 1 (unit elasticity) as shown in Fig. 26.

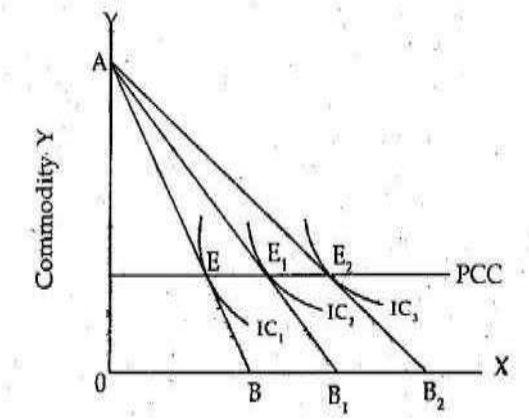


Figure 8

PCC with Varying Slope

Generally, PCC has different slopes at different price ranges. At higher price levels, it usually slopes downwards. Thereafter, it may have a horizontal shape for some price ranges. Ultimately it slopes upwards for further falling in price. APCC with different shapes and slopes is illustrated figure

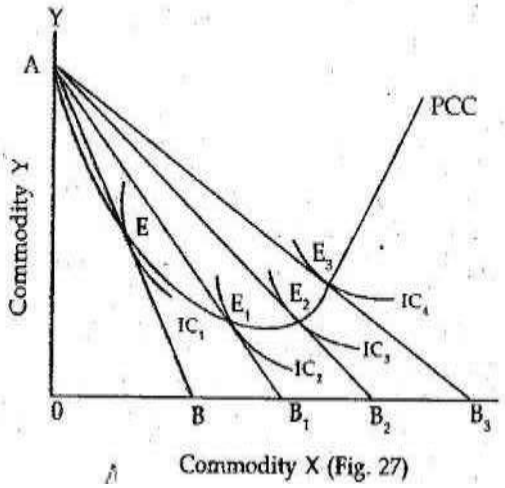


Figure 9

Substitution Effect

Another important factor responsible for the changes in consumption of goods is the substitution effect. The substitution effect shows the change in the quantity of a good purchased due to a change in the relative price of a good, while the income of the consume remains constant. When the price of a good falls, the real income of the consumer increases. Suppose, a customer has Rs. 100 with him. He buys 4 oranges at the rate of Rs.25. Suppose, the price of oranges falls to Rs.20. If he buys 4 oranges as before, he can save Rs.20. Or he can buy 5 Oranges falls to Rs.20. If he buys 4 oranges now I i.e., he can spend more with the amount saved).

Thus, when the price of a commodity falls, the real income of the consumer increases and vice versa. The increase in real income induces the consumer to buy more of it under the income effect. But with fall in the price of a good, it becomes relatively cheaper. This induces the consumer to buy more to substitute it in place of the costlier good. This is called substitution effect.

Thus, a change in price has both income as well as substitution effects. More of X instead of buying Y. Second, when the price of the commodity falls, the real income of the consumer increases (the price of Y is constant). With this increased real income (surplus money) the consumer can buy more of X or both the goods. This is the income effect (if the price of a good increases, the reverse happens). In both cases, the quantity demanded of the good(s) increases as a result of fall in the price of one good. This is the price effect. The total increase in demand happens in two ways-due to substitution effect and due to income effect. Thus, price effect is the result of substitution effect and income effect. In other words, price effect=substitution effect + income effect.

Now the question arises, if PE = SE + IE, then how these should be separated or decomposed. There are two methods for decomposing the price effect into substitution effect and income effect. One method was given by J.R. Hicks. The other was suggested by E. Slutsky. Here we discuss only the method proposed by Hicks (Hicksian approach).

Hicks has separated the substitution effect and income effect from the price effect through "compensating variation in income" by changing the relative price of a good while keeping the real income of the consumer constant. According to Hicks, "when the relative prices of the goods change, the money income of the consumer is altered in such a manner that his real purchasing power remains constant and he is neither better off nor worse off than before, i.e., he continues to remain on the same indifference curve". Let us now discuss Hicksian approach.

Inferior Commodity Case (HIcksian Method)

Substitution effect for any commodity raises its demand due to fall in its relative price. Suppose there are only two goods in the market -A and B. Whenever relative price of A falls compared to that of B, A becomes cheaper compared to B. Therefore, demand for A increases and that of B decreases as a result of substitution effect. Thus, substitution effect is always positive. This is because both real income and demand move in the same direction, i.e., both increases. Moreover, when the price of a commodity falls relative to other commodity, consumers will substitute the cheaper commodity in place of costlier one. Thus, demand increases. Since the demand increase, substitution effect will always be positive. But the income effect is not like that. It depends on the nature of the commodity. It is positive for normal good. This is because when the price of good A falls, the real income of the consumer increases. Hence, he will demand more of that good. Thus, the demand for A increases. Hence, the price effect for A is positive. On the other hand, the income effect for inferior good is negative. Suppose there were two goods - X and Y. X is inferior while Y is normal (superior). Suppose the price of X (inferior good) falls. The price of Y and the income of the consumer remain the same. Then the substitution effect of X is positive. This is because X become cheaper and its demand will increase. But when its price falls, real income increases. The consumer no longer wants to purchase the inferior good. He reduces his demand for inferior good and purchases normal good in its place. Thus, for inferior good the income effect is negative. As already pointed out, the price effect is the sum of substitution effect and income effect. In case of inferior good, the substitution effect and income effect work at the opposite direction. However, the positive substitution effect is stronger or more than the negative income effect. Put it in another way, the negative income effect is smaller or weaker than the positive substitution effect. Hence,

the net result of purchase of inferior good will increase. This is because substitution effect is stronger. In other words, as pricefalls, demand for inferior good increases to a lesser extent than that of any normal good.This may be explained with the help of a graph.

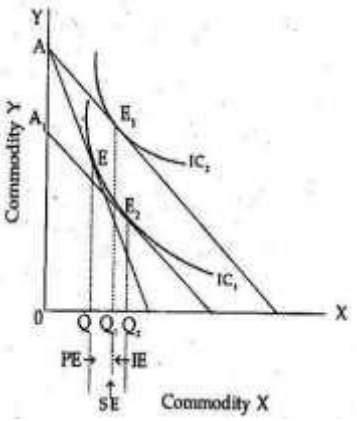


Figure 10

There are two commodities - x and Y. X is inferior good, while Y is normal good. X is taken on OX axis and Y on OY axis. The prices of X and Y and the money income of the consumer are given. The budget line is AB. It is tangent to the indifferent curve IC, with the equilibrium point E. The consumer is purchasing OQ quantity of X. When the price of X falls, the budget line AB shifts its position to AC. His equilibrium shirts to the point E_1 on IC_2 (higher indifference curve) where AC is tangent. Now he purchases QQ, more quantity of X due to fall in its price. This is price effect. This is positive because he purchases more, i.e., movement is forward. From the price effect, we have to separate substitution effect and income effect. In order to break up income effect and substitution effect, the money income of the consumer is reduced (money is taken away from him) to keep his real income as constant by compensating variation in income in such a way that the budget line AC shirts parallel to its left and the new budget line is A_1 C_1. This is tangent to the initial indifference curve IC_1 is substitution effect (i Hicksian sense). This would induce the consumer to buy QQ_2 more quantity of X. Now if the amount of money which was taken away from him is given back to him, he will move from E_2 on IC_1 to E_1 on IC_2. The movement from E_2 to E_1 is the income effect. This is in the opposite direction. This induces the consumer to purchase Q_1 Q_2 less quantity of X. This is due to negative income effect. We can see that negative income effect Q_1 Q_2 is smaller or weaker than the stronger positive substitution effect (QQ_2). Hence the net purchase of X increases only by an amount of QQ_1. This is means that the consumer will buy only QQ_1 more of X.

Thus, price effect is equal to positive substitution effect + negative income effect and the result is positive price effect (buying more). That is $QQ_1 = QQ_2 - Q_1 Q_2$.

Giffen Commodity Case (HIcksian Approach)

All Giffen goods are inferior goods but all inferior goods are not Giffen goods. This implies that there is similarity as well as difference between inferior goods and Giffen goods. The similarity is that the income effect is negative on both. The dissimilarity is that in case of inferior goods substitution effect is stronger than income effect, while income effect is stronger than substitution effect in case of Giffen goods. A Giffen good is a good whose demand increases with the increase of its price and vice versa. A fall in price leads to a reduction in the quantity purchased. Thus, for a Giffen good, price and quantity purchased move in the same direction. In the case of Giffen good, the income effective is negative. This means that the consumer buys less of this good with the increase in his income. As the price of a Giffen good falls the real income of the consumer increases. However, he purchases less of this good because it is an inferior good. Thus, its income effect is negative. But its substitution effect is positive. He purchases more of this good due to substitution effect as its relative price falls. Thus, the income effect and substitution effect work in the opposite direction as in the case of inferior good. But the income effect is stronger than substitution effect.

Now let us see how the price effect of a Giffen good is decomposed into substitution effect and income effect with the help of a graph

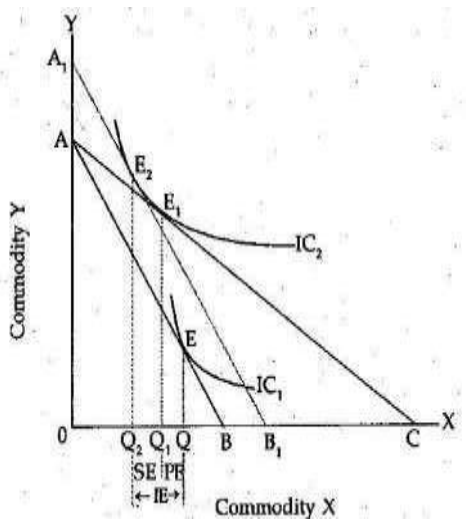


Figure 11

Initially the consumer is at equilibrium point E when the price line (budget line) AB is tangent to the difference curve IC_1. He purchases OQ quantity of commodity

X (Giffen good) Suppose the price of X falls. As a result, the price line shifts to AC. The new equilibrium point is at E_1 on the high difference curve IC_2. At this point the consumer purchases OQ_1 quantity of

X. We can see that QQ_1 quantity is less than OQ quantity. This shows that X is a Giffen good. This is because its quantity demanded decreases with fall of price. Hence movement from E to E_1 is called price effect. The movement is backward

(i.e., purchase is less). Hence price effect is negative. Due to the price effect, the consumer purchases QQ_1 less of X. If we assume that price of Y does not fall and the income of consumer is increased in such a way so as to keep him on the same indifference curve IC_2. Then the price line shifts parallel to the right of original price line AB. The new price line is A_1B_1. This is tangent to the indifference curve IC_2 at point E_2. Thus, the new equilibrium point E_2. At this point the consumer purchases OQ_2quantity of X. The movement from E to E_2 is called income effect. This movement is backward (purchase is less). Hence income affect is negative. It is bigger or stronger as well. Due to income effect, the consumer purchases QQ_2 less of X. Again, movement from E_2 to E_1 is called substitution effect. The movement is forward (i.e., purchase is more). Hence substitution effects positive. Due to substitution effect, he purchases Q_2 Q_1 more of X. Here QQ_2 (negative income effect) is greater than Q_2 Q_1 (positive substitution effect). HereQQ_2 is bigger than Q_2 Q_1. It means that negative income effect is stronger than positive substitution effect. Hence, price effect is negative. Thus, the consumer purchases QQ_1 less of X as its price falls.

Demand Curve for Normal, Inferior and Giffen goods

The shape of the demand curve depends on the nature of the goods. Goods may be normal, inferior or Giffen goods.

Demand Curve for Normal Goods

When price falls, demand for normal good increases and vice versa. It happens so because of price effect. Thus, change in price and change in demand moves in opposite direction. That is, demand curve for normal good is negatively slopped. To be more clearly, demand curve for normal goods slopes downward from left to right. It is shown in the given graph

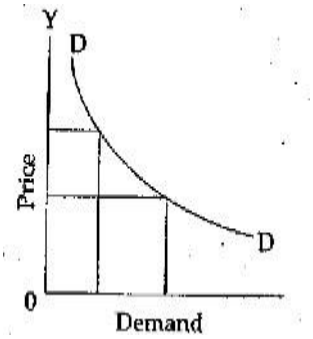


Figure 12

Normal good may be luxury or comfort or necessity. If demand is not very sensitive to income, the product is known as a 'necessity product'. If demand is very sensitive to income, the product is known as a 'luxury product'. As income increases, the demand curve for a luxury good shirts more to the right than a necessity good.

Demand Curve for Inferior Goods

When price falls, demand for inferior good increases as a result of price effect. However, it increases only to a lesser extent than that of any normal good. Hence, demand curve will be steep. In the case of inferior good also there is an inverse relationship between price and demand. That is, demand curve for inferior good has negative slope. It is shown in the following graph:

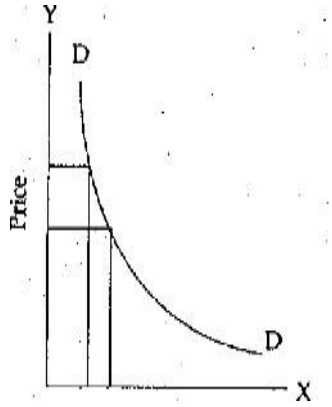


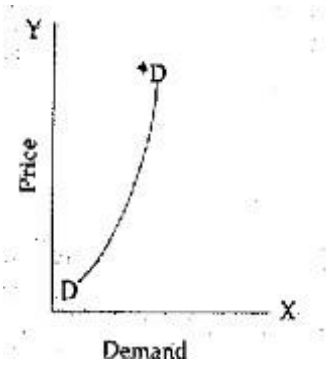
Figure 13

Demand Curve for Giffen Goods

When price falls, demand for Giffen goods decreases as a result of price effect. Thus, in the case of Giffen goods, the law of demand does not held good. Here, the change in price and change

in demand move in the same direction. In other words, the demand curve for Giffen good is positively slopped. This is shown below:

Figure 14



Now the three demand curves may be shown in one graph as below:

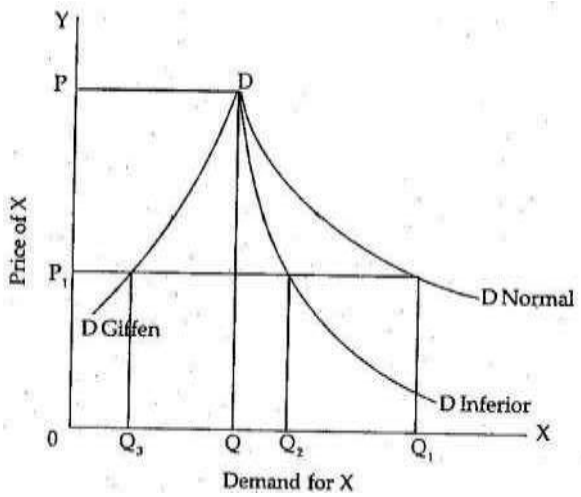


Figure 15

In the above graph we can see that at initial price level P, demand for X is OQ for all types of commodities. As price level decreases from P to P_1, if X is a normal good, demand increases from QO to OQ_1. Joining the two combinations (P, Q) and (P_1 Q_1) we get the negatively sloped demand curve for normal good, DD normal. Again, as price level decreases from OP to OP_1, if X is an inferior good, demand increases from OQ to OQ_2. Joining the two combinations (P, Q) and (P_1, Q_2) we get the negatively slopped demand curve for inferior good, DD inferior. We can see that both DD normal and DD inferior are negatively sloped, but DD inferior is steeper than DD normal. Again, as price

level decreases from OP to OP_1, if X is Giffen good, demand decreases from the OQ to OQ_3. Joining the two combinations (P, Q) and (P_1 Q_3), we get the positively sloped demand curve for Giffen good, DD Giffen.

CASE STUDIES

Case Study 1: The Shift from Inferior to Normal Goods

Context: A rural household consumes maize and local coarse cloth due to budget constraints.

Event: With an increase in household income due to a new government job, the family begins purchasing wheat and branded clothes.

Analysis: This transition reflects a positive income effect for normal goods (wheat, branded clothes) and a negative income effect for inferior goods (maize, coarse cloth).

Conclusion: The income consumption curve initially rises but bends backward as more superior goods replace inferior ones, showing income sensitivity in consumption behaviour.

Case Study 2: Giffen Goods in Low-Income Communities

Context: In a remote area, rice is a staple and cheaper food, while vegetables are more expensive.

Event: When rice prices fall, surprisingly, demand for rice decreases and demand for vegetables increases.

Analysis: Rice acts as a **Giffen good**. The fall in rice prices raises real income, prompting people to reduce rice intake and opt for more desirable food items. The **negative income effect outweighs the substitution effect**, causing a drop in rice demand.

Conclusion: This real-world anomaly validates the backward-sloping demand curve for Giffen goods.

8.3 LET US SUM UP

- At given money income and prices consumer satisfaction is maximized at a point at which the budget line is tangent to the highest indifference curve. The budget line can be tangent to only one indifference curve. Therefore, at equilibrium the marginal rate of substitution is equal to the price ratio (MRSX,Y = PX/PY).
- If the convex indifference curves are either steeper or flatter than the budget line everywhere, then the consumer's equilibrium is represented by a corner solution.

- In case of concave indifference curves and perfect substitutes the consumer satisfaction is maximized at a corner point.

KEYWORDS

1. **Consumer Equilibrium**
   The point at which a consumer achieves maximum satisfaction given their income and the prices of goods.
2. **Price Consumption Curve (PCC)**
   A curve showing how the consumption of goods changes when the price of one good changes, with income and other prices constant.
3. **Income Effect**
   The change in quantity demanded of a good resulting from a change in consumer income, holding prices constant.
4. **Substitution Effect**
   The change in demand due to a good becoming cheaper or more expensive relative to another, causing substitution in consumption.
5. **Giffen Good**
   A type of inferior good where the income effect is so strongly negative that a price drop leads to a decrease in quantity demanded.

ASSIGNMENT QUESTIONS

1. Explain how the concept of consumer equilibrium is determined using an indifference curve and budget line.
2. Differentiate between normal, inferior, and Giffen goods with suitable graphs.
3. Discuss the role of the substitution effect and income effect in determining consumer behavior when prices change.
4. Describe the characteristics and implications of a backward sloping Price Consumption Curve.
5. Using Hicksian approach, explain how the price effect is broken into substitution and income effects.

OBJECTIVE TYPE QUESTIONS

1. Consumer equilibrium occurs when the indifference curve is _____ to the budget line.

2. The _____ effect refers to changes in demand due to changes in real income.
3. A Giffen good has a demand curve that is _____ sloping.
4. Substitution effect is always _____.
5. PCC shows the effect of change in _____ on consumer equilibrium.
6. Income consumption curve bends backward when the good is _____.
7. The shape of the demand curve for normal goods is _____ sloping.
8. Goods with zero income effect are called _____ goods.
9. When price falls and demand rises less than proportionately, the demand is _____ elastic.
10. The ICC for two normal goods is _____ sloping.
11. If both substitution and income effects are positive, the price effect will be _____.
12. Inferior goods have a _____ income effect.
13. When the income elasticity of demand is zero, the ICC is a _____ line.
14. Giffen goods violate the _____ of demand.
15. In Hicksian method, consumer is kept on the same _____ curve to isolate substitution effect.

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UNIT-9: ROLE AND SCOPE OF THEORY OF PRODUCTION

STRUCTURE

9.1 Introduction

9.2 Some Basic Concepts of Production Theory

9.3 Production Function

9.3.1 Cobb-Douglas Production Function

9.4 Production Function with one Variable Input

9.5 Importance of Production Theory

9.6 Isoquants

9.7 Let Us Sum UP

9.8 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Know about the concepts of production theory.
- Discuss production function.
- List the importance of production theory.
- Derive the relationship between AP, MP& TP

BLOOM'S LEVEL OF LEARNING:

The Bloom's level of learning for this unit on production theory can be classified primarily under the Understanding and Applying levels. After studying the concepts, students should be able to comprehend various production theories and their underlying principles, such as the relationship between inputs and outputs, and how they affect production processes. Additionally, learners are expected to apply this understanding to analyze production functions, distinguish between fixed and

variable factors, and recognize the significance of concepts like Total Product, Average Product, and Marginal Product. This includes deriving relationships between these concepts and explaining them in the context of real-world production scenarios. The practical application of these concepts equips students to address questions about optimizing production processes, managing costs, and improving efficiency in business settings, thus reflecting both the understanding and application levels of Bloom's taxonomy.

9.1 INTRODUCTION

Today, one of the key concerns of the business managers is to achieve optimum efficiency in production or minimizing cost for a given output. In a competitive market, a firm can survive only if it is able to produce goods and services at a competitive cost. Hence, business managers should make an effort to minimize the cost of production or maximize output for a given level of inputs. There are some basic questions which managers are confronted with while endeavoring to minimize the production cost. These are:

1. How does output change with the increase in quantity of inputs?
2. How can production be optimized to achieve reduction in cost?
3. How does technology help in minimizing the cost of production?
4. How to achieve the least cost combination of inputs for a given output?

The theory of production seeks to explain answers to these questions with the help of economic models that are built under hypothetical conditions. The production theory provides various tools and methods to analyze the relationship between input and output which provide directions to find solutions to practical business problems.

9.2 SOME BASIC CONCEPTS OF PRODUCTION THEORY

Production

In economics, the transformation of resources (men, material, time and so on) into a different & more useful commodity through a process is called production. In general, the production function is limited to only ‘manufacturing’ in which inputs (labour, machine, raw material, time & so on) are transformed into an output.

In economics sense, there are different forms of production other than manufacturing. When a commodity is transferred from one place to another for consumption in the production process is production. For eg. When sand is transferred from the river bank to a construction site by the sand

dealer, this activity, too, is production. When a commodity is stored for future consumption or sale, this is also called production. In a broader sense, *production can be defined as the creation of utility*. It is not necessary that only physical conversion of inputs into tangible goods take place in the production process. Some types of production involve conversion of an intangible input into intangible output. For e.g. both input and output are intangible in the production of medical, legal, social and consultancy services.

Fixed and Variable Factors of Production

A variety of goods and services are used by a firm in the process of production called factors of production or inputs such as land, labour, raw material, plant and machinery, factory premises, tools and equipment etc. These factors of production can be classified into two parts:

1. Fixed factors
2. Variable factors

Fixed factor is one whose quantity remains the same irrespective of the level of output. Its quantity cannot be readily changed in the short run. On the other hand, a variable factor is one whose quantity changes with the changes in the level of output. In the short run, a larger quantity can be employed by the users of such factors. For example: a firm wants to increase the production of books from 1000 to 2000 daily. To do so, it will need more factors of production. But, there are some factors whose supply cannot be increased immediately e.g. printing press, building etc. Therefore, the firm will have to use those factors whose quantity can be increased immediately such as labour, raw material etc. In the given example, the printing press and building are fixed factors of production, labour and raw material are variable factors of production.

Short run and Long run

Short run refers to that time period in which supply of certain factors is fixed i.e. it cannot be increased or decreased. For e.g. plant, machinery, building etc. Therefore, a firm can increase the production of a commodity in the short run, by increasing the use of variable factors such as labor and raw materials.

Long run refers to that time period in which supply of all factors i.e. fixed and variable is elastic, but not sufficient to allow a change in technology. In the long run, no factor is a fixed factor, all factors are variable. Therefore, a firm can increase the production of a commodity in the long run by increasing the use of both variable and fixed factors of production.

There is another term 'very long run' used by economists which refers to that time period in which technology of production can also be changed. The production function also changes in the very long time period.

9.3 PRODUCTION FUNCTION

Production function is a mathematical statement used to explain the technical relationship between input and output in physical terms over a given period of time.

According to Koutsoyiannis, "the production function is purely a technical relation which connects factor inputs and output.

The general form of production function tells that output of any commodity depends on the quantity of inputs and techniques of production. Given the state of technology, several methods using different combinations of factors are used in the production of a commodity. For e.g. the production of cloth requires cotton or polymer or silk as raw material with power loom, handloom or even computerized machines. The same product can be produced in several possible ways using different kinds of raw materials and technology options. Thus, there can be various technically efficient methods of production. Production function incorporates all such technically efficient methods.

Production function is a schedule showing maximum output of a commodity produced per unit of time with each combination of inputs and with the given state of technology. A different set of inputs are used to produce a given level of output and each of these sets may be technically efficient.

Suppose, a manufacturing firm employs only two factors- Labour (L) and Capital (K) in manufacturing steel chairs. Then, general form of its production function can be written as:

$$Q = f(L, K)$$

Where Q= Quantity of steel chairs produced

L= Labour

K= Capital

The above production function shows that the quantity of steel chairs produced depends on the quantity of Labour (L) and Capital (K) employed. The production of steel chairs can be increased by increasing K and L. The time period (i.e. short run or long run) considered for increasing production is used to decide whether both K and L or only L will be increased.

In the short run, production of steel chairs can be increased by employing more labour only. The production in the long run can be increased by employing more of both capital and labour as all factors are variable in the long run.

Accordingly, production function can be of two kinds:

1. Short run production function
2. Long run production function

The short run production function can be written as: Q=

f (K, L)

Where K means that capital is fixed in amount.

The long run production function can be written as:

Q= f (K, L)

The production function is based on some assumptions. These are:

1. There is a perfect division of both inputs and output.
2. Only two factors of production are employed- capital and labour.
3. Imperfect substitution between labour and capital.
4. Technology is given and constant.
5. In the short run, supply of fixed factors is inelastic.

The production function will have to be changed, if there is some change in these assumptions.

9.3.1 Cobb-Douglas Production Function:

Production function of the linear homogenous type is invested by Junt wicksell and first tested by C. W. Cobb and P. H. Dougles in 1928. This famous statistical production function is known as Cobb-Douglas production function. Originally the function is applied on the empirical study of the American manufacturing industry.

Cabb – Douglas production function takes the following mathematical form.

$$Y= (AK^{X} L^{1-X})$$

Where Y=output

K=Capital

L=Labour

A, ∞=positive constant

Assumptions:

It has the following assumptions

1. The function assumes that output is the function of two factors viz. capital and labour.
2. It is a linear homogenous production function of the first degree
3. The function assumes that the logarithm of the total output of the economy is a linear function of the logarithms of the labour force and capital stock.
4. There are constant returns to scale
5. All inputs are homogenous
6. There is perfect competition
7. There is no change in technology

9.4 PRODUCTION FUNCTION WITH ONE VARIABLE INPUT

In the short run, producers can change the output by changing variable factors only. Let us suppose a firm uses two factors, labour and capital where labour is variable and capital is the fixed factor. If labour is increased to increase the output keeping the amount of capital unchanged, then there will be a change in the ratio in which these factors are used. Thus, output can be changed through a change in labour only. This type of production function is also known as variable proportion production function. It is essentially a short run production function which shows maximum output that can be produced by changing the quantity of variable factors keeping other factors fixed.

The short run production function can be written as:

$$Q= f (K, L)$$

Total Product (TP)

It can be defined as the total quantity of goods that can be produced by a firm during a specified time period. TP of a firm can be raised by utilizing more and more units of the variable factor i.e. labour. As can be seen in fig. 1.1, TPL increases at an increasing rate in the beginning with the employment

of more and more of variable factors. After a point, it starts increasing at a decreasing rate with further increase in the use of the factor. It is not necessary that the total product will always increase with the increase in the quantity of variable factors. For example, if a firm employs labours that are beyond the capacity of the factory then they will not be in a position to work efficiently. Thus, the TPL curve slopes steeply upward at first, reaches a maximum and then starts declining.

Marginal product (MP)

Marginal product is defined as change in total production due to use of an extra unit of a variable factor. For example, when 10 labours were employed in a firm, total production of chairs was 1000. Now, if one extra labour is employed, total production increases to 1050 chairs attributable to 11 labours. Since 11th labour has added 50 chairs to the total production, the marginal product of 11th labour is 50 chairs.

The marginal product of labour can be written as:

$$MP_L = \frac{\text{Change in Total Product}}{\text{Change in Labour}}$$

$$= \frac{\Delta TP_L}{\Delta L}$$

The formula for calculating MP_L is

MP_L for nth unit $= TP_n - TP_{n-1}$ n= number of labour units

In fig 1.1, marginal product rises in the beginning. When the total product is maximum, it becomes zero. Here total product is not increased with the use of extra units of variable factor. Ultimately, marginal product becomes negative when total product declines.

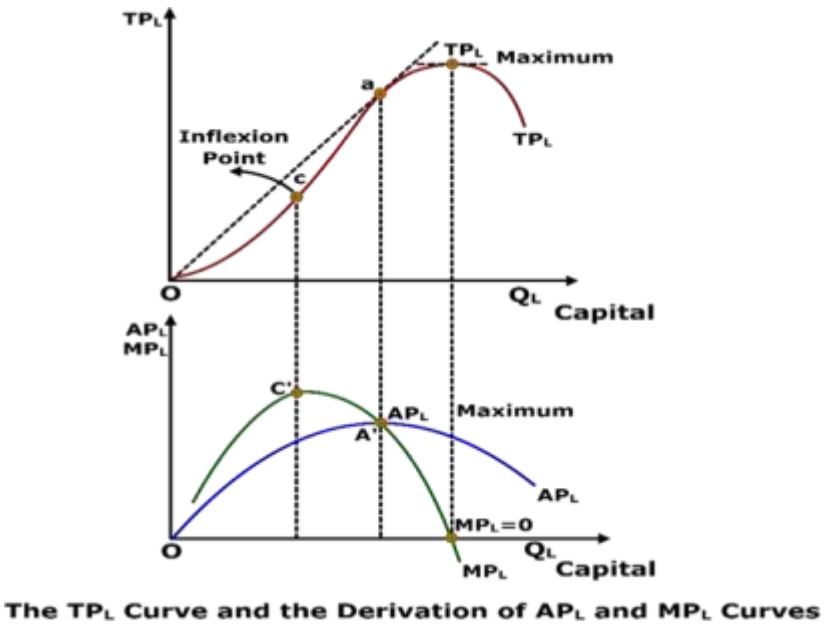

Fig 1.1: The TPL curve and the derivation of APL and MPL curves

Average Product

Average product of an input is the total product divided by the total quantity of labour employed to produce this output. It measures on an average how much output is produced by each labour. Thus,

$$APL = \frac{\text{Total Product}}{\text{Labour Input}}$$

As can be seen in fig. 1.1, like marginal products, average product also rises initially and then falls. It is an inverted U-shape curve and can never be zero or negative.

9.4.1 RELATIONSHIP BETWEEN TPL, APL, AND MPL CURVES

With the help of TPL curve, we can derive APL and MPL curve as graphically illustrated in fig. 1.1. TPL is the total product of the labour curve. Here it is assumed that the only variable factor is labour and the amount of capital is fixed (K). TPL rises when the quantity of labour increases but the increase becomes smaller and smaller.

Relationship between TPL and APL Curves

1. APL at a point is given by the slope of the straight line from the origin upto a point on the TPL curve. Till point a, the value of the slope increases and after that it declines.
2. In the beginning, APL curve increases, becomes maximum and then declines.

3. APL is positive as long as TPL is positive.
4. It is always greater than zero.

Relationship between TPL and MPL Curves

1. MPL at any point on the total product curve is the slope of the tangent at that point or MPL between two points is the slope of the line joining the two points on the TPL curve. Till point c, the value of the slope increases, then declines until TPL is maximum. MPL is zero at that point and after that it becomes negative.
2. In the beginning, MPL increases, becomes maximum when the slope of the tangent is steepest and then falls.
3. MPL is zero, when TPL is maximum and becomes negative when TPL declines.
4. Area under the MPL curve gives TPL.

Relationship between APL and MPL Curves

1. In the beginning, when both APL and MPL curves are increasing, MPL increases at a greater rate than APL curve. Both curves increase till fixed input (K) is underutilized.
2. Both APL and MPL curves decline, but MPL curves decline at a higher rate than APL curves.
3. MPL is equal to APL, when the APL curve is maximum.
4. When MPL starts falling, APL continues to increase

9.5 IMPORTANCE OF PRODUCTION FUNCTION

1. When inputs are specified in physical units, production function helps to estimate the level of production.
2. It becomes is equates when different combinations of inputs yield the same level of output.
3. It indicates the manner in which the firm can substitute on input for another without altering the total output.
4. When price is taken into consideration, the production function helps to select the least combination of inputs for the desired output.
5. It considers two types' input-output relationships namely 'law of variable proportions' and 'law of returns to scale'. Law of variable propositions explains the pattern of output in the short-run as the units of variable inputs are increased to increase the output. On the other hand law of returns to scale explains the pattern of output in the long run as all the units of inputs are increased.

6. The production function explains the maximum quantity of output, which can be produced, from any chosen quantities of various inputs or the minimum quantities of various inputs that are required to produce a given quantity of output.

Production function can be fitted the particular firm or industry or for the economy as whole. Production function will change with an improvement in technology.

9.6 ISOQUANTS

The term Isoquants is derived from the words 'iso' and 'quant' – 'Iso' means equal and 'quent' implies quantity. Isoquant therefore, means equal quantity. A family of iso-product curves or isoquants or production difference curves can represent a production function with two variable inputs, which are substitutable for one another within limits.

Iqoquants are the curves, which represent the different combinations of inputs producing a particular quantity of output. Any combination on the isoquant represents the some level of output.

For a given output level firm's production become,

$$Q= f (L, K)$$

Where 'Q', the units of output is a function of the quantity of two inputs 'L' and 'K'.

Thus, an isoquant shows all possible combinations of two inputs, which are capable of producing equal or a given level of output. Since each combination yields same output, the producer becomes indifferent towards these combinations.

Assumptions:

There are only two factors of production, viz. labour and capital.

The two factors can substitute each other up to certain limit

The shape of the isoquant depends upon the extent of substitutability of the two inputs.

The technology is given over a period.

An isoquant may be explained with the help of an arithmetical example.

Table 1:

Combinations	Labour (units)	Capital (units)	Output (quintals)
A	1	10	50
B	2	7	50
C	3	4	50
D	4	4	50
E	5	1	50

Combination 'A' represent 1 unit of labour and 10 units of capital and produces '50' quintals of a product all other combinations in the table are assumed to yield the same given output of a product say '50' quintals by employing any one of the alternative combinations of the two factors labour and capital.

CASE STUDIES:

Case Study 1: Optimizing Production in a Furniture Manufacturing Unit

Background: A medium-sized furniture company in Jaipur uses labour and capital to manufacture wooden chairs. The manager observes that despite increasing the number of workers, output is not increasing proportionately.

Application: Using the law of variable proportions, the manager analyses the Total Product (TP), Marginal Product (MP), and Average Product (AP). It is revealed that the MP becomes negative after a certain point due to overcrowding and insufficient machines.

Conclusion: The firm adjusts the labour-to-capital ratio to reach the point where APL is maximized and MPL = APL, leading to optimal production and reduced costs.

Case Study 2: Implementing Cobb-Douglas Production in a Tech Start-Up

Background: A Bangalore-based tech start-up utilizes automated tools (capital) and software engineers (labour) to develop SaaS products.

Application: By applying the Cobb-Douglas Production Function ($Y = A K^{\alpha} L^{(1-\alpha)}$), the firm assesses how changes in inputs impact output. It notices that increasing labour without upgrading technology results in diminishing returns.

Conclusion: The company scales both labour and capital proportionately, achieving constant returns to scale and maintaining high productivity.

9.7 LET US SUM UP

- The transformation of resources (men, material, time and so on) into a different & more useful commodity through a process is called production.
- Fixed factor is one whose quantity remains the same irrespective of the level of output. Its quantity cannot be readily changed in the short run.
- A variable factor is one whose quantity changes with the changes in the level of output.
- Short run refers to that time period in which supply of certain factors is fixed i.e. it cannot be increased or decreased. For e.g. plant, machinery, building etc.
- Long run refers to that time period in which supply of all factors i.e. fixed and variable is elastic, but not sufficient to allow a change in technology.
- Production function is a mathematical statement used to explain the technical relationship between input and output in physical terms over a given period of time.
- Production function can be of two kinds: (a) Short run production function (b) Long run production function.

KEYWORDS:

1. **Production Function:** A mathematical representation showing the relationship between inputs and the resulting output.
2. **Isoquant:** A curve that shows all input combinations that produce the same level of output.
3. **Marginal Product (MP):** The additional output generated by using one more unit of a variable input.
4. **Law of Variable Proportions:** Describes how output changes when only one input varies, keeping other inputs constant.
5. **Cobb-Douglas Function:** A specific production function form that shows the output as a function of capital and labour with constant returns to scale.

9.8 REVIEW QUESTIONS

1. Define and explain the concept of the production function with suitable examples.
2. Differentiate between short-run and long-run production functions.
3. Discuss the significance of the Cobb-Douglas production function in managerial decision-making.
4. Illustrate the relationship between Total Product, Marginal Product, and Average Product with the help of a diagram.
5. What is an isoquant? Explain its properties and relevance in production planning.

OBJECTIVE TYPE QUESTIONS

1. The _____ function expresses the relationship between inputs and outputs.
2. In the short run, _____ factors cannot be changed.
3. _____ is the additional output from employing one more unit of input.
4. Total Product increases at a _____ rate initially.
5. When Marginal Product becomes zero, Total Product is at its _____.
6. An _____ represents equal levels of output for different input combinations.
7. The Cobb-Douglas function assumes _____ returns to scale.
8. In the long run, all factors of production are _____.
9. The shape of the isoquant depends on the _____ between inputs.
10. _____ Product is calculated by dividing total product by the number of labour units.
11. When MP is greater than AP, AP is _____.
12. When TP is declining, MP is _____.
13. The variable proportion production function applies in the _____ run.
14. Technology is assumed to be _____ in the production function model.
15. A firm achieves least-cost input combination by analyzing its _____.

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- Managerial Economics and Business Strategy by Michael Baye

UNIT-10: LAW OF VARIABLE PROPORTIONS

STRUCTURE

10.1 Introduction

10.2 Law of Production 10.3 Law of Variable Proportions

10.4 Graphical Representation:

10.5 Applications of the Law of Variable Proportions

10.6 Let's Sum Up

10.7 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

1. Understand the concept of the Law of Variable Proportions and its significance in production theory.
2. Explain the relationship between the variable and fixed inputs in the production process and how this affects total output in the short run.
3. Identify and describe the three stages of the Law of Variable Proportions: Increasing Returns, Diminishing Returns, and Negative Returns.
4. Illustrate the Law of Variable Proportions with the help of graphs showing Total Product (TP), Marginal Product (MP), and Average Product (AP).
5. Discuss the applications of the Law of Variable Proportions in production decisions, cost analysis, labor and capital allocation, and short-run vs. long- run production.
6. Evaluate the impact of diminishing returns on cost behavior and resource allocation in a firm's production process.

BLOOM'S LEVEL OF LEARNING:

The content on the Law of Variable Proportions primarily aligns with the "Understand" and "Apply" levels of Bloom's Taxonomy. It emphasizes the conceptual understanding of how varying one

production input affects output, while other factors remain constant. Students are expected to comprehend the theory and its significance in production decisions, cost analysis, and resource allocation in the short run. Additionally, they must apply this knowledge by illustrating different stages of the law—Increasing Returns, Diminishing Returns, and Negative Returns—using graphical representations of Total Product, Marginal Product, and Average Product. Moreover, the text encourages students to connect the theory to real-world production scenarios, making informed decisions about labour and capital allocation based on diminishing returns. The explanations help learners grasp theoretical concepts and enable them to apply these ideas in practical production contexts, bridging the gap between abstract theory and real-life applications in economics.

10.1 INTRODUCTION

The law of variable proportion is a key principle in production theory, describing how variations in one production input impact the overall output, while other inputs remain unchanged. This law is especially relevant in the short run, where one factor of production, such as machinery or land, remains fixed, and only the variable input, like labor or raw materials, can be modified. In the short run, businesses can alter certain inputs, like the number of workers, but cannot change the fixed resources, such as factory space or machines. The law examines how increasing the quantity of the variable input impacts production, assuming that the fixed inputs remain unchanged.

1. Understanding the Law in the Context of Short-Run Production:

In the short run, the law of variable proportion aids businesses in understanding the impact of increasing a variable input, such as labor, while fixed inputs, like machinery, remain unchanged. At first, as more units of the variable input are introduced, output rises at an accelerated rate because the fixed inputs are not being fully utilized.

However, as additional units of the variable input are added, the rate of output growth begins to diminish. Eventually, the extra units of the variable input lead to inefficiencies, as the fixed inputs become overwhelmed or less efficient in contributing to the production process.

Why This Happens:

- Fixed Inputs: In the short run, some factors of production (like machinery) cannot be altered. These are referred to as fixed inputs.
- Variable Inputs: “The inputs that can be adjusted in the short run, such as labor, are considered variable”.

As more workers are hired, the efficiency of using the fixed capital (like machines) starts to diminish. At some point, adding more labor will result in only marginal increases in output, and in extreme cases, could lead to a reduction in overall productivity.

2. Stages of the Law of Variable Proportions:

1. **Stage of Increasing Returns**: "In the initial stage, adding more of the variable input (e.g., labor) leads to greater efficiency, and output increases at an increasing rate because the fixed resources are being used more effectively".
2. **Stage of Diminishing Returns**: After a certain point, the addition of more units of the variable input results in progressively smaller increases in output. This is because the fixed resources are becoming fully utilized, and further additions to the variable input are less productive.
3. **Stage of Negative Returns**: If too many units of the variable input are added, the output may begin to decline as the fixed resources are overwhelmed, and productivity decreases.

10.2 LAW OF PRODUCTION

The law of production in economics explains "the process of converting inputs (such as labor, capital, and land) into outputs (goods or services)". Production is the activity that combines various resources to create products that satisfy consumer needs. The law helps businesses determine how to combine different inputs in the most efficient way to maximize output.

Concepts of the Law of Production:

1. **Inputs**: These are the resources required for production. They are divided into three main categories:

- **Land**: Natural resources like raw materials, water, and space that are used in production.
- **Labor**: The human effort, including physical and mental work, that contributes to the creation of goods and services.
- **Capital**: "The tools, machinery, equipment, and infrastructure used in the production process".

2. **Outputs**: These are the finished goods or services produced using the inputs. The goal of production is to transform these inputs into valuable outputs that can be consumed or sold in the market.

The law of production refers to how different inputs are combined to produce output. It involves understanding the relationship between the amount of input used and the resulting output, as well as determining the most efficient combination of inputs to achieve maximum production.

One important aspect of the law is the concept of marginal productivity, which looks at how much additional output is produced when an extra unit of an input is added while keeping other inputs constant.

In the short run, where some inputs are fixed and others are variable, this law helps businesses determine how much labor to hire or how much capital to invest, depending on the availability of other resources.

Optimizing the Use of Inputs:

A key goal in production is to optimize the mix of inputs to generate the highest possible output with the least waste. For example:

- If a company has a fixed amount of machinery (capital), it may need to adjust the number of workers (labour) it employs to maximize output.
- However, there is a limit to how much the addition of labour will improve output if the machinery is already being fully utilized.

The law of production guides businesses in making these decisions by analysing the effects of changing input quantities and understanding diminishing returns—when adding more of a variable input (like labour) leads to progressively smaller increases in output.

Factors Affecting Production Decisions:

- **Technology**: Technological advancements can change the efficiency with which inputs are combined, potentially improving productivity and reducing costs.
- **Costs of Inputs**: The prices of labour, capital, and land influence how a business chooses to allocate resources. For instance, if labour costs rise, a firm might invest in automation to reduce dependence on human workers.
- **Economies of Scale**: "As production increases, businesses may experience economies of scale, meaning that larger production volumes lead to lower costs per unit".

10.3 THE LAW OF VARIABLE PROPORTIONS OR DIMINISHINGMARGINAL RETURNS

The Law of Variable Proportions, "often referred to as the Law of Diminishing Returns, is an essential concept in production theory. It explains how the output generated by a combination of variable inputs, like labor, and fixed inputs, such as machinery or land, changes as more units of the variable input are added to the fixed input". **Concept of the Law:**

The law states that, as more units of a variable input (e.g., labor) are added to a fixed input (e.g., capital), the total output initially increases but at a diminishing rate after a certain point. This happens because, in the beginning, the addition of more labor helps to utilize the fixed inputs more efficiently. However, once a certain threshold is reached, the additional labor becomes less effective, leading to smaller increases in output.

Phases of the Law of Variable Proportions:

1. **Increasing Returns to the Variable Input (Stage I)**:

- "In the early stages of production, adding more units of the variable input (like labor) leads to a larger increase in output.
- This happens because the fixed inputs (e.g., machines or land) are underutilized, and the addition of labor or other variable factors leads to better and more efficient use of these resources.
- In this phase, the marginal product (the additional output generated by each additional unit of labor) increases. Workers are able to work together more effectively, leading to a higher total output".

2. **Diminishing Returns to the Variable Input (Stage II)**:

- "As more units of the variable input are added, a point is reached where each additional unit of the variable input adds less to total output. This is known as diminishing returns".
- The marginal product starts to decrease because the fixed input becomes crowded or overused, leading to inefficiencies. For instance, too many workers may be trying to operate the same amount of machinery, resulting in less productive output.
- While the total output continues to increase, it does so at a slower rate compared to the previous stage.

3. **Negative Returns to the Variable Input (Stage III)**:

- At this stage, adding more units of the variable input results in a decrease in total output.
- The marginal product becomes negative because the excess variable input creates overcrowding or excessive strain on the fixed input, leading to inefficiencies and even a drop in production.
- For example, too many workers in a factory or too much fertilizer on crops can lead to negative productivity, where additional input actually reduces output.

Example:

Imagine a factory that produces widgets using machines (fixed input) and labor (variable input). Initially, adding more workers allows the factory to operate more efficiently because the machines are being used more effectively. However, after reaching a certain number of workers, the factory becomes

crowded, and additional workers start to get in each other's way, leading to a slower increase in output. Eventually, if too many workers are added, output may even decline because the workers are unable to effectively use the machines.

Why It Happens:

- In the initial stage (increasing returns), there are many unused or underused fixed resources, so adding more variable inputs (like labour) helps to fully utilize these resources, leading to greater efficiency.
- In the diminishing returns stage, the fixed resources (capital, machinery, land) become saturated. As more labour is added, each worker has less equipment to work with, leading to inefficiency and a slower rate of increase in output.
- In the negative returns stage, the overcrowding or excess of the variable input (labour) creates such inefficiencies that it actually reduces the total output produced.

10.4 GRAPHICAL REPRESENTATION

The Law of Variable Proportions can be graphically represented through three important curves: "Total Product (TP), Marginal Product (MP), and Average Product (AP)". These curves help to visualize how output changes as more units of a variable input are added, while other inputs remain fixed.

Explanation of the Curves:

1. **Total Product (TP) Curve**:

- This curve represents the total output produced with varying levels of the variable input, keeping the fixed inputs constant.
- In the initial stage (Stage I), the TP curve increases at an increasing rate because the fixed input is underutilized, and the addition of the variable input leads to a more efficient use of resources.
- In Stage II, the TP curve increases, but at a decreasing rate, as diminishing returns begin to set in. The additional output from each additional unit of the variable input becomes smaller.

2. **Marginal Product (MP) Curve**:

- The MP curve represents the additional output generated by each additional unit of the variable input.
- In the first stage, the MP increases as more variable inputs are added, reflecting increasing returns.

- In Stage II, the MP curve starts to decline because of diminishing returns; each additional worker, for example, contributes less to the overall output.
- In Stage III, the MP becomes negative as the addition of more variable inputs leads to inefficiency and a reduction in total output.

3. **Average Product (AP) Curve**:

- The AP curve shows the average output produced per unit of the variable input.
- The AP initially increases as the number of variable inputs increases, reflecting higher productivity.
- As more inputs are added, the AP reaches a peak and then begins to decline, following the trend of the MP curve.
- The AP curve typically peaks before the MP curve starts to fall, as the average product is influenced by the marginal product but will increase as long as the MP is above it.

Example Graph:

Below is a conceptual diagram representing the relationship between Total Product (TP), Marginal Product (MP), and Average Product (AP):

- X-axis represents the quantity of variable input (e.g., labor).
- Y-axis represents the output (either total product, marginal product, or average product).

```
|
|          TP
|         /
|        /
|       /
|      /
|--------/ ---------------------------- > Variable Input (e.g., Labor)
|     /
|    /
|   /
|  /
| /
```

| /

|/

MP, AP

- Total Product (TP): Initially, it increases at an increasing rate and then at a decreasing rate in Stage II.
- Marginal Product (MP): MP starts at a high value, increases, reaches a peak, and then declines as more units of the variable input are added.
- Average Product (AP): The AP curve rises until it peaks and then starts to decline, following the behavior of the MP curve.

Example Scenario:

Let's consider a factory producing shirts, with a fixed number of sewing machines (capital) and a variable number of workers (labor):

- **Stage I (Increasing Returns)**: Initially, adding more workers allows the factory to use the available machines more efficiently. Each additional worker makes better use of the machines, leading to a greater increase in the number of shirts produced per worker (increasing returns). Here, both TP and AP increase, while MP rises.
- **Stage II (Diminishing Returns)**: As more workers are added, the machines start becoming crowded. Each new worker contributes less to total production because there is not enough machinery for everyone to work effectively. Thus, TP still increases, but at a decreasing rate, MP begins to fall, and AP reaches its peak.
- **Stage III (Negative Returns)**: After a certain point, adding more workers creates overcrowding, and workers start to get in each other's way. In this stage, additional workers decrease the total number of shirts produced, so TP begins to fall, and MP becomes negative.

10.5 APPLICATIONS OF LAW OF VARIABLE PROPORTIONS

The Law of Variable Proportions plays a vital role in guiding production decisions, cost analysis, and the allocation of resources in a firm. "It provides valuable insights into how output changes as more units of a variable input are added to fixed inputs, helping businesses optimize their production process and make more informed decisions". Let's explore its applications with examples.

1. Production Decisions:

- Explanation: Understanding the Law of Variable Proportions helps firms decide the optimal combination of labour and capital in their production process. "By analysing how the addition of labour (a variable input) affects output, firms can determine the most efficient combination of labour and capital to maximize production while minimizing costs".
- Example: Imagine a bakery with a fixed number of ovens (capital) and a variable number of bakers (labour). Initially, adding more bakers allows the bakery to produce more loaves of bread because the ovens are not being fully utilized. However, after a certain point, overcrowding occurs, and each additional baker contributes less to total production. The bakery must recognize the point where additional bakers no longer increase output efficiently and stop hiring more workers. Thus, understanding the law helps the bakery decide the optimal number of workers for its available ovens, minimizing labour costs and maximizing output.

2. Cost Analysis:

- Explanation: "The Law of Variable Proportions plays a crucial role in understanding cost behaviour". As diminishing returns set in, firms can anticipate that the costs associated with using additional units of the variable input (such as labour) will increase. "This is because each additional unit of the variable input yields less additional output, leading to higher average costs".
- Example: Consider a construction company using a fixed number of bulldozers (capital) and a variable number of workers (labour). In the initial stages, adding more workers to the site increases the efficiency of bulldozer use, leading to a reduction in the cost per unit of construction (lower average cost). However, as more workers are added, the site becomes overcrowded, and the workers become less effective, leading to rising costs per unit of construction (higher average cost). The company can use the Law of Variable Proportions to predict when adding more workers will no longer be cost-effective and adjust its labour hiring decisions accordingly.

3. Labor and Capital Allocation:

- Explanation: The law helps firms allocate resources (labour and capital) effectively. By understanding the point at which diminishing returns set in for labor, firms can determine the most

efficient amount of labour to add to their existing fixed capital (such as machinery or land). This prevents the inefficiency that arises from overusing one type of resource while underutilizing another.

- Example: A manufacturing company has 10 machines (capital) and initially employs 5 workers (labor). The company is able to produce 100 units of product efficiently. If the company decides to hire 10 more workers, the productivity of each worker may begin to decline due to overcrowding, and the additional workers may produce fewer units than the first 5. The company can use the Law of Variable Proportions to evaluate how many workers should be hired without overwhelming the available machines. This ensures that the allocation of labor and capital is balanced to achieve maximum efficiency.

4. Short-Run vs. Long-Run Production: • Explanation: "The Law of Variable Proportions applies in the short run when one or more factors of production (such as capital) are fixed. It helps businesses understand the relationship between input quantities and output levels in the short run. In the long run, however, all factors of production can be varied (e.g., labour and capital), and firms can achieve economies of scale", which allows them to reduce average costs by increasing the scale of production.
   - Example: In the short run, a car manufacturing plant may have a fixed number of assembly lines (capital) and can only hire a certain number of workers (labour). As the plant hires more workers, the production increases, but after a point, the addition of more workers leads to diminishing returns. If the company keeps hiring workers beyond this point, the increase in production will become smaller and costs will rise.

However, in the long run, the company can invest in more assembly lines (increasing capital) or improve its production technology. By expanding its capacity and investing in better technology, the company can achieve economies of scale, where increasing the number of workers or expanding capital leads to proportionally greater increases in output, thus reducing costs per unit of production. The transition from short-run to long- run production involves moving beyond the constraints of the Law of Variable Proportions and adopting strategies that allow for more flexible resource allocation.

Case Studies:

Case Study 1: Labor Allocation in a Garment Factory

A medium-sized garment factory employs 15 sewing machines and starts production with 5 workers. As demand rises, the manager hires 5 more workers. Initially, the output per worker increases because the machines are better utilized. However, when 5 additional workers are hired again, crowding around

machines starts to occur. Productivity increases at a decreasing rate, marking the stage of diminishing returns. Eventually, hiring 5 more workers leads to confusion, longer wait times for machine usage, and even a drop in total output — a clear case of negative returns. The factory uses the Law of Variable Proportions to cap hiring and optimize resource use.

Case Study 2: Agricultural Land and Fertilizer Use

A farmer owns a fixed plot of 5 acres and applies fertilizer to improve crop yield. In the first few applications, the output increases significantly as the nutrients enhance plant growth. This represents increasing returns. As the farmer continues to add fertilizer beyond the optimal level, the increase in yield begins to taper off, indicating diminishing returns. Eventually, over-fertilization damages the soil and reduces crop productivity, a textbook case of negative returns. Understanding the Law of Variable Proportions helps the farmer apply just the right amount of fertilizer for maximum yield and cost efficiency.

10.6 LET US SUM UP

The Law of Variable Proportions is crucial in understanding "how the addition of a variable input (like labour) to a fixed input (like capital)" affects total output in the short run. It is characterized by three distinct phases:

1. **Increasing Returns** – As more units of the variable input are added, output increases at an increasing rate.
2. **Diminishing Returns** – After a certain point, output increases at a decreasing rate.
3. **Negative Returns** – Beyond a certain point, adding more of the variable input leads to a reduction in total output.

This law helps firms optimize the allocation of resources, such as labor and capital, to maximize output and minimize costs in the short run. It also forms the foundation for decision-making in production planning, cost control, and labor management. In the long run, all factors of production can be varied, and firms can achieve economies of scale.

KEYWORDS

1. **Variable Input**

   An input that can be changed in the short run, such as labor or raw materials, to adjust the level of production.

2. **Fixed Input**

   A production resource, like land or machinery, that remains unchanged in the short run.

3. **Marginal Product (MP)**

   The additional output produced by adding one more unit of a variable input, keeping other inputs constant.

4. **Total Product (TP)**

   The overall quantity of output produced from a given set of inputs in the short-run production process.

5. **Diminishing Returns**

   A stage in production where each additional unit of the variable input contributes less to total output than the previous one.

10.7 ASSIGNMENT QUESTIONS

1. What is the Law of Variable Proportions, and why is it important in production theory?
2. Explain the three stages of the Law of Variable Proportions.
3. How does the Law of Variable Proportions apply to a firm's decision-making process in the short run?
4. Illustrate the relationship between Total Product, Marginal Product, and Average Product with a graph.
5. How does the Law of Variable Proportions help in cost analysis and resource allocation?

OBJECTIVE TYPE QUESTIONS

1. The Law of Variable Proportions applies to the _____ run.
2. _____ inputs remain constant in the short run.
3. The additional output from an extra unit of input is called _____ product.
4. In Stage I of the Law of Variable Proportions, total output increases at an _____ rate.
5. The point at which marginal product starts to decline marks the beginning of the _____ returns stage.
6. When marginal product becomes negative, it indicates the stage of _____ returns.
7. The Law of Variable Proportions is also known as the Law of _____ Returns.
8. As more variable input is added to a fixed input, the total product initially _____.
9. Average product peaks before the marginal product becomes _____.

10. In the stage of diminishing returns, marginal product is _____ but falling.
11. Production efficiency in Stage I is due to underutilization of _____ inputs.
12. In the short run, firms can only vary their _____ inputs.
13. The TP curve eventually becomes flat or declines in Stage _____.
14. The concept of the Law of Variable Proportions is based on the assumption of one input being _____.
15. Overuse of fertilizer leading to lower yield is an example of _____ returns.

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- Economic Theory – Bo Sodersten
- Principles of Economics – N. Gregory Mankiw

UNIT-11: LAW OF RETURN TO SCALE

LEARNING OBJECTIVES

After studying this unit, you will be able to:

- Understand the meaning and returns to scale through isoquants
- Discuss about product lines and isoclines
- Explain the economies and diseconomies of scale
- Derive the long run cost curve

STRUCTURE

11.1 Introduction

11.2 Returns to scale through Isoquants

11.3 Economies and Diseconomies of Scale

11.3.1 Economies of Scale

11.3.2 Diseconomies of Scale

11.4 Long-run cost Curves

11.5 Let us Sum up

11.6 Review Questions

BLOOM'S LEVEL OF LEARNING:

The Bloom's level of learning can be assessed at various stages depending on the depth of understanding. The unit encourages students to progress through multiple cognitive levels, starting from remembering (basic recall of definitions like returns to scale, isoquants, and cost curves), then advancing to understanding (explaining concepts such as economies and diseconomies of scale). As students work through the content, they engage in applying (using the knowledge of cost curves and returns to scale to solve real-world problems), analyzing (differentiating between internal and external economies and analyzing shifts in isoclines), and evaluating (judging the effectiveness of long-run cost curves and their

implications for business expansion). At the highest level, students will engage in creating by integrating their understanding of these concepts to form comprehensive strategies for production planning and optimization, predicting potential changes in a firm's cost structure over time.

11.1 INTRODUCTION

In the long run, all factors of production are variable. A firm can increase its output by increasing the quantity of all inputs i.e. the 'Scale of Operation'. The returns that are generated are called 'Returns to Scale'. The law of return to scale examines the behavior of output. It explains input output relationship in the long run.

11.2 RETURNS TO SCALE

In the long run the study of production function is concerned with the law of returns to scale. When all factors are changed in the same proportion keeping their proportions constant, the scale of output will also change. The response of the output to these changes in inputs is called returns to scale. Technology is assumed to remain constant while studying the returns to scale. There are three possibilities of these output response, when the producers increase the inputs in a given proportion i.e. the output may increase more than proportionately, just proportionately or less than proportionately.

Suppose in the production function there are two variable inputs i.e. capital (K) and labour (L), then the production function can be expressed as: $Q = f(K,L)$

Q denotes the quantity of output produced. Now suppose both the factor inputs i.e.

K&L are increased in proportion 'm' and the total output increases in proportion 'n', the new production function can be expressed as: $nQ= f(mK, mL)$

The proportion 'n' i.e. the response of the output may be equal to, greater than or less than 'm'

i.e. the change in factor proportion. Accordingly, the stages of the law of returns to scale can be:

1) If $n>m$, the increase in quantity of commodity produced is greater than the proportional increase in the input units, this is a situation of increasing returns to scale. Thus, if factors of production are doubled and the quantity of commodity produced i.e. output is more than doubled then there is increasing returns to scale. 2) If $n=m$, the increase in the quantity of commodity produced is equal to the proportional increase in the input units, this is a situation of constant returns to scale. Thus, if factors of production are doubled and the quantity of commodity produced i.e. output is also doubled then there is constant returns to scale.

3) If n<m, the increase in the quantity of commodity produced is less than the proportional increase in the input units, this is a situation of diminishing returns to scale. Thus, if factors of production are doubled and the quantity of commodity produced i.e. output is less than doubled then there is diminishing returns to scale. Here the three possibilities of the response of output to a proportionate change in all the inputs are expressed by the concept of Isoquants. Graphically the returns to scale may be expressed by the shifts of the Isoquants on a product line.

Product Line and Isocline

A product line shows the physical movement from one Isoquant to another in response
to changes in quantity of input(s). It is independent of the prices of factors of production. It does not shows any actual choice of expansion which is based on the prices of factors, rather it shows the technically feasible paths of expanding output.

If all the factors are variable i.e. in the long run, the product line passes through the origin. If only one factor is variable and all other factors are constant i.e. in the short run, the product line is a straight line parallel to the axis of the variable factor.

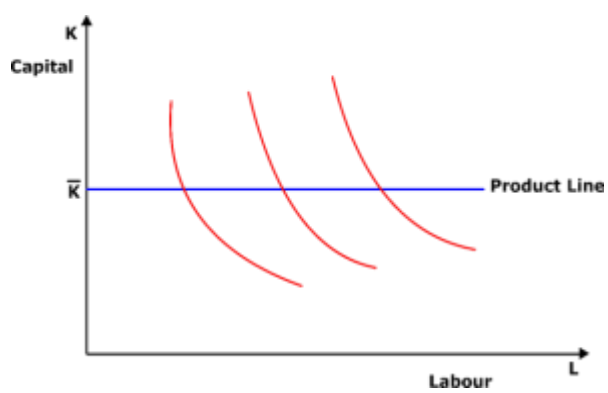


Figure 1.4

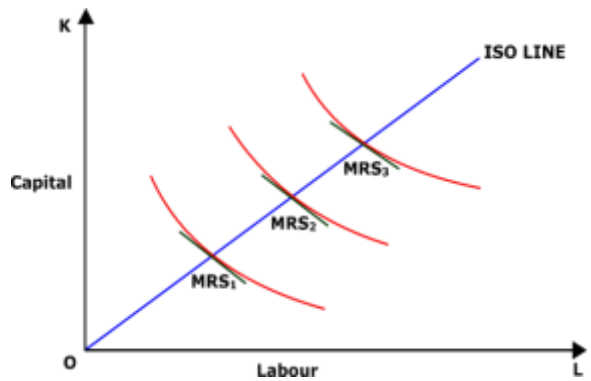


Figure 1.5

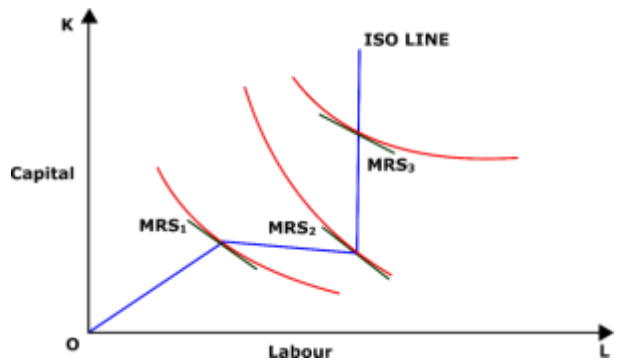


Figure 1.6

Fig 1.4 depicts an isocline for a family of isoquants that are homogenous such that the locus of the point at which the product line touches the respective isoquant has MRS which is equal at all points of tangency. Thus, in case of homogenous isoquants isocline will be a straight line arising from the origin of uniform slope.

In fig 1.5 we have depicted an isocline which is based on a family of isoquants which are non-homogenous. In this case the point at which the product line touches the respective isoquant should have the MRS. However, because the isoquants are non- homogenous, the same MRS is not achieved along the same straight line arising from the origin. The slope of such an isoquant would be varying and hence isocline itself would be twiddly.

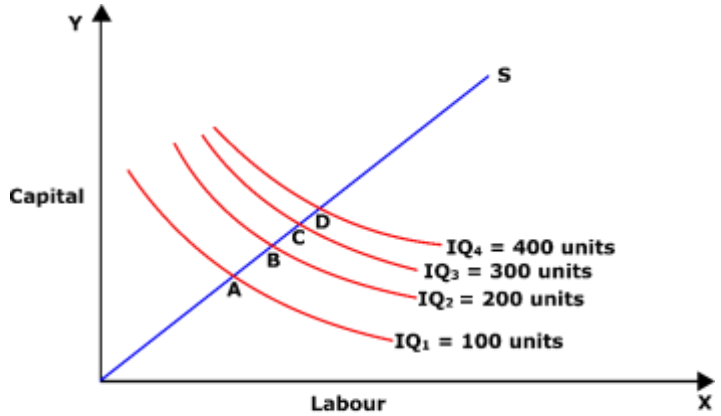
If the product line subtends equal angles with the axes i.e. at which the MRS of factors is constant then it is called an Isocline. The Isocline are straight line through the origin, if the production function is homogeneous and if the production function is non-homogeneous the shape of the Isocline will be twiddly

RETURNS TO SCALE FOR A HOMOGENEOUS PRODUCTION FUNCTION

OS, a ray from the origin represents increase in the scale. Along Isocline the ration between labour and capital remains the same, though their absolute amount keep rising, the technically possible alternative paths of expanding output is shown by this ray.

INCREASING RETURNS TO SCALE

The proportion of increase in output is more than the proportion of increase in all inputs. The distance between consecutive multiple Isoquants along the Isocline, OS decreases, in case of increasing returns to scale. It means an equal increase in output is obtained by smaller and smaller increments in inputs. Here AB>BC>CD. In case of increasing returns to scale if the firm wants to double its output, it requires less than double inputs.



Increasing Returns to Scale

Figure 1.7

DIMINISHING RETURNS TO SCALE

The proportion of increase in output is less than the proportion of increase in all inputs. The distance between consecutive multiple Isoquants along the Isocline OS increases, in case of diminishing returns to scale. It means an equal increase in output is obtained by larger and larger increments in inputs. Here

AB<BC<CD. Thus, in case of diminishing returns to scale if the firm wants to double its output, it requires more than double inputs.

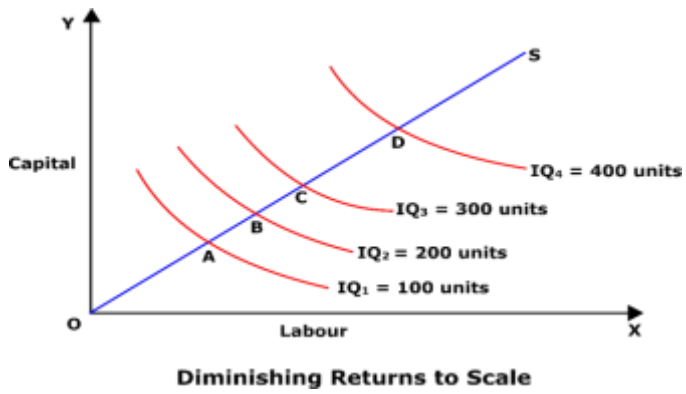


Figure 1.8

CONSTANT RETURNS TO SCALE

The proportion of increase in output is same as the proportion of increase in all inputs. The distance between consecutive multiple Isoquants along the Isocline OS remains same, in case of constant returns to scale. It means an equal increase in output is obtained by equal increments in inputs. Here AB=BC=CD. Thus, in case of constant returns to scale if the firm wants to double its output, it requires double inputs.

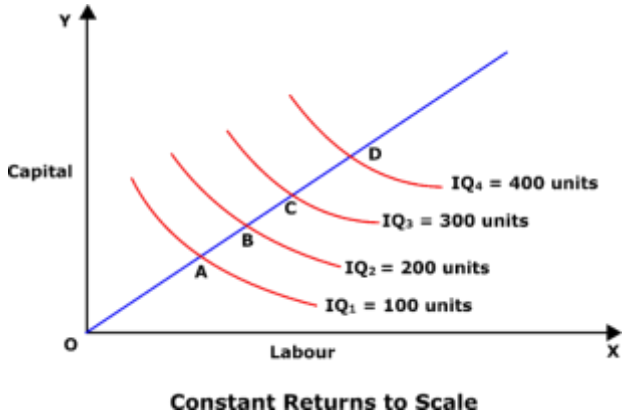
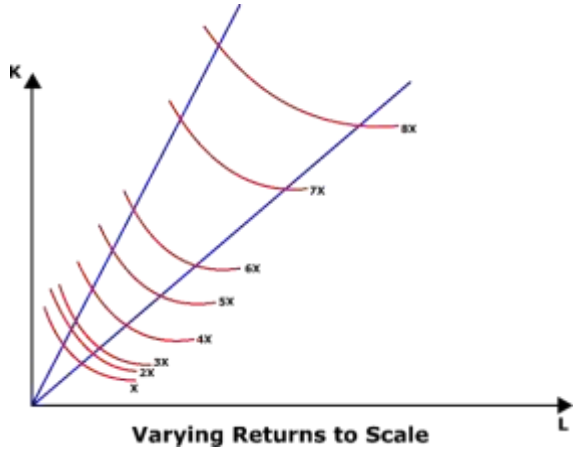


Figure 1.9

In a single production function the three phases of returns to scale occurs in the order of increasing, constant and diminishing returns to scale. In the long run, as firm



continues its production eventually diminishing returns to scale sets in after passing through the phases of increasing and constant returns to scale.

Figure 1.10

11.3 ECONOMIES AND DISECONOMIES OF SCALE

11.3.1 ECONOMIES OF SCALE

In order to earn more profits, a business firm expands its scale of operations. The large scale of production provides many economies to the firm by helping in reducing the cost of production and increasing its productive efficiency. Such economies that occur to a firm due to increase in all factors of production or increase in number of firms in the industry is called the economies of scale. Internal economies and external economies are the two types of such economies.

INTERNAL ECONOMIES

The economies that accrue to a firm due to increase in its plant size or due to increase in the number of its plants are referred to as the internal economies of scale. These economies are independent of the actions of other firms and accrue to a firm largely because of its own efforts. It means that the firm do

not share the advantage of these economies with its rival firms. Intra-plant economies of scale arise due to increase in the size of a single plant whereas inter-plant economies of scale arise due to increase in the number of plants. This shows that the internal economies of scale are further categorizes as the intra-plant and inter-plant economies of scale. In internal economies of scale the economic advantage to a firm is proportionately larger than the increments in factor inputs thus causing the increasing returns to scale. These economies are different from firm to firm and are peculiar to each firm. Such economies of scale determine the shape of long run average cost curve by reducing it and the process of cost reduction continues till such economies are fully exhausted. Some of the internal economies of scale are as follows:

1) Technical economies

Technical economies are the economies which accrue due to the capacity of larger firms being able to use more efficient techniques of production and the fact that larger plants are often cheaper to run. It includes

a) Economies due to superior technique- Larger firms are able to install specialized machines. Though the fixed cost of these machines may be very high but the average cost will be much lower as the total cost will be spread over a large scale of production. Therefore, the larger firm enjoys the benefit of superior techniques of production.

b) Economies due to linked processes-Instead of purchasing the materials and other inputs from outside suppliers the firm can decide to produce these inputs itself. Similarly, by taking up the marketing function of selling and distribution of its own products the firm can save the expenses on intermediaries. Thus, by doing this a large firm can enjoy the benefits from backward and forward integration of processes.

c) Economies by using the by- products-A large firm can achieve more economies by using the waste of one product as the by- product of another product. For example, the waste left after manufacturing the sugar can be used as the by- product for producing paper by installing a plant for this purpose.

d) Economies of reserve capacity-When the firms operate at large scale they have built in reserve capacity in their plants, machinery and other equipments. By this these firms are in a better position to meet the changes that arises in demand and avoid disruption due to break down of machinery etc.

2) Marketing Economies

Marketing economies refers to advantages acquired by a firm due to purchase of inputs at cheaper prices and sale of finished goods at the highest prices. These arise due to:

(a) **Economies of Purchase**- A large firm is able to buy raw materials and other factors of production in bulk quantities. Because of this, it can obtain concessions from transport and railways companies. Also, large firms generally hire experienced persons to deal with buy of raw material.

(b) **Economies of Sale**- A large firm can reduce its selling cost by employing highly experienced staff who purchase and sell on behalf of the firm in favourable conditions of the market. Also, large firm is able to take up more advertising activity to increase the sales.

3) Financial Economies

A large firm with a large asset base and high creditworthiness in the market is able to obtain loans from the banks and financial institutions at easy and cheaper rates. It can also obtain loans from private sources. Small firms are not able to get such opportunities.

4) Risk Bearing Economies

There are general and particular risks which every firm has to face for its existence. Small firms are not in a position to bear such risks and thus go into liquidation. On the other hand, large firms are not only able to absorb such risks but also diversify their output, sources of supply, market and manufacturing process.

5) Labour Economies

Higher degree of specialization and division of labour becomes possible due to increase in scale of production. This specialization and division of labour makes the labour perfect in their work which in turn helps in reducing the wastage of time in moving from one job to another and also in changing tools, which results in higher efficiency and productivity in the firm causing the increasing returns to scale. In other words the reduce in labour cost per unit helps in achieving the increasing returns to scale in the organization.

EXTERNAL ECONOMIES

External economies are the advantages that are generated outside the firm. These economies arise to a firm when there is an expansion of an industry. The firms are not required to make any individual

cost reduction efforts. All firms of an industry can enjoy these benefits. The following are the different kinds of external economies:

1) *Economies of Concentration*- Economies of concentration arises when a number of firms producing the same products are concentrated in a particular area. The following economies arises:

a) Availability of more and better skilled labourers saves the cost of training.

b) Availability of better financial facilities at cheaper cost.

c) Availability of better and cheaper transportation and communication facilities.

d) Availability of better marketing facilities.

2) *Economies of Information*-Economies of information arise to a firm as all the necessary and pivotal information regarding output, labour, profit etc are easily available without doing any efforts. If a firm is situated in a remote area, it is very difficult to know the conditions of the market. But, when large number of firms are concentrated in a particular area, they start a bulletin or information paper that gives all essential statistics. This leads to reduction in time and costs.

3) *Economies of disintegration*-Economies of disintegration arise due to localization of industries. A single firm cannot produce adequate wastage or by products to enable specialized firm to use them. But, if a number of firms are concentrated at one place, then it is possible that more specialized firms are using by- products.

11.3.2 DISECONOMIES OF SCALE

Diseconomies of scale are the disadvantages or difficulties that arise to a firm when it expands the scale of operations beyond the optimum capacity. It results in increased cost of production which gives rise to decreasing returns to scale. There are two types of diseconomies:

INTERNAL DISECONOMIES

Internal diseconomies refer to the disadvantages that increases the production cost when output level increases beyond the certain limit. These diseconomies are: 1) ***Technical Diseconomies*** – When the production is increased beyond the optimum capacity, then technical diseconomies will emerge out. Further, high maintenance costs, heavy losses in case of an accident and lack of availability of technical experts are some restraining factors that may cause decreasing return to scale.

2) ***Financial Diseconomies***- When the finance is secured beyond an optimum scale, then financial diseconomies will emerge out. It results in concentration of income and wealth and pressure on firms to show their credit-worthiness.

3) ***Risk Bearing Diseconomies***- The risks of the firm increase with the expansion in the scale of production. An error in the decision making by the sales manager or production manager may adversely affects sales or production, resulting in losses. 4) ***Managerial Diseconomies***- Managerial diseconomies arise due to lack of skilled and efficient management. When the output level exceeds the optimum point, then manager faces the problem of control and coordination. This adversely affects the operational efficiency.

EXTERNAL DISECONOMIES

When an industry expands, the price of various factors of production like raw materials, capital, labour etc rises because of its growing demand. There may be a shortage of power, transport, labour, raw materials and equipments due to localization of industries. The per unit cost rises due to such diseconomies. The following are the examples of external diseconomy:

1) The concentration of an industry in a particular area pollutes the environment which creates health hazards for consumers.
2) The localization of an industry puts heavy pressure on transport system. This results in delay in transportation of raw materials and finished goods.
3) The prices of factors of production rise due to intense competition among the firms.

11.4 LONG RUN COST CURVES

According to the traditional theory, the short run cost curves are u-shaped. In the short run many factors of production are fixed but in the long run all factors are assumed to turn variable. The long run cost curves are also called the planning curve as it guides the entrepreneurs to guide their future expansion of output. Long run cost curves are derived from the short run cost curves. The long run average cost curve is the envelope of many short run average cost curves with each short run average total cost curve tangent to the long run average cost curve at a single point corresponding to a single output quantity.

Assume that a firm at a particular point of time, with a given state of technology can adopt three methods of production. These three production method corresponds to different plant size-small plant, medium plant and large plant.

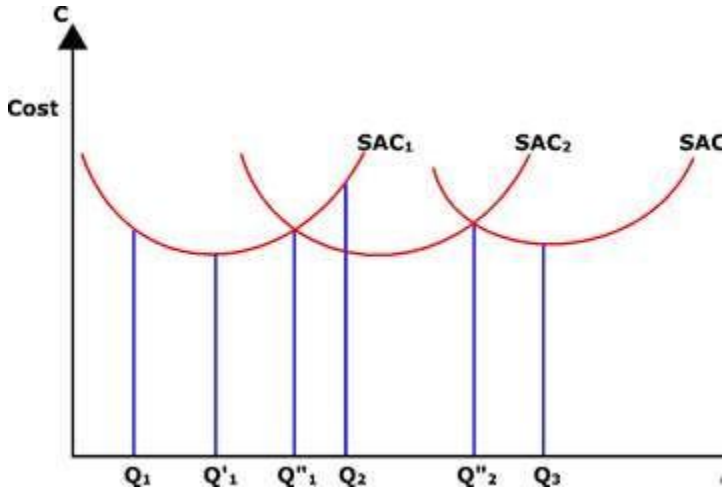


Figure 1.11

The small plant operates with cost denoted by SAC1, the medium plant cost denoted by SAC2 and the large plant cost is denoted by SAC3.

The firm will choose the small plant if it plans to produce output Q1. Similarly, if the firm wishes to produce Q2 units of output it will choose SAC2 and with Q3 units of output the firm will choose SAC3. Suppose the firm is producing Q1 with small size plant but its demand increases gradually, the cost will be lower up to the level of Q1. But beyond this point the cost starts increasing.

Now if the firm reaches the level of Q1" , the firm has two options either continue to produce with the small plant or can resort to the medium size plant. At this point the decision of the entrepreneur depends not only on the costs but also on the firm's expectations about its future demand. If the firm expects that its demand will expand beyond Q1" then it is better for the firm to install medium size plant because the output larger than Q1" can be produced with lower cost only with the medium size plant.

Similar decision holds at the output level of Q2".

Now we assume that there is not only three plant but infinite number of plant sizes, with given state of technology, each suitable for a certain level of output. With this we get a continuous curve which is the planning long run average cost curve of the firm. Each point on this curve shows the minimum cost of producing the corresponding level of output. On the basis of this curve the firm takes the decision about what plant to set up in order to produce the expected level of output at the minimum cost that is why

this curve is known as the planning curve of the firm. In the traditional theory of the firm the LRAC curve is often called the envelope curve because it envelopes the SRC curves.

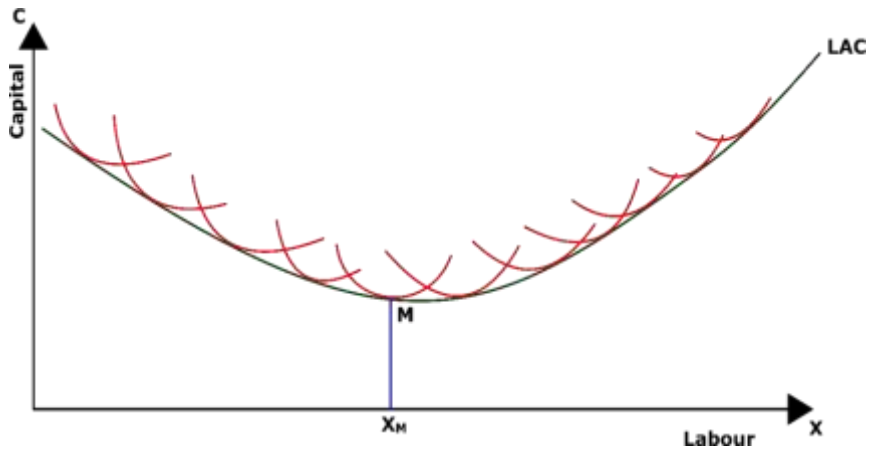


Figure 1.12

The shape of this curve reflects the law of returns to scale. Initially due to the economies of scale as the plant size increases the unit costs of production decreases. But the economies of scale exists only upto a certain size of plant which is known as the optimum plant size.

As per the traditional theory with the optimum plant size all the possible economies of scale are fully exploited. If the size of plant increases beyond this optimum size the diseconomies of scale starts arising because of the managerial inefficiencies. The turning up of the LAC curve is due to the managerial diseconomies of scale.

On the LAC curve each point shows the least unit cost for producing the corresponding level of output. Any point above the LAC curve shows the higher cost of producing the corresponding level of output. Any point below the LAC curve is desirable because it shows the lower unit cost but it is not achievable with the current state of technology and the prevailing prices of the factors of production.

LMC shows the change in long run total cost per unit due to change in output. The long run marginal cost curve (LMC) is derived from short run marginal cost curve (SMC). It is derived from the intersection point of the SMC with the vertical lines drawn from the tangency point of corresponding SAC and LAC curves.

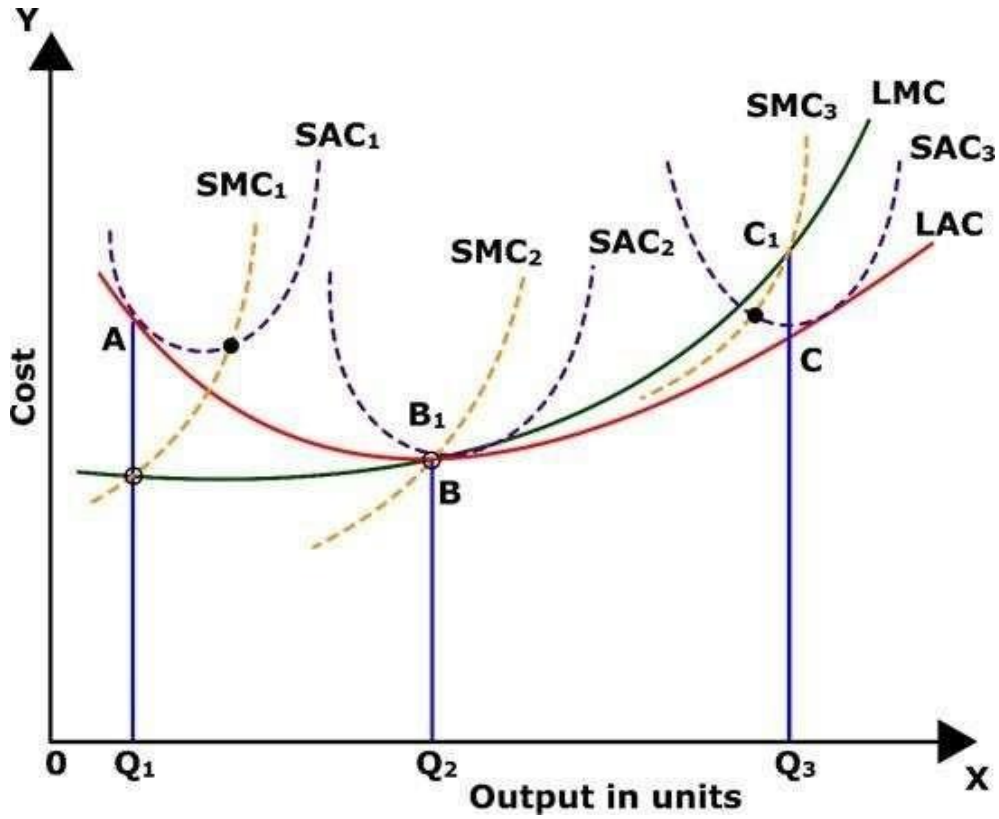


Figure 1.13

If the firm wants to produce Q1 level of output, it will operate at point 'A', the point of tangency of LAC and SAC1. From this point, a vertical line is drawn which intersect the SMC1 at point A1. A1 will show one point of the LMC curve because for the output level Q1, LAC is tangent to SAC1, then LMC must be equal to SMC1.

If Q2 level of output is produced, it will operate at point 'B', the point of tangency of LAC and SAC2. From this point, a vertical line is drawn which intersect the SMC2 at point B1. B1 will show another point of the LMC curve because for the output level Q2, LAC is tangent to SAC2, then LMC must be equal to SMC2.

Finally, if Q3 level of output is produced, it will operate at point 'C', the point of tangency of LAC and SAC3. From this point, a vertical line is drawn which intersect the SMC3 at point C1. C1 will show another point of the LMC curve because of the same reason explained above.

The LMC curve is obtained by joining points 'A1', 'B', 'C1'.

In the modern theory of the firm the cost curve, is roughly l- shaped. This is due to the reduction in cost with increases in output. In long run the costs are divided into two heads:

1) Production costs
2) Managerial costs

1) Production cost

As the scale of production increases, the production costs depict a tendency to fall steeply in the beginning and gradually thereafter. This is due to the technical economies of large scale production. These economies are substantial in the beginning but stabilize thereafter at a certain minimum scale as soon as the technical economies get exhausted.

If a new technology is introduced in production it must be cheaper to operate at a higher level of output. Similarly same economies can be achieved with the existing technology at the higher level of output. For example when inputs are purchased in bulk quantity they may leads to substantial savings for the firm due to quantity discounts on bulk purchases. Similarly if the firm reaches a certain size lower repair costs may be attained.

2) Managerial costs

Managerial costs vary with the size of the plant as there are specific organizational administrative set up appropriate for the smooth operating of that plant. Also there are various levels of management, each with its appropriate kind of management techniques. Each management technique is appropriate for a range of output. When the size of the plant is small, the managerial costs falls with expansion of the output but when the size of the plant is large the managerial costs falls up to a certain plant size and thereafter it may rise very slowly at a very large scale of output. Hence at very large scale of production the production costs falls smoothly while the managerial costs rises but very slowly. The fall in production costs is more than offsets the rise in the managerial cost, therefore at very large scale of output the LRAC curve falls smoothly or remains constant, and the chances of long run average cost to rise is completely ruled out.

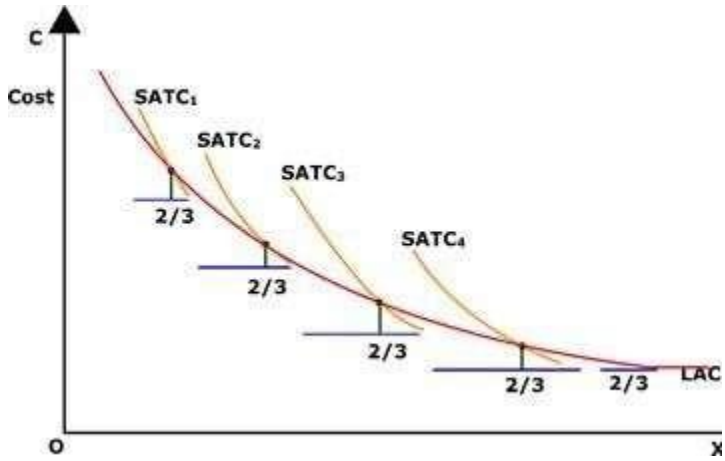


Figure 1.14

CASE STUDIES:

Case Study 1: Expansion of ABC Textiles – A Study of Economies of Scale

Background:

ABC Textiles, a mid-sized fabric manufacturing company, decided to expand its operations to meet growing market demand. Initially operating a single production unit, the firm invested in advanced machinery and established two additional units in nearby locations.

Analysis:

The expansion resulted in technical economies as the firm could now use more specialized machinery that was previously unaffordable. Marketing economies were also realized through bulk purchasing and increased negotiation power with suppliers. Moreover, the centralized distribution system reduced transportation costs per unit.

Outcome:

As a result of these internal economies, ABC Textiles experienced increasing returns to scale, lowering their per-unit production cost and improving overall profitability. Eventually, the firm encountered managerial diseconomies due to coordination issues across units, highlighting the importance of organizational efficiency during scale-up.

Case Study 2: XYZ Electronics – When Expansion Backfires

Background:

XYZ Electronics, a successful manufacturer of consumer gadgets, decided to aggressively expand by opening multiple plants and increasing output volume significantly, expecting higher profits through economies of scale.

Analysis:

While initially benefiting from production cost savings and labour specialization, the company soon faced external diseconomies such as rising competition in the region and internal diseconomies like poor communication among departments. Additionally, technical diseconomies emerged when high-capacity machines became underutilized due to mismanaged supply chains.

Outcome:

The firm's long-run average costs increased, demonstrating diminishing returns to scale. XYZ Electronics had to restructure and scale down operations to regain control over production efficiency.

11.5 LET US SUM UP

- All the factors of production are variable in the long run.
- When all factors are changed in the same proportion keeping their proportions constant, the scale of output will also change. The response of the output to these changes in inputs is called returns to scale.
- There are three possibilities of these output response when the producers increases the input in a given proportion.
- When output increases more than proportionately it is called increasing return to scale, when output increases less than proportionately it is called decreasing returns to scale and when output increase just proportionately it is called constant returns to scale.
- A product line shows the physical movement from one Isoquant to another in response to changes in quantity of input(s).
- If the product line subtends equal angles with the axes i.e. at which the MRS of factors is constant then it is called an Isocline.
- Economies that occurs to a firm due to increase in all factors of production or increase in number of firms in the industry is called the economies of scale.

- The economies that accrue to a firm due to increase in its plant size or due to increase in the number of its plants are referred to as the internal economies of scale. These economies are independent of the actions of other firms and accrue to a firm largely because of its own efforts.
- External economies arise to a firm when there is an expansion of an industry. The firms are not required to make any individual cost reduction efforts.
- Diseconomies of scale are the disadvantages or difficulties that arise to a firm when it expands the scale of operations beyond the optimum capacity.
- Long run cost curves are derived from the short run cost curves.
- The long run average cost curve is the envelope of many short run average cost curves with each short run average total cost curve tangent to the long run average cost curve at a single point corresponding to a single output quantity.
- The long run cost curves are also called the planning curve as it guides the entrepreneurs to guide their future expansion of output.
- In the modern theory of the firm the cost curve, is roughly l- shaped. This is due to the reduction in cost with increases in output.

KEYWORDS

1. **Returns to Scale:**
   The rate at which output changes as all inputs are increased proportionately in the long run.
2. **Isoquants:**
   Curves that represent all combinations of two inputs that produce the same level of output.
3. **Economies of Scale:**
   Cost advantages a firm experience as it increases its scale of production, resulting in lower per-unit costs.
4. **Diseconomies of Scale:**
   The rise in per-unit costs as a firm expands beyond its optimum capacity due to inefficiencies.

5. **Long-Run Average Cost (LRAC):**
   A curve that shows the minimum average cost of production at each level of output when all inputs are variable.

11.6 ASSIGNMENT QUESTIONS

1. Explain the concept of returns to scale with the help of isoquants and product lines.
2. Discuss internal and external economies of scale with real-life examples.
3. How does the long-run average cost curve relate to the short-run cost curves? Explain with a diagram.
4. What are the primary causes of diseconomies of scale in large organizations?
5. Differentiate between homogeneous and non-homogeneous isoquants with suitable diagrams.

OBJECTIVE TYPE QUESTIONS:

1. In the long run, all factors of production are _____.
2. Returns to scale describe the relationship between input and _____ in the long run.
3. Isoquants represent different combinations of inputs that yield the same level of _____.
4. If output increases more than proportionately with inputs, it is called _____ returns to scale.
5. When isoquants are homogeneous, isoclines appear as _____ lines.
6. The _____ curve is derived from the envelope of all short-run average cost curves.
7. Technical economies arise due to the use of _____ machinery in large firms.
8. Financial economies allow large firms to get credit at _____ interest rates.
9. In the short run, the cost curves are usually _____ shaped.
10. Internal economies result from the expansion of _____ operations.

11. External economies are shared by _____ firms in an industry.
12. Risk-bearing economies allow large firms to absorb market _____.
13. The turning up of the long-run cost curve is due to _____ inefficiencies.
14. Diseconomies of scale cause the average cost of production to _____.
15. A product line that originates from the origin implies all inputs are _____.

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FURTHER READINGS

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- Production and Cost Functions by Ronald W. Shephard
- Applied Microeconomics by Andrew Schotter
- Industrial Organization by Jeffrey Church and Roger Ware

UNIT-12: MODERN THEORY OF COST ANALYSIS

STRUCTURE

12.1 Introduction

12.2 Types of costs

12.3 Concept of cost

12.4 Concept of long run cost

12.5 Modern theory of cost curves

12.6 Long run cost under modern theory of cost

12.7 Let us Sum up

12.8 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the types of cost concepts
- Discuss the significance of opportunity costs in business decision- making.
- Explain the relations among various cost curves.
- Differentiate between cost elements that are relevant for managerial decision-making.
- Understand the concept and types of long run costs.
- Explain the modern Theory of cost.

BLOOM'S LEVEL OF LEARNING:

The content provided covers cost concepts in managerial economics, including types of costs, short-run and long-run cost analysis, and graphical representations of cost curves. This unit primarily aligns with Bloom's Taxonomy levels of Remembering, Understanding, and Applying.

At the Remembering level, students are expected to recall definitions (e.g., money cost, opportunity cost, real cost), formulas (e.g., AC = TC/Q, MC = ΔTC/ΔQ), and distinctions (e.g., between fixed and variable costs). At the Understanding level, learners are required to explain these concepts in their own words, describe the behavior of cost curves, and interpret tables and graphs that illustrate relationships between different types of costs. Lastly, the Applying level is evident when learners use formulas to calculate total cost, average cost, and marginal cost, and when they analyze cost behavior in different scenarios (e.g., calculating costs given data on output and inputs). The inclusion of graphs, examples, and mathematical calculations further supports these three levels of cognitive learning.

12.1 INTRODUCTION

In managerial economics, cost is bethought from the producer's or firm's point of view. In producing a commodity or service, a firm has to employ various factors of production such as land, labour, capital and entrepreneurship. These factors are to be rewarded by the firm for their contribution in producing that commodity. This reward is the cost, usually it terms as factor price. Thus, the cost of production of a commodity is the aggregate of price paid for the factors of production used in producing that commodity. The term cost has different meaning. Accountant view of cost is different from that of economics the accountant tend to focus on the explicit and historical cost. On the other hand, economists emphasize that for the efficient decision making by the firm it is opportunity cost rather than explicit and historical cost that must be considered.

12.2 TYPES OF COSTS

The term cost has various concepts. These are: (a) Money cost. (b) Opportunity cost.

(c) Real cost.

(a) Money Cost: When a producer pays in term of money or cash in order to produce a commodity is called money cost. The payment made in cash or money to the owner of factors of production, raw material or any other resource used in production is called money cost. This cost may be in form of payment in money for the purchase of Raw material, payment of Interest for the use of Capital, payment of Rent for the use of Land or Building, Electricity bill, Publicity, Packing charges, Transport and Insurance premium etc.

Money cost has two constituents: (1) Explicit cost (2) Implicit cost

(1) Explicit cost--- Explicit cost refers to all expenses made by a firm to purchase good directly. It include, payments for raw material, wages, rent, interest, depreciation charges, transportation, power, high fuel, advertising, taxes and so on. In nut shell we can say that, explicit costs are those payments which firms make to outsiders for contributing their services and goods.

(2) Implicit cost--- Implicit costs may be defined as the earning of those employed resources which belongs to the owner itself. In fact these costs refer to the implied or unnoticed costs. Implicit cost includes the Rent for entrepreneur's own building or land ,Interest on his own capital, Wages for his own labour etc. in short we can say that, implicit cost are the costs of self owned, self employed resources.

Total cost= Explicit cost+ Implicit cost

(b) Opportunity cost: The concept of opportunity costs was first developed by Austrian School of Economics. But afterward, it was propounded by American economist named Devenport. It is also known as Transfer cost or Alternative cost. In simple words, it is the comparison between the policy that was chosen and the policy that was rejected. So cost of production of any unit of commodity for the value of factors of production used in producing other unit is called opportunity cost. As the resources of the society are scarce and have alternative uses. When these resources are put to use for the production of particular type of commodity, then they are not available to produce other goods at that point of time. The productions of some of the other goods have been sacrificed. For example, you have 100 rupee. You can buy pen or register with this amount. The price of both is the same. If u buy pen, you have to sacrifice register and if you buy register, you have to sacrifice pen. So the opportunity cost of pen is register and opportunity cost of register is pen. So the opportunity cost is the next best alternative that you have to sacrifice. It is the cost of next best alternative gone.

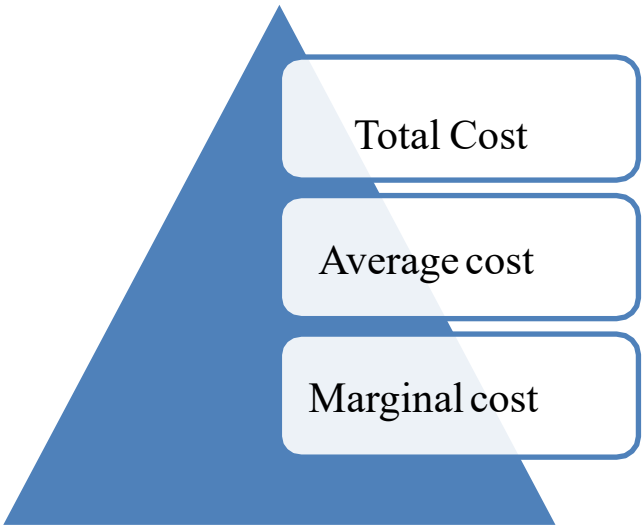
(c) Real cost: Another concept of cost is the Real costs. It is a philosophical concept which refers to all those efforts and sacrifices undergone by various members of the society to produce a commodity. It is also called 'Social Costs of Production'. In simple words, payment made to factors of production to compensate for the total sacrifice and efforts for the services they rendered is known as real cost. According to Marshall "The exertion of all the different kind of labour that are directly or indirectly involved in making it, together with the abstinence or rather. The waiting required, for saving the capital used in making it, all these sacrifices together will be called the real cost of production of commodity". So real cost of production signifies toils, troubles, sacrifice on account of loss of consumption for saving, social effects of pollution caused by factory smoke, automobiles, etc

12.3 CONCEPTS OF COST

Time plays an important role in the theory of cost. On the basis of time period, costs are divided into two parts i.e. costs in short run and cost in long run.

Costs in short period: In short period costs are mainly of the following types:

1. **Total cost 2. Average cost 3. Marginal cost**



1. **TOTAL COST**: Total cost of production is the sum of all expenditure incurred in producing a given volume of output. In other words, the amount of money spent on the production of different levels

of a good is called total cost. Conceptually, total cost includes all types of money costs, explicit as well as implicit.

Total cost = Fixed cost + Variable cost

i. Fixed cost - The cost which remain fixed at any level of output is known as fixed cost. Fixed costs, in short run, remain fixed because the firm does not change its size and amount of fixed factors employed. Fixed costs or Supplementary costs usually include:

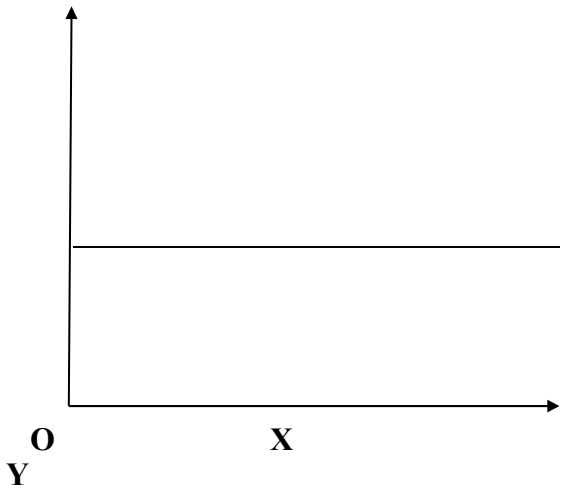
- Payments of rent for building
- Insurance premium
- Depreciation and maintenance
- Administrative expenses--- salaries of office and managerial staff
- Interest paid on capital
- Business taxes
- License fees etc

It is also called Indirect cost, Overhead cost or Supplementary cost. It can be explained with the help of following table no. 1 and figure

FIXED COST

Quantity of output	Fixed cost (Rs.)
0	100
10	100
20	100
30	100
40	100
50	100
60	100
70	100
80	100

This table shows that at various quantities of output fixed cost remains fixed at Rs.100. Even a producer produces no output. It can also be shown in following figure. In figure output is shown on OX axis and cost is taken on OY axis. At any level of output fixed cost is fixed.



Cost

Output

ii. **Variable cost:** variable cost refers to those costs which change with the change in the volume of output. So the cost which is incurred on variable factors of production such as raw material, wages, and transport etc. variable cost is also known as direct costs, special costs or prime costs.

The short run variable costs include:

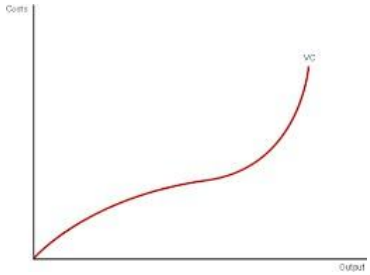
- Prices of raw materials**,**
- Fuel and power charges,
- Excise duty,
- Sales tax,
- Transport expenditure
- Wages of labour , etc

It can be explained with the help of following table no. 2 and figure

VARIABLE COST

Quantity of output	Variable cost	Change in variable cost
0	0	0
10	100	100
20	180	80
30	240	60
40	280	40
50	320	40
60	380	60
70	460	80
80	620	160

In table as the output increases from 0-40 unit variable cost is increasing at a decreasing rate. When production increases from 50-60 units then variable cost increases at a constant rate and if firm produces more output i.e. from 60-80 units variable cost start to increase at an increasing rate. But when size of production is zero at that time variable cost will also be equal to zero.



In figure output is measured on OX axis and variable cost is measured on OY axis.

Variable cost is increasing first at decreasing then constant and then increasing rate.

FIXED, VARIABLE AND TOTAL COSTS IN THE SHORT –RUN

In order to determine the total cost of a firm, we aggregate fixed as well as variable costs at different levels of output.

TC = FC + VC

It can be shown with the help of following table no. 3 and figure.

Relationship between fixed, variable, and Total cost

Output (1)	Fixed cost (2)	Variable cost (3)	Total cost (2+3)
0	100	0	100
10	100	100	200
20	100	180	280
30	100	240	340
40	100	280	380
50	100	320	420
60	100	380	480
70	100	460	560
80	100	620	720

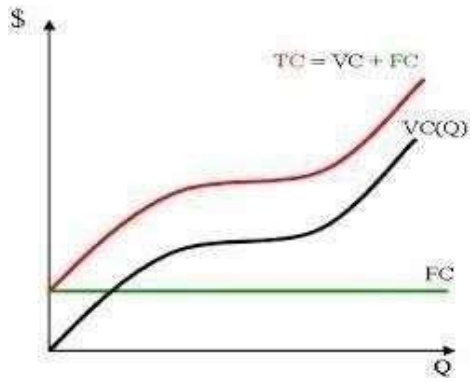
In table 3

TC =TFC + TVC

TFC =TC – TVC

TVC =TC – TFC

In table 3, when output is zero, variable costs are also zero. But fixed costs as well as total costs are 100. As the output increases to 80 units, total costs go up to 720. It means as the output increases fixed costs remain the same, but variable costs increase at diminishing rate and then increasing rate. This relationship has been shown in following figure.



In figure quantity is measured on OX axis and cost on OY axis. FC is fixed cost curve which is parallel to horizontal axis which signifies the fact that at all level of output, fixed cost remain the same. VC is variable cost curve. It is of the shape of reverse S. It means as the output zero variable costs are also zero. TC shows the total cost of production.

2. AVERAGE COST: Per unit cost of a commodity is called as Average cost.

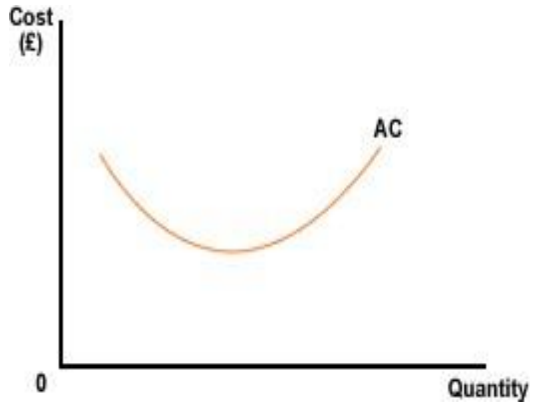
According to Dooley, "The average cost of production is the total cost per unit of output." In other words, average cost of production is the total cost of production divided by the total number of units produced. For example, cost of producing 100 units is Rs. 20,000, the average cost will be:

AC =TC/Q AC = Average cost TC = Total cost Q= quantity

Ac = 20,000/100 = Rs. 200

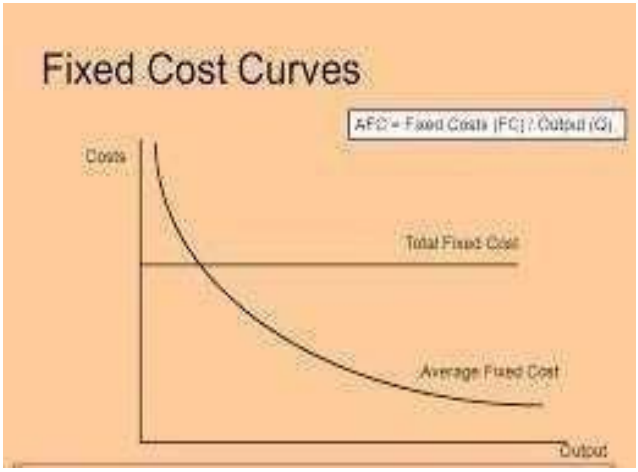
Another way to calculate average cost is:

AC = AFC + AVC **AC =Average cost, AFC = Average fixed cost, AVC = Average variable cost.**



(a) Average Fixed cost: Average fixed cost is the total fixed cost divided by the number of units of output. It is per unit cost of fixed factors of production. Thus:

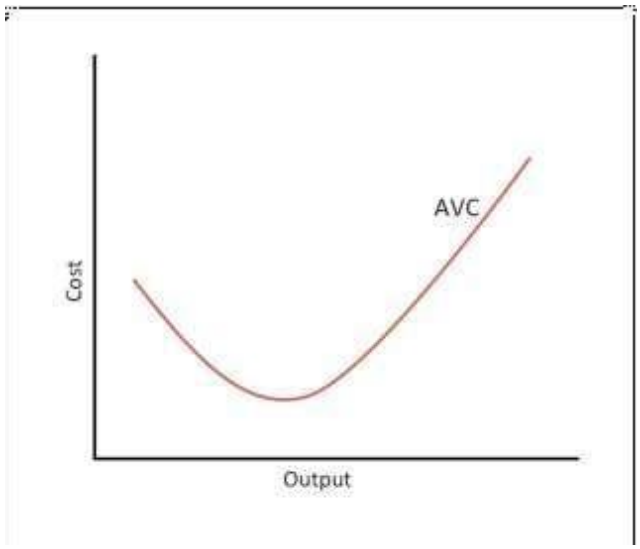
AFC = TFC/Q, AFC = Average fixed cost, TFC = Total Fixed cost, Q = Quantity



Since, total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases, and thus, the average fixed cost curve slopes downward with a view to touch the horizontal axis. But it will not be so because AFC can never be zero.

(b) Average Variable cost: Average variable cost is the total variable cost divided by the number of units of output produced. Thus

AVC = TVC/Q, AVC = Average variable cost, TVC = Total variable cost, Q = Output



Normally, the AVC falls as output increases from zero to the normal capacity output due to the law of increasing returns. But afterward, the AVC will rise because of the operation of the law of diminishing returns. It is always U-shaped.

RELATIONSHIP AMONG AVERAGE COST, AVERAGE FIXED COST AND AVERAGE VARIABLE COST

Average cost is the sum total of both Average fixed cost and Average variable cost.

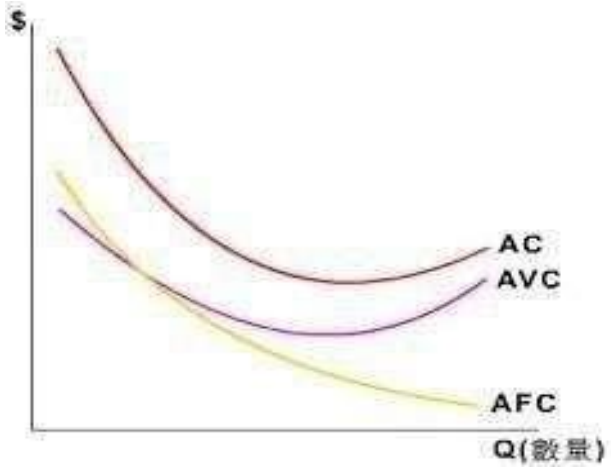
This is shown with the help of following table and figure.

Relation among AC, AFC, and AVC

Output	**Fixed cost**	**Average fixed cost (2/1)**	**Variable cost**	**Average variable cost (4/1)**	**Average cost (3+5)**	**Total cost (2+4)**
(1)	(2)	(3)	(4)	(5)	(6)	(7)
0	100	--	0	0	0	100
1	100	100	100	100	200	200
2	100	50	180	90	140	280
3	100	33.33	240	80	113.33	340

4	100	25	280	70	95	380
5	100	20	320	64	84	420
6	100	16.67	380	63.33	80	480
7	100	14.28	460	65.71	79.99	560
8	100	12.5	620	77.5	90	720
9	100	11.11	800	88.89	100	900

This table shows that AC decreases as the volume of output increases from 1 to 7 units and then it start to increase when volume of output increases to 8 to 9 units. Relation among AC, AFC and AVC can be shown in following figure.



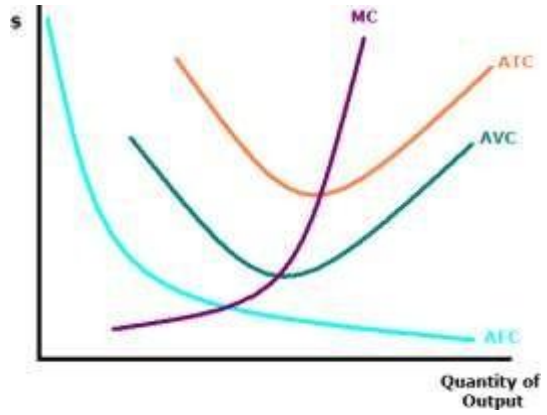
It is observed from the figure:

- Average cost is a sum total of both AFC and AVC.
- AVC and AFC both falls with the increase in output.
- AVC starts to rise with the increase in output and AFC will fall continuously. AC also falls up to a certain stage.
- With the increase in output AVC start to increase beyond a certain point and AFC remain decreasing but AC also start to increase remaining minimum at some stage.
- AC curve is ‘U’ shaped.

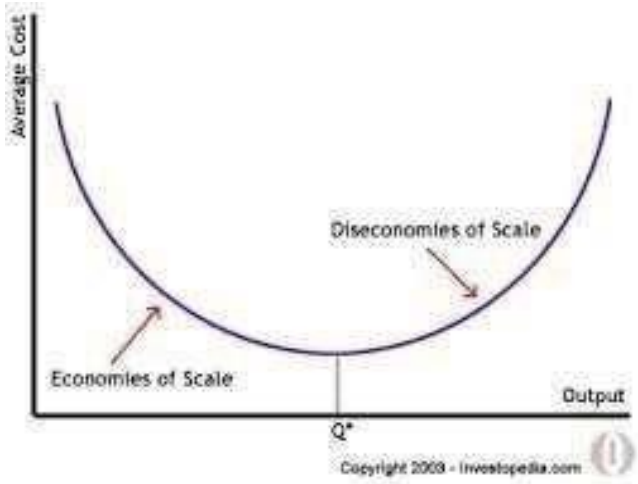
Why AC curve is 'U' shaped

In short period average cost curves are of U shaped, which shows, initially it falls and after reaching the minimum point it starts rising upwards. Following points may justify the U shaped AC curve.

- Due to Averages fixed cost and Average variable cost: it is well known, that average cost is the aggregate of average fixed cost and average variable cost (AFC+AVC).in the initial stages as production increases both AFC and AVC falls. When AFC and AVC added then the falling rate of AC will be more. If production continuously increases then AFC will fall and AVC will be minimum at one stage at that time AC will also be minimum. In the last stage with the increase in production AFC goes on falling but AVC start to increase. Adding both the increasing rate of
AC will be more than AVC. If we join all three stages of AC then we will get 'U' shaped AC curve. It can be seen in following diagram.



- Due to Law of Variable proportion: The law of variable proportion states that when variable factors are combined with fixed factors in short run then production in the initial stage will increase at an increasing rate then constant and in last stage production will increase at diminishing rate.
- Due to indivisibilities of the factors of production: Another reason due to which the average cost curve forms U shape is the indivisibilities of factors. When in the short period a firm increases its production, due to indivisibilities of fixed factors, it gets various internal economies. In these economies which cause the average cost curve to fall in the initial stage. When these fixed factors are fully utilized by the firm then cost of production began to rise again so AC increases. It can be seen in



following figure.

3. MARGINAL COST: The marginal cost is the addition made to total cost by producing one more unit of output. In other words, marginal cost may be defined as the change in total cost associated with a one unit change in output. It is also an 'extra unit cost' or incremental cost, as it measures the amount by which total cost increases when output is expended by one unit. It can also be calculated by dividing the change in total cost by the one-unit change in output.

$$\text{Marginal Cost (MC)} = \frac{\Delta TC}{\Delta Q}$$

Where

$\Delta =$ Change
$TC =$ Total Cost
$Q =$ Quantity

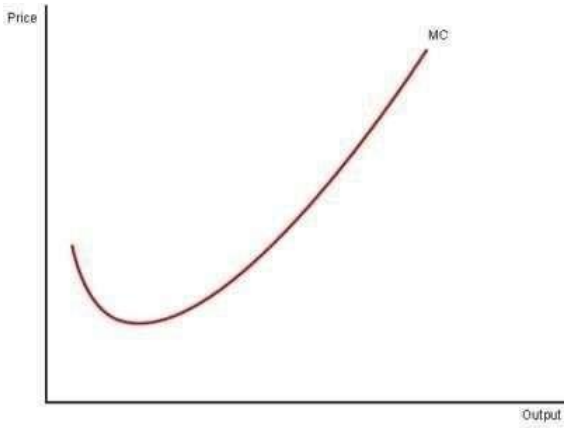
It can be shown with the help of following table and figure.

Marginal cost

Output(Units)	Total cost (Rs.)	Marginal cost (Rs.)
(1)	(2)	(3)

0	100	--
1	200	100
2	280	80
3	340	60
4	380	40
5	420	40
6	480	60
7	560	80
8	720	160
9	900	180

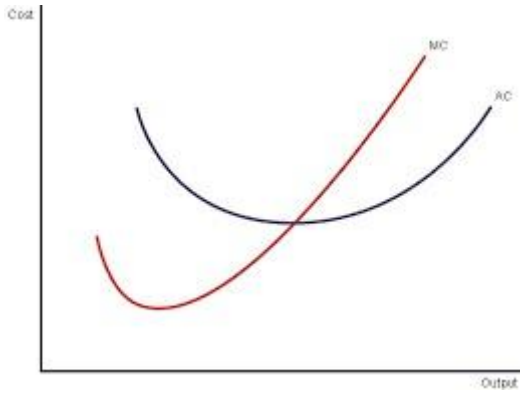
It is clear from the table that when 1 unit is produced then total cost is Rs.200 and total cost increases to Rs.280when extra one unit produced. Marginal cost of producing one extra unit is Rs 80 (280-200). It shows MC will fall initially with the increase in production then remains constant and then it starts to increase.



In figure output is taken on OX axis and Marginal cost on OY axis. It is observed that MC id decreasing with the increase in the quantity of output. Then it starts increasing with the increase in the size of production. It is also U- shaped.

Relationship between Average cost and Marginal cost

1. Both Average and Marginal cost are calculated from Total cost. As is known, average cost is the ratio of total cost to total output. In the same way marginal cost is also calculated from total cost. It refers to an addition made to total output by producing one more unit of output.
2. Initially, both MC and AC curves are sloping downward. When AC curve is falling, MC curve lies below it, as can be seen in figure.



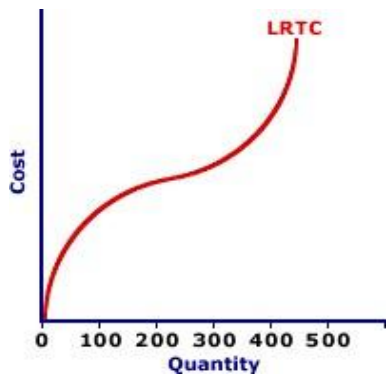
3. When AC is rising, after the point of intersection, MC lies above it
4. When AC is constant MC becomes equal to AC
5. MC cuts AC at its lowest point.

12.4 CONCEPT OF LONG RUN TOTAL COST

In the context of production theory, long-run is defined as a period in which all the inputs become variable. The variability of inputs is based on the assumption that in the long run, supply of all the inputs, including capital equipment and factory building, becomes elastic. So the long run is a period in which there is sufficient time to alter the scale of production or equipment or the organization with a view to produce different quantities of output. The firms are therefore in a position to expand the scale of their production by hiring a more quantity of all the inputs. In the long period all factors of production are variable and there is no distinction between fixed and variable costs. All the costs are variable costs. All the factors, building, machines and even employees and organization are changeable. In the long run, the firm has enough time to bring about change in fixed factor of production. So in the long run we have three types of costs:

1. Long Run Total Cost
2. Long Run Average Cost
3. Long Run Marginal Cost

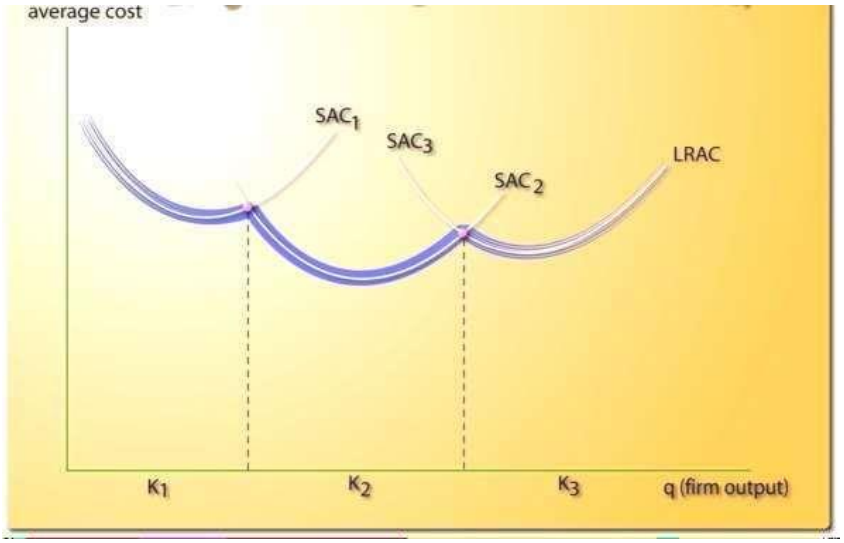
1. Long run Total cost: Long run total cost is always equal or less than the short run total cost, but it is not more than the short run total cost. Long run total cost curve



represents the least cost of different quantities of output.

The shape of the long-run total cost curve is S-shaped, which is very similar to a short- run total cost curve. When there is small scale of output, the slope begins to flatten. Then for large scale the slope makes a turn-around and becomes steeper. The flattening portion of this long-run total cost curve is due to economies of scale or Increasing returns to scale. The steepening portion is then due to diseconomies of scale or decreasing returns to scale.

2. Long run Average cost: Long run Average cost is the long run total cost divided by the level of output. Long run average cost curve shows the least possible average cost for producing different levels of output. So Average cost is the minimum cost which a producer faces by producing different quantities of output in a long run. In order to understand, consider the three short term Average cost curves. These short run Average cost curves are also called Plant curves. Suppose a firm has three plants with different sizes, plant one with small, plant two with minimum and plant third with large size. The cost curve of small plant is presented by SAC1 and medium size by SAC2 and large size with SAC3. This is shown in the following figure.

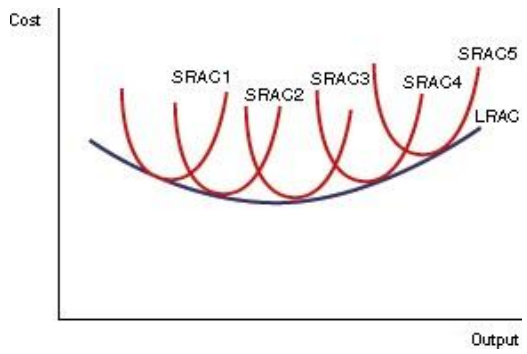


So in the short run average cost curve the firm will increase or decrease its output by varying the amount of the variable outputs. But, in the long run, the firm can choose among the three possible sizes of plant as depicted by short run average cost curves SAC1, SAC2 and SAC3. In the long run the firm will examine with which size of plant or on which short run SAC2 curve it should operate to produce a given level of output at the minimum possible cost.

Let us take another assumption now, there are large number of plants, with different sizes, then the curve which will reflect the minimum cost of production of each plant will be the long run average cost curve of the firm. Eachpoint lying on this curve shows the least cost for producing the corresponding level of output. The long run average cost is the locus of point denoting the least cost or minimum cost of producing the corresponding output. This LAC helps a producer to decide what plant to set up in order to produce at a minimum cost at the expected level of output of a commodity.

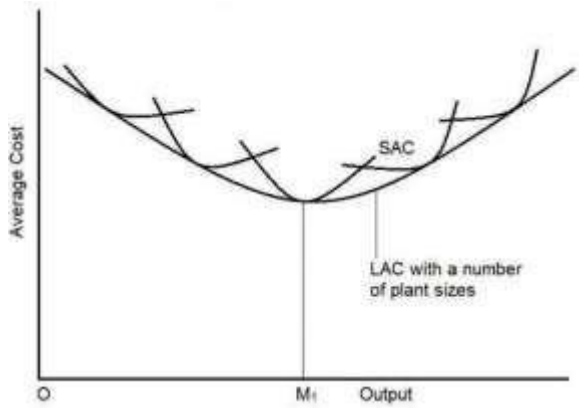
This is also known as Planning curve as it serves as a guide to the producer in his plan to expand production, with its help, a firm can plan as to which plant, it should use to produce different quantities of output so that production is obtained at minimum cost. So the long run average cost curve reveals to the firm that how large should be the plant for producing a certain output at the least possible cost.

It is also known as Envelope curve, because it is the sum of all short run average cost curve in it or it envelopes the short run average cost curves. It shows that average cost in the long run cannot exceed the short run average cost. This is shown in following figure:



RELATIONSHIP BETWEEN LAC AND SAC

LAC and SAC both are cost curves. LAC represents the cost of different plants whereas SAC represents the cost of single plant. The relationship between LAC and



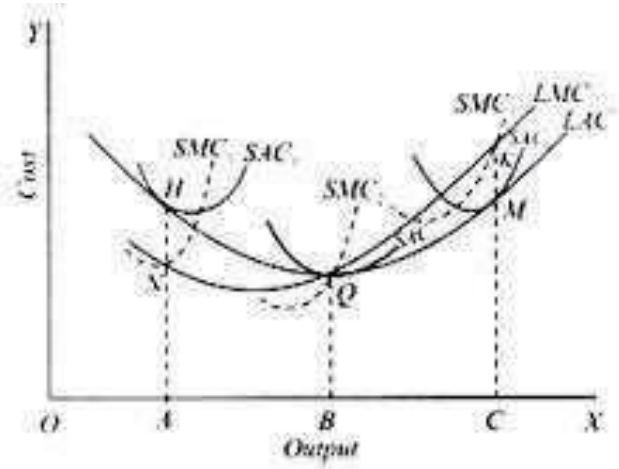
SAC can be explained with the help of following figure:

By considering this figure, we can draw a following relationship between LAC and SAC.

1. Like SAC, LAC is also U-shaped but it is relatively flatter. It shows that, in the long run, increase or decrease in costs relatively less.
2. LAC cannot be more than SAC because, LAC is tangent to SAC.
3. SAC represents the cost of single plant whereas LAC represents the costs of different plants.

3. Long run Marginal Cost: Change in total cost due to change in output by producing one more unit is marginal cost. The marginal cost is the addition made to total cost by producing one more unit of output. In other words, marginal cost may be defined as thechange in total cost associated with a one unit change in output. It is also termed as 'Cost of extra unit' or incremental cost, as it measures the

amount by which total cost increases when output is expended by one unit. The relation of LMC with LAC will be the same as the relation of SMC with SAC. LMC curve also cut LAC curve at its lowest

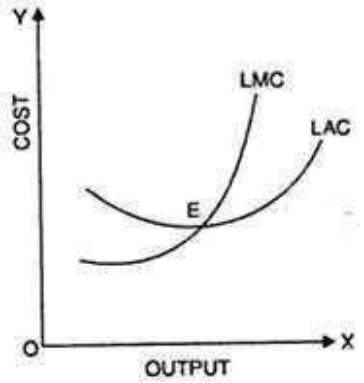


point. LMC curve is flatter than SMC curve. It can be shown with the help of followingfigure.

It is clear from this figure that the long run marginal cost curve LMC, like the long run average cost curve LAC, is U- shaped. But the long run LMC is flatter than the short run marginal cost curves.

Relationship between LMC and LAC

Basically, the relation between the long run marginal cost and the short run average cost is almost the same but the only difference in LAC and LMC is that long run marginal and average costs are more flatter than SAC and SAC and SMC. The reason behind is the variability of all factors of production. It should be remembered that the relationship between the long run marginal cost curve LMC and the long run average cost curve LAC is the same as that between the short run marginal cost curve and the short run average cost. It can be shown with help of following figure.

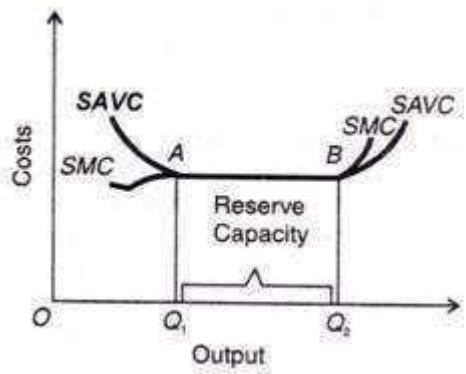


12.5 THE MODERN THEORY OF COST CURVES

Earlier it was believed that both the short run and the long run cost curves have a U shape. But in recent years, has been observed that in real life, the cost curves are Saucer- shaped or L- shaped. The theory which explains its new shape is called the modern theory of cost curves. The Modern theory of cost curves has been propounded by economists like Andrew, Stigler, Sergeant Florence and Friedman. Modern theory is based on empirical evidences, the short run SAVC and SMC coincide with each other and a horizontal straight line over a wide range over output. So the nature of short run and the long run cost curves under modern theory are given below:

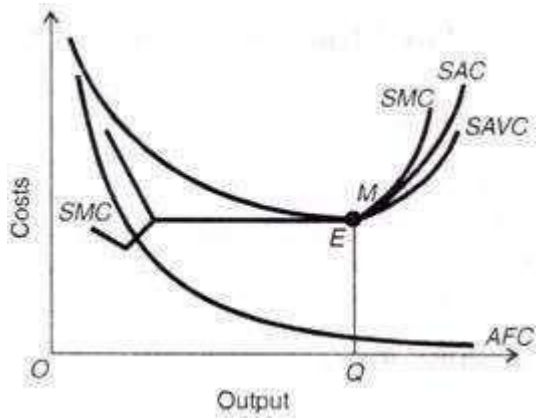
- **Short run cost curves under modern theory of cost:** The short run cost curves are derived from the total cost which is the sum total of fixed cost and variable cost. So like traditional theory, modern theory of cost curve also have the short run cost curves such as AFC, SAVC, SAC and SMC.
- AFC: Average fixed cost is the cost of indirect factors of the firm. In simple words we can say that, the cost of physical and personal organization of the firm. Costs such as the salaries of the staff involved directly in production but paid on a fixed term basis, depreciation of machinery, and maintenance expenses of factory building etc.
- SAVC: The average variable cost includes wages of labour employed, cost of raw materials, and day today expenses of machinery. The SAVC curve in modern theory of cost is Saucer-shaped. That is broadly U shaped. It has a flat stretch over a range of output. This flat stretch represents the built in reserve capacity of plant. The falling portion of SAVC shows reduction in cost due to better utilization of fixed factors, improvement in the skill and efficiency of labour and less wastage of raw material. On the other hand, rising portion of the SAVC curve shows the inefficiency of labour, frequent break down of machinery and the wastage of raw material.
- SAC: According to the modern theory of cost SAC is continuously falling up to given level of output. This given level of output represents the reserved capacity of output.
- SMC: In initial stags SMC in modern theory of cost, falls and then becomes horizontal. The main difference is about the shapes of the curves. In simple words we can say that in modern theory these short run costs curves have a saucer shape or bowlshape instead of U shaped. In modern theory, short run cost curves have been investigated by economists on the basis of empirical studies behaviour pattern. According to this theory, the firm chooses only that plant which can be run with the available variable direct factors and the one which has a great flexibility with some reserve capacity. The basic

objective of installation such type of plant is to produce maximum output in order to meet any increase in demand forits product. The SAVC and SMC curves are shown below in the figure. It shows falling of both the curves up to point A and SMC curve lies below the SAVC curve. Falling SAVC shows the optimum utilization of fixed factors and increase of productivity of the variable factorthat causes the reduction in cost.



Plants can attain that level with better utilization of available resources, better skills and less wastage in raw material. The Saucer- shaped SAVC curve has a flat stretch over Q1 –Q2 as far as output concerned, it shows application of constant returns to scale. Divisibility of fixed factors is the main reason behind the saucer shape of SAVC curve. After reaching to the point B, both curves have started rising, which shows increase in costs. This increased cost may be due to increase in wastage of raw material, frequent breakdowns, lack in labour productivity and overtime payment to labour . When SAVC is rises after the full utilization of the reserve capacity of the plant, SMC curve lies above it.

SAC is obtained by adding the average fixed cost to the average variable cost. As shown in the following figure AFC falls as the output is extended. The SAC falls up to the level of output at which the reserve capacity is fully exhausted i.e. up to the OQ level. Afterward SAC rises with the increase in output. At point M, SAC intersect SMC, this is because of steeply rising SAVC from point E and on the other side AFC is falling at low rate.



12.6 LONG-RUN COST CURVES UNDER MODERN THEORY OF COST

As the output increases, the firm's average cost of producing that output is likely to decline, at least to a point. This can happen for the following reasons:

Modern theory of cost explains that long term cost curves have an L shape instead of U shape. First the LAC curve falls, then remains flat or may slope downward to its right hand side. Some economists have given the following reasons forthe L-shape of the LAC.

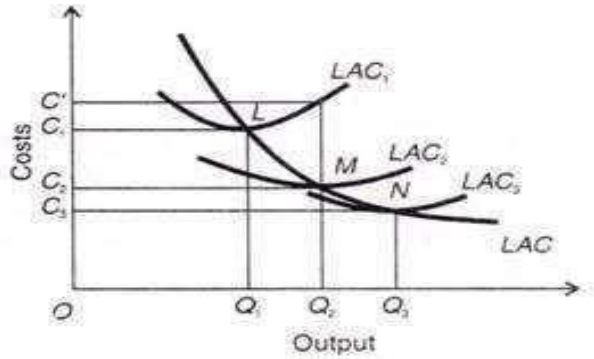
1. Production or technical economies: As we know, in the long period all the costs are variable costs. Whenever the firm thinks about the effect of expansion of output on the average cost, the production and the managerial costs are taken in to consideration. With increase in the scale of production, its production cost falls in the beginning and then gradually keeps on falling. That happens because of technical economies of large scale. Therefore, a firm producing on large scale can enjoy the economies by the use of superior techniques. Modern technology is highly specialized. The advanced technology makes it possible to conceive the whole process of production of a commodity in one composite unit of production. For example, the production of cloth in a textile mill may comprise of such plants as Spinning, Weaving, Printing, Packing etc. The Technical economies are of following types:

- *Economies of dimension:* When a firm increases its scale of production, the average cost of production falls but its average return will be more.

- *Economies of by-product:* When a firm increases its scale then it is in a position to use its by-products and waste material to produce another material.
- *Economies of linked process:* As the scale of production increases, firm carries all productive activities. These activities get economies. These linked activities save time and transport cost of the firm.
- *Labour Economies:* When a firm's scale of production expands, more and more workers of varying skill and qualification are employed. With the employment of large number of workers, it becomes increasingly possible to divide the labour according to their qualifications and skills and assign them the function to which they are best suited. The workers are skilled in their operation which reduces cost, increases the productivity and saves the time. But after achieving all the economies, the firm reaches at MES that is to the minimum optimal scale.

2. **Managerial cost:** The Managerial Economies arise from specialization in managerial activities and mechanization of managerial functions. For a large firm, it becomes possible to divide its management into specialized departments under specialized personnel, such as the production manager, the sales manager, the personnel manager, etc. This increases the efficiency of management at all levels of management because of decentralization of decision making. It increases production, given the cost, large scale firms have the opportunity to use advanced techniques of communication, telephones and telex machines, computers, and their own means of transport. All these lead to quick decision making, which thereby helps in saving the valuable time of the management and hence improves the managerial efficiency.
3. **Marketing Economies:** In the marketing economies, we include the advertisement economies, the opening up of showrooms, the appointment of sole distributors etc. Moreover, a large firm can conduct its own research to affect improvement in the quality of product and to reduce the cost of production. With the expansion of the firm, the total production increases. But the expenditure on advertising the product does not increase proportionately. Similarly, selling through the wholesale dealers reduces the cost of distribution of the firm's production.
4. **Technical progress:** Another major reason for L-shaped cost curve is the technical progress. As a result of technological progress in the real world, the long run average cost curve will shift downward over the period. It is assumed in the traditional theory that there is no technical progress, but empirical result confirms the existence of technological progress in firms. It is shown with the

help of following diagram. In this figure, it is shown that when technological progress takes place the long run average cost curve shows a falling trend.



It is shown in the diagram that if firm introduces new technology then its cost will be lower.

5. Learning process: The learning process also contributes a lot in L- shaped cost curves. People always learn while doing something. The greater the amount of work done, the lower will be its cost. It is well known fact that the experienced people will do the work in better way and thus it reduces the cost of production and increases the productivity. A firm learns to produce a commodity more efficiently as the aggregate amount of output produced by it increases over time. The best example of learning by doing is a new firm which produces their product for the first time. After producing the first unit, they may be able to reduce the time required for second unit. And if they are able to do so it will reduce their cost of production.

So along with the technological progress, learning process also provides us with the reason that why long run average cost curve is L shaped rather than U shaped. To quote stonier and Hague, " Even with a given technology, a firm can learn to produce at a lower unit cost the longer the period of time that has elapsed since a previous observation and the greater the aggregate amount of that product that has consequently been made. It may be therefore concluded that technological change is not the only reason why long- run average cost curves are L- shaped rather than having U- shape".

CASE STUDIES:

Case Study 1: Cost Optimization in a Small Manufacturing Firm

Background: A small-scale textile firm in Gujarat operates under increasing competition. Initially, it incurs high average costs due to underutilized capacity.

Application of Cost Concepts:

By analyzing its Short Run Average Variable Cost (SAVC) curve, the firm identifies a flat portion where additional units can be produced without increasing cost. It adjusts production from 500 to

800 units weekly, staying within the reserve capacity.

Result: Per-unit costs stabilize, profit margins improve, and pricing becomes more competitive.

Learning Outcome: Proper understanding and application of modern cost curves like SAVC and SMC help in making optimal production decisions.

Case Study 2: Long-Run Cost Planning in an E-commerce Startup

Background: An e-commerce startup is expanding rapidly and wants to plan its infrastructure for the next five years.

Application of Cost Concepts:

By examining Long Run Average Cost (LAC), the firm evaluates various plant sizes and selects the most efficient scale for long-term growth. It uses the LAC as a planning curve to minimize average cost across different output levels.

Result: The startup successfully avoids diseconomies of scale by choosing optimal capacity, maintaining efficiency even as operations expand.

Learning Outcome: LAC helps businesses determine the most cost-effective way to scale production in the long run.

12.7 LET US SUM UP

The module 'Theory of Cost' has focused on the basic concepts of costs. Money cost is the money expenditure on different factors of production. Money cost includes cost of raw material, machinery, and equipment, wages, salaries and supplements, rent on business premises, interest paid on capital, power-fuel charges, insurance premium etc. The implicit cost is the earnings of those resources which belong to owner itself. For decision making both explicit as well as implicit costs are important. The opportunity cost is the return from the second best use of a resource, which the Firm foregoes when taking up the opportunity of its best use. Fixed cost do not vary with the level of output whereas, variable cost change with changes in the level of output. In short period cost curves are always U shaped. It suggests initially falling and eventually rising. Marginal cost is the ratio of change in total cost to a change in output.

Long-run is defined as a period in which all the inputs become variable. The variability of inputs is based on the assumption that in long run, supply of all the inputs, including the capital equipment and the factory building, becomes elastic. So the long run is a period in which there is sufficient time to alter the scale of production or equipment or the organization with a view to produce different

quantities of output. The firms are therefore in a position to expand the scale of their production by hiring a more quantity of all the inputs. In long period all factors of production are variable and there is no distinction between the fixed and the variable costs. All the costs are variable costs. In recent years, it is observed that in the real life, the cost curves are Saucer- shaped or L- shaped. The theory which explains its new shape is called the modern theory of cost curves. The Modern theory of cost curves has been propounded by economists like Andrew, Stigler, Sergeant Florence and Friedman. The Modern theory is based on the empirical evidences, the short run SAVC and the SMC coincide with each other and a horizontal straight line over a wide range of output.

KEYWORDS

1. **Opportunity Cost:**
   The value of the next best alternative foregone when a decision is made; important in resource allocation.
2. **Marginal Cost (MC):**
   The additional cost incurred from producing one more unit of output.
3. **Average Cost (AC):**
   The total cost divided by the number of units produced; includes both fixed and variable costs.
4. **Saucer-Shaped SAVC:**
   A modern cost curve shape indicating constant average variable cost over a range due to reserve capacity.
5. **Long Run Average Cost (LAC):**
   The lowest possible cost of producing each level of output when all inputs are variable.

12.8 ASSIGNMENT QUESTIONS

1. Differentiate between short-run and long-run costs with suitable examples.
2. Explain the significance of opportunity cost in managerial decision-making.
3. How do modern cost curves differ from traditional U-shaped curves? Discuss with a diagram.
4. Discuss the concept of reserve capacity and its relevance to modern SAVC.
5. What is the relationship between LAC and SAC, and why is LAC also called the planning curve?

OBJECTIVE TYPE QUESTIONS

1. _____ cost is the sum of explicit and implicit costs.
2. Opportunity cost is also known as _____ cost.
3. Real cost includes the _____ and efforts made by all members involved in production.
4. In the short run, fixed cost _____ change with output.
5. Variable cost _____ with the level of production.
6. Total Cost = Fixed Cost + _____ Cost.
7. Average Cost is calculated by dividing total cost by _____.
8. AFC _____ as output increases.
9. The AVC curve is usually _____ shaped.
10. Marginal Cost is the _____ in total cost due to producing one extra unit.
11. LAC is also referred to as the _____ curve.
12. In the long run, all factors of production are _____.
13. Modern SAVC curve has a _____ shape.
14. The intersection point of MC and AC occurs at the _____ point of AC.
15. LAC is flatter than SAC because all inputs are _____ in the long run.

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- Lipsey, R.G., & Chrystal, K.A. (2007). Economics. Oxford University Press.
- Baumol, W.J. (1982). Economic Theory and Operations Analysis. Prentice Hal

FURTHER READINGS

- Managerial Economics – D.N. Dwivedi
- Principles of Economics – N. Gregory Mankiw
- Microeconomic Theory – H.L. Ahuja
- Cost Accounting – M.N. Arora
- Economics: Theory and Practice – G.K. Kapoor & Sanjay G. Dharmani

UNIT-13: MEANING AND SCOPE OF PRICING STRATEGY

STRUCTURE

13.1 Introduction

13.2 Types of pricing strategies

13.3 Let us Sum up

13.4 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the types of pricing strategies.
- Explain the determination of different types of prices strategies
- Discuss the advantages of prices strategies
- Differentiate between cost elements that are relevant for managerial decision making.
- Understand the concept and types of Long run costs.
- Explain the modern Theory of cost.

BLOOM'S LEVEL OF LEARNING

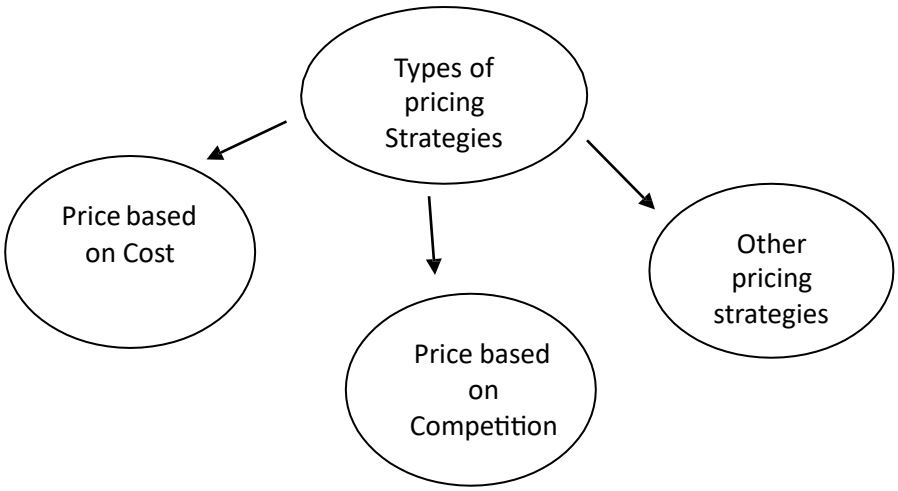
Based on the learning outcomes and content structure provided for this unit on pricing strategies in business economics, the material primarily aligns with the Understanding, Applying, and Analyzing levels of Bloom's Taxonomy. Learners are expected to understand key concepts (e.g., types of pricing strategies, modern cost theory), explain and discuss their applications and implications in managerial contexts (which falls under applying), and differentiate between various elements like cost structures and pricing decisions (which reflects analyzing). This unit emphasizes comprehension of theoretical concepts, the application of those concepts in real-life business scenarios, and the ability to compare and critique different pricing strategies, which collectively reflect the middle tiers of cognitive complexity in Bloom's taxonomy.

13.1 INTRODUCTION

In the last chapter we studied the different objectives of the pricing policies and the different factors that affect the pricing decisions of the firms including the impact of market structure on its pricing decisions. In the present chapter we shall discuss the different types of pricing strategies that a firm can have. In economic theory, it is believed, that the firms follow the MC and MR rule. This implies that the price is decided at a point where marginal cost of the product is equal to the marginal revenue earned from the product. In real life, firms deviate from this principle and base their pricing strategies differently. These are discussed in the next section.

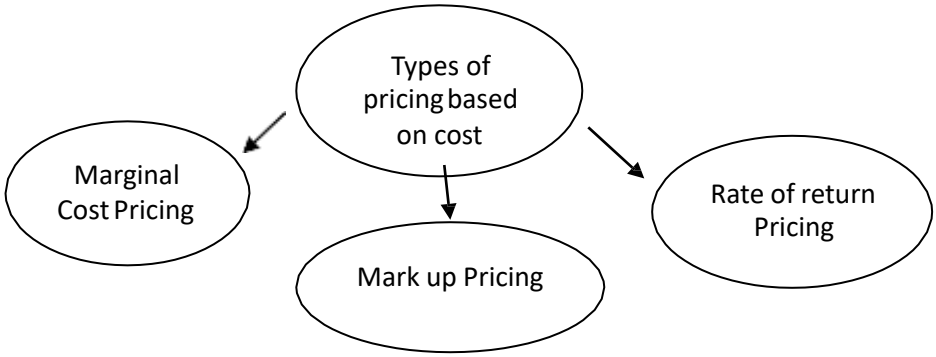
13.2 TYPES OF PRICING STRATEGIES

The different types of pricing strategies are given below:



Let us discuss these one by one.

a) **Pricing based on cost**: The following strategies are covered under this category:



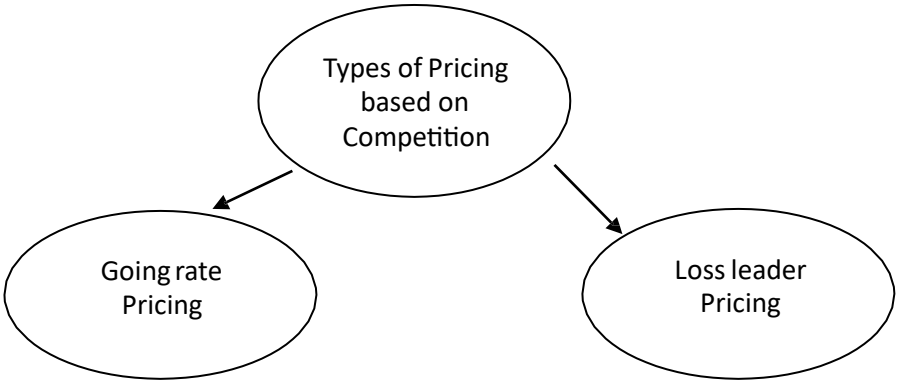
i) **Marginal cost pricing**: This is a concept of short period. This principle covers variable costs only. In economic theory, we use the word, 'marginal' but in practice, entrepreneur use the word, 'incrementalism'. In daily life, entrepreneurs do not estimate the change in costs with the change in per unit of output rather they consider the changes in the costs due to the larger proportions of output. Principle of marginalism says that the price of the firm is decided by the equality of the marginal cost and marginal revenue. The theory says that if the price falls to the level where it is just covering variable costs then the firms shut down, however, the principle of incrementalism shows that the firms do not shut down at this level and continue with production. Such a pricing strategy is helpful when the firms have stocks of unsold products and can sell this stock by reducing the price to just cover the variable costs. This strategy also helps the firms when they introduce new products or face a stiff competition in the market. However, such a strategy cannot be continued in the long run as fixed costs are needed to be covered.

ii) **Mark up pricing:** Many economists like Hall and Hitch showed that in real life firms followed mark up pricing instead of marginal cost and marginal revenue approach. In this method, the firms add the variable costs and the fixed costs of a product and then add a mark up to it. If the cost of a cup of coffee is Rs 100 to the producer and the mark up is Rs 20 then the sale price of coffee would be Rs 120 (Rs 100 + Rs 20). So, here, the mark up is 20%. Fixing a mark- up can be rigid or flexible. It is rigid when the mark – up is added to the variable cost only. In such a case mark – up is so large that it covers up the fixed costs. However, in the case of flexible mark- up, different mark ups are fixed for fixed and variable costs. Mark – up pricing is followed when the business firms do not have a correct estimate of the market demand of their product and the pricing strategies of the rivals are unknown.

iii) **Rate of return pricing:** Such a strategy is used when the firms want to revise the prices of their products. It helps the firm to fix a mark up based on rate of return on capital invested. The first step is to find the expected rate of return by dividing the total earnings by the amount invested by

the firm. The firm then estimates its full cost and divide the total amount invested by the firm by the cost. This is called as capital turnover ratio. Mark up percentage is estimated by multiplying capital turnover ratio with the expected rate of return which essentially comes out to be total earnings divided by the costs. Then the rate of return is estimated by adding the mark up to the costs. Thus, here, the guiding principle for the firm is to get returns from the money invested.

b) **Pricing based on competition:** This type of pricing strategy includes the following:



i) **Going Rate Pricing**: In this pricing strategy, the firms, first, observe the prices of their rivals. Thus, costs are not the guiding principle. This type of strategy is used when the firm do not want to disturb the existing situation in the market. Hence, the firm fixes its own price based on the prices of their rivals. Such a strategy is also adopted when the firms feel that the product market has reached a certain level of maturity and the customers also prefer the products of the firm and do not want any hiccups in the price pattern. In such cases firms try to minimize their costs rather than disturbing their customers with frequent price changes. Such a type of the strategy may also be followed by a new entrant in the industry. Being a new entrant, the firm, first, observes the prices of its rivals and then fixes its own price based on the rate going in the market.

ii) **Loss leader pricing:** 'Loss leader' refers to the product of the industry. This is usually applied in retail selling. If a seller is selling more than one product (which most of the modern sellers do) and one of its product has higher sales as compared to the others, then the seller may intentionally lower the price of its more selling product. This is done because the seller feels that when consumer comes to buy this product, may also buy other products. Therefore, the seller intentionally keeps its main product at low price. Therefore, it is called as 'loss leader' as the product is the leader in terms of attracting the consumers but does not give profits to the seller.

c) **Other pricing strategies**: It includes the following types of pricing strategies:

Other Pricing strategies	Administered Prices
	Dual Pricing
	Predatory Pricing
	Multiple Pricing
	Skimming Pricing
	Differential Pricing
	Penetration Pricing
	Product Pricing
	Strategy based on the prices of new firms

i) **Administered prices:** The term administered prices was originally given in context with monopoly. The monopolist intentionally administers the prices of its products and does not go by the principle of demand and supply for price fixation. Now, administered prices refer to the fixation of the price of a product by the government. Such a price fixation is intentional and is done to benefit a particular section of the economy. In India, price of petrol, gas etc has been administered since long. Such type of pricing strategy has legal support and is used to correct market inequalities. Administered prices are needed when there are huge inequalities of income in the economy i.e. the number of poor is very high and the commodity under consideration is a necessity. Therefore, the government needs to fix the prices of the necessary commodities so as to make these available for the poor and the needy. Such a policy is also needed to control the prices of essential raw materials that are highly scarce. If this is not done then the process of

industrialisation gets affected by bending towards a few entrepreneurs who can use this as a strategy to restrict the entry of new firms or may cause the exit of rival firms. This type of strategy is also

used in case of agricultural produce. Since, agricultural produce is heavily dependent on weather, therefore, to give some assured incomes to the farmers, the government may fix either the prices of the agricultural produce or may fix a limit below which the prices may not fall. This type of intervention is needed as, these days, the states are not just performing their basic functions but are also expected to provide welfare functions. Therefore, it is the duty of the state to provide basic commodities at affordable prices. Hence, the need for administered prices arises. The prices are administered either fully or partially. Full administered prices means the government administers the prices of the total output produced. The partial prices are discussed in the next category.

ii) **Dual Pricing**: This pricing strategy is followed when for the same commodity two different prices are charged. This happens in monopoly. A monopolist may charge two different prices for the same product in the two different markets.

However, his demand and supply curves for the two markets would be different. This type of pricing strategy is also followed by the government in case of electricity or in case of essential commodities.

(Source:https://www.snapdeal.com/product/power-india-meters-single phase/637647068928)

In India, government charges different rates of electricity from the households and from commercial enterprises. Similarly, the government may force two different prices for the same commodity. The private sector may continue to produce the commodity but the government

procures a part of it and sells at lower prices to the poor and the remaining output can be sold by the private producers at the market rate. In such a case, the demand and supply curves of the commodity would show a kink indicating the two different markets and two different prices. In India, this kind of pricing strategy was adopted by the government from 1950's. Later on, the distribution of the commodities at concessional rates came to be known as public distribution system (PDS)

(Source:

https://www.google.com/search?q=electricity+pricing,+indian&source=lnms&tbm=isch&sa=X&ved=0ahUKEwiYhO_v1_jTAhVEpY8KHVNoAHQQ_AUIBygC&biw=1024&bih=494#tbm=isch&q=public+distribution+depots,+india&imgrc=KFvRZ5ONRaYuFM:)

At the different depots of PDS, the commodities of essential nature are made available to the general public. However, in such a system, the burden of filling up the costs of the private sector falls on the government and secondly, the identification of the target beneficiaries is faulty because of which many actual beneficiaries do not get the real benefit. Many a time's wrong commodities also, enter into the list of essential commodities. It has also been observed that due to the corruption in depots, the traders buy the commodities at lower rates from the depots and sell these at higher rates in the open market. However, in a welfare system, the government cannot overlook the marginalized sections of the economy.

iii) **Predatory pricing**: This type of pricing policy is followed by a discriminating monopolist. The market of the monopolist can be divided into two categories and the monopolist charges two different prices of the product. If, in one market, there is a threat of entry then the monopolist may charge a price below the variable costs

so as to block the entry of its rivals. The losses of this market are funded by the profits of the other market where the price charged is relatively high. The difference between dual pricing and predatory pricing is that in the latter market price in one market is intentionally reduced below the cost and profits are funded from the other market. This type of strategy may also be followed in an oligopolistic market where one of the firm may indulge in lowering the prices below the costs. For example, in India, Ola and Uber cabs have entered into virtual price wars and have slashed the rates much below the costs.

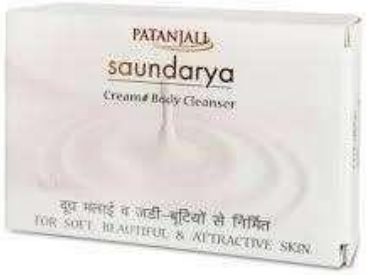
(Source: http://www.ndtv.com/delhi-news/ola-uber-cabs-back-on-delhi-roads-as-drivers-strike-crumbles-1660955)

(Source: http://www.india.com/topic/uber-delhi-rape-case/)

This is done to wipe out the other from the market. The loss in one market is filed up with the profits from the other market.

iv) **Multiple pricing**: These days, most of the firms produce multiple products. In multiple pricing strategy, the firm pursues more than one price strategy for its different products. For example, Patanjali Ayurved has entered the Indian fast moving consumer goods sector and has claimed a considerable share in the total market. It has two types of products: one which is low priced and the other which is highly priced. For example: it offers two types of soaps: the first is Patanjali Saundarya and the second one includes: Patanjali Aloe vera kanti, Rose kanti etc. Both these types

are for different set of customers. The latter one is for those who want to pay less for a good brand and the former one is for those who can pay more for a good brand.



(Source: http://www.amazon.in/Patanjali-Saundarya-Cream-Body

Cleanser/dp/B01FTXN5UE)

(Source: http://www.amazon.in/Patanjali-Kanti-Aloe-Vera-Cleanser/dp/B01CGF85IA)

v) **Skimming pricing:** Skimming pricing refers to the strategy of fixing a high price when a new product is introduced in the market. Such a product is different due to innovation and technological superiority. Since, the product is new, so the firm has to create a market for the product which calls for incurring certain expenditure. As a result, the firm may have to spend heavily on advertising. This increases the costs of the product and becomes one reason for fixing a high price for it. Another factor in deciding the price strategy in this case is the number of rivals in the market. Since, the product is new, therefore, it may not have any rivals. So, the firm fixes a high price initially. Eventually, as the new firms enter into the market, the market for the product may expand due to the increased acceptability of the product in the consumers. This may lead to the decline in the price of the product. Another factor is the elasticity of demand of the product. If the product has inelastic demand then

the firm may fix a higher price. In skimming strategy, the concerned firm fixes, initially, a high price due to high costs on advertising for creating a market and due to inelastic demand but once, the product is accepted and new firms enter into the market then the firm under consideration moves on to the other customers by reducing its price. Such a strategy is basically used for technological products. This strategy is quite common in the smart phone market. Whenever, a new phone is introduced, if its features are technically superior or different from the existing products in the market then the price of the new phone is fixed at a high level. Once, the rivals introduce their products with similar features then the concerned firm reduces the price of its phone. This is typically true of iphone by Apple Incl. Whenever a new model of iphone is launched; Apple fixes a high price for its products. The company knows that there are consumers in the market who are ready to pay a higher price for the latest technology. Such, first time buyers are, usually, called as 'innovators'.

(Source: https://www.snapdeal.com/product/apple-iphone-5s-16-gb/347830397) Gradually, the rivals also launch new phones at par with the technological level of the latest iphone. Also, by that time, Apple has captured most of the 'innovators' segment of the market and then it reduces the price of the iphone to cater to the next segment of consumers who want to buy the iphone at reduced prices.

'Reverse' skimming pricing is the opposite of skimming pricing. In such a strategy, the initial price is low and the firm increases its price after some time. Reverse skimming pricing is followed in the bookings of air tickets. Airlines offer different rates of the tickets depending on the date of the scheduled flight.

(Source:https://www.thebalance.com/last-minute-airline-specials-and-deals_4058256) The price of the tickets is low, if the bookings are done many days prior to the scheduled flight. The price of the tickets is enhanced once the date of the scheduled flight approaches.

vi) **Differential pricing:** Differential pricing implies charging different prices for the same unit of a product from the different customers. This may happen at different points of time or may continue over a period of time.

vii) **Penetration pricing:** Penetration pricing is pursued when there are many sellers of the product and the new entrant wants to penetrate and carve a space for itself in the market. Since, the product of most of the companies is the same, therefore, the new entrant distinguishes itself by offering a low price or by giving special 'offers' to the customers. This leads to the deviation of the customers of the existing firms to the new entrant and once, it has established itself then the prices are increased. The entry of Reliance Jio in the Indian telecom market is an apt example here.

(Source:https://chandigarhmetro.com/reliance-jio-new-4g-phone-cost-less-than-rs-2000/)

For penetrating into the market it offered free services for a limited time period.

This increased its customer base and shook its rivals. However, through this strategy, it has been able to create a big share in the market for itself.

viii) **Product line pricing:** Product line pricing is used when the firm manufactures more than one product and all are related to each other. For example: Britannia offers many types of biscuits like good day biscuits, nutri choice etc. Likewise, ITC offers farmlite, dark fantasy, delishus etc. The decisions of fixing the prices depend upon the cross price elasticity of demand and other factors. The firm may either choose to have same price for all the products that fall under the same category or it may go for different prices.

(Source: http://britannia.co.in/)

In shoe industry for the same type of shoes, usually, firms fix the same price irrespective of the size of the shoe. In other industries, however, some firms may reduce the price of the product when it is offered in large amounts. For example, the price of the one kilograms of oil is more than the price of five kilograms of oil.

ix) **Strategy based on the prices of the new firm:** Such a strategy is adopted by the already established firms in the market. When it is expected that a financially strong firm is expected to enter into the markets, the already established firms may not reduce their price rather these firms wait for the new firm to reveal its price strategy and then adjust their price accordingly.

CASE STUDIES:

Case Study 1: Skimming Strategy – Apple Inc.

Apple is known for its innovative products and premium pricing. When it launches a new iPhone model, it sets a high price due to the product's advanced technology and brand loyalty. This pricing allows Apple to target early adopters (called "innovators") who are willing to pay a premium for the latest technology. Over time, as competitors release similar phones and demand from the high-end market stabilizes, Apple lowers its prices to attract a wider audience. This represents a classic use of the skimming pricing strategy.

Case Study 2: Penetration Strategy – Reliance Jio

When Reliance Jio entered the Indian telecom market, it offered free data and voice services for an initial period. This aggressive pricing strategy aimed to attract users from existing players like Airtel

and Vodafone. After gaining significant market share and disrupting the telecom industry, Jio began to introduce charges, making it one of the top telecom service providers in India. This is a textbook example of the penetration pricing strategy.

13.3 LET US SUM UP

In this chapter, we have discussed the different types of pricing strategies that are followed by the firms in the real markets. The strategies are opted on the basis of the cost structure and the prevailing market conditions. Although there are many strategies, however, it must be kept in mind that the firms may not always pursue same strategy throughout the life cycle of the product. Also, apart from the other factors, the size of the firm and the brand name of the firm also play a significant role in deciding the pricing strategies. Large firms may not prefer to respond quickly to the market upheavals.

(Source: http://gibsonandryan.co.uk/portfolio-item/industrial/)

These firms may continue to charge the same price or may alter their prices with significant time interval. Such firms know that their products are sold on the basis of their reputation and brand name and thus, do not want to alter their customer base. When Kellogg entered India, its penetration was very slow. However, they did not immediately lower their price to attract customers. It, rather, waited and established customer base on the basis of its brand. However, small firms may not follow the same strategy as the large firms. They may be quick to respond to price challenges posed by

their rivals or by new firms. Thus, pricing strategies varies according to size, rivals, costs and many other factors.

KEYWORDS

- **Marginal Cost Pricing** – A pricing approach where a firm sets the price based only on variable costs, especially useful during surplus or initial market entry.
- **Mark-up Pricing** – A cost-based pricing strategy where a fixed percentage is added to the total cost to determine the selling price.
- **Loss Leader Pricing** – A tactic in which one popular product is sold at a loss to attract customers who might buy other profitable items.
- **Predatory Pricing** – A competitive strategy where a firm lowers prices below cost to eliminate competition and dominate the market.
- **Dual Pricing** – A pricing mechanism where two different prices are charged for the same product in different markets or consumer segments.

13.4 REVIEW QUESTIONS

1. Explain the difference between skimming pricing and penetration pricing with suitable examples.
2. Discuss the role of cost elements in determining pricing strategies for a new product.
3. What is predatory pricing? How can it affect market competition in the long run?
4. Analyze how government-administered prices can help control market inequalities.
5. Compare and contrast marginal cost pricing and mark-up pricing strategies.

OBJECTIVE TYPE QUESTIONS:

1. Marginal cost pricing considers only _____ costs.
2. In mark-up pricing, the final price is obtained by adding _____ to the cost.
3. _____ pricing is used to attract customers by offering a well-known product at a very low price.
4. Penetration pricing is often used by _____ firms entering a competitive market.
5. Skimming pricing initially sets a _____ price for a new product.
6. Predatory pricing aims to _____ competitors from the market.
7. _____ prices are fixed by the government to benefit specific groups.
8. Dual pricing involves charging _____ prices for the same product.

9. Rate of return pricing is based on the expected _____ on capital invested.
10. Going rate pricing follows the pricing pattern of _____.
11. Multiple pricing is used when firms offer _____ products.
12. Product line pricing is influenced by the _____ elasticity of demand.
13. Administered prices are often applied to _____ goods.
14. A monopolist may use predatory pricing to block _____.
15. In reverse skimming, the price initially set is _____ and later increased.

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FURTHER READINGS

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- Microeconomics and Behaviour – Robert H. Frank
- Pricing Strategy: Setting Price Levels, Managing Price Discounts, & Establishing Price Structures – Tim J. Smith
- Monetary Economics – S.B. Gupta

UNIT-14: PRICING EQUILIBRIUM UNDER PERFECT COMPETITION

STRUCTURE

14.1 Introduction

14.2 Understand the significance of market forms

14.3 Understand price taker assumption

14.4 Derive short-run equilibrium

14.4.1 Short-run equilibrium price determination

14.4.2 Short-run profitability of a firm

14.5 Derive Long-run equilibrium

14.6 Let us Sum up

14.7 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Understand the significance of different market forms.
- Discuss the importance of price taker assumption in perfect competition.
- Explain how price determination is done in short run?
- Understand long run equilibrium of a firm in perfect competitive market.

BLOOM'S LEVEL OF LEARNING

The unit on market structures and perfect competition primarily aligns with the lower to mid-levels of Bloom's Taxonomy, specifically encompassing Remembering, Understanding, Applying, and Analyzing. Learners are first expected to remember key concepts such as types of market forms and the assumptions of perfect competition. They then understand the economic implications of price-taking behaviour and the characteristics of different market structures. The tasks also involve

applying theoretical models to derive short-run and long-run equilibrium conditions, using graphical and algebraic analysis. Additionally, analysing firm behaviour under different profit scenarios and market dynamics demonstrates a higher cognitive level, though it stops short of more advanced levels like evaluating or creating. Overall, the unit is structured to build foundational comprehension and analytical thinking rather than higher-order critical evaluation or synthesis.

14.1 INTRODUCTION

The degree to which a firm is able to alter the price of their product in the market by its own action depend upon market forms. To understand how market behaves as a whole, there is need to understand, what are the different kinds of markets forms? The answer begins with the two sides to every market- supply and demand. These two components need to be combined to understand how the whole market behaves. It is also necessary for each and every market form. In this module, we are focusing on the perfectly competitive market form. The competitiveness of the market depends on individual firm's power to influence market prices. The more is the power an individual firm to influence the market, the more competitive is the market. Market form is based on the market structure.

14.2 UNDERSTAND THE SIGNIFICANCE OF MARKET FORMS

The term market structure is the concept that rests upon the type of market in which firms operate. Market can be distinguished by the number of firms in the market and the type of product that they are selling. The firms that produce a product, or a closely related set of products, constitute an industry. The market demand curve for any particular product is collective curve gathered from the industry.

When firms take their production and sales decision, they need to know about what quantity they can sell at various prices. Their concern is, therefore, not with the market demand curve of whole industry, but rather with the demand curve of their own product. If a firm's managers know the demand curve of their own firm faces, they know the sales that their firm can make and the revenue it will earn at each possible price. If they also know their cost of production, they can calculate the profits that would be associated with rate of output. With this information, they can choose the output that maximizes their profits. The structures of the market in which a firm operate determine the relationship between the market demand curve for the product and the demand curve facing each individual firm in that industry.

Broadly, there are three types of market form:-

- Perfect competition
- Imperfect competition: - Monopolistic competition and oligopoly
- Monopoly

Monopoly and perfect competition lie at the two extremes of the market structures. In monopoly the industry contains only one firm that can, therefore, set its price without concern about how competing firms in the industry will react. In prefect competition there are so many firms in the industry that no one of them has power to influence the market price for its product. Most of the real markets are somewhere in between these two extremes cases. Firms operate in the market structures that are either monopolistically competitive or oligopolistic.

Table 1

TYPES OF MARKET FORMS				
Structure	**Number of Producers and Degree of Product Differentiation**	**Part of the Economy Where Prevalent**	**Firm's Degree of Control Over Price**	**Methods of Marketing**
Perfect Competition	Many producers; identical products	Financial markets and agriculture products	None	Market exchange or auction
Imperfect Market				
Monopolistic	Many	Retail trade		

Competition	producers; many real or perceived differences in product	(pizzas, beer etc.)	Some	Advertising and quality rivalry; administered prices
Oligopoly	Few producers; little or no difference in product	Steel, chemical		
Monopoly	Single producer; product without close substitutes	Franchise monopoly (electricity, water); Microsoft windows; Patented drugs	Considerable	Advertising

Basic features of different market forms:-

Table 2

Market forms	Seller entry barriers	Seller number	Buyer entry barriers	Buyer number
Perfect competition	NO	MANY	NO	MANY
Monopolistic competition	NO	MANY	NO	MANY

Oligopoly	YES	FEW	NO	MANY
Monopoly	YES	ONE	NO	MANY

Assumptions of perfect competition

1. **Large number of sellers and buyers**: The industry includes a large number of firms so that each individual firm however large supplies only a small part of the total quantity offered in the market.
2. **Product homogeneity**: The industry is defined as a group of firms producing a homogenous product. This assumption imply that the individual firm in pure competition is a price taker: its demand curve is infinitely elastic.
3. **Free entry and exit of firms**: There is no barrier to entry or exit from the industry. Entry or exit may take time, but firms have freedom of movement in and out the industry.
4. **No government regulation**: there is any government regulation in the market. Tariffs, subsidies and rationing of product are ruled out
5. **Perfect knowledge**: It is assumed all the sellers and buyers have complete knowledge of the market. Information is free and costless. There is :-

   ASYMMETRIC INFORMATION

   It is defined as a situation in which a buyer and a seller have different information about a transaction. It is not possible to have complete knowledge of all relevant facts.

   It is characteristic of many business or institutional set ups.

14.3 UNDERSTAND PRICE TAKER ASSUMPTION

Perfect competition is the world of price- takers. A perfectly competitive firm sells a homogenous product. The firm is so small relative to its market that it cannot affect the market price; it simply takes the prices as given. Just as consumers must generally accept the prices that are charged by internet access providers or movie theatres, so must competitive firms accept the market prices of the goods that they produce., We can depict a price-taking prefect competition by examining the way demand looks to a perfect competitive firm. In Figure 1 shows the contrast between the industry demand curve (the DD curve) and the demand curve facing a single competitive firm (the dd curve). Because a competitive industry is populated by firms that are small relative to the market, the firm's segment of the demand curve is only a tiny segment of the industry's curve. Graphically the competitive firm's portion of the demand curve is so small that the firm's dd demand curve looks completely horizontal

or infinitely elastic. Because competitive firm cannot affect the price, the price for each unit sold is the extra revenue that the firm will earn. For example, at a market price of Rs 40 per unit, the competitive firm can sell all its wants at Rs 40. If it decides to sell 101 units rather than 100 units, its revenue goes up by exactly by Rs 40.

Here are the major points to remember:

1. Under perfect competition, there are many firms, each producing an identical product and each too small to affect the market price.
2. The prefect competitor faces a completely horizontal demand curve.
3. The extra revenue gained from each extra unit sold (MR) is therefore the market price (AR).

14.4 DERIVE SHORT RUN EQUILIBRIUM

14.4.1 SHORT-RUN EQUILIBRIUM PRICE DETERMINATION

The price of the product sold in a perfectly competitive market is determined by the interaction of the industry's short run supply curve and the market demand curve. Although no one can influence the market price significantly, the collective actions of all the firms in the industry and collective action of consumers together determine the equilibrium price. This occurs at the point where the market demand curve and the industry supply curve intersect.

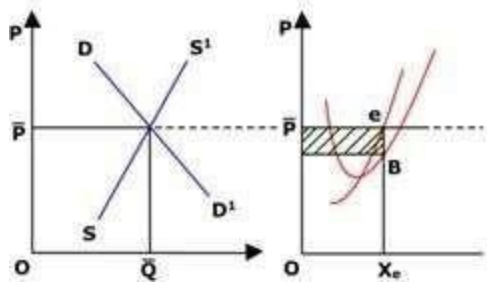
At the equilibrium price each firm is producing and selling a quantity for which its marginal cost (MC) equals marginal revenue (MR). But because price equals AR and MR also equals AR, under perfect competition, under equilibrium:

AR= ***MR*= *MC***= Price

While ***MR*= *MC*** is the equilibrium under any market form, under perfect competition, in addition they are both equal to price as well.

Given their fixed inputs, all the firms are maximising their profits and so have no incentives to alter output in the short run. Because total quantity demanded equals total quantity supplied, there is no reason for market price to change. Thus, the market and all the firms in the industry are in the short-run equilibrium.

Figure 2



14.4.2 SHORT-RUN PROFITABILITY OF A FIRMS

We know that when an industry is in short run equilibrium, each firm is maximising its profit. However, we do not know how large these profits are. It is one thing that a firm is doing as well as it can, given its particular circumstances; it is another thing to know how well it is doing.

Figure 2 shows three possible positions for a firm in short-run equilibrium. In all cases the firm in maximizing its profit by producing where price equals marginal cost, but the size of the profits is different in each case. In part (i) it is just covering all of its costs- it is just breaking even. In part (ii) the firm is suffering losses. In part (iii) it is making pure profits because average revenue exceeds average total cost.

14.5 DERIVE LONG RUN EQUILIBRIUM

In the long run firms are in equilibrium when they have adjusted their plant so as to produce at the minimum point of their long-run AC curve, which is tangent to the demand curve defined by the market price. In the long run the firms will be earning just normal profits, which are included in the LAC. If they are making excess profits new firms will be attracted in the industry; this will lead to a fall in the price (a downward shift in the individual demand curve) and a upward shift of the cost curves due to the increase of the prices of factors as the industry expands. The changes will continue until the LAC is tangent to the demand curve defined by market price. If the firms make losses in the long run they will leave the industry, price will rise and cost may fall as the industry cover their total costs inclusive of the normal rate of profit. Figure 4 shows how firms adjust to their long-run equilibrium position. If the price is P1, the firm is making excess profit working with the plant whose cost is denoted by SAC1. It will therefore have an incentive to build new capacity and it will move along its LAC. At the same time new firms will be entering the industry attracted by the excess profits.

As the quantity supplied in the market increases (by the increased production of expanding old firms and by the new established ones) the supply curve in the market will shift to the right and price will fall until it reaches the level of P*(Figure 3) at which the firms and the industry are in the long-run equilibrium. The LAC in figure 4 is the final cost curve including any increase in the prices of factors that may have taken place as the industry expanded.

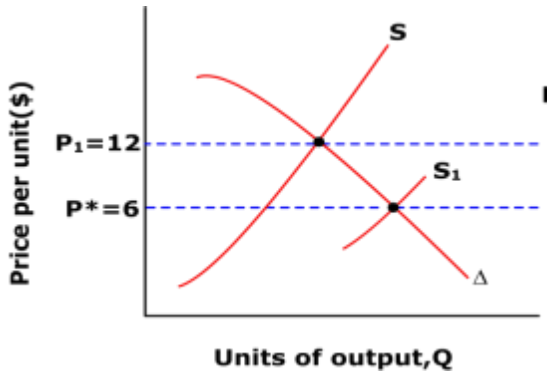


Figure 3

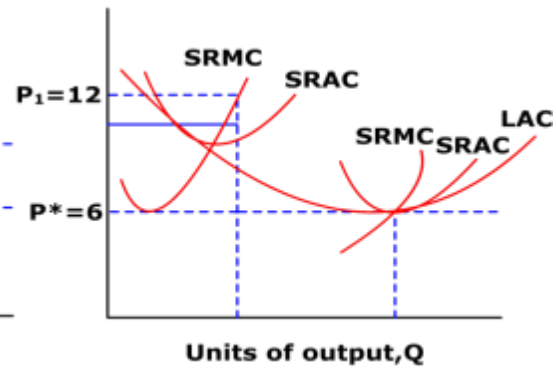


Figure 4

The condition for the long-run equilibrium of the firm is that the marginal cost be equalto the price and to the long-run average cost

LMC= LAC= P

The firm adjusts its plant size so as to produce that level of output at which the LAC is the minimum possible, given the technology and the prices of the factor of production. At the equilibrium the short-run marginal cost is equal to long-run marginal cost and the short-run average cost is equal to long-run average cost.

Thus, given the above equilibrium condition, we have

SMC= LMC= LAC= LMC= P= MR

This implies that at the minimum point of the LAC the corresponding (short-run) plant is worked at its optimal capacity, so that the minima of the LAC and SAC coincide. On the other hand, the LMC cuts the LAC at its minimum point and the SMC cuts the SAC at its minimum point. Thus at the minimum point of the LAC the above equality between short-run and long-run costs is satisfied.

At the industry level a constant influx and out flux of firms takes place. In perfect competition since the firm is a price taker the producer cannot influence the market price either in the short-run or the

long-run. They simply have to react to the conditions in the market in the long run. The result is that when few firms reap supernormal profitit leads to an influx.

Contrarily, when some firms make losses there is an out flux. Finally, industry equilibrium is achieved when there is no incentive to entry and no disincentive to leave.At this point stable equilibrium occurs in industry and so does a firm remain in long run equilibrium. Theoretically speaking all firms make normal profits, in this manner, in the long run. Incidentally, at this stage they are all operating at the most efficient level of output.

Perfect competition and Pareto- optimality

A situation is defined as Pareto- optimal if it is Impossible to make anyone better off without making someone worse off. Pareto efficiency is reached when:

it is not possible to improve the welfare of one agent without reducing that of another agent, i.e. there is an optimal allocation of resources Every competitive economy is Pareto efficient because the perfectly competitive market mechanism allocates scarce resources efficiently. Pareto Efficiency is associated with economic efficiency, but not necessarily with equity or fairness. A Pareto Optimum may not be socially desirable (for example an economy based on slavery may reach a Pareto optimum, but it is undesirable). However:

Society may achieve any Pareto efficient resource allocation by an appropriate initial redistribution of resources and free trade ***Pareto efficiency requires:***

- ***Production efficiency***: production is maximized given the resources and technology available. This means that given the resources available, the production of one good cannot be increased without reducing the production of another.
- ***Exchange efficiency***: social welfare is maximized given the available resources. This means that *whatever the goods produced, they go to the individuals who value them most(are willing to pay more for them).*
- ***Product mix efficiency (or total efficiency)*:** the goods and services produced correspond to those desired by individuals.

In perfectly competitive markets:

- **Firms**: have to be competitive in buying all their inputs (production factors) at the lowest possible price, they use these inputs to maximize production, and sell outputs at the lowest possible price to remain in business. Competition minimizes costs and maximizes production (production efficiency)

- **Consumers:** acquire goods and services at the lowest possible prices and are thus able to maximize their utility and consumption given their preferences and budget constraints (exchange efficiency)
- Hence both Production and Consumption are maximized, within the constraints imposed by the available resources.
- If perfect competition exists everywhere, and then economy wide production and consumption - will be maximized and will not be possible to increase either one by reallocating resources (General equilibrium).
- Since all producers and consumers face the same set of prices no rearrangement of production and consumption is possible that will increase economic welfare, for a given set of production conditions and consumers' preferences.

Example of Chicago wheat market

Consumers believe that all firms in perfectly competitive markets *sell identical (or homogeneous) products*. For example, in the wheat market, it is not possible to determine any significant and consistent qualitative differences in the wheat produced by different farmers. Wheat produced by Farmer Jones looks, feels, smells, and tastes like that produced by Farmer Smith. In short, a bushel of wheat is a bushel of wheat. The products of all the firms are considered to be perfect substitutes. Perfectly competitive market is approximated most closely in highly organized markets for securities and agricultural commodities, such as the New York Stock Exchange or the Chicago Board of Trade. Wheat, corn, soybeans, cotton, and many other agricultural products are sold in perfectly competitive markets. Although all the criteria for a perfectly competitive market are rarely met, a number of markets come close to satisfying them. Even when all the assumptions don't hold, it is important to note that studying the model of perfect competition is useful because many markets resemble perfect competition—that is, markets in which firms face highly elastic (flat) demand curves and relatively easy entry and exit. The model also gives us a standard of comparison. In other words, we can make comparisons with the perfectly competitive model to help us evaluate what is going on in the real world.

At the Chicago Board of Trade (CBOT), prices are set by thousands of buyers interacting with thousands of sellers. The goods in question are typically standardized (e.g., grade A winter wheat), and information is readily available. Every buyer and seller in the market knows the price, the quantity, and the quality of the wheat. Transaction costs are negligible. For example, if a news story breaks on

an infestation in the cotton crop, the price of cotton will rise immediately. CBOT price information is used to determine the value of a particular commodity all over the world.

CASE STUDIES

Case Study 1: Chicago Wheat Market – A Classic Example of Perfect Competition

The Chicago Board of Trade (CBOT) provides a real-world example of how a perfectly competitive market function. Thousands of buyers and sellers interact simultaneously to trade standardized agricultural commodities like wheat. All the wheat traded is of uniform quality (e.g., Grade A Winter Wheat), and no single farmer or buyer has the power to influence the market price. Prices are determined by supply and demand, with participants being price takers. Transparency in price, easy availability of market information, and negligible transaction costs make this market highly efficient and a textbook case of production and exchange efficiency.

Case Study 2: Short-Run Profitability in a Dairy Cooperative

A regional dairy cooperative consisting of multiple small dairy farmers functions under near-perfect competition. Each farmer produces homogenous milk that is pooled and sold collectively. During a temporary increase in milk demand, the price per litre rises. In the short run, some farmers experience supernormal profits as price exceeds average total cost. However, due to the absence of entry barriers, more farmers join the cooperative. Over time, the supply increases, prices fall, and profits normalize. This dynamic showcases the transition from short-run profits to long-run equilibrium in a perfectly competitive setting.

14.6 LET US SUM UP

The term market structure refers to the type of market in which firms operate. Market can be distinguished by the number of firms in the market and the type of product that they are selling. Broadly, there are three types of market form:-

- Perfect competition
- Imperfect competition: - Monopolistic competition and oligopoly
- Monopoly
- Perfect competition market is a most basic form of market structure. It is theoretical and hypothetical, but the most ideal form of market. It is a market structure characterized by a complete absence of rivalry among the individual firms.

- The market demand curve for the whole industry is a standard downward sloping curve, which shows alternative combination of price and output available to buyers, such that an individual buyer is able to get maximum amount of output at existing price at a given time.
- The demand curve of individual firm is a horizontal straight line showing that the firm can sell infinite volume of output at the same price and it is price taker.
- The goal of a competitive firm is to maximize profit. Profit maximization occurs at the quantity where *marginal revenue equals marginal cost*. In the short run, an individual firm may either earn super normal profit or normal profit or incur losses.

  This depends on position of short run cost curves.
- In the long run perfect competitive firm only earn normal profit. This is due to unrestricted entry into and exit of firms from the industry in the long run.
- If some of the existing firm earns super normal profit, this attracts new firm into the industry to gain profit. With the entry of new firms, the supply of commodity in the market increases, assuming no change in the market price, this lowers market price. This process of adjustment continues till the price becomes equal to the long run average cost (AR = AC =MR =MC).As such, supernormal profits of existing firms are squeezed until all the firms in the industry earn normal profit.

KEYWORDS

1. **Perfect Competition**

   A market structure characterized by a large number of buyers and sellers, identical products, and free entry and exit, where no firm can influence the market price.
2. **Price Taker**

   A firm or consumer that must accept the prevailing market price; it has no power to influence the price due to its small size relative to the market.
3. **Marginal Cost (MC)**

   The additional cost incurred by producing one more unit of a good or service.
4. **Long-Run Equilibrium**

   A situation where firms in a market earn normal profits, and there is no incentive for entry or exit, typically where price equals marginal cost and minimum average cost.

5. **Pareto Efficiency**

   A state of resource allocation where it's impossible to make one party better off without making someone else worse off, signifying optimal efficiency.

14.7 ASSIGNMENT QUESTIONS

1. Explain the significance of the price taker assumption in perfect competition.
2. Discuss how short-run and long-run equilibrium differ under perfect competition.
3. How does free entry and exit of firms affect profits in a perfectly competitive market?
4. Describe the concept of Pareto efficiency and its relationship with perfect competition.
5. Using a graph, explain how a firm determines equilibrium output in the short run under perfect competition.

OBJECTIVE TYPE QUESTIONS:

1. Under perfect competition, each firm is a _____.
2. In the short run, firms may earn _____ profits.
3. The industry supply and demand curves determine the _____ price.
4. In the long run, firms earn only _____ profits.
5. A firm's demand curve under perfect competition is _____.
6. _____ competition exists when many firms sell identical products.
7. When MC = MR, a firm is said to be in _____.
8. The long-run average cost curve is tangent to the _____ curve at equilibrium.
9. _____ efficiency ensures maximum utilization of production resources.
10. In perfect competition, products are _____.
11. Entry barriers in a monopoly are _____.
12. Under perfect competition, MR is equal to _____.
13. In the long run, no firm has an incentive to _____ or _____ the market.
14. Pareto efficiency does not imply _____ in distribution.
15. The demand curve faced by a competitive firm is perfectly _____.

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- Price Theory and Applications by Steven E. Landsburg
- Managerial Economics by D. Salvatore
- The Theory of Industrial Organization by Jean Tirole
- Markets and Market Failure by Stephen Munday

UNIT-15: PRICING UNDER OLIGOPOLY

STRUCTURE

15.1 Meaning of Oligopoly

15.1.1 Conditions of Oligopoly

15.1.2 Classification of Oligopoly

15.2 Various Models of Oligopoly

15.2.1 Cournot's Model

15.2.2 Kinked Demand Model

15.3 Let us Sum up

15.4 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Know the meaning of Oligopoly
- Identify the conditions of Oligopoly
- Discuss the basis of classification of Oligopoly
- Understand the models of Oligopoly.

BLOOM'S LEVEL OF LEARNING

The content in the unit "Pricing Under Oligopoly" primarily aligns with the lower to middle levels of Bloom's Taxonomy, specifically the "Remembering," "Understanding," and "Applying" levels. The learning outcomes such as knowing the meaning of oligopoly, identifying conditions, and discussing classifications and models correspond to "Remembering" (recalling facts and definitions) and "Understanding" (explaining concepts, summarizing models). The inclusion of model-based explanations like Cournot's and Sweezy's Kinked Demand Curve also introduces "Applying" level skills, as learners are expected to interpret diagrams and apply theoretical assumptions to understand market behaviour. However, there is limited emphasis on higher-order cognitive skills like

"Analysing," "Evaluating," or "Creating", which would involve comparing models critically, justifying pricing strategies, or constructing new theories based on given data.

15.1 MEANING

Oligopoly is an imperfect market where there are few sellers, producing homogeneous or differentiated product but the products are not perfect substitutes. Oligopoly has some of the elements of both monopoly and monopolistic market.

Definition: -

According to McConnell: - “Oligopoly is a market situation in which number of firms in an industry is, so small that each must consider the reactions of rivals in formulating its price policy”.

15.1.1 CONDITIONS OF OLIGOPOLY

1. **Few sellers and large number of buyers:-**Oligopoly is a market structure where there are few sellers which have direct competition .The product in oligopoly can be either differentiated or homogeneous which are sold to large number of buyers.
2. **Characteristics of Monopoly:-** Under oligopoly, as there are only few firms so these firms create brand loyalty among the large number of buyers. Thus because of brand loyalty among consumers firms enjoy monopoly and earn more profits.
3. **Homogeneous or differentiated products:-**Types of oligopoly depends upon what type of the product the oligopolistic firms are selling. If the firms are selling homogeneous product then it is known as pure oligopoly whereas if the firms are selling Differentiated products, it is known as differentiated oligopoly or imperfect oligopoly.
4. **Advertisements:-**Advertisement or selling costs plays an important role in oligopoly. Advertisement constitutes the major chunk of the total cost of the product. It can make ormar a firm. So for a handling cut throat competition or for attracting consumers, advertising is necessary in oligopoly.

5. **Lack of uniformity: -** There can be lack of uniformity in size of the firm. One firm could be large and others can be small and similarly can hold more or less of the market share.
6. **Price rigidity**: - In an oligopoly, if firm tries to reduce the price, its competitors also reduces its price. This will create no advantage to the firm. In other situation, if the firm increases the price, its competitors do not do so, as a result the firm loses its consumers. As a result, in both the situations the firm is likely to lose, thus there remains price rigidity.

15.1.2 BASIS FOR THE CLASSIFICATION OF OLIGOPOLY

1. **On the basis of product differentiation: -**If the all firms of a market produces homogeneous products, then that is known as pure or perfect oligopoly. On the other hand if firms are having differentiated products, then it is known as differentiated or imperfect oligopoly.
2. **On the basis of agreement: -**In oligopoly where firms follow a common price policy through an agreement or mutual understanding is known as collusive oligopoly and an oligopoly where firms act independently is known as non-collusive oligopoly.
3. **On the basis of entry of the firms: -**If there is an open entry of the firms in market. It is known as open oligopoly and the market where free entry is not allowed, is known as closed oligopoly.
4. **On the basis of Price Leadership: -** In oligopoly where dominant firm or a price leader fixes the price and the other firms follow the price, this kind of oligopoly is known as Partial oligopoly and the oligopoly where no firm is dominant and price is fixed by each firm for itself is known as Full oligopoly.

15.2 VARIOUS MODELS OF OLIGOPOLY

15.2.1 COURNOT'S MODEL OF OLIGOPOLY

Augustin Cournot was a French economist who develop the model of Oligopoly which is based on following assumptions: -

(a) There are two firms in a market, both of them owns a mineral water well

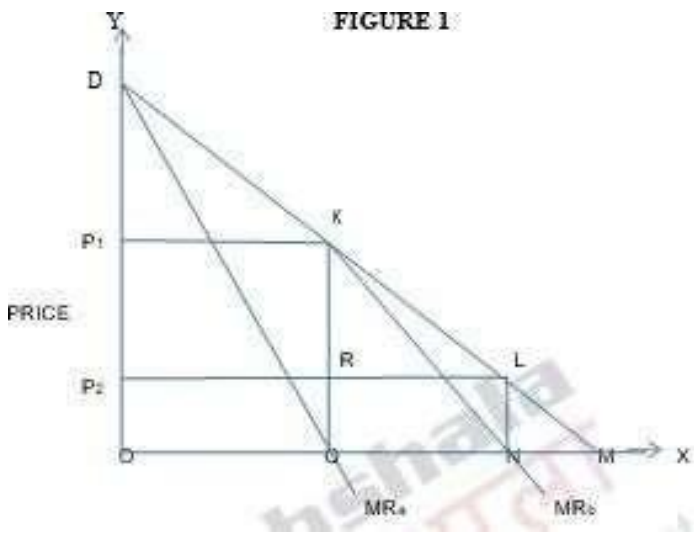
(b) Both operate their wells at zero marginal costs

(c) Both firms had constant negative slope demand curve

(d) Each seller takes a decision keeping in mind that the other firm will not react to hisdecision to change his output and price.

Cournot, on the basis of this model concluded that each seller supplies one third of the market demand and charges the same price and one third of the market remains unsupplied.

Understand the model with the help of a diagram: -



Source: https://epgp.inflibnet.ac.in

QUANTITY

In the Figure 1. DM is the market curve and MR is the marginal revenue curve for mineral water.OM is the total quantity which the two firms can supply. Now, suppose there are two firms in the market A and B but initially A enters into the market, it will sell mineral water where it's MC=MR, thus will sell OQ quantity at OP1 price, thus its profit is OP1KQ. Now B enters into the market. Now B has QM quantity to sell (half of the total market).B assumes that A will continue selling at the same price, the same output. Thus the market available to B is QM, where KN where it intersects at N so the quantity sold by firm B is ON at price OP2. The total revenue is QRLN. Due to the entry of B price is reduced to OP2, thus reducing the profit of A to OP2RQ. Now atries to adjust the output and price in the light of changed circumstances, assuming B will not change its output and price. Thus A half of the three fourth of the market i.e. 3/8 making his share reduced to 3/8 from ½. Now B assumes that A will not change its 3/8 share and market available to him is 1-3/8=5/8. Thus B will serve half of the available market to maximize its profits, making his share as 5/16. Now it is the turn of firm A and

therefore this process will continue until both of the firms end up at supplying 1/3 each of market share and 1/3 remains unsupplied.

15.2.2 KINKED DEMAND MODEL OF OLIGOPOLY

This Model is also known as Sweezy's Kinky Demand Curve Model. This model is regarded as the most important model of Non –Collusive Models of Oligopoly. This Model does not explain the price determination but it explains why the price rigidity prevails in the oligopoly. This Model is based on the following assumptions:- (a) If one firm decreases its price the other firms will also decrease their prices.

(b) If one firm increases its price the other firms will also increase their prices.

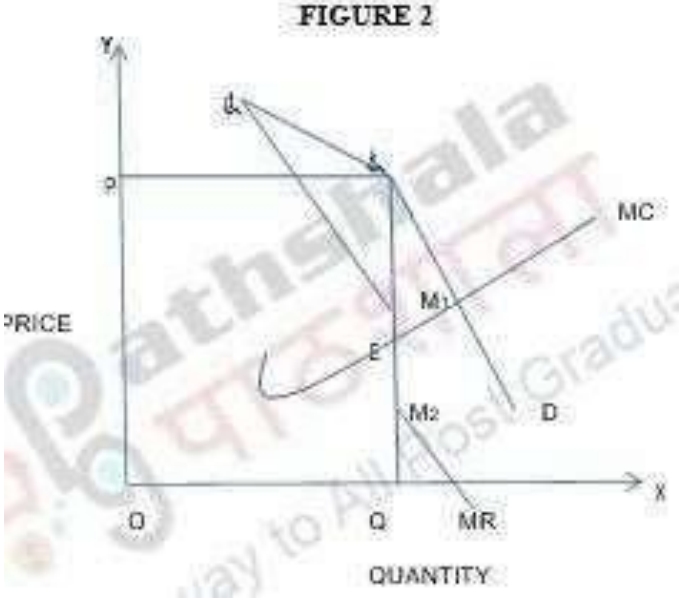
(c) There is an established existing price.

(d) The Marginal cost curve will pass through the dotted portion of marginal revenue curve.

According to Paul Sweezy, the demand curve in oligopoly is not continuous but it is kinked. Similarly Marginal Revenue curve is also in discontinuity (shown by the dotted line).

Explanation of the Model: - The Model can be explained with the help of diagram:-

FIGURE 2

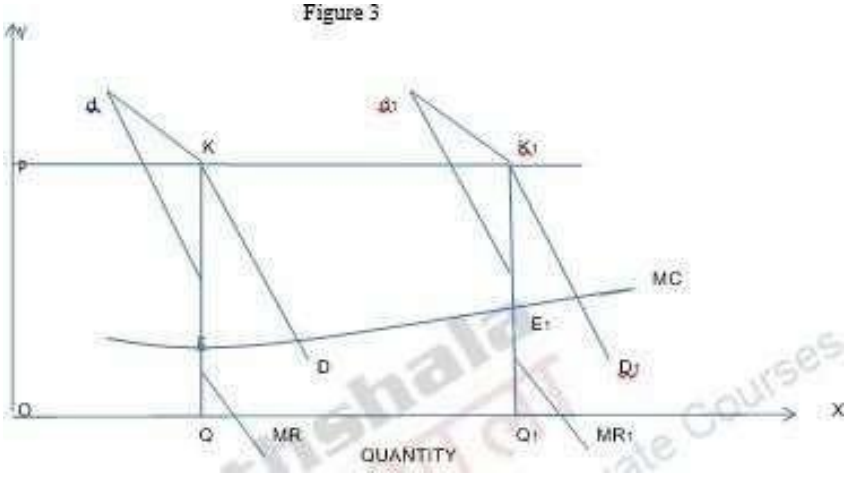


Source: https://epgp.inflibnet.ac.in

In the Figure 2 , at point E ,MC=MR and MC is cutting MR in the discontinued portion , thereby

fixing an output at OQ and price at OP and reason for price rigidity is that so long as MC will continue to cut MR at its discontinued portion Price will remain unchanged and thus there comes price rigidity in oligopoly. This scenario happens even if cost of production increases or cost of production decreases Price rigidity also prevails in case of **change in demand** too. This could be explained with the help of diagram:-

Figure 3



Source: https://epgp.inflibnet.ac.in

In the diagram when dD was the demand curve, the equilibrium price was E and demand was OQ whereas price was OP. When the demand increased the demand curve shifted tonew demand curve d1D1 where same cost curve MC cuts at the discontinued portion making a new equilibrium at point E1 where the new quantity is

OQ1 but the price remains at OP. Thus it is clear from above figure that even if the demand changes, the price remains the same. Hence, the Price Rigidity prevails.

Criticisms of Kinky Demand Curve Analysis:-

1. This model is unable to explain the Price and Output Determination.
2. This model is not applicable where products are not homogeneous.
3. This model is not applicable in case of collusive oligopoly.
4. This model ignores the non price competition

CASE STUDIES

1. **The Indian Automobile Industry: A Case of Oligopoly** The Indian automobile industry is a classic example of an oligopolistic market structure. Dominated by a few major players such as Tata Motors, Maruti Suzuki, and Mahindra & Mahindra, these firms produce both homogeneous and differentiated products. The market exhibits characteristics like price rigidity, where companies are hesitant to change prices due to the interdependence of firms. For instance, when one company introduces a new model or alters its pricing strategy, competitors often respond swiftly to maintain their market share. This interdependence leads to a stable pricing environment, with firms focusing on non-price competition strategies like advertising and product differentiation to attract consumers.
2. **The Airline Industry: Collusive Oligopoly in Action** The airline industry in many countries operates as a collusive oligopoly, where a few dominant carriers control the majority of the market. These firms often engage in tacit collusion, setting prices at a level that maximizes collective profits while avoiding explicit agreements that could attract regulatory scrutiny. For example, major airlines may coordinate schedules and pricing indirectly, ensuring that price wars are avoided and profits are maintained. This behavior is facilitated by high entry barriers, such as significant capital investment and regulatory approvals, which prevent new competitors from entering the market easily. Consequently, consumers face higher prices and fewer choices, while the incumbent firms enjoy sustained profitability.

15.3 LET US SUM UP

In this module, we have learnt about:-

- Oligopoly is an imperfect market where there are few sellers, producing homogeneous or differentiated product but the products are not perfect substitutes.

Oligopoly has some of the elements of both monopoly and monopolistic market**.**

- The conditions of oligopoly are Few sellers and large number of buyers, Homogeneous or Differentiated products, Advertisements, Lack of uniformity ,Price rigidity

• Oligopoly can be classified on the basis of on the basis of product differentiation , on the basis of agreement, on the basis of entry of the firms and on the basis of Price

Leadership

- Augustin Cournot develop the model of Oligopoly in which there are two firms in a market both operate their wells at zero marginal costs, both firms had constant negative slope demand curve**,** each

seller takes a decision keeping in mind that the other firm will not react to his decision to change his output and price.

- Sweezy's Kinky Demand Curve Model is regarded Non –Collusive Model of Oligopoly does not explain the price determination but it explains why the price rigidity prevails in the oligopoly based on the assumptions that if one firm decreases its price the other firms will also decrease their prices, If one firm increases its price the other firms will also increase their prices, There is an established existing price and the Marginal cost curve will pass through the dotted portion of marginal revenue curve.Kinky Demand Curve is unable to explain the Price and Output Determination, is not applicable where products are not homogeneous, is not applicable in case of collusive oligopoly and ignores the non price competition.

KEYWORDS

1. **Oligopoly:** A market structure characterized by a small number of firms whose decisions are interdependent, leading to limited competition.
2. **Price Rigidity:** A situation where prices remain stable over time despite changes in demand or cost conditions, often due to the interdependent nature of firms in an oligopoly.
3. **Collusive Oligopoly:** An oligopoly where firms cooperate, either explicitly or tacitly, to set prices and output levels, maximizing joint profits at the expense of consumer welfare.
4. **Non-Collusive Oligopoly:** An oligopoly where firms independently set prices and output levels, often leading to competitive behaviors like price wars and non-price competition.
5. **Kinked Demand Curve:** A model explaining price rigidity in oligopolies, where firms face a demand curve with a discontinuity, leading to a marginal revenue curve with a gap that makes price changes unprofitable.

15.4 ASSIGNMENT QUESTIONS

1. Explain the concept of oligopoly and discuss its key characteristics.
2. Differentiate between collusive and non-collusive oligopolies with suitable examples.
3. Illustrate and explain Cournot's model of oligopoly. How does it predict the behavior of firms in an oligopolistic market?

4. Describe the kinked demand curve model. How does it account for price rigidity in oligopolistic markets?

5. Analyze the impact of advertising and product differentiation on competition in an oligopolistic market.

OBJECTIVE TYPE QUESTIONS

1. Oligopoly is a market structure where there are _____ sellers and many buyers.
2. Products in an oligopoly can be either _____ or differentiated.
3. The kinked demand curve model was developed by _____.
4. In Cournot's model, each firm assumes the _____ of the rival remains constant.
5. Price rigidity means that prices remain _____ despite changes in demand or cost.
6. In a collusive oligopoly, firms may form a _____ to avoid competition.
7. Cournot's model assumes that marginal cost is _____.
8. Advertising plays a _____ role in oligopolistic competition.
9. Pure oligopoly involves _____ products.
10. Differentiated oligopoly involves products that are _____ but not identical.
11. Price leadership occurs when one _____ firm sets the price for the industry.
12. Non-collusive oligopoly involves firms acting _____.
13. In the kinked demand model, marginal revenue curve is _____.
14. Entry barriers in oligopoly markets are generally _____.
15. Firms in oligopoly often engage in _____ competition rather than price wars.

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- "Managerial Economics" – D.N. Dwivedi
- "Microeconomic Theory" – Andreu Mas-Colell, Michael Whinston, Jerry Green
- "Modern Economic Theory" – K.K. Dewett
- "Industrial Economics" – R.R. Barthwal

UNIT-16: PRICING UNDER MONOPOLY AND MONOPOLISTIC

STRUCTURE

16.1 Meaning of Monopoly

16.1.1 Conditions of Monopoly

16.2 Relationship between Demand and Revenue

16.3 Short-run equilibrium under Monopoly

16.4 Long-run equilibrium under Monopoly

16.5 Concept of Monopolistic

16.6 Assumption of Monopolistic Competition

16.7 Product differentiation and Demand curve

16.8 Concept of 'Industry' and 'Product group'

16.9 Derive long-run equilibrium

16.9.1 Short-run equilibrium under Monopolistic competition

16.9.2 Long-run equilibrium under Monopolistic competition

16.10 Let us Sum up

16.11 Review Questions

LEARNING OUTCOMES

After studying this unit, you will be able to:

- Know the meaning of Monopoly.
- Identify the conditions of Monopoly.
- Discuss the relationship between Demand and Revenue.
- Evaluate short-run and long-run equilibrium under Monopoly.
- Understand the Assumption of Monopolistic Competition.

- Know about the importance of product differentiation and demand curve faced by Monopolistic firm.
- Learn about the short run and long run equilibrium of a firm in monopolistic competition.

BLOOM'S LEVEL OF LEARNING:

The content you've provided on monopoly and monopolistic competition in business economics primarily aligns with Bloom's Taxonomy levels of "Understanding" and "Applying." The material explains key economic concepts such as monopoly, its features, revenue relationships, and equilibrium conditions in both the short and long run, which requires learners to comprehend and interpret economic models, definitions, and graphical representations—this corresponds to the "Understanding" level. Additionally, when the unit delves into the use of formulas (e.g., degree of monopoly power) and applies theoretical concepts to different market situations like supernormal profits or excess capacity, it engages the "Applying" level, where learners must use knowledge in specific contexts. If the learner is required to interpret graphs or solve numerical examples based on elasticity and marginal cost relationships, it may also touch the "Analyzing" level, though most of the unit content is not yet requiring deep comparison or evaluation. Overall, the unit is primarily situated in the Understanding to Applying range of Bloom's learning levels.

16.1 MEANING OF MONOPOLY

Monopoly is a market situation in which there is one seller of the good with no close substitutes. A Monopolist is the sole seller of the product and he have full control over the supply and the price of the product. Thus a monopolist is a price maker as he does not have to think in the light of actions taken by its rivals.

DEFINITION

According to McConnell: - "Pure or Absolute Monopoly exists when a single firm is the sole producer for a product for which there are no close substitutes."

16.1.1 CONDITIONS OF MONOPOLY

One seller and large number of Buyers: - In Monopoly there is only one seller. A monopolist could be an individual or a group of partners or a Joint Stock company. This is the reason that in monopoly there is no difference between firm and industry. The demand of Monopolist is demand of industry.

1. **Restriction in the entry of new firms:-**In Monopoly there is a strict restriction on the entry of the new firms because it won't be a Monopoly if there are more than one seller in the industry.
2. **No close substitutes: -** The other condition of monopoly is that there should not be close substitute of the product sold by Monopolist. It means that if there is any close substitute of the product available then the monopoly ceases to exist.
3. **Nature of the demand curve: -** The aggregate of the demand by the buyers will be the demand of the monopolist and the demand of the market.
4. **AR and MR Curves: -** AR is the average revenue curve of the monopolist which slopes downwards whereas MR is Marginal revenue curve which is also negatively sloping and lies below AR curve.

16.2 RELATIONSHIP BETWEEN DEMAND AND REVENUE

The relationship between Demand and Revenue can be understood by analyzing the nature of the Demand curve and the Average Revenue and Marginal Revenue curve.

Nature of the Demand Curve: - In Monopoly as there is only one producer, the aggregate demand of all the buyers will form the demand for a monopolist. As we know that the slope of individual demand curve is negative. Given the fact that monopolist's demand is the aggregate of the individual demand so the demand curve ofthe monopolist is also negative sloping. It means that a monopolist can sell more at a lower price. On the other hand if the monopolist raises its price then his sales will be reduced.

AR and MR curves: - The downward sloping demand curve means that average revenue or price falls as the sales increases. Average Revenue slopes downward, marginal revenue slope lie below the average revenue. In other words MR curve of monopolist also slopes downward from left to right.

The relationship can be summarized as:

Under monopoly,

$$P = d = AR > MR$$

And $MR = MC$

Therefore, $\boxed{P > MC}$

The P-MC difference is called degree of monopoly. An alternative index or monopoly power has been proposed by Abba Learner who pointed out that the firm being a price maker under monopoly, always set a price (P) which is higher than marginal costs (MC). Monopoly power depends upon the ability to sell its product at a higher price than the marginal costs. Larger the gap between price and marginal costs, greater the monopoly power. The degree of monopoly power depends upon the price elasticity of demand(e).

$$MR = P(1-1/e)$$

And $MR = MC$

Therefore $MC = P(1-1/e)$

Or $MC = \frac{P.e-P}{e}$

or $P-MC = P/e$

or $\frac{P-MC}{P} = \frac{1}{e}$

Where e is elasticity of firm's demand curve

So, the last equation is called degree of monopoly power.

16.3 EQUILIBRIUM UNDER SHORT RUN

Price –output Determination in the short period can be seen in three situations:-

1) Supernormal profits
2) Normal profits
3) Minimum loss

1) **Supernormal profits: -** In the monopoly the equilibrium where MC=MR. The equilibrium output is OM and PM is the equilibrium price. In this situation the Monopolist firm is earning supernormal profits as PM i.e. average revenue exceeds QM i.e. average cost. The supernormal profits are shown by shaded area in figure 1

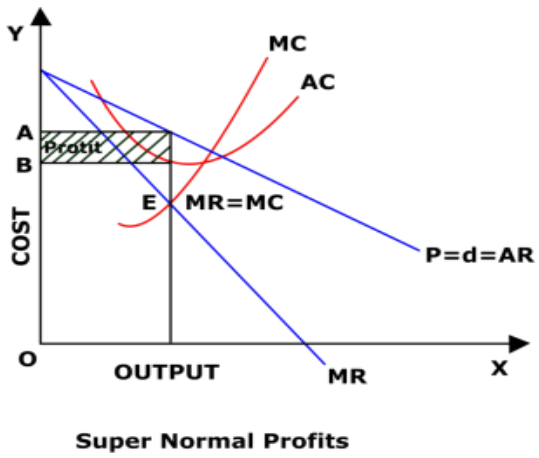


Figure 1

2) **Normal Profits**: - E is the point of equilibrium as there MC=MR. Om is the equilibrium output. PM is the price which is equal to average cost. Thus making a firm earn normal profits in figure 2.

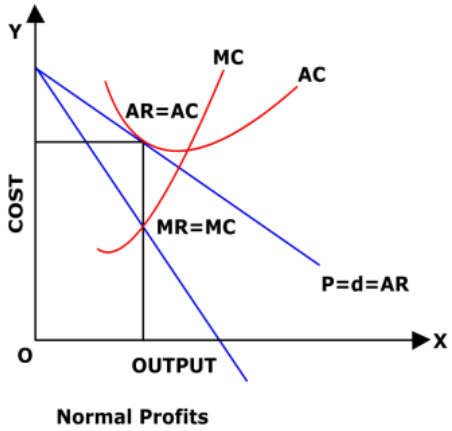


Figure 2

3) **Minimum loss: -** In the short run the firm may incur minimum loss if monopolist price is at least equal to average variable costs in figure 3.

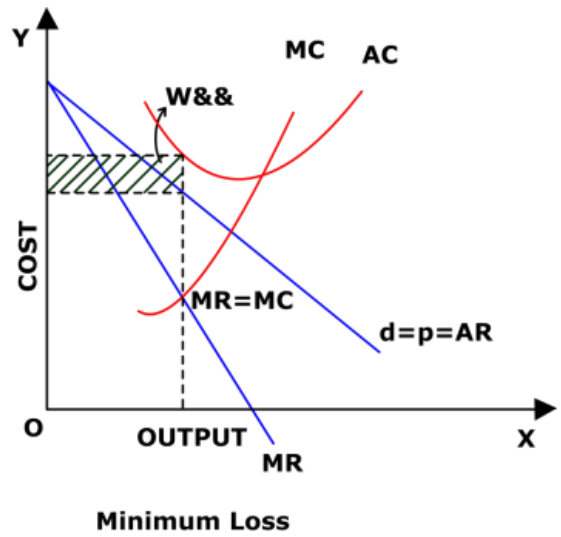


Figure 3

Case of no supply curve

We now examine that there is no supply curve for the monopolist derived from his MC. Given his MC, the same quantity may be offered at different prices depending on the price elasticity of demand. Graphically this is shown in figure 4. The quantity X will besold at price P1 if demand is D1 while the same quantity X will be sold at price P2 if demand is D2. Thus there is no unique relationship between price and quantity.

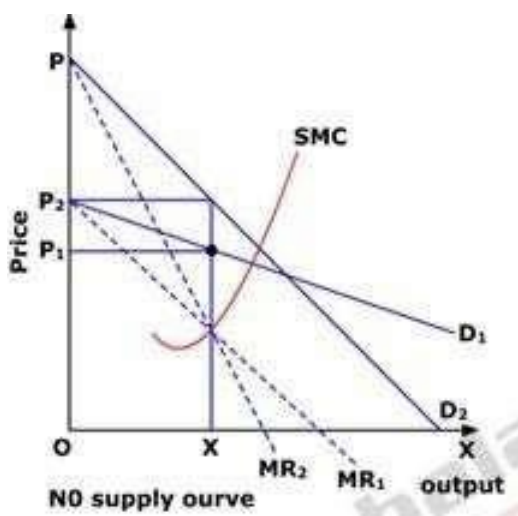


Figure 4 *Source: https://epgp.inflibnet.ac.in*

16.4 EQUILIBRIUM UNDER LONG RUN

Long period is the period in which supply can be adjusted according to the demand. In long period all the factors of production are variable. Both Volume and capacity of production can be changed. Monopolist in equilibrium in long run at a point where MR is equal to Long run Marginal cost. In the short run Monopolist can earn both profits as well as loses, but in long run monopolist can earn only profits because if the in long run monopolist's price is less than the average variable cost, he would close down the unit and hence will not incur a loss. On the other hand, Monopolist in long run are not contended with the normal profits, rather they aim to have super normal profits as shown in Figure 5 below: -

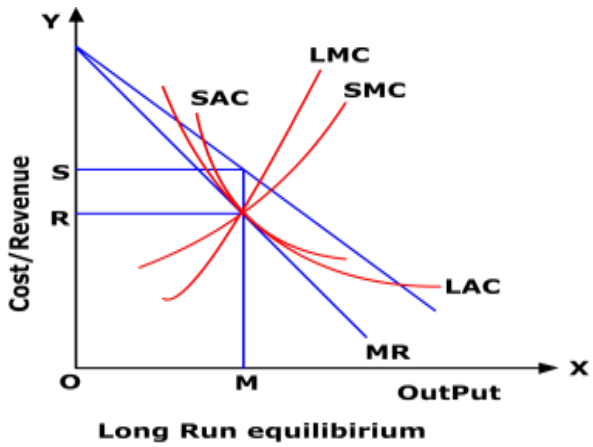


Figure 5

In the above figure Monopolist is earning supernormal. Profit of SPER, as at point E he is in equilibrium where LAC is equal to LMC thus making OM an equilibrium output and PM equilibrium price. In figure 6 we depict the case in which the market size does not permit the monopolist to expand the minimum point of LAC. In this case not only is his plant of sub-optimal size but also the existing plant is under- utilized. This is because to the left of the minimum point of the LAC the SRAC is tangent to the LAC at its falling part, and also because the short-run MC must be equal to the LRMC. This occurs at e, while the minimum LAC is at b and the optimal use of the existing plant is at a. Since it is utilized at the level e', there is excess capacity.

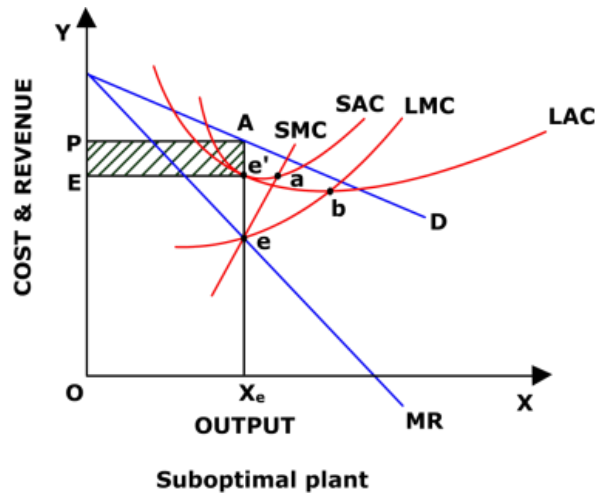


Figure 6

In figure 7 we depict the case where the size of the market is so large that the monopolist, in order to maximize his output, must build a plant larger than the optimal and over utilize it. This is because to the right of the minimum point of the LAC the SRAC and the LAC are tangent at a point of their positive slope, and also because the SRMC must be equal to the LAC. Thus the plant that maximizes the monopolist's profits leads to higher costs for two reasons: firstly it is larger than the optimal size and secondly, it is over utilized.

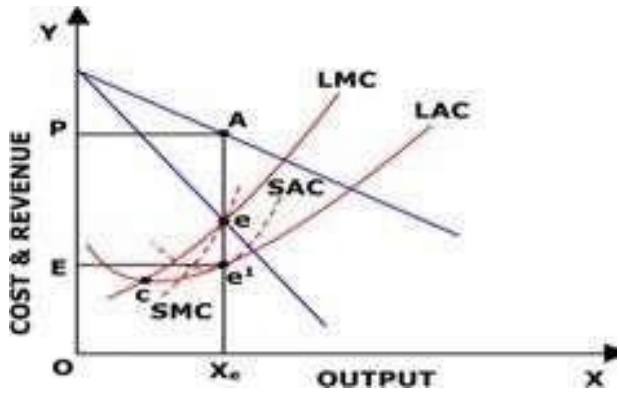


Figure 7

16.5 CONCEPT OF MONOPOLISTIC

Perfect competition and monopoly are extreme cases that do not involve active competition at all. In the former, firms have no influence over the market price, while in the latter there is only one producer. Most real firms operate under intermediate market structures. To begin with, they are not monopolies because their industries contain several firms, which often compete actively against each other. On the other hand,

these firms do not operate in perfectly competitive markets because the number of competing firms is often small and, even when the number is large; the firms are not price taker.

As in real world, most of the firm operate in intermediate market forms, where they do have some market power but also have competition. Therefore, there is need to study the firm behaviour in intermediate market structures. They are also called *imperfectly competitive.* The word competitive emphasizes that we are not dealing with monopoly and the word imperfect emphasizes that we are not dealing with perfect competition. In this module, we are focusing on one of the prominently prevailing intermediate market structure that is monopolistic competition. This form of market is close to perfect competition with one important difference. Firm do not sell a homogenous product.

16.6 UNDERSTAND THE ASSUMPTION OF MONOPOLISTIC COMPETITION

One of the simple models of imperfect competition is known as monopolistic competition. Tractability of this model is achieved by limiting the form of the interdependence between producers. Monopolistic competition refers to a market in which there are many firms and each sells a single differentiate product. Since each firm's is somewhat different from those of competitor, each faces a negatively sloped demand curve for its product.

The theory based on four key assumptions:

1. *Each firm produces one specific variety, or brand, of the industry's generic product.* Each firm thus faces a demand curve that is negatively sloped, yet is elastic, because many close substitutes are sold by other firms.
2. *Within the industry there are a number of firms each one ignores the possible outcome of its competing firm when it decides its own price and output decisions.*

   Each firm's decision making is based on its own demand and cost conditions and does not take any account for the potential reaction by other firms.
3. *Free Entry and Exit Exists.* If existing firms are earning profits, new firms have an incentive to enter. When they do, the demand must be shared among more brands.
4. *There is symmetry*. When a new firm enters selling a new differentiated version of the generic product, it takes customers equally from all the existing firms. For example, a new entrant that captured 5 percent of the existing market would do so by capturing 5 percent of the sale each existing firm.

16.7 PRODUCT DIFFERENTIATION AND THE DEMAND CURVE

Product differentiation is intended to distinguish the product of the one producer from that of the other producers in the industry. The products is not homogenous of the products are while the basic features may be the same, yet the consumer is persuaded, via advertising or the other selling activities, that the products are different. Product differentiation exists when differences in the factor inputs, or the location of the firm, which determines the convenience with which the product is accessible to the consumer, or service offered by the producer. Fancied differentiation happens due to by advertising, difference in packaging, difference in design, or simply by brand name. Whatever the case, the aim of the product differentiation is to make the product unique in the mind of the consumer. Yet differentiated products are closely related if they are included in the same 'Product Group': the products should be close substitutes with high prices and cross- elasticities.

The effect of differentiated products is that the producer has some discretion in the determination of the price. Here the producer is a price- taker to some extent, but has some degree of monopoly power . However, he faces the keen competition of the close substitutes offered by the other firms. Hence the discretion over the price is limited. Product differentiation creates brand loyalty of the consumer and this is responsible for a negatively sloping demand curve. Product differentiation finally, provides rationale for selling price. Perfectly competitive firms face a perfectly elastic demand curve for their product because all firms in their industry produce the exact same product. A monopolistic firm faces a downward-sloping firm demand curve. This type of curve happens due to notion that the firm can change its price without losing all of its business because buyers do not see any perfect substitute. The fewer the substitutes (i.e., the more product differentiation), the less elastic the demand curve will be.

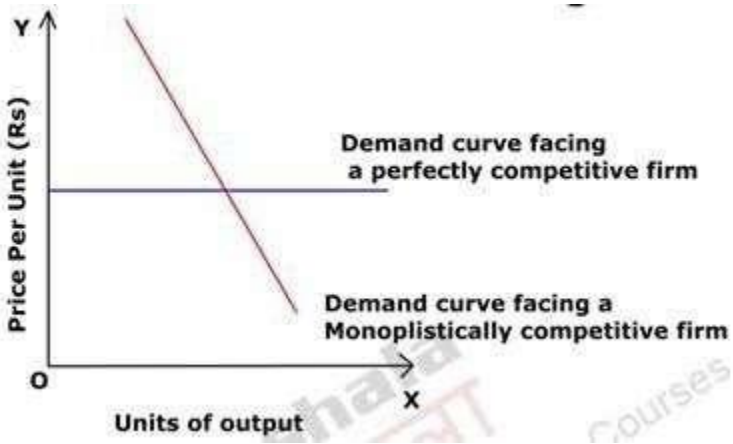


Figure 8 *Source: https://epgp.inflibnet.ac.in*

16.8 THE CONCEPT OF 'INDUSTRY GROUP' AND 'PRODUCT GROUP'

Product differentiation creates difficulties in the analytical treatment of the industry. Heterogeneous products cannot be added to form the market demand and supply schedules as in the case of homogeneous products. Therefore the concept of the industry needs redefinition. Chamberlin uses the concept of 'Product Group', which includes products which are 'Closely Related'. The products should be close *technological and economical substitutes*. Technological substitute are products which can technically cover the same want. For example, all motor cars are technological substitute in the sense they provide transport. Economic substitute are products which cover the same want and have similar prices. For example, a Hillman Avenger and a Morris 1300 can be considered as economic substitutes. An operational definition of the 'product group' is that the demand of each single product is highly elastic and that it shifts appreciably when the price of other products in the group changes. In the other word, product forming the 'group' or 'industry' should have high price and cross-elasticites.

Summation of the individual demand and cost curves to form the 'industry' demand and supply requires the use of common denominators. Even so, with the product differentiation there is no such unique price except under Chamberlin's Heroic assumption. As per Chamberlin 'Heroic assumption', is that cost and demand functioning is common throughout the group. This requires that consumers' preferences be evenly distributed among seller, and that differences between the products be such as not give rise in differences in cost. Simultaneously as product differentiation allow each firm to charge a different price. There will be no unique equilibrium price, but an equilibrium cluster of prices, reflecting the preferences of consumers for the product of the various. firms in the group. When the market demand shifts or cost condition change in a way all firms, then the entire cluster of prices will rise or fall simultaneously.

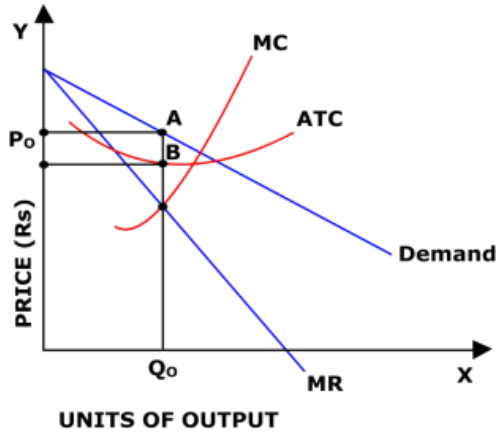
16.9 DERIVE LONG RUN EQUILIBRIUM

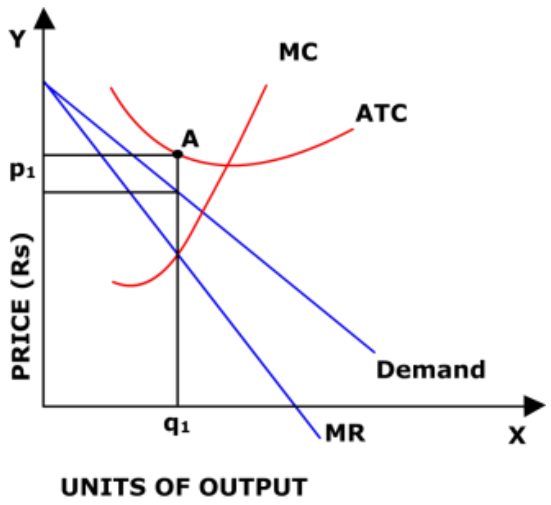
Monopolistic competition refers to the market with a fairly large number of firms which sell differentiated products. A single firm in the product group (industry) hardly any impact on the market price. However, if it reduces price, it can expect a considerable increase in its sales. Each firm is capable of drawing consumers from other firms by conjuring make belief differences or real difference with the help of a range of non price competition like advertisement and sales promotion. Under these circumstances the firm can raise the price and yet not lose out on customer demand. This is because of the fact that the product is differentiated from competing firms due to price and non-price factors.

The demand curve (AR curve) of the monopolistic firm is therefore, highly elastic and is downward sloping. The marginal revenue curve, also slopes downward and lies below the demand curve because there is a fixed relationshipbetween AR and MR.

16.9.1 SHORT RUN EQUILIBRIUM UNDER MONOPOLISTIC/IMPERFECT COMPETITION:

During the short period, the number of firms in the 'group' remains the constant. The size of the plant of each firm remains constant. As the firm whether operating under perfect competition, or monopoly wants to maximize profits. In order to achieve this objective, it goes on producing a commodity so long as the MR is greater than MC. When MR = MC, equilibrium is established and produces the best level of output. If a firm produces less than or more than the MR = MC output, it will then not be making maximum of profits. In the short-run, a monopolistically competitive firm may be realizing abnormal profits or suffering losses. If it is earning profits, no new firms can enter the industry in the short-run. In case, it is suffering, losses but covering full Variable cost, the firm will continue operating so that the losses are minimized. If the full variable cost is not met, the firm will close down in the short-run. The short-run equilibrium with profits and short run equilibrium with losses of a monopolistically competitive firm are explained with the help of two separate diagrams as under.



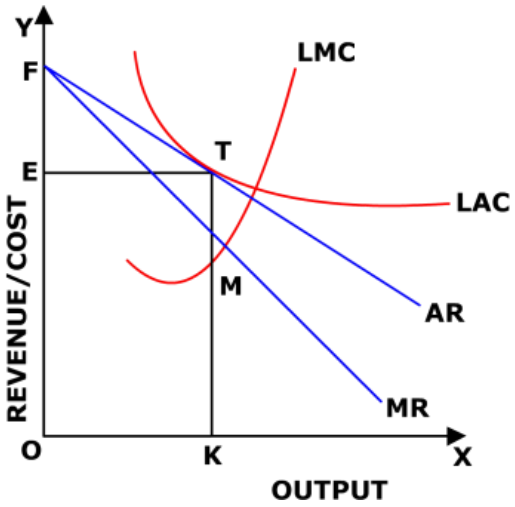


Since the firm has a downward-sloping demand curve, it will also have a downward- sloping marginal revenue (MR) curve. A profit-maximizing firm produces where marginal cost (MC) equals marginal revenue (q0 in the above graph) and charges the price determined by demand (P0). In panel (a) of the figure, the monopolistic competitor will make a profit. However, like a monopoly, a monopolistic competitor is not guaranteed to make a profit in the short run. The firm may make a loss in the short run; its profitability will depend on the demand. This is shown in panel (b).

16.9.2 LONG RUN EQUILIBRIUM UNDER MONOPOLISTIC/IMPERFECT COMPETITION:

In the long run, the firms are able to alter the scale of plant according to the changed conditions of demand for a product in the market. They can also leave or enter the industry. If the firms reap subnormal profits in the short run, then new firm would enter the 'product group'(industry). The tendency of the new firms to enter the industry continues till the abnormal profits are competed away and the firms economic profits are zero. Monopolistically competitive firms realize losses in the short-run, thensome of the firms will leave the industry. The exit of the firm continues till zero economic profits are restored with the operating firms.

In the long-run, there are no entry barriers for the new firms. The incoming try to differentiate their products . The old firms operating with .the used machinery try to match up with the new entrants by improved variety of products in their group. They incur expenditure on advertisement and on sales promotional . They employ better quality staff for making technical improvement in their products. All firms incur additional expenditure on account of the quality improvements, the cost curves of all

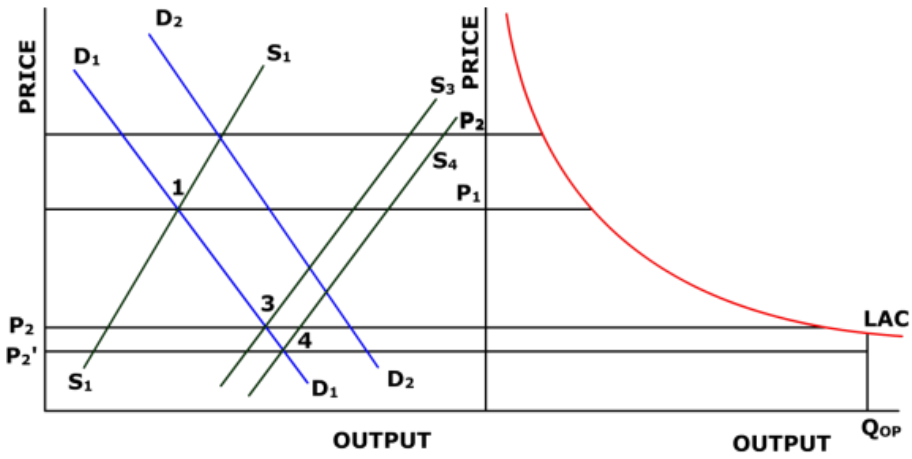


the firms move up. Due to entry of new firms in the industry and higher costs of production, the output of each competing firm is reduced. Monopolistic competition leads to, a waste in the economic resources of the country. Long run equilibrium of the monopolistic firm is now explained with the help of Figure. In this figure the long run marginal cost curve (LMC) cuts the marginal revenue curve at point M. The demand curve is tangential to the ATC curve at point T. The equilibrium firm produces OK output and sells it at OE price. *a)* Since price equals LAC at point T, no firm is making positive profit. There is no incentive for the new firms to enter the industry nor for the existing firms to leave it. All firms are Fig. making normal profit, so the industry too is in the long run equilibrium each existing firm.

Unlike a perfectly competitive firm, a monopolistically competitive firm ends up choosing a level of output that is *below* its minimum efficient scale. When the firm produces below its minimum efficient scale, it is under-utilizing its available resources. In this situation, the firm is said to have **excess capacity** because it can easily accommodate an increase in production. This excess capacity is the major social cost of a monopolistically competitive market structure.

Price taker assumption has certain advantage because a competitive producer can produce as much as the market produces. One of the advantages is that atomistic producer does not bother about product. Under monopolistic competition, when entry take place and competition intensify then every producer has to devise their own method to deal with competition. Some of them used advertising where as other use product differentiation and sales promotion technique because of which firstly, AC goes on increasing and secondly as entry took place demand curve become elastic. When demand curve becomes elastic then producer cannot change price so much.

On the one hand price cannot increase and other hand cost increases. So those producers who succeeded in product differentiation, branding and sales promotion etc are benefited because all these additional costs are effective in generating more demand. Therefore, their demand curve expands. As their demand curve expands there is greater possibility of production and profit increases but at the same time there is inefficient producer who does not cope with increases in cost and competition. So in long run there is influx of producer those who can make profit and out flux of those who have burned their finger. Since entry and exist condition make this possible. So, in monopolistic competition in the long run, market is not in a state of stable equilibrium.



Above figure shows different possible equilibrium point for industry operating in monopolistic competition, due to expansion of demand and supply curve. However due to different possible equilibrium position, it is quite feasible for a firm to earn super normal profit and incur losses in long run but it can be sustained in longevity. As, above diagram shows different possible equilibrium point for industry operating in monopolistic competition, due to expansion and contraction of demand curve and supply curve. However due to different possible equilibrium position, it is quite feasible for a firm to earn abnormal profit and incurred losses even in the long run but it cannot sustain in overall long run. In long run, when in initial year some firm earn abnormal profit, it gives incentives to new firm to enter the industry and produce a differential product. This tendency continues until abnormal profit wiped out normal profits. In long run under monopolistic competition there is no stable equilibrium due to influx and out flux of firms.

Initially there are relatively fewer producers in the market. Therefore the supply is less (S1S1). Also, since the producers have not been able to influence demand through advertisement, selling costs and product differentiation, the demand curve is low

(D1D1). At the equilibrium point '1' sufficient expansion has not taken place in the market and the supply (MC) is high the price is OP1. Some of the producers become capable of exploiting their monopolistic tendency and are not able to raise demand

(D2D2). Thus equilibrium in the market happens to be at '2'. The prices rise to OP2. So far other producers from outside the industry have not realized the potential for profit (P2- P2). The moment they do there is a flux into the market, ushering in the outset of long run. Supplycurve shifts to the right (S3S3)., but simultaneously since there are many more producers the demand curve also falls. This is because not all producers are equally efficient in selling their product. The new equally efficient in selling their product. The new demand has shifted back to D1D1. Ordinarily equilibrium would take place at '3'. Here the normalprofits obtain. This ideal situation continues under perfect competition,. But under monopolistic competition each new producer thinks that he is better than the existing ones. So he continues to stick it out. This way there are too many producers. The supply curve slides down to S4S4. The new equilibrium is now at '4'. Here the firm make loses (P2'-P2). Finally some producers leave and the supply function rise again to S1S1 and price rise to OP1.

CASE STUDIES

Case Study 1: Monopoly in the Indian Railways

The Indian Railways operates as a government-owned enterprise and is the sole provider of railway transport for passengers and goods across most of the country. As there are no close substitutes for long-distance rail travel at comparable prices, Indian Railways represents a classic example of a monopoly. The firm sets ticket prices based on government policies and demand elasticity, not competition. In the short run, Indian Railways may incur losses or generate supernormal profits depending on operating costs and subsidies. In the long run, due to government support, the monopoly persists and adjusts its operations to maintain service efficiency without the threat of market entry.

Case Study 2: Monopolistic Competition in the Fast-Food Industry

India's fast-food sector is a textbook case of monopolistic competition. Brands like McDonald's, KFC, and local outlets like Haldiram's offer differentiated products targeting similar customer needs. While all sell food, product variations (e.g., taste, ambiance, menu diversity, branding) provide firms

with a degree of price control. Despite advertising and promotions, new firms can enter the market, causing shifts in demand and profitability. In the short run, firms may earn supernormal profits, but in the long run, profits normalize due to competition, resulting in excess capacity and increased promotional expenditures.

16.10 LET US SUM UP

The P-MC difference is called degree of monopoly. Perfect competition

- Price –output Determination in the short period can be seen in three situations:- ○ Supernormal profits ○ Normal profits ○ Minimum loss
- Imperfect competition refers to those market structures that fall between perfect competition and pure monopoly. Markets that have some features of competition and some features of monopoly. One of the intermediate market structures is monopolistic competition.
- Monopolistic competition refers to a market in which there are many firms and each sells a single differentiate product. Since each firm's is somewhat different from those of competitor, each faces a negatively sloped demand curve for its product.
- Product Differentiation ○ Each firm produces a product that is at least *slightly different* from those of other firms.

  ○ Rather than being a price taker, each firm faces a *downward-sloping demand curve*.
- As heterogeneous products cannot be added to form the market demand and supply schedules as in the case of homogeneous products. Therefore the concept of the industry needs redefinition. Chamberlin uses the concept of 'Product Group', which includes products which are 'Closely Related'. The products should be close *technological and economical substitutes*.
- The goal of a monopolistic firm is to maximize profit. Profit maximization occurs at the quantity where *marginal revenue equals marginal cost*. In the short run, an individual firm may either earn super normal profit or normal profit or incur losses.

  This depends on position of short run cost curves.
- In the long run perfect competitive firm only earn normal profit. This is due to unrestricted entry into and exit of firms from the industry in the long run.
- If the firms are earning abnormal profits in the short run, then new firm will enter the 'product group'(industry). The tendency of the new firms to enter the industry continues till the abnormal profits are competed away and the firms economic profits are zero. Similarly if monopolistically

competitive firms realize losses in the short-run, then some of the firms will leave the industry. The exit of the firm continues till zero economic profits are restored with the operating firms.

- Unlike a perfectly competitive firm, a monopolistically competitive firm ends up choosing a level of output that is *below* its minimum efficient scale. In this situation, the firm is said to have **excess capacity.** This excess capacity is the major social cost of a monopolistically competitive market structure.

KEYWORDS

1. **Monopoly** – A market structure where a single seller controls the entire supply of a product with no close substitutes.
2. **Monopolistic Competition** – A type of imperfect competition where many firms sell similar but not identical products, allowing for brand differentiation.
3. **Product Differentiation** – The process of distinguishing a product from others in the market through features, branding, or quality.
4. **Degree of Monopoly Power** – The extent to which a firm can raise prices above marginal cost, often measured using price elasticity of demand.
5. **Excess Capacity** – A situation in monopolistic competition where firms produce below the level that minimizes average total cost, leading to resource underutilization.

16.11 ASSIGNMENT QUESTIONS

1. Define monopoly and explain the key conditions under which it exists.
2. Differentiate between short-run and long-run equilibrium in monopolistic competition.
3. Analyze the role of product differentiation in influencing the pricing power of firms.
4. Explain the relationship between price elasticity and monopoly power using relevant formulas.
5. Discuss the impact of free entry and exit on long-term profits in monopolistic competition.

OBJECTIVE TYPE QUESTIONS

1. A _____ exists when there is a single seller in the market with no close substitutes. (Monopoly)
2. In monopolistic competition, products are similar but not _____. (identical)

3. The demand curve under monopoly is _____ sloping. (downward)
4. The gap between price and marginal cost under monopoly is called _____. (degree of monopoly)
5. AR stands for _____. (Average Revenue)
6. MR stands for _____. (Marginal Revenue)
7. In the long run, monopolistic firms earn only _____ profits. (normal)
8. Under perfect competition, the demand curve is perfectly _____. (elastic)
9. The _____ cost curve shows the cost per unit in the short run. (Average)
10. In monopolistic competition, the number of firms is _____ and each firm sells a differentiated product. (large)
11. Product differentiation leads to _____ loyalty. (brand)
12. In the short run, a monopolist can earn _____ profits or incur losses. (supernormal)
13. Under monopolistic competition, each firm faces a _____ demand curve. (negatively sloped)
14. Excess capacity leads to underutilization of _____. (resources)
15. Free entry and exit of firms result in _____ economic profits in the long run. (zero)

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FURTHER READINGS

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- Managerial Economics – D.N. Dwivedi
- Principles of Economics – N. Gregory Mankiw
- Theory of Industrial Organization – Jean Tirole

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ODISHA STATE OPEN UNIVERSITY

SAMBALPUR

ADDRESS

ODISHA STATE OPEN UNIVERSITY
At- Badsinghari, Po- Goshala Chowk,
Dist- Sambalpur, Pin-768024, Odisha, India
EMAIL: info@osou.ac.in

TOLL FREE NUMBER: 1800-34567-58